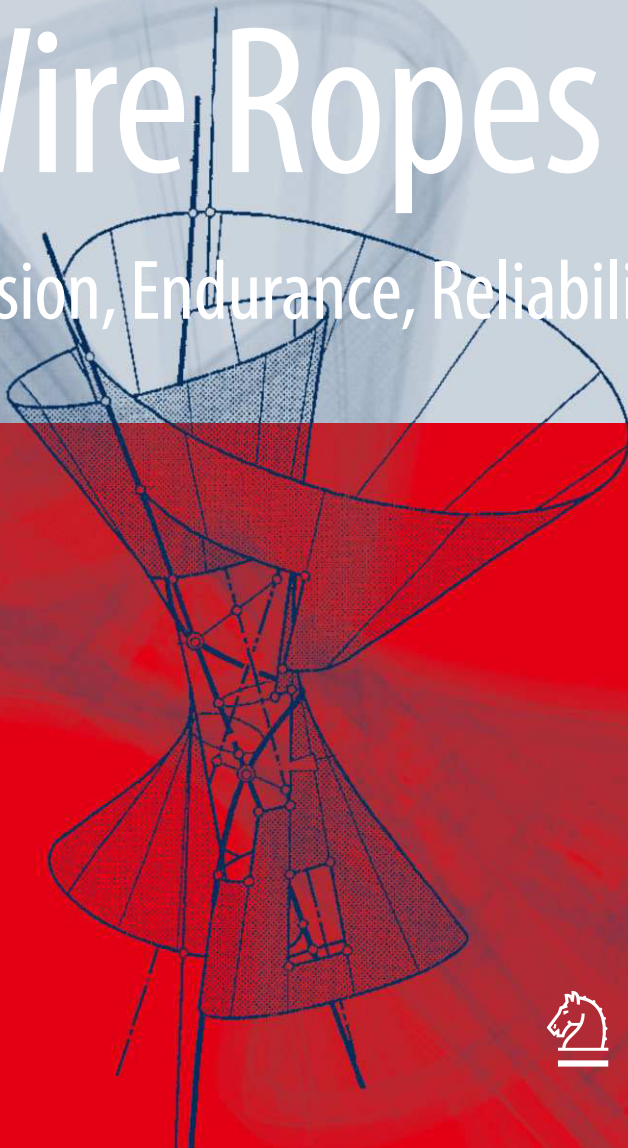


Klaus Feyrer

Wire Ropes

Tension, Endurance, Reliability



Springer

Wire Ropes

K. Feyrer

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Tension, Endurance, Reliability

With 171 Figures and 49 Tables

 Springer

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Preface

This book on wire ropes is dedicated mainly to all users of wire ropes – the construction engineers, operators and supervisors of machines and installations running on wire ropes – and it is divided into three main sections. The first section deals with the different types of wire rope and their component parts, the second looks into the effects of wire ropes under tensile forces (stationary ropes), while the third takes a look at the wire ropes under bending and tensile force (running ropes).

In addition to deriving the various tensile, bending and twisting stresses, the compression and the extensions, the book includes the findings and descriptions of a great number of detailed experiments, in particular those concerning rope endurance under fluctuating tension and bending and rope discard criteria. As far as possible, the test findings have been evaluated statistically so that not only the mean value but also the scattering is apparent.

It has been the main concern of this book to present the methods used to calculate the most important rope quantities (rope geometry, wire stresses in the rope under tension, bending and twist, rope elasticity module, rope torque, rope efficiency, the bearable number of load cycles or bending cycles and the discard number of wire breaks, etc.) as well as to explain how they are applied by means of a large number of calculations as examples.

Essentially, this book is a translation of the first three chapters of the book *Drahtseile*, which was also published by Springer-Verlag in the year 2000. The translation has been supplemented in many places by more recent findings and examples of calculations. My translation has been revised and polished by Mrs. Merryl Zepf, who has a good understanding of the project. I am extremely grateful for all her efforts.

It would not have been possible to create this book without the support and encouragement of the head of Stuttgart University's Institut für Fördertechnik und Logistik (Institute for Mechanical Conveying and Handling and Logistics), Professor Dr.-Ing. K.-H. Wehking. He welcomed the opportunity to present in English the diverse findings originating from almost 80 years of rope research at the institute combined with more general knowledge about

wire ropes. I am extremely grateful to Professor Wehking for his advice and support and for being able to use the infrastructure of the institute.

There have also been many enlightening discussions held with Prof.em. Dr. techn. Prof. E.h. Franz Beisteiner, the former head of the institute. I would like to thank him very much indeed for his constant willingness to have a discussion and for his sound advice that helped to clarify many a point in question. Also, I would also like to thank all members of staff at the institute for their readiness to discuss details and especially for their willingness to help in solving promptly any computer problems that arose. Thanks also go to Dr. Christoph Baumann at Springer Verlag for his pleasant cooperation.

Putting together this compilation of what we know about wire ropes today – even though there are probably some gaps – was certainly made easier by the OIPEEC (Organisation pour L’Étude de L’Endurance des Câbles). During the past few decades, the OIPEEC has developed into the most important forum for discussing questions in connection with wire ropes. I am very grateful to my colleagues at the OIPEEC for their very stimulating discussions. The same is true for the members of the DRAHTSEIL-VEREINIGUNG e.V. (Wire Rope Association, Germany). Furthermore, I would also like to thank those wire rope manufacturers who have been interested in and supportive of wire rope research from the very beginning and have been responsible for a great deal of technical information and practical assistance.

Even though extreme care is always taken, it is hardly possible to print a book that has absolutely no errors. This is true for this book as well. Because of this, I would like to point out that a list has been created where any printing errors or inaccuracies can be entered. The latest version of this list of corrections can be found in the internet under:

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To make the list of corrections as comprehensive as possible, I would like to ask all readers for their assistance to report any mistakes found to the following address:

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Wire Ropes, Elements and Definitions

1.1 Steel Wire

The very high strength of the rope wires enables wire ropes to support large tensile forces and to run over sheaves with relative small diameters. Very high-strength steel wires had already been existence for more than a hundred years when patenting – a special heating process – was introduced and the drawing process perfected. Since then further improvements have only occurred in relatively small steps.

There are a number of books about the history of wire ropes and wire rope production beginning with its invention by Oberbergrat Wilhelm August Julius Albert in 1834 and one of these is by Benoit (1935). Newer interesting contributions on the history of wire ropes have been written by Verreet (1988) and Sayenga (1997, 2003).

A voluminous literature exists dealing with the manufacture, material and properties of rope wires. In the following, only the important facts will be presented, especially those that are important for using the wires in wire ropes.

1.1.1 Non-Alloy Steel

Steel wires for wire ropes are normally made of high-strength non-alloy carbon steel. The steel rods from which the wires are drawn or cold-rolled are listed in Table 1.1 as an excerpt of a great number of different steels from the European Standard EN 10016-2. The rods for rope wires have a high carbon content of 0.4–0.95%.

The number in the name of the steel gives the mean content of carbon in weight percent multiplied with the factor 100. For example, the steel name C 82 D means that the steel has a mean carbon content of 0.82%. Steels with high carbon content close to 0.86% with eutectoid fine perlite – a mix of cementite (Fe_3C) and ferrite – are preferred for rope wires.

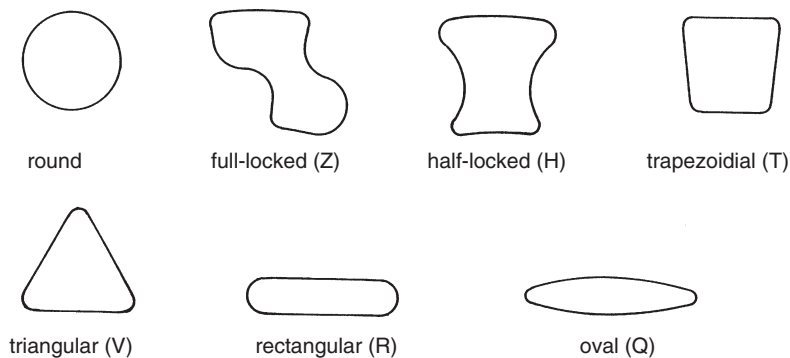
Table 1.1. Non-alloy steel rod for drawing (excerpt of EN 10 016-2)

Steel name	steel number	Heat analysis carbon content (%)
C 42 D	1.0541	0.40–0.45
C 48 D	1.0517	0.45–0.50
C 50 D	1.0586	0.48–0.53
.....		
C 82 D	1.0626	0.80–0.85
C 86 D	1.0616	0.83–0.88
C 88 D	1.0628	0.85–0.90
C 92 D	1.0618	0.90–0.95

Carbon steels only contain small quantities of other elements. EN 10016-2 gives the following limits for the chemical ingredients of carbon steel rods used for rope wires: Si 0.1–0.3%, Mn 0.5–0.8%, P and S <0.035%, Cr <0.15%, Ni <0.20%, Mo <0.05%, Cu <0.25% and Al <0.01%. The strength increases with an increasing carbon content and the breaking extension decreases if all other influences are constant. Higher contents of sulphur S, phosphorus P, chrome Cr and copper Cu reduce the steel's ductility, Schneider (1973).

Usually, wires for wire ropes have a round cross-section. In special cases, however, wires with other cross-sections – called profile wires – are used. The different cross-sections are to be seen in Fig. 1.1. The profile wires in the upper row are inserted in locked coil ropes. The wires below are used for triangular and oval strands.

In wires with a high carbon content which had been aged artificially, Unterberg (1967) and Apel and Nünninghoff (1983) found a distinct decrease in the breaking extension and the number of turns from the torsion test. The

**Fig. 1.1.** Wire cross-sections for wire ropes

number of test bendings is slightly reduced and the strength slightly increased. The finite life fatigue strength is partly increased or decreased.

Bending tests were repeated with three wire ropes after they had been in storage for 22 years. The original tests were well documented and the new tests were done in the same way with the usual lubrication. There was virtually no difference in the rope bending endurance documented for the original tests and the new tests. For two of these wire ropes, the mean strength of the wires was reduced during the long period of storage by a maximum of 3%. For one rope, the mean strength of the wires increased by 2.7%.

1.1.2 Wire Manufacturing

After the rod has been patented in a continuous system, the wire diameter is reduced in stages by cold drawing or cold rolling, rolling especially for profile wires. Patenting is a heating process. First the wire is heated in an austenising furnace at about 900°C . Then the temperature is abruptly reduced to about 500°C when the wire is put through a lead bath. After remaining there for a while, the wire then leaves the bath and enters the normal temperature of the surroundings. Figure 1.2 shows the course of the temperature during the patenting process. In recent times, the patenting process has partly been replaced by cooling in several stages while drawing or rolling the rod, Marcol (1986).

By patenting, the steel rod gets a sorbite structure (fine stripes of cementite and ferrite) which is very suitable for drawing. In the following drawing process, the wire cross-section is reduced in stages, for example in seven stages from 6 to 2 mm in diameter. After the wires have been patented, they can be drawn again. The quality of the wire surface can be improved by draw-peeling the wire rod, Kieselstein (2005).

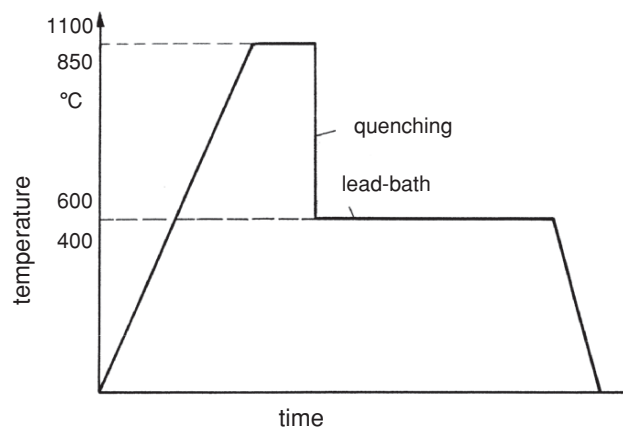


Fig. 1.2. The course of the temperature in the patenting process

The principle of the wire drawing was described at an early date by Siebel (1959). The strength increases with the growing decrease of the cross-section by drawing and at the same time the breaking extension also decreases. The higher the carbon content of the wires, the stronger they are. For wires with small diameters below 0.8 mm, the strength can reach about 4,000 N/mm², for thicker wires about 2,500 N/mm², and in all cases the remaining ductility is low. The standardised nominal strengths of rope wires are

- $R_0 = 1,370 \text{ N/mm}^2$ (in special cases)
- $R_0 = 1,570 \text{ N/mm}^2$
- $R_0 = 1,770 \text{ N/mm}^2$
- $R_0 = 1,960 \text{ N/mm}^2$
- $R_0 = 2,160 \text{ N/mm}^2$ (with a smaller wire diameter)
- $R_0 = 2,450 \text{ N/mm}^2$ (with a smaller wire diameter).

The nominal strength is the minimum strength. The deviation allowed above the nominal strength is about 300 N/mm². However, the real deviation is usually much smaller.

1.1.3 Metallic Coating

Rope wires needing to be protected against corrosion are normally zinc coated. Zinc coating provides reliable protection against corrosion. Even if the zinc layer is partly damaged, the steel remains protected as the electro-chemical process results in the zinc corroding first. With zinc, the wires can be coated by hot zincing or a galvanizing process. With hot zincing, the outer layer consists of pure zinc. Between this layer and the steel wire there is a boundary layer of steel and zinc compounds. With zinc galvanized wires, the whole layer of the coating, which can be relatively thick, consists of pure zinc and has a smooth surface.

In most cases the wires are covered by hot zincing. The layer of FeZn-compounds should be avoided or at least kept thin as they are relatively brittle which can lead to cracks when the wire is bent. To keep the FeZn layer thin, the wires should only be left in the zinc bath (with a temperature 440–460°C) for a short time.

During the hot zincing, the strength of the wires is somewhat reduced Wyss (1956). Because of this, and also because of the rough surface resulting from the zincing, the wires are often drawn again. This process increases the strength of the wire again and the zinc surface is smoothed. Before drawing, the zinc layer should be thicker than required as part of the zinc layer will be lost during the drawing process.

Blanpain (1964) found that during the re-drawing the brittle Fe–Zn layer may tear especially if the Fe–Zn layer is relatively thick. The resulting gaps will be entered from inside by a steel arch and are not visible from outside as they are closed with zinc. The fatigue strength of these wires is reduced due to the sharp edges of the gaps.

As an alternative to zinc, the wires can be coated with galfan, an eutectoid zinc–aluminium alloy Zn95Al5 (95% zinc, 5% aluminium). Nünninghoff and Szczepanski (1987) and Nünninghoff (2003) found that this Zn–Al alloy offers better protection against corrosion than pure zinc. The Zn95Al5 coating also has the further advantage that the brittle Fe–Zn-layer is avoided. However, the Zn95Al5-layer is not as resistant to wear as the pure zinc layer which means that Zn95Al5-coated wires are not as suitable for running ropes.

In Table 1.2, the surface-related mass of zinc coating is listed as an excerpt of Table 1.1 of EN 10244-2 in different classes. For a very thick coating, a multiple of class A can be used, as for example $A \times 3$. A surface-related zinc mass of 100 g/m^2 means that the thickness of the zinc layer is about 0.015 mm . For the Zn95Al5 coating, EN 10244-2 provides nearly the same surface-related mass for the classes A, B and AB. Unlike EN 10244-2, in Table 1.2 and in the following the symbol δ is used for the diameter of the wire.

1.1.4 Corrosion Resistant Wires

In exceptional cases corrosion resistant wires (stainless steel) have been used as rope wires. Some corrosion resistant steels for wires are listed in Table 1.3 from prEN 10088-3: 2001. The steel names of these high alloy steels begin with the capital letter X. The following number gives the carbon content in % multiplied with the factor 100. Then the symbols and the contents in % of the alloy elements are given. For example, for the steel X5CrNiMo17-12-2 the contents are 0.05% carbon, 17% chromium, 12% nickel and 2% molybdenum.

Table 1.2. Surface-related mass of zinc coating (excerpt of EN 10 244-2)

Wire diameter (mm)	class				
	A (g/m^2)	AB (g/m^2)	B (g/m^2)	C (g/m^2)	D (g/m^2)
$0.20 \leq \delta < 0.25$	30	20	20	20	15
$0.50 \leq \delta < 0.60$	100	70	50	35	20
$1.00 \leq \delta < 1.20$	165	115	80	60	25
$1.85 \leq \delta < 2.15$	215	155	115	80	40
$2.8 \leq \delta < 3.2$	255	195	135	100	50
$4.4 \leq \delta < 5.2$	280	220	150	110	70
$5.2 \leq \delta < 8.2$	290	–	–	110	80

Table 1.3. Strength of drawn wires out of corrosion resistant steel (excerpt of prEN 10 088-3:2001, Table 1.8)

Steel name	steel number	strength range (N/mm^2)
X10CrNi18-8	1.4310	600–800
X5CrNiMo17-12-2	1.4401	900–1100
X3CrNiMo17-13-3	1.4436	1,000–1,250
X1CrNiMoCuN20-18-7	1.4547	1,400–1,700
X1CrNi25-21	1.4335	1,600–1,900

Corrosion resistant wires for ropes have an austenite structure. Because of this structure they cannot be magnetized which means that the highly effective magnetic method of testing cannot be used to inspect the ropes. It should also be taken into consideration that these steels are not corrosion resistant in all environments.

Like non-alloy carbon steel wires, corrosion resistant steel wires are produced by drawing. The strength range stated in Table 1.3 is valid for drawn wires with a diameter ≥ 0.05 mm. From the different corrosion resistant steels available usually those of medium strength are used. Corrosion resistant wire ropes running over sheaves are not usually as durable as those made of non-alloy carbon steel.

1.1.5 Wire Tensile Test

The tensile test is standardised according to EN 10002-1. The main results of the tensile test provide the measured tensile strength R_m and the total extension ε_t . It is not possible to detect precisely the limit where the yielding of the wire begins. However, the yield strength is defined for a small residual extension. Here, the most frequently used extension is $\varepsilon = 0.2\%$ and the stress at this point is the yield strength $R_{p0.2}$. The elasticity module can be evaluated with a special tensile test.

If only the tensile strength R_m has to be evaluated, it can be done without straightening the wire. However, if the different extensions and the yield strength have to be evaluated too, the wire has to be straightened prior to testing. The measurement starts at a stress of about 10% of the tensile strength R_m . Under this stress, the height of the wire bow at a distance measured of 100 mm should be smaller than 0.5 mm.

A typical stress-extension diagram of a straightened wire is shown in Fig. 1.3. It is possible to take the tensile strength R_m , the total extension ε_t and the residual extension ε_r directly from this figure. To determine the elasticity module E and the yield strength $R_{p0.2}$, the following method has to be used. After a certain yielding, the wire has to be unloaded and loaded again. As a result, a hysteresis loop occurs as seen in Fig. 1.4. A middle line of this hysteresis defines the elasticity module $E = \Delta\sigma/\Delta\varepsilon$. To evaluate the yield strength $R_{p0.2}$, a parallel to the middle line of the hysteresis has to be drawn through the residual extension $\varepsilon_r = 0.2\%$ on the abscissa. Then the yield strength $R_{p0.2}$ is found as an ordinate where the parallel meets the stress extension line.

To determine stresses, strengths and elasticity modules, the cross-section A of the unloaded wire has to be measured very precisely. (Unlike EN 10002-1, the symbol A is used here for the cross-section.) The error in measurement of the cross-section should be 1% at the most. For round wires the cross-section

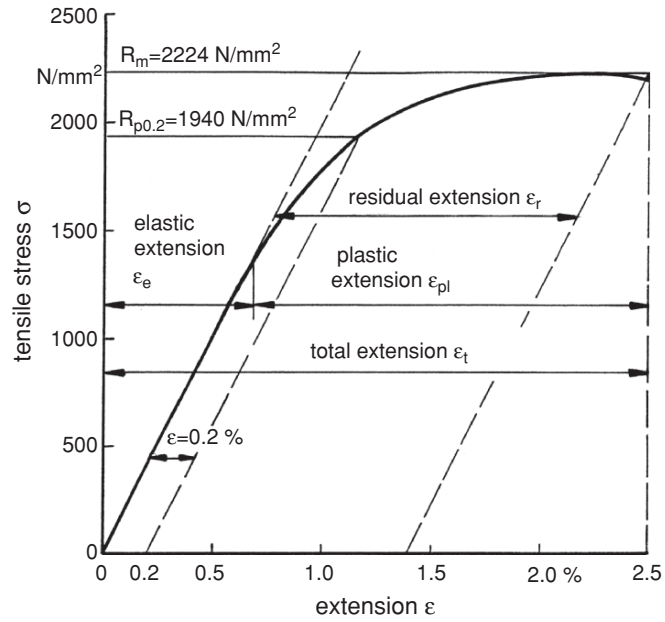


Fig. 1.3. Stress extension diagram of a straightened wire, $\delta = 1.06 \text{ mm}$

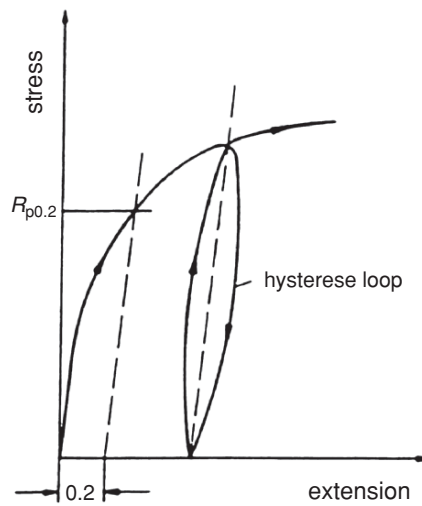


Fig. 1.4. Evaluation of the yield strength $R_{p0.2}$ according to EN 10002-1

has to be calculated from two wire diameters δ measured perpendicular to each other.

To fulfil this accuracy requirement for the cross-section, the wire diameter δ should be measured with a maximum deviation of 0.5%. With commonly used measuring instruments this accuracy requirement can only be achieved for thicker wires. For thin wires and for profile wires, the cross-section can be evaluated by weighing. With the wire weight m in g, the wire length l in mm and the density ρ , the cross-section is then

$$A = \frac{m}{l \cdot \rho}. \quad (1.1)$$

The density for steel is normally $\rho = 0.00785 \text{ g/mm}^3$. However, because of the great carbon content of wires used for wire ropes it is to use $\rho = 0.00780 \text{ g/mm}^3$.

The total extension of steel wires for ropes amounts to about $\varepsilon_t = 1.5\text{--}4\%$ and the yield strength $R_{p0.2}$ is about 75–95% of the measured tensile strength R_m . For wires taken out of ropes and straightened, the total extension is about $\varepsilon_t = 1.4\text{--}2.9\%$ and the yield strength $R_{p0.2}$ is about 85–99% of the tensile strength R_m , Schneider (1973).

Because of early yielding in parts of the cross-section of the non-straightened wires and even if the wires are straightened in the normal way, the elasticity module can only be evaluated precisely enough if the wire has yielded before over the whole cross-section. However, that does not mean that the two parts of a broken wire resulting from a tensile test can be used for the evaluation of the elasticity module. These wire parts cannot be used because they may have new inherent stresses due to buckling from the wire breaking impact, Unterberg (1967).

For straightened wires from wire ropes, Wolf (1987) evaluated a mean elasticity module $E = 199,000 \text{ N/mm}^2$. For new wires, Häberle (1995) found the mean elasticity module $E = 195,000 \text{ N/mm}^2$. Together with other measurements – after loading the wires close to the breaking point – a mean elasticity module has been evaluated for the stress field of practical usage. This mean elasticity module of rope wires made of carbon steel in the following is $E = 196,000 \text{ N/mm}^2$.

The elasticity module decreases a little with larger upper stresses. For drawn corrosion resistant wires with the steel number 1.4310 and 1.4401, Schmidt and Dietrich (1982) evaluated the elasticity module $E = 160,000 \text{ N/mm}^2$, respectively, $E = 150,000 \text{ N/mm}^2$.

1.1.6 Wire Endurance and Fatigue Strength

Test Methods, Definitions

The wires in wire ropes are stressed by fluctuating tension, bending, pressure and torsion. For a long time wires have been tested in different testing

machines under one or a combination of these fluctuating stresses. The tests with combined stresses, especially bending and pressure, have been done with the aim to imitate the stresses in a wire rope. Such tests have been done by Pfister (1964); Lutz (1972), Pantucek (1977) and Haid (1983). However the test results do not come up to expectations, or only imperfectly. The wire endurance for example has even been increased when the wires – loaded by fluctuating bending – are loaded in addition by fluctuating pressure. This effect can probably be attributed to a strain hardening of the wire surface. An overview of the test methods with single or combined stresses has been described by Wolf (1987).

Nowadays, wire fatigue tests are normally tests with only one fluctuating stress – mostly a longitudinal stress. The test methods with fluctuating longitudinal stresses are:

- Tensile fatigue test (wire under fluctuating tensile force)
- Simple bending test (fluctuated bending of the wire over one sheave)
- Reverse bending test (fluctuating bending of the wire over two sheaves or sheave segments)
- Rotary bending test (wire bending by rotating the wire).

For these test methods, the principle wire arrangement in the test machines is shown in Fig. 1.5. The fluctuating longitudinal stress affects different zones of the wire cross-sections. The wire cross-sections with the zones of the highest fluctuating longitudinal stress are shown in Fig. 1.5 below the wire arrangements. The highest stressed zones are shaded. The highest fluctuating longitudinal stress is taken as the nominal fluctuating stress.

For fatigue strength (infinite life), instead of the stress the symbols σ are written with indices as capital letters.

In Fig. 1.5, the stress amplitude σ_a and the middle stress σ_m are listed for general cases in fatigue tests. Below them the stresses are listed for the special cases alternate stress σ_{alt} and repetitive stress σ_{rep} . Figure 1.6 shows the stress course over one load cycle with the stress amplitude σ_a and the middle stress σ_m for general cases. In a fatigue test, the endurance of the wire is counted by the number of load cycles N it takes.

Figure 1.7 shows a Haigh-diagram with the abscissa for the constant middle stress σ_m and the ordinate for the fluctuating strength σ_A as amplitude around the middle stress σ_m . The special cases alternate and repetitive stresses are inserted. The alternate strength σ_{Alt} is the amplitude for a middle stress $\sigma_m = 0$. The repetitive strength $\sigma_{Rep} = 2\sigma_A$ is the strength range for a middle stress $\sigma_m = \sigma_A$. That means for the repetitive strength σ_{Rep} the lower stress is $\sigma_{lower} = 0$.

The two basic stresses are tensile stress

$$\sigma_t = S/A \quad (1.1a)$$

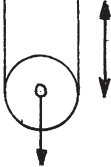
	tension	simple bending	reverse bending	rotary bending
wire arrangements for testing				
zone of maximum fluctuating stress				
stresses				
stress amplitude	$\sigma_a = \sigma_{t,a}$	$\sigma_a = \sigma_b / 2$	$\sigma_a = \sigma_b$	$\sigma_a = \sigma_b$
middle stress	$\sigma_m = \sigma_{t,m}$	$\sigma_m = \sigma_{t,m} + \sigma_b / 2$	$\sigma_m = \sigma_{t,m}$	$\sigma_m \approx \sigma_b$
certain fluctuating stresses				
alternate stress	$\sigma_{t,alt}$ with $\sigma_{t,m} = 0$	—	$\sigma_{b,alt}$ with $\sigma_{t,m} \approx 0$	σ_{rot} with $\sigma_{t,m} \approx 0$
repetitive stress	$\sigma_{t,rep}$ with $\sigma_{t,m} = \sigma_{t,a}$	$\sigma_{b,rep}$ with $\sigma_{t,m} \approx 0$	$\sigma_{b,rep}$ with $\sigma_{t,m} = \sigma_{b,rep} / 2$	—

Fig. 1.5. Wire arrangement for the fatigue tests, zone of maximum stress amplitude in the wire cross-section, stress amplitudes and middle stresses

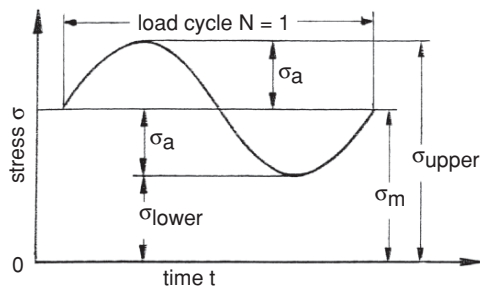


Fig. 1.6. Stress course during a load cycle

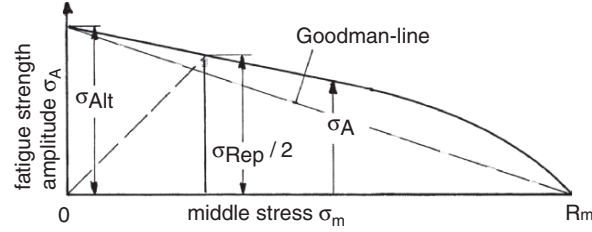


Fig. 1.7. Haigh-diagram with some definitions

and bending stress according Reuleaux (1861)

$$\sigma_b = \frac{\delta}{D} \cdot E. \quad (1.1b)$$

In these equations S is the tensile force, A the wire cross-section and δ the wire diameter. D is the curvature diameter of the wire centre on the sheave, which means $D = D_0 + \delta$, with the contact diameter D_0 between wire and sheave. E is the elasticity module.

Testing Machines

Tensile fatigue test. Testing methods with fluctuating tensile forces for the testing of materials and components are very commonly used. For rope wires, such tests were started as early as those from Pomp and Hempel (1937). The wire terminations are the main problem in carrying out these tests. If a normal press clamp is used, the wire would mostly break in the clamp.

In order to find out the real endurance or the real tensile fatigue strength of the wire, the wire has to be fastened in such a way that the wire breaks in the free length. To do this, a lamella clamp is used where the tensile force is gradually transferred from lamella to lamella. In addition the wire ends – which are fastened in the clamps – are strain-hardened by a rolling process. During this process, the fatigue tensile strength of the wire ends is increased slightly over that of the wire in the free length. This hardening of the wire ends together with the lamella clamp provides a high probability that the wire breaks in the free length.

In the fluctuating tensile tests the stress is constant over the whole cross-section as shown in Fig. 1.5. The total stress is composed of a constant middle stress σ_m and a stress amplitude σ_a ,

$$\sigma = \sigma_m \pm \sigma_a.$$

Because of the risk of buckling, the stress amplitude σ_a normally should be smaller than the middle stress σ_m . Tensile fatigue tests with a compressive section can only be done with very short wires.

Simple bending test, one sheave. In this method the wire moves over one sheave forwards and backwards, Woernle (1929), Donandt (1950) Müller (1961) etc. The wire is loaded by a constant tensile force S and a fluctuating bending. For this test,

$$\text{the middle stress is} \quad \sigma_m = \sigma_{t,m} + \sigma_b/2$$

$$\text{and the stress amplitude is} \quad \sigma_a = \sigma_b/2.$$

The fluctuating bending stress σ_b exists only in one small segment of the wire cross-section in the outside wire bow, as shown in Fig. 1.5. In the other small wire segment lying on the sheave, the bending stress is compressive. There is no wire breakage to be expected from this stress, especially if this stress is reduced – as is normal – by a tensile stress $\sigma_{t,m}$.

Reverse bending test, two sheaves. In this method the wire first moves over one sheave and then over a second sheave with reverse bending forwards and backwards, Schmidt (1964). In another method the wire is bent and reverse bent over sheave segments, Unterberg (1967). For both methods, the wire is loaded by a constant tensile force S and a fluctuating bending. For the wire reverse bending test

$$\text{the middle stress is} \quad \sigma_m = \sigma_{t,m}$$

$$\text{and the stress amplitude is} \quad \sigma_a = \sigma_b.$$

The fluctuating bending stress σ_b exists only in two small segments of the wire cross-section, as shown in Fig. 1.5.

The pressure between the wire and the single sheave or two sheaves is small and can be neglected. The advantage of this bending test method is that the tensile stress σ_m , and the bending stress can be chosen quite freely. There should be only a tensile force chosen that is large enough to ensure that the wire lies securely on the sheave in contact with the bow.

Rotary bending test. In a rotary bending machine, the wire is bent in a free bow around its own axis. By turning the wire, the stress in an outer fibre of the bent wire changes from compressive to tensile stress and back again. In one turn of the wire around the wire axis, each of the outside fibres of the wire is stressed by a complete cycle of longitudinal stress. The stress amplitude $\pm \sigma_a$ decreases linearly from the outside of the wire to the wire axis. Therefore – as shown in Fig. 1.5 – the maximum (nominal) stress exists only in a small ring zone. The advantage of rotary bending tests is that it can be done very quickly with a frequency of 50 and more turns/second.

Older bending machines which rotate the wire are the Haigh/Robertson machine, NN (1933), the Schenck machine, Erlinger (1942) and the Hunter testing machine, Votta (1948). These machines have the disadvantage that of the whole wire length only a small part is bent with the maximal (nominal) bending stress σ_a . The newer Stuttgart rotary bending machine, Fig. 1.8, avoids this disadvantage, Wolf (1987). In this machine the wire has almost the same bending stress for the whole bending length. The wire bow between the two parallel axes of the rotating wire terminations with the distance C is nearly a circular arc. One of the two wire terminations is driven.

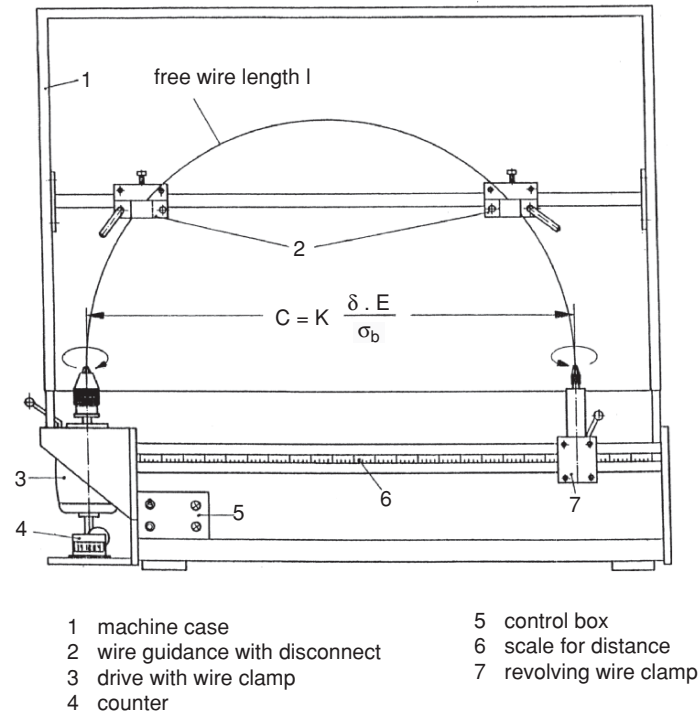


Fig. 1.8. Stuttgart rotary bending machine, Wolf (1987)

One slight disadvantage of both the Hunter testing machine and the Stuttgart testing machine is that the bending length is determined by the chosen bending stress. The great advantage of these two machines is, however, that the bending stress is determined by geometric dimensions only and these can be measured very simply.

For the Stuttgart rotary bending machine, the bending length (free wire length between the terminations) l is only slightly larger than the circle bow length $C\pi/2$. Therefore the bending stress on the terminations is only slightly smaller than in the middle of the bending length. This means that wire breakage in or close to the terminations is almost certainly avoided and the bending stress is nearly constant over the whole of the bending length. Figure 1.9 shows the bending stress along the bending length in the Schenck machine and the Stuttgart machine.

The bending stress amplitude in the middle of the bending length l (free wire length) is

$$\sigma_a = \sigma_b = \frac{k \cdot \delta \cdot E}{C} \quad (1.1c)$$

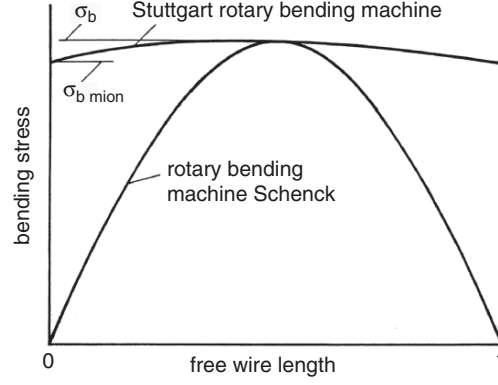


Fig. 1.9. Bending stress along the wire bending length, Wolf (1987)

Table 1.4. Constants k , k_0 and the ratio of wire bending stresses $\sigma_{b, \min}/\sigma_b$ in the Stuttgart rotary bending machine

l/C	$\pi/2$	1.58	1.59	1.60
k	1.0	1.008	1.017	1.026
k_0	1.0	0.969	0.936	0.906
$\sigma_{b, \min}/\sigma_b$	1.0	0.961	0.921	0.883

This bending stress, the maximum stress, is taken as the nominal bending stress of the wire in the Stuttgart rotary bending machine. The minimum bending stress on both of the wire terminations is

$$\sigma_{b, \min} = \frac{k_0 \cdot \delta \cdot E}{C}. \quad (1.1d)$$

For both equations: k and k_0 are constants in Table 1.4, δ is the wire diameter, E is the elasticity module and C is the distance between the parallel axes of the wire terminations.

Furthermore, in Table 1.4, the ratio of the minimum and the nominal bending stress $\sigma_{b, \min}/\sigma_b$ is listed. For his tests Wolf (1987) used the ratio $l/C = 1.6$ instead of $\pi/2$ with the minimum stress $\sigma_{b, \min} = 0.883 \cdot \sigma_b$ or a 11.7% smaller stress at the wire terminations. Later on, it was shown in a great number of tests that in practically all cases where the ratio is $l/C = 1.58$ the wires break in the free wire length. Since 1990, therefore, all the tests with the Stuttgart rotary bending machine have been done with the ratio $l/C = 1.58$. Because of the very small stress reduction of only 4% at the wire terminations, the maximum bending stress amplitude can be considered as stress over the whole bending length.

The middle stress is practically $\sigma_m \approx 0$. As an example for the ratio $l/C = 1.58$ and the bending stress $\sigma_b = 600 \text{ N/mm}^2$, the compressive stress is only $\sigma_m = 965 \cdot (\delta/C)^2 = 0.0089 \text{ N/mm}^2$.

Wöhler Diagram

Wolf (1987) did a great number of fatigue tests with the simple Stuttgart rotary bending machine using wires taken from wire ropes. He straightened all the wires in the same way with a special device before conducting the fatigue tests. Fig. 1.10 shows the numbers of bending cycles N resulting from a series of tests with wires of 1 mm diameter taken from Seale ropes for varying amplitude of the rotary bending stress (alternate bending stress on the whole circumference) $\sigma_{\text{rot}} \approx \sigma_{b,\text{alt}}$. For the logarithm normal distribution of the bending cycles N , the standard deviation increases in the usual way with decreasing bending stress amplitude σ_{rot} .

Wolf (1987) transferred this number of bending cycles to a Wöhler diagram as shown in Fig. 1.11. In the Wöhler diagram, he drew a line for the mean number of bending cycles and lines for 5 and 95% of the breaking probability. The mean rotary bending strength (infinite life fatigue strength) for wires in 12 Seale ropes is $\bar{\sigma}_{\text{Rot}} = \pm 640 \text{ N/mm}^2$. In the Wöhler diagram shown in Fig. 1.12, the number of rotary bending cycles for 0.95 mm diameter wires

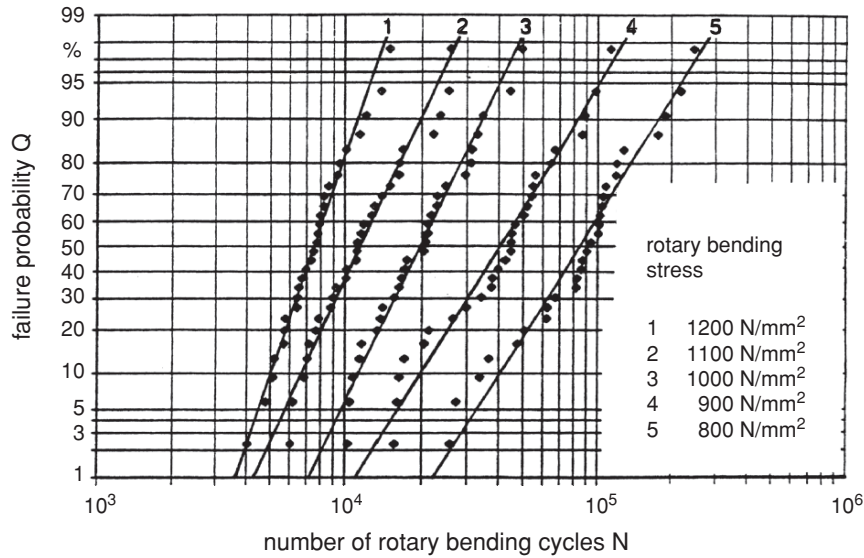


Fig. 1.10. Number of bending cycles for straightened wires, diameter $\delta = 1 \text{ mm}$ from Seale ropes, Wolf (1987)

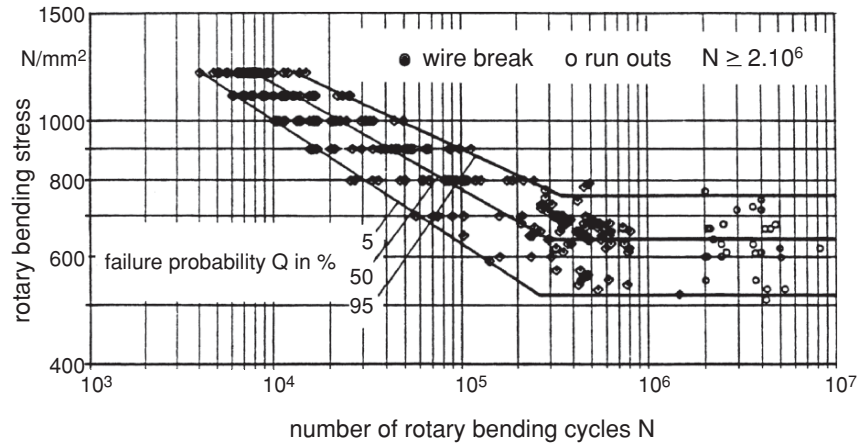


Fig. 1.11. Wöhler-diagram for wires, diameter $\delta = 1$ mm, from Seale ropes, Wolf (1987)

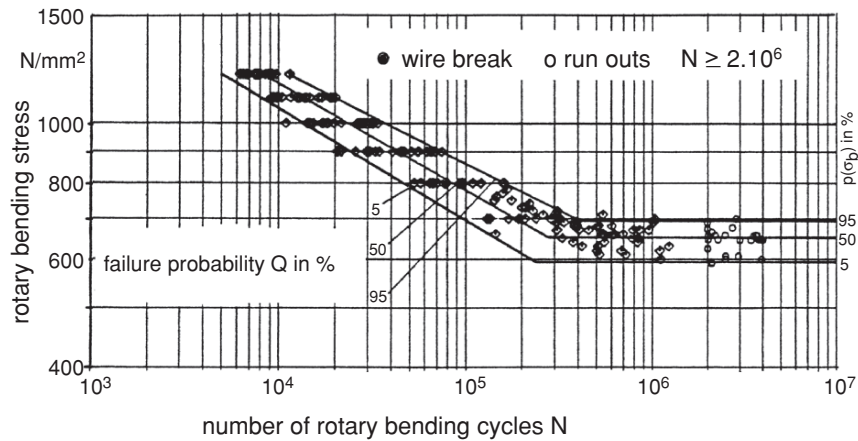


Fig. 1.12. Wöhler-diagram for wires, diameter $\delta = 0.95$ mm, from Warrington ropes, Wolf (1987)

taken from 20 Warrington ropes has been transferred in the same way. The mean rotary bending strength is $\bar{\sigma}_{\text{Rot}} = \pm 640 \text{ N/mm}^2$. The deviation for the number of rotary bending cycles N and for the rotary bending strength σ_{Rot} is much smaller than in the Wöhler-diagram in Fig. 1.11.

For both wires, the transition from the finite to the infinite life strength lies at the number of bending cycles of about $N = 300,000$. This is situated in the range between $N = 150,000$ and $N = 500,000$ that Hempel (1957) and Unterberg (1967) previously found in rotary bending and fluctuating tensile tests.

Finite Wire Endurance

For straightened wires taken from wire ropes, Wolf (1987) evaluated a mean number of rotary bending cycles for wires with diameter $\delta = 0.8\text{--}1.0\text{ mm}$

$$\lg \bar{N} = 21.708 - 5.813 \cdot \lg \sigma_{\text{rot}}. \quad (1.2a)$$

Briem (2000) Ziegler, Vogel and Wehking (2005) have also done a great number of fatigue tests with a Stuttgart rotary bending machine. In both series of fatigue tests the wires were new (not taken from a rope). They were only straightened before the tests. The following endurance equations were found by regression calculation using the test results in the finite life region: Briem (2000)

$$\lg \bar{N} = 13.74 - 3.243 \cdot \lg \sigma_{\text{rot}} - 0.30 \cdot \lg \delta - 0.74 \cdot \lg \frac{R_0}{1770} \quad (1.2b)$$

for wire diameters $\delta = 0.8 - 2.2\text{ mm}$ and

for nominal strength $R_0 = 1,770; 1,960; \text{ and } 2,160\text{ N/mm}^2$.

Ziegler, Vogel, Wehking (2005) found

$$\lg \bar{N} = 12.577 - 3.542 \cdot \lg \sigma_{\text{rot}} - 0.072 \cdot \lg \delta - 0.612 \cdot \lg R_m \quad (1.2c)$$

for wire diameters $\delta = 0.8\text{--}1.8\text{ mm}$ and

for nominal strength $R_0 = 1,370\text{--}2,160\text{ N/mm}^2$.

The influence of the diameter of the wire and its tensile strength is different in the two equations. Briem (2000) even found that the number of rotary bending cycles is reduced when the tensile strength is increased. Only the bending stress as a main influence may be considered as a common result because of the relatively small range of wire diameters and tensile strength tested. Thus, the mean number of rotary bending cycles for new wires with a diameter $\delta = 1\text{ mm}$ and tensile strength $R_0 = 1,770\text{ N/mm}^2$ is

$$\lg \bar{N} = 14.152 - 3.393 \cdot \lg \sigma_{\text{rot}}. \quad (1.2d)$$

According to these equations with rotary bending stress $\sigma_{\text{rot}} = \pm 900\text{ N/mm}^2$ as an example, the mean number of bending cycles for new wires is $\bar{N} = 13,000$, (1.2d) and for wires taken from ropes $\bar{N} = 34,000$, (1.2a). As an additional test result Briem (2000) found a 19% smaller endurance for zinc-coated wires than for bright wires.

The endurance of the wires depends on the size effects of the two parameters, the wire diameter and the wire length where the wire is stressed (bending length or stressing length). The wire diameter cannot change in isolation. That means, with the wire diameter, the other parameters which influence endurance will always also be changed. Therefore, to find out the influence of the wire diameter, the other parameters which influence wire endurance should be kept as similar as possible and there should be a wide range of different wire diameters. As already mentioned, the influence of the wire diameter on the wire fatigue endurance is only known in a first form using the given results represented by equations (1.2b) and (1.2c).

The influence on wire endurance of the wire length, which is the other parameter affecting the size, can be evaluated reliably by conducting tests with parts of one and the same wire and theoretically with the help of the reliability theory. A series of wire fatigue tests done by bending over one sheave have been used to evaluate the influence of the bending length, Feyrer (1981). The wire diameter is $\delta = 0.75$ mm, the measured tensile strength is $R_m = 1,701$ N/mm². The wire bending diameter over the sheave is 115.75 mm; with these conditions the wire bending stress is $\sigma_b = 1,270$ N/mm². The constant tensile stress from a loaded weight is $\sigma_{t,m} = 400$ N/mm². For the test bending over one sheave, the wire is loaded by

$$\begin{aligned} \text{the middle stress } \sigma_m &= \sigma_{t,m} + \sigma_b/2 = 1,035 \text{ N/mm}^2 \\ \text{and the stress amplitude } \sigma_a &= \sigma_b/2 = 635 \text{ N/mm}^2. \end{aligned}$$

The test results are shown in Fig. 1.13. Together with the points taken from the test results, the figure shows the curves calculated for the mean number of bending cycles $\bar{N}$ and the limiting number of bending cycles for 10 and 90% probability. The calculation of these curves is based on the reliability theory.

The survival probability is the smaller the larger the bending length l (as a string of bending lengths l_0) of the wire being considered is. For a given survival probability R_0 of the wire bending length l_0 , the survival probability $R(l)$ of the wire with the bending length l is

$$R(l) = R_0^{(l-\Delta l)/(l_0-\Delta l)}. \quad (1.2e)$$

The bending lengths l and l_0 are the theoretical lengths without considering the bending stiffness of the wire. These lengths would occur for bending limp yarn. For the wire near the sheave, the fluctuating bending stress is small. The short bending length Δl is introduced to take this into account. Δl is the shorter part of the bending length having the smaller radius difference of the rope curvature than 90% of the total one. The curves in Fig. 1.13 are calculated

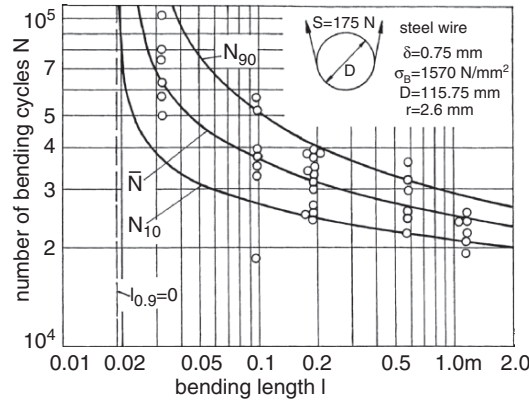


Fig. 1.13. Number of bending cycles of a wire with different bending lengths, Feyrer (1981)

using standard deviation $\lg s = 0.086$ of the logarithm normal distribution derived from the 13 bending cycles found for the wire bending length $l_0 = 2 \times 96$ mm. More information on this method of calculation is presented for wire ropes in Sect. 3.2.2 where the influence of the bending length is of practical interest.

In principle, the findings of Benjin Luo's (2002) tensile fatigue tests produced the same result. For his tests with different stressing lengths, he used a wire with diameter $\delta = 2$ mm, made of material X5CrNi18-10, No. 4301 and having tensile strength $R_m = 840$ N/mm². He did 60 tensile fatigue tests for each of the wire lengths $l = 25, 125$ and 250 mm with the middle stress $\sigma_{t,m} = 356.5$ N/mm² and the stress amplitude $\sigma_{t,a} = \pm 290$ N/mm² only a little above the infinite tensile fatigue strength. For the short wire length of 25 mm there are five run-outs with more than $N = 2 \times 10^6$. For the wire length $l = 250$ mm, the parameters for the logarithm normal distribution are the mean number of tensile cycles $\bar{N} = 238,000$ and the standard deviation $\lg s = 0.136$.

Infinite Wire Endurance

The fatigue strengths (infinite life fatigue strengths) have been evaluated using Wöhler-diagrams. As before, the fatigue strengths are characterized by indices in capital letters and the stresses by indices with small letters.

For his tensile fatigue tests, Unterberg (1967) used short wire pieces with a length between 15 and 35 mm so that he could start – without the risk of buckling – with the middle stress $\sigma_m = 0$. The test results are shown in Fig. 1.14. The mean relative tensile strength amplitude is

$$\bar{\sigma}_{t,A} (\bar{R}_0 = 1,770; \bar{\delta} = 2.7) = 0.313 \cdot R_m - 0.249 \cdot \sigma_{t,m} \quad (1.3)$$

and related to the lower stress $\sigma_{t,lower}$

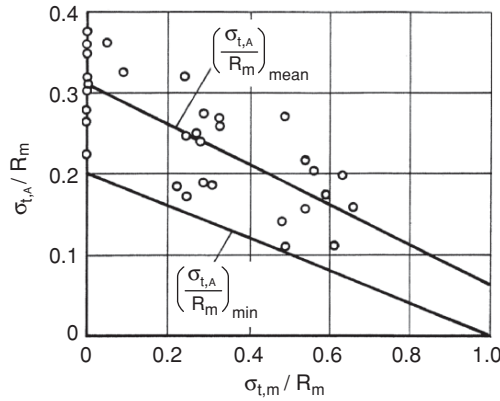


Fig. 1.14. Tensile strength amplitude for wires with diameters $\delta = 1.17\text{--}4.2$ mm and nominal tensile strength $R_0 = 1570\text{--}1960$ N/mm², Unterberg (1967)

$$\bar{\sigma}_{t,A} (\bar{R}_0 = 1,770; \bar{\delta} = 2.7) = 0.251 \cdot R_m - 0.199 \cdot \sigma_{t,lower}.$$

The standard deviation is large. As a lower limit for this tensile strength amplitude Unterberg gave the Goodman-line

$$\sigma_{t,A,min} (\bar{R}_0 = 1,770; \delta = 2.7) = 0.2 \cdot R_m - 0.2 \cdot \sigma_{t,m}.$$

From his reverse bending tests, Unterberg (1967) found the mean relative bending strength amplitude

$$\bar{\sigma}_{b,A} (\bar{R}_0 = 1,770; \delta = 2.7) = 0.271 \cdot R_m - 0.170 \cdot \sigma_{t,m}. \quad (1.3a)$$

The bending strength amplitude, (1.3a), is a little smaller than the tensile strength amplitude, (1.3). The deviation of the bending strength is also a little smaller than that of the tension strength to be seen in Fig. 1.14. The bending length $l = 80$ mm in relation to the smaller tensile stressing length $l = 15\text{--}35$ mm in case of tensile fatigue tests may be the reason for that. For the middle stress $\sigma_m = 0$ and the tensile strength $R_m = 1,770$ N/mm², the mean bending strength amplitude (alternate bending amplitude) is $\bar{\sigma}_{b,Alt} = 0.271 \cdot R_m = 480$ N/mm² and the mean tensile strength amplitude is $\bar{\sigma}_{t,Alt} = 0.313 \cdot R_m = 554$ N/mm².

In any case, if the theory of stress gradient effect is valid, there should be an advantage for the bending strength. However there is no such advantage to be found in the test results. According to the theory of stress gradient effect, Faulhaber (1933), Hempel (1957) and Siebel (1959), the fatigue strength of the wire should be the greater, the greater the stress gradient in the wire cross-section is. The theory of stress gradient means that if the stress gradient is large, the outer highly stressed lay can be supported by the less stressed layer below. However the bending strength amplitude according to (1.3a) is not at all greater than the tensile strength amplitude according to (1.3) although there the stress gradient is 0. Because of this and also as a result of other observations, Unterberg (1967) stated that the theory of stress gradient does not exist for rope wires.

In Fig. 1.15 the influence of the wire diameter δ is shown using the results from different authors. In this figure the mean repetitive fatigue strength $\bar{\sigma}_{t,Rep} = 2 \cdot \bar{\sigma}_{t,A}$ is used. As a reminder, repetitive strength means that the middle stress is $\sigma_{t,m} = \sigma_{t,A}$ and the lower stress is 0. As an equation using the results in Fig. 1.15, the repetitive strength is expressed as

$$\sigma_{t,Rep} = 2 \cdot \sigma_{t,A} (\sigma_{lower} = 0) = 1,200 \cdot e^{-0.122 \cdot \delta}. \quad (1.3b)$$

The influence of the other size parameter, the stressed wire length l , can be evaluated for the fatigue strength amplitude in the same way as for a number of load cycles N if the standard deviation of the fatigue strength amplitude for one and the same wire were known.

The influence of the tensile strength R_m on the rotary bending strength σ_{Rot} is shown in Fig. 1.16 from Wolf (1987). In this diagram, Wolf put in the results gained by Buchholz (1965) wires with lower tensile strength to get an

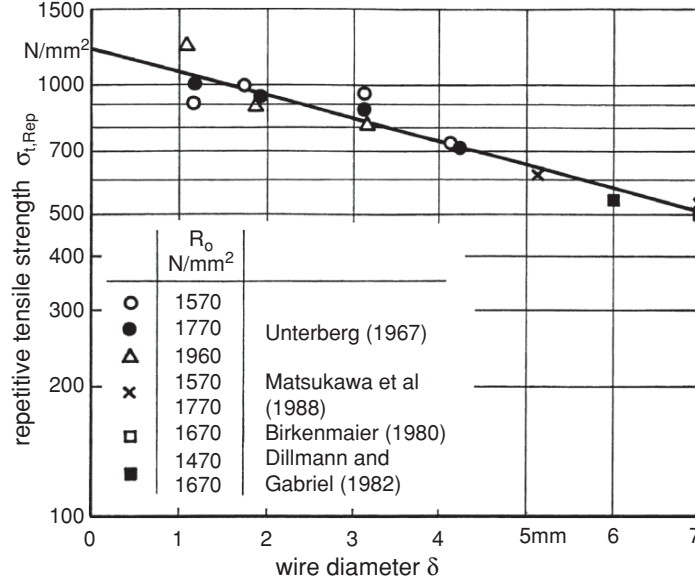


Fig. 1.15. Repetitive tensile strength for different wire diameters δ for a mean nominal tensile strength $\bar{R}_0 = 1,720 \text{ N/mm}^2$

overview for a greater strength range. For small tensile strengths, the rotary bending strength increases almost proportionally with tensile strength R_m . The rotary bending strength does not increase as much for rope wires (wires with tensile strength between 1,300 and 2,200 N/mm²). According to Wolf (1987), it is

$$\sigma_{\text{Rot}} = 0.334 + 0.173 \cdot R_m. \quad (1.3c)$$

That means that with increasing tensile strength R_m , the relative rotary bending strength σ_{Rot}/R_m will be reduced as seen in Fig. 1.17, Wolf (1987).

Ziegler, Vogel and Wehking (2005) and Wehking (2005) evaluated the rotary bending strength from tests with the Stuttgart rotary bending machine to be

$$\lg \sigma_{\text{Rot}} = 1.411 + 0.396 \cdot \lg R_m - 0.128 \cdot \lg \delta. \quad (1.3d)$$

According to (1.3d), the rotary bending strength for new wires with tensile strength $R_m = 1,770 \text{ N/mm}^2$ and wire diameter $\delta = 1 \text{ mm}$ is $\sigma_{\text{Rot}} = 500 \text{ N/mm}^2$. According to (1.3c), Wolf (1987) found the mean rotary bending strength for different wires to be between $\bar{\sigma}_{\text{Rot}} = 510 \text{ N/mm}^2$ and 730 N/mm^2 for wires with diameters between 0.8 and 1.0 mm taken from wire ropes and tested under the same conditions.

The fatigue strength is reduced for wires with zinc coating, Reemsnyder (1969), and especially for wires with a thick zinc coating, Apel and Nünninghoff

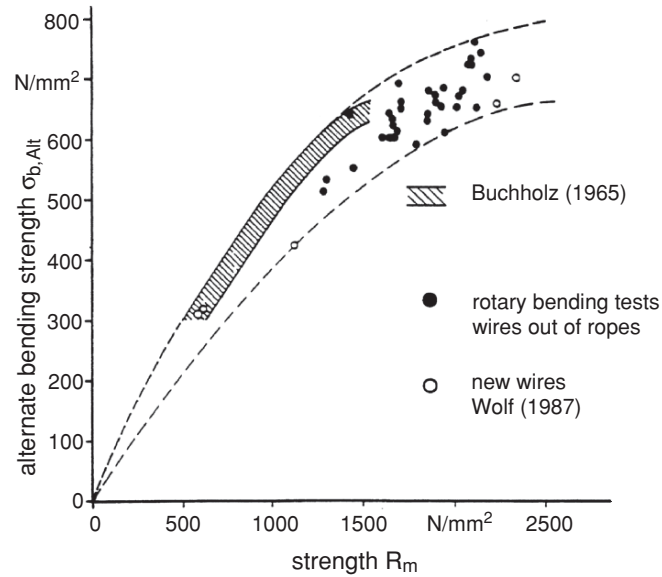


Fig. 1.16. Alternate strength σ_{Alt} for wires with a great range of measured tensile strength R_m , Wolf (1987)

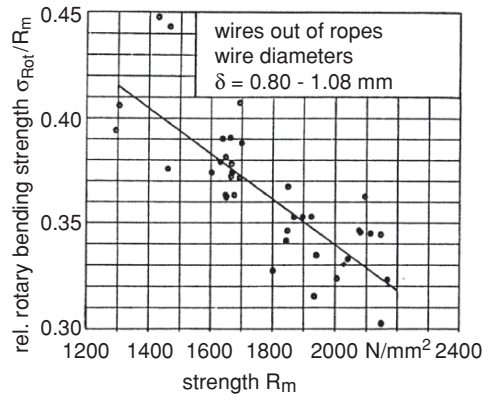


Fig. 1.17. Relative rotary bending strength σ_{Rot}/R_m for rope wires, Wolf (1987)

(1979). For normal zinc-coated wires, Briem (2000) also found that fatigue endurance is reduced in relation to bright wires. On the other hand, the corrosion that should be prevented by the zinc coating. Corrosion reduces the fatigue strength enormously, as can be seen in Fig. 1.18, Jehmlich (1969).

The fatigue strength of wires depends on their various contents and method of manufacture. With the loss of cross-section during repeated wire drawing, the repetitive tensile strength $\sigma_{t,Rep}$ first increases and then decreases. The

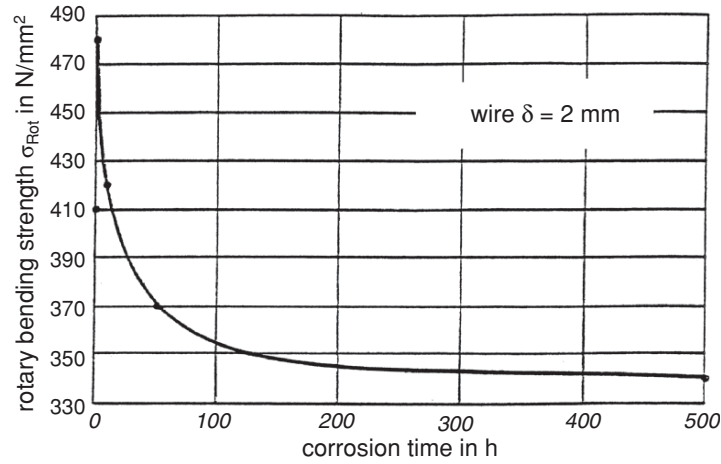


Fig. 1.18. Rotary bending strength for different corrosion times, wire diameter $\delta = 2$ mm, Jehmlich (1969)

maximum repetitive strength exists for a cross-section loss between about 60 and 80%, Becker (1977).

Unterberg (1967) found that aged wires have increased fatigue strength. After being artificially aged for three weeks at a temperature of 90°C , a wire (diameter $\delta = 3.1$ mm, measured tensile strength $R_m = 1,760$ N/mm²) has an 11% higher rotary bending strength σ_{Rot} .

For wires made of corrosion resistant steel X5CrNi18-10, Nr 4301, Benjin Luo (2002) found the mean tensile strength amplitude to be $\bar{\sigma}_{t,A} = 290$ N/mm² with the middle stress $\sigma_m = 356$ N/mm². The diameter of the wire is $\delta = 2$ mm and the stressed wire length $l = 125$ mm. The tensile strength is $R_m = 840$ N/mm².

The influence of the tensile strength and the fluctuating strength of wires on the tensile endurance of wire ropes is still unknown. For the bending endurance of wire ropes the influence of tensile strength R_0 is presented in detail in Sect. 3.2.2. The complete results – without considering the influence of wire fatigue strength – is given by (3.51b) and for a part result in Fig. 3.36.

The influence of the rotary bending strength of the wires on the endurance of ropes made from these wires has been evaluated by Wolf (1987) and Ziegler et al. (2005). Wolf (1987) evaluated the rotary bending strength σ_{Rot} of wires taken out of the wire ropes tested. The nominal tensile strength of these wires is $R_0 = 1,570 - 2,160$ N/mm². Because of the enormous influence of the fibre core mass on rope endurance N , he only used the results of ropes with a relative fibre core mass between 70 and 80%. This relative mass is related to the required core mass for ropes used for rope ways, BO-Seil, 1982. The wires used by Ziegler et al. (2005) all have the nominal tensile strength $R_0 = 2,160$ N/mm². The rotary bending strength σ_{Rot} for these wires has

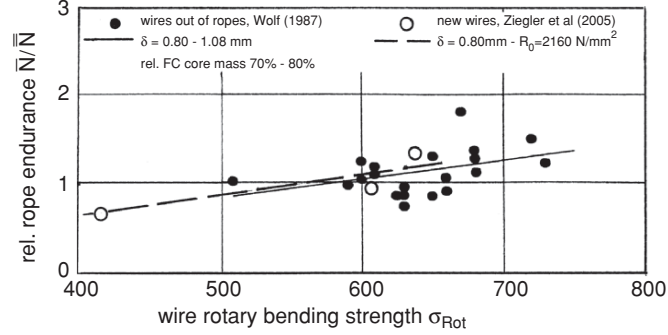


Fig. 1.19. Relative number of rope bending cycles $\bar{N}/\bar{N}$ for different rotary bending strengths of rope wires

been evaluated before the wires have been twisted into ropes. The three Seale 6x19-FC test ropes were manufactured with the same fibre core and as far as possible in the same way.

The effect of the rotary bending strength σ_{Rot} of the wires used on the rope's number of bending cycles N can be seen in Fig.1.19. The results of Ziegler, Vogel and Wehking (2005) have been inserted into this diagram drawn by Wolf (1987). The increase of the relative rope endurance $\bar{N}/\bar{N}$ with the rotary bending strength σ_{Rot} of the wires from both series of tests is practically the same.

1.2 Strands

1.2.1 Round Strands

Lay length, lay angle. In the simplest case, the strand consists of three or four twisted wires. The first wire rope made in 1834 by Albert (1837) has three of these simple strands with four wires each. In practice, however, such simple strands are no longer used. Nowadays the simplest strand has one layer of wires laid helically around a centre wire. Such a strand with six outside wires around a centre wire is shown in Fig. 1.20.

In Fig. 1.20 the important values are clearly defined: the wire lay length h_W , the lay angle α and the wire winding radius r_W . The wire lay length h_W is the length of the strand in which an lay wire makes one complete turn. The wire lay angle α is given by the equation

$$\tan \alpha = \frac{2 \cdot \pi \cdot r_W}{h_W}. \quad (1.4)$$

The lay direction of the lay wires in the strand can be right (symbol z) or left (symbol s). Figure 1.21 illustrates the origin of these symbols. Strands with more than one wire layer have very different constructions.

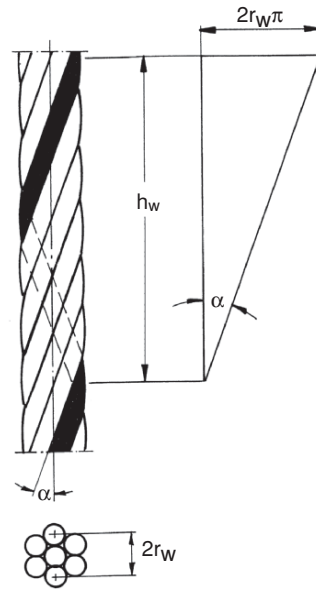


Fig. 1.20. Simple strand

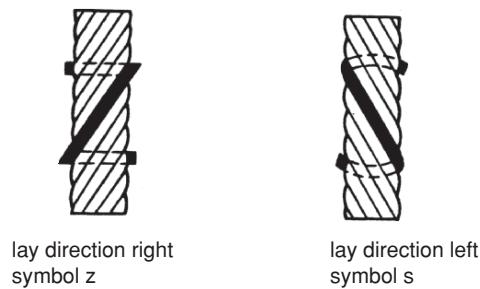


Fig. 1.21. Lay direction of wires in a strand

Cross lay strands. In the so-called cross lay strands (symbol M), the wires in the different layers do not have the same lay length. Therefore the wires of the layers cross each other. The cross lay strands normally have the same lay angle and the same lay direction for all layers. As they all have the same lay angle, in principle the wires of all layers transfer the same tensile stress. The advantage gained by having the same tensile stress has less influence than the disadvantage arising from the pressure between the crossing wires. Therefore wire ropes with cross lay strands are seldom used. The cross-section of both of the cross lay strands still used $-1 + 6 + 12$ or 19M and $1 + 6 + 12 + 18$ or

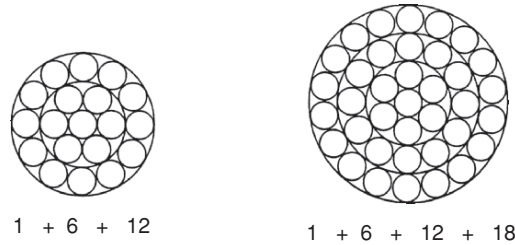


Fig. 1.22. Cross lay strands (multiple operation lay)

37M – are shown in Fig.1.22. Apart from the centre wire, which is slightly thicker, all other wires have the same diameter.

Parallel lay strands. In parallel lay strands, the lay length of all the wire layers is equal and the wires of any two superimposed layers are parallel, resulting in linear contact. The wire of the outer layer is supported by two wires of the inner layer. These wires are neighbours along the whole length of the strand. Parallel lay strands are made in one operation. The endurance of wire ropes with this kind of strand is always greater than of those with cross lay strands.

Parallel lay strands with two wire layers have the construction Filler, Seale or Warrington. The cross-section of this type strand is shown in Fig.1.23 in the most frequently used version with 19 wires – not counting the six very thin Filler wires.

The Filler strand was invented by the American James Stone in 1889. The Seale strand is named after the inventor Tom Seale (1885). The inventor of the Warrington strand is unknown. The strand presumably gets its name from the British town of Warrington, Verreet (1988, 1989).

Parallel lay strands with three – they are rarely found with more – layers are also laid in one operation. Once again, the wires of all layers have the same lay length. Of all parallel lay strands with three layers, the one used most is the Warrington-Seale strand. The Warrington-Seale strand consists of a central Warrington strand construction and an outside Seale wire layer.

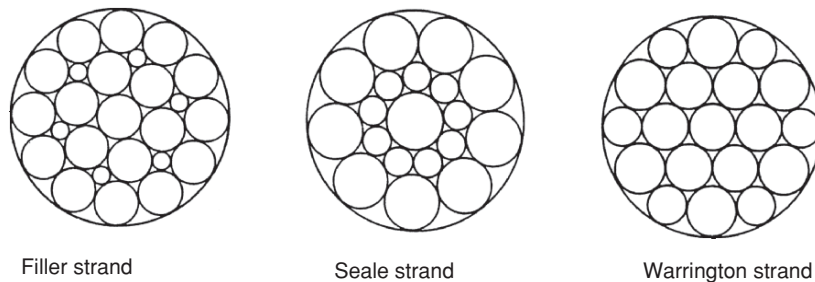


Fig. 1.23. Parallel lay strands with two wire layers

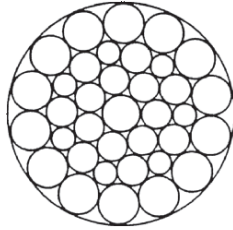


Fig. 1.24. Warrington-Seale strand

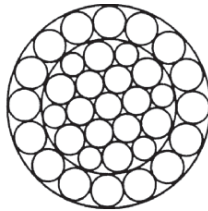


Fig. 1.25. Compound lay strand

Warrington-Seale ropes mostly have $1 + 6 + (6 + 6) + 12 = 36$ wires (36WS) as shown in Fig. 1.24.

Compound lay strands. Compound strands (symbol N) contain a minimum of three layers of wires where a minimum of one layer is laid in a separate operation, but in the same direction, over a parallel lay construction forming the inner layers. In Fig. 1.25, a Warrington compound strand with 35 wires (35WN) is shown as an example.

1.2.2 Shaped Strands

Strands which are not round are called shaped strands, Fig. 1.26. The triangular strand (symbol V) has a perpendicular cross-section which is approximately the shape of a triangle. The oval strand (symbol Q) has a perpendicular cross-section which is approximately the shape of an oval. Triangular strands and oval strands are inserted in low-rotating or non-rotating ropes.

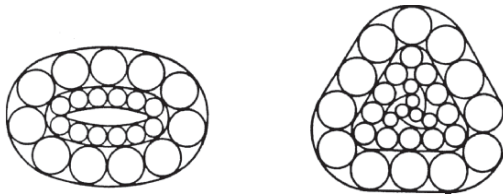


Fig. 1.26. Shaped strands (oval or triangular)

1.2.3 Compacted Strands

The compacted strand, Fig. 1.27, (symbol K) has been subjected to a compacting process such as drawing, rolling or swaging whereby the shape of the wires and the dimensions of the strand are modified. In the compacting process the mass M_S of the strand (with the actual strand diameter $d_{S,m}$, the strand length l_S and the strand mass factor W_S) remains constant while the strand diameter, the strand length and the length-related strand mass change

$$M_S = d_{S,m}^2 \cdot l_S \cdot W_S = m_{S,comp} \cdot l_{S,comp}.$$

Compacting Grade (proposal for the definition). The compacting grade Γ - however - is only based on the metal portion of the strand cross-section, compacted or not. The compacting grade is defined by

$$\Gamma = 1 - \left(\frac{d_{S,comp}}{d_{S,m}} \right)^2. \quad (1.4a)$$

In this equation $d_{S,m}$ is the measured diameter for the not compacted and $d_{S,comp}$ for the compacted strand, both with the same length-related strand mass $m_{S,comp}$. That means that the diameter of the not compacted strand is given by

$$d_{S,m}^2 = m_{S,comp} / W_S.$$

Introduced in equation (1.4a) the compacting grade is

$$\Gamma = 1 - \frac{W_S \cdot d_{S,comp}^2}{m_{S,comp}}. \quad (1.4b)$$

Strand Mass Factor. The strand mass factor W_S can be evaluated by measurements or calculated with the ratio of the effective strand cross-section $A_{s,e}$ multiplied by the wire mass density ρ and divided by the square of the strand diameter $d_{S,m}$

$$W_S = \frac{A_{s,e} \cdot \rho}{d_{S,m}^2} = \frac{\rho}{d_{S,m}^2} \cdot \frac{\pi}{4} \cdot \sum_{i=0}^n z_i \cdot \delta_i^2 / \cos \alpha_i.$$

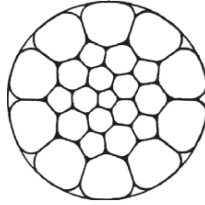


Fig. 1.27. Compacted strand

With small deviation the length-related strand mass factor is

$W_S = 0,00485 \text{ kg}/(\text{mm}^2\text{m})$	for	Filler strand 19 + 6F wires
$W_S = 0,00475 \text{ kg}/(\text{mm}^2\text{m})$	for	Seale strand 19 wires
$W_S = 0,00483 \text{ kg}/(\text{mm}^2\text{m})$	for	Warrington strand 19 wires
$W_S = 0,00485 \text{ kg}/(\text{mm}^2\text{m})$	for	Warr.-Seale strand 36 wires

Example 1.1: Compacting grade

Data:

For a Filler rope $6 \times (19 + 6F)$ piece of the length $L = 400 \text{ mm}$ is
the mass of one strand piece $M_{S,\text{comp}} = 121 \text{ g}$
the strand diameter $d_{S,\text{comp}} = 7,27 \text{ mm}$
the strand lay angle $\beta_{\text{comp}} = 18.6^\circ$

Results:

strand length $l_{S,\text{comp}} = L / \cos \beta_{\text{comp}} = 422 \text{ mm}$
length-related strand mass $m_{S,\text{comp}} = \frac{M_{S,\text{comp}}}{l_{S,\text{comp}}} = \frac{121}{422} = 0.287 \text{ g/mm} =$
 0.287 kg/m

The compacting grade is according equation (1.4b)

$$\Gamma = 1 - \frac{W_S \cdot d_{S,\text{comp}}^2}{m_{S,\text{comp}}} = 1 - \frac{0.00485 \cdot 7.27^2}{0.287} = 0.107 \rightarrow 10,7\%.$$

1.3 Rope Cores

The rope core is the central element of a round rope around which are laid helically the strands of a stranded rope. The different types of cores are listed in Table 1.5. They are usually made of fibre or wires. For dimensioning of fibre cores see Sect. 1.6.3 and of steel cores 1.6.4.

Fibre cores (FC) can be made either natural fibres (NFC) or synthetic fibres (SFC). They are normally produced in the sequence: fibres to yarns, yarns to strands and strands to fibre rope. In Fig. 1.28 different fibre cores are shown, Singenstroth (1984). Fibre cores have the advantage that they can store a relatively large amount of lubricant. The strands are supported softly. The fibre core should be well rounded and without any knots.

During the life of a wire rope, the diameter of the fibre core will be reduced. According to Singenstroth (1984) and Sivatz (1975), the diameter loss is about 3–5% for natural and polypropylene fibre cores and about 0.5–1% for polyamide fibre cores. Therefore, the clearance between the strands must be large enough to prevent any strong pressure arising between them. The endurance of the wire rope is influenced to a great extent by the dimension and the form of the fibre core, see Sect. 3.2.3, Wolf (1987).

Table 1.5. Symbols for rope cores based on ISO 17893

Fibre core	fibre core	FC
	natural fibre core	NFC
	synthetic fibre core	SFC
Steel core	steel core	WC
	wire strand core	WSC
	wire rope core	WRC
	independent wire rope core	IWRC
	independent wire rope core with compacted strands	IWRC(K)
	independent wire rope core covered with a polymer	EPIWRC
	wire rope core enveloped with fibres	EFWRC *
	wire rope core enveloped with solid polymer	ESWRC *
	parallel steel core with strands	PWRC
Steel core in parallel-closed rope	parallel wire rope core with compacted strands	PWRC(K)
Multi-strand rope (rotation-resistant)	fibre centre	FC
	wire strand centre	WSC
	compacted wire strand centre	KWSC

*Supplement, not listed in ISO 17893

Steel cores (WC) are made from steel wires arranged as a wire strand (WSC) or normally as an independent wire rope (IWRC). A strand core is only used for very small ropes or for multi-strand ropes. The steel wire rope core can either be covered with fibres (EFWRC) or solid polymer (ESWRC). Wire rope cores can also be parallel-closed with the outer strands (PWRC), and these parallel-closed ropes are often simply called “double parallel ropes”. The different types of steel cores are shown in Fig. 1.29.

In contrast to wire ropes with fibre cores, wire ropes with independent steel cores should only have very slight clearance between the outer strands to prevent lateral strand movements when rope is running over sheaves. After completing a large number of bending tests, Hugo Müller found that the endurance can only be expected to be good if this is the case. The results were published by Greis (1979) as Müller’s employer.

Wire ropes with steel cores covered with solid polymer (ESWRC) and those with steel cores enclosed parallel with the outer strands (PWRC) have better bending endurance than wire ropes with a not covered independent wire rope core. Here, it is not necessary to have only small clearance between the outer strands, Wolf (1987). The reason is that in this case the outer strands are very well bedded by the core as in the case of fibre cores.

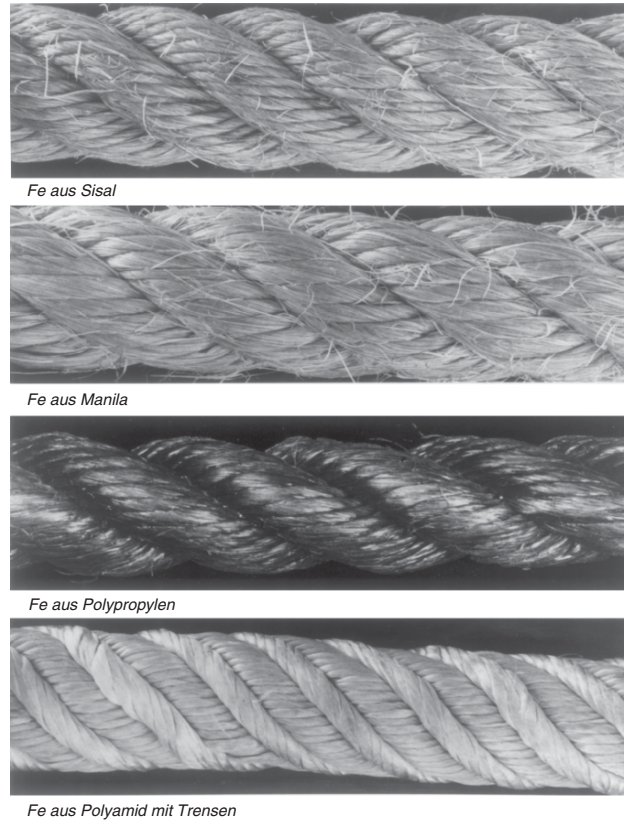


Fig. 1.28. Fibre cores, Singenstroth (1984)

1.4 Lubrication

1.4.1 Lubricant

When a wire rope is bent considerably, the wires and the strands move against each other. Relative movements also occur between the wires in stranded ropes changing the tensile forces by friction, Schmidt (1964). There are also movements between wire ropes and sheaves, especially in the case of wire rope side deflection (relative to the flank of the sheave groove), as well as in traction sheaves, Gräbner (1993).

The duty of the lubrication is to reduce the friction between the wires and strands and between the wire ropes and sheaves. This means that there will be a reduction in the wear and the friction-induced secondary tensile stresses, Schmidt (1964). However, the lubricant used to reduce the friction

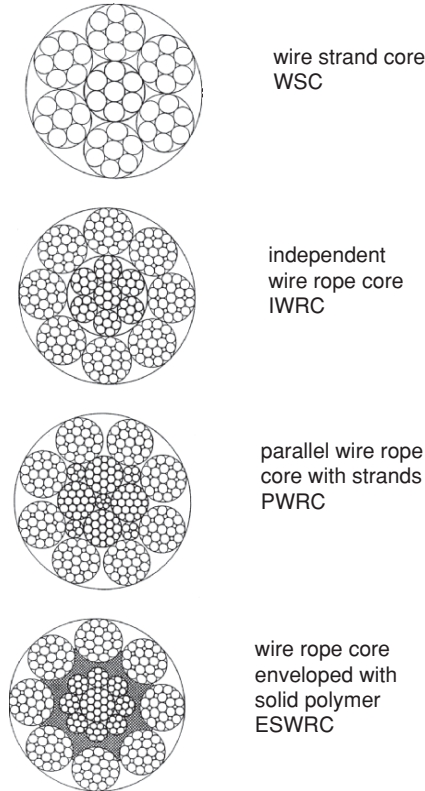


Fig. 1.29. Steel cores in wire ropes

only provides minimal protection from corrosion. On the other hand, preservation agents used as a protection against corrosion are of hardly any use for reducing the friction. Zinc-coated wires prevent corrosion but do not reduce the friction forces.

Relative movements of the wires in the rope take place in the state of boundary lubrication, but here the lubricant can only be effective if it sticks to the wire surface by adhesion, Donandt (1936). Therefore the lubricant has to be very adhesive and after being displaced by pressing, it must seep back to the contact points. Furthermore, the lubricant must have sufficient viscosity so that it will not be centrifuged away from wire ropes when they are running at relatively high speeds over sheaves.

The lubricant should not contain water or acids and should not produce acids over time. Vegetable or animal grease should also be avoided, Meebold (1959). The lubricants should not exert any negative influence on the wires or the fibre cores of the rope, Naumann and Gedecke (1971). Vaseline or mineral oil with high viscosity will fulfil these requirements quite well.

The friction coefficient of wire ropes lubricated with mineral oil in traction sheaves made of cast iron or steel is sufficient for elevator regulations requirements even if they are swimming in oil, Molkow (1982). In combination with plastic traction sheaves, a special lubricant has to be used that does not reduce the friction coefficient too much.

1.4.2 Lubricant Consumption

During manufacture, the interstices of the rope and the core are filled with lubricant. The ropes, especially those with fibre cores, should not be filled with too much lubricant as it will otherwise seep out of the rope when it is first loaded. Normally, the first lubrication will be sufficient for some time.

Long – and therefore expensive – wire ropes should be re-lubricated after being in service for some time. However, only a small amount of lubricant gets used up. To find out how much lubricant is used and to set a limit, the quantity of lubricant in a series of bending tests was reduced from test to test. In the last test before the endurance was reduced, the consumption of lubricant was only 1.8 g/m for 100,000 bendings of the 16 mm wire rope. The lubricant – transported by a pump and dripped onto the rope – was a mineral oil without any additives and with a viscosity of 1,370–1,520 cSt.

Some methods of lubrication for ropes in service are described by Winkler (1971). It is better to have continuous lubrication than to do it at long intervals. A good method is to let it fall drop by drop from a pump onto a slightly opened rope (where it is bent over a sheave). Lubricants with very high viscosity can only be inserted into the inner rope with lubricating devices. Verreet (1989) describes such a device with a pressure sleeve. Oplatka (1984) presents another device which injects the lubricant with pressure between the strand lanes.

1.4.3 Rope Endurance

Of all the influences on the endurance of running ropes provided by the rope itself, lubrication has the greatest effect. To evaluate this influence, Müller (1966) made bending tests with a lubricated and a de-lubricated Filler rope. In these tests, the endurance of the de-lubricated rope only reached 15–20% of that reached by the lubricated rope, Sect. 3.2.2.

Müller (1977) also found that lubricated wire ropes had an advantage in the case of tensile fatigue tests. For different stranded ropes, he found that de-lubricated wire ropes have an endurance of about 75% of that found with lubricated ropes, Sect. 2.6.3. Lubrication reduces the friction between the wires in strands periodically bent by fluctuating tensile forces, Andorfer (1983). With the reduction in friction, the secondary tensile stress in the wires will also be reduced and the endurance will thus be increased.

In most cases, the lubrication provided by the manufacturing of the wire rope will be enough for the entire life of the rope. Bending tests showed

that, up to a breaking number of bending cycles of about $N = 80,000$, giving the wire rope an additional lubrication during the test did not increase its endurance, Feyrer (1998). When re-lubricated, the endurance of the wire ropes will be increased if the endurance of the wire rope without this re-lubrication is greater than about number of bending cycles $N = 80,000$. For example, in a series of bending tests, the mean breaking number of bending cycles increases from $N = 246,000$ to 392,000 when re-lubricated, Sect. 3.2.2.

1.5 Wire Ropes

1.5.1 The Classification of Ropes According to Usage

Depending on where they are used, wire ropes have to fulfil different requirements. The main uses are depicted in Fig. 1.30. Running ropes are bent over sheaves and drums. They are therefore stressed mainly by bending and secondly by tension. Stationary ropes (stay ropes) have to carry tensile forces and are therefore mainly loaded by static and fluctuating tensile stresses. Track ropes have to act as rails for the rollers of cabins or other loads in aerial ropeways and cable cranes. In contrast to running ropes, track ropes do not take on the curvature of the rollers. Under the roller force, a so called free bending radius of the rope occurs. This radius increases (and the bending stresses decrease) with the rope tensile force and decreases with the roller force. Wire rope slings are used to harness various kinds of goods. These slings are stressed

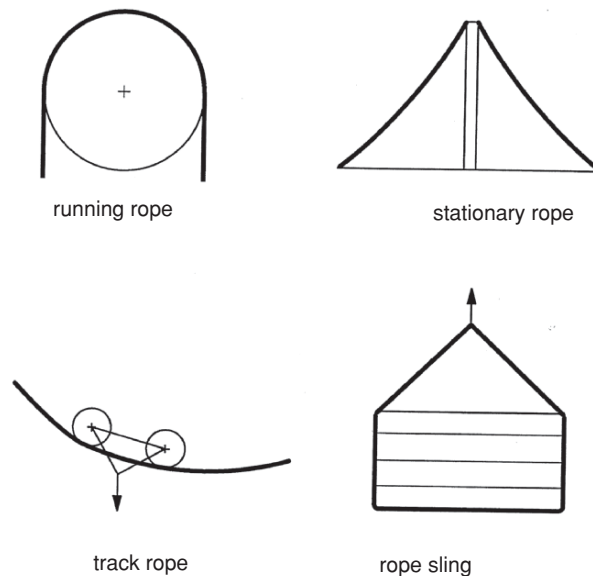


Fig. 1.30. Rope usage classification

by the tensile forces but first of all by bending stresses when bent over the more or less sharp edges of the goods.

1.5.2 Wire Rope Constructions

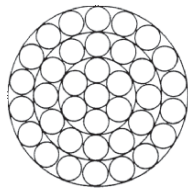
There are many different constructions existing for wire ropes. An overview of the different types is presented in Table 1.6.

Spiral ropes. In principle, spiral ropes are round strands as they have an assembly of layers of wires laid helically over a centre with at least one layer of wires being laid in the opposite direction to that of the outer layer. Spiral ropes can be dimensioned in such a way that they are non-rotating which means that under tension the rope torque is nearly zero. The centre of the spiral rope is usually a wire but it can also be a parallel lay strand. Examples are shown of the three basic types of spiral ropes in Fig. 1.31.

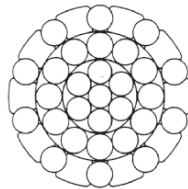
The open spiral rope consists only of round wires. The cross-section of a spiral rope 1×37 is shown as an example in Fig. 1.31 with the numbers of wires $1 + 6 + 12 + 18$ in the three wire layers. The half-locked coil rope and the full-locked coil rope always have a centre made of round wires. The half-locked

Table 1.6. Overview of the wire rope types

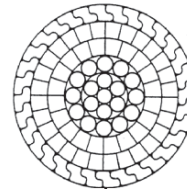
Round ropes	spiral rope	open spiral rope (strand with round wires) half-locked coil rope full-locked coil rope
	stranded rope (round strands)	single-layer rope (one layer of strands) multi-strand rope (several layers of strands)
	shaped strand rope	triangular strand rope oval strand rope
	cable-laid rope	round stranded ropes around a core
Braided rope		round strands interlaced or plaited together
Flat rope		four strand ropes stitched or riveted together



open spiral ropes



half-locked coil rope



full-locked coil rope

Fig. 1.31. Spiral ropes

coil rope has an outer layer of half-locked (H-shaped) wires and round wires. The full-locked coil rope has one or more outer layers of full-locked (Z-shaped) wires.

Half-locked coil ropes and full-locked coil ropes have the advantage that their construction prevents the penetration of dirt and water to a greater extent and it also protects them from loss of lubricant. In addition, they have one further very important advantage as the ends of a broken outer wire cannot leave the rope if it has the proper dimensions.

Open spiral ropes are mainly used as stay ropes in simpler uses. Half-locked coil ropes are not often used. Full-locked coil ropes are installed in bridges and important steel constructions. They are also used as track ropes for aerial rope ways and cable cranes.

Stranded ropes. Stranded ropes are an assembly of several strands laid helically in one or more layers around a core. Most types of stranded ropes only have one strand layer over the core. The lay direction of the strands in the rope can be right (symbol Z) or left (symbol S) and the lay direction of the wires can be right (symbol z) or left (symbol s). This kind of rope is called ordinary lay rope if the lay direction of the wires *in* the outer strands is in the *opposite* direction to the lay of the outer strands themselves. If both the wires *in* the outer strands *and* the outer strands themselves have the *same* lay direction, the rope is called a lang lay rope (formerly Albert's lay or Lang's lay), Fig. 1.32. The strand lay direction normally used is right (Z).

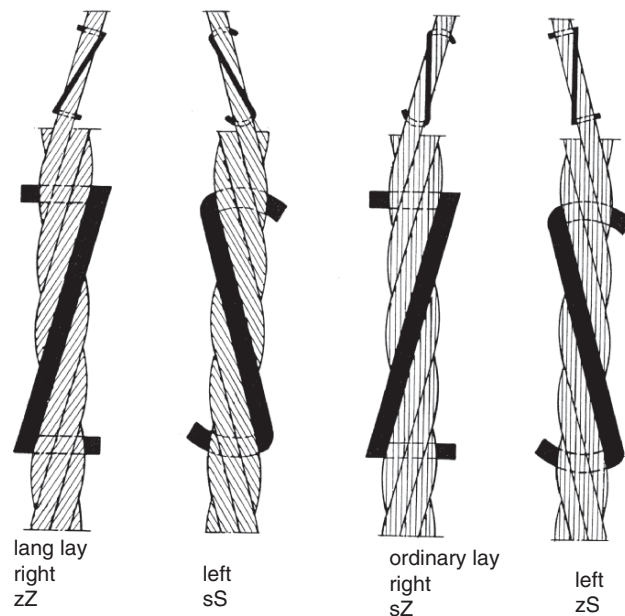


Fig. 1.32. Lay directions of stranded ropes

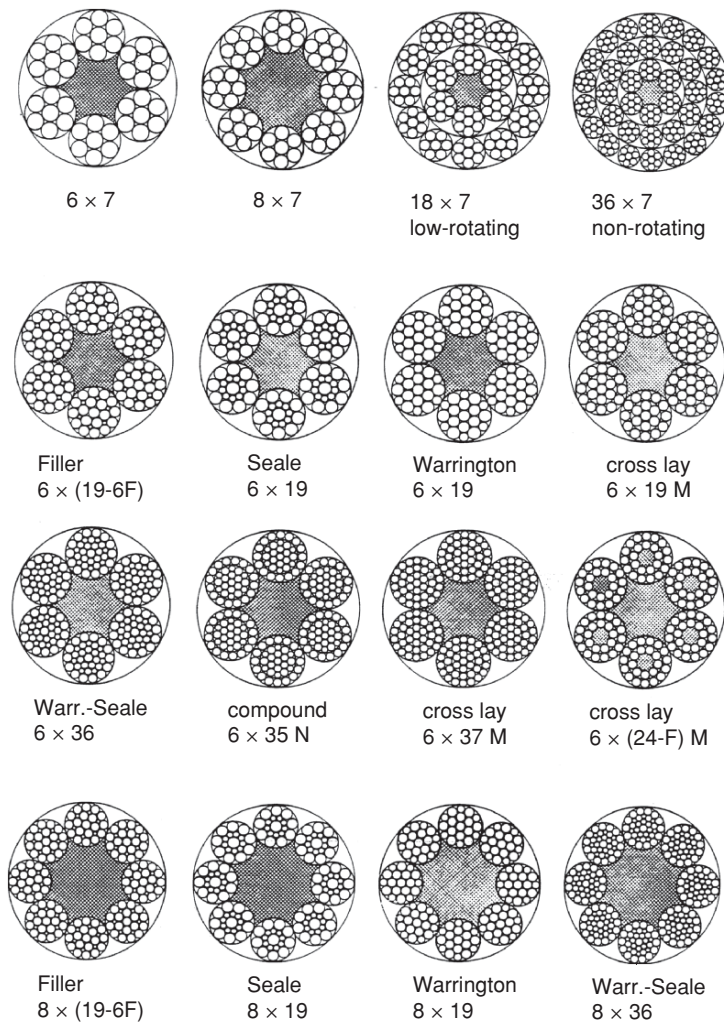


Fig. 1.33. Round stranded ropes

Figure 1.33 shows the cross-sections of various kinds of stranded ropes (with fibre cores). The ropes most commonly used are single-layer stranded ropes. Multi-strand ropes are all more or less resistant to rotation and have at least two layers of strands laid helically around a centre. The direction of the outer strands is opposite to that of the underlying strand layers. Ropes with three strand layers can be nearly non-rotating. Ropes with two strand layers are mostly only low-rotating. A proposal for the definition of the limits and characteristics of non-rotating ropes is to be found in Sect. 2.4.2.

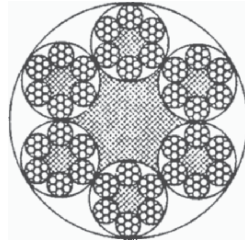


Fig. 1.34. Cable-laid rope

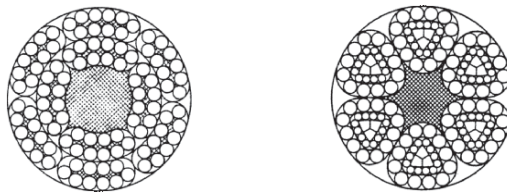


Fig. 1.35. Oval strand rope and triangular strand rope

Cable-laid ropes consist of several (usually six) round stranded ropes (referred to as unit ropes) closed helically around a core (usually a seventh stranded rope). Such a cable-laid rope is shown in Fig. 1.34 – in this case with fibre cores in the stranded ropes as well as in the whole rope. Cable-laid ropes are only used as rope slings.

Oval strand ropes are usually constructed with two oval strand layers as shown in Fig. 1.35. The direction of the outer strands is opposite to that of the underlying strand layers. These ropes are rotation-resistant. Triangular strand ropes are often only constructed with a single-layer as shown in Fig. 1.35.

Special rope types. Braided ropes consist of several round strands that are interlaced or plaited together, Fig. 1.36a. The lay direction for one half of the strands is right and for the others left. This means that braided ropes really are non-rotating. However, when braided ropes are bent over sheaves they do not last very long and so these ropes are only used in exceptional cases.

Flat ropes consist of an assembly of unit ropes known as “reddies”, each with four strands, whereby usually six, eight or ten reddies are laid side by side with alternating left and right direction of lay (therefore non-rotating) and are then held in position by stitching wires, strands or rivets. In Fig. 1.36b a flat rope is to be seen from the top and in cross-section. Flat ropes are normally used as balance ropes in pit hoistings.

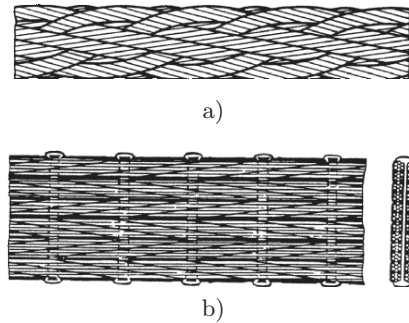


Fig. 1.36. Special rope types **a)** Braided rope. **b)** Flat rope

1.5.3 Designation of Wire Ropes

The designation of wire ropes follows ISO17893:2003(E). The system for designating steel wire ropes details the minimum amount of information that is required to describe a rope. The system is capable of accommodating most rope constructions, strength grades, wire finishes and layers of steel wire ropes. The symbols for the designation of wire ropes are given in Table 1.7 with the symbols for the core in Table 1.5. The following examples show how this designation works.

The symbol sequence

22-6×36WS-IWRC-1770-B-sZ

describes a 6-strand Warrington-Seale rope with 22 mm nominal diameter, independent wire rope core, nominal strength 1,770 N/mm², bright, ordinary lay.

A Filler rope may be described by the symbol sequence

16-8×25F-NFC-1770-B-zZ or 16-8×(19+6F)-NFC-1770-B-zZ

to make it quite clear that six of the 25 wires are very small Filler wires.

An example of the symbol sequence for a multi-strand rope is

20-18×7-WSC-1960-zn-sZ.

For an open spiral rope (round wires) the symbol sequence is

32-1×61-1770-zn-Z or 32-1×61-1770-zn-SZSZ

to describe the lay directions of all the wire layers.

For a full-locked coil rope the symbol sequence is

50-1×(37-22Z-28Z)-1570-zn-Z or 50-1×(1-6-12-18-22Z-28Z)-1570-zn-Z.

Table 1.7. Designation of round wire ropes, ISO 17893

designation	symbol
a) nominal diameter (mm)	d
b) rope construction (number of strands $t \times$ number of strand wires z), strand constr.	
spiral rope with round wires	$1 \times z$
spiral rope with round/Z-wires	$1 \times (1 - z_1 - z_2 \dots z_n Z)$
single lay rope	$t \times z$
Seale	$t \times zS$
Warrington	$t \times zW$
Filler	$t \times zF$
Warrington-Seale	$t \times zWS$
cross lay	$t \times zM$
compound lay	$t \times zN$
compound lay, Warrington	$t \times zWN$
triangular strand rope	$t \times zV$
multi-strand rope, rotation resistant rope	$t \times z$
multi oval strand rope	$t \times zQ$
c) core construction from Table 1.5	
d) nominal wire strength, grade in N/mm ²	R_0
e) wire finish	
bright	B
coated	U
zinc coated	zn
f) lay type and direction (Fig. 1.32)	
ordinary lay right	sZ
ordinary lay left	zS
lang lay right	zZ
lang lay left	sS

1.5.4 Symbols and Definitions

The symbols of nominal values are given without indices. The measured values always have the index m .

Diameters. The rope diameter is the diameter of the circle circumscribing the rope cross-section in mm.

d is the nominal rope diameter d .

d_m is the measured wire rope diameter (actual rope diameter) that can be at most 5% more than the nominal rope diameter.

d_S is the diameter of a strand in mm and

δ is the diameter of a wire in mm.

Clearance. s_W is the wire clearance, the distance between two adjacent wires in the same wire layer and s_S is the strand clearance, the distance between two adjacent strands in the same strand layer.

Lay length. h_W is the lay length of a wire in a strand in mm and h_S is the lay length of a strand in a rope in mm.

Factors. f is the fill factor, that means the ratio between the sum of the nominal metallic cross-section areas A of all wires in the rope and the circumscribed area A_u of the rope based on its nominal diameter d

$$f = A/A_u.$$

Jenner (1992) calculated fill factors f and the ratios of wire diameters and rope diameters δ/d for round stranded ropes under uniform conditions. These values are listed in Table 1.8. The listed fill factors f and diameter ratios δ/d are related to the nominal rope diameter d . However, with the given values, the resulting rope diameter is 2.5% greater than the nominal rope diameter d . In practice this is the mean actual rope diameter.

A is the nominal metallic cross-section area (sum of the wire cross-sections)

$$A = C \cdot d^2. \quad (1.5)$$

with nominal metallic cross-sectional area factor C , Table 1.9

$$C = f \cdot \frac{\pi}{4}. \quad (1.5a)$$

The nominal rope length-related mass m in kg/m contains the mass of the strands, the core and the lubricant. Because M is normally the mass of a single load, m is used for the rope length mass in this book in contrast usage norms. The nominal rope length-related mass m in kg/m will be calculated with the rope mass factor W listed in Table 1.9 and d in mm

$$m = \frac{W \cdot d^2}{100}. \quad (1.5b)$$

Rope breaking forces. $F_{\min}$ is the minimum breaking force of the rope in kN. It is normally obtained by calculation from the product of the square of the nominal diameter d in mm, the rope grade R_o in N/mm² and the breaking force factor K

$$F_{\min} = \frac{d^2 \cdot R_o \cdot K}{1000}. \quad (1.6)$$

In this R_o is the rope grade in N/mm², i.e. the nominal tensile strength of the wires. K is the minimum breaking force factor, the empirical factor used in the determination of the minimum breaking force and obtained from the product of fill factor f , the spinning loss factor k and the factor $\pi/4$

$$K = \frac{\pi \cdot f \cdot k}{4} = k \cdot C. \quad (1.6a)$$

F_m is the measured breaking force. This breaking force is obtained by using a prescribed measuring method.

Table 1.8. Ratio of wire diameter δ to nominal rope diameter d for ropes with measured diameter $d_m = 1.025 d$, and fill factor f , Jenner (1992)

		6-strand ropes				8-strand ropes			
		Filler	Seale	Warrington	Warr.-Seale	Filler	Seale	Warrington	Warr.-Seale
Wire ropes with fibre core									
Fill factor	f	0.506	0.492	0.502	0.508	0.456	0.443	0.452	0.458
Strand	δ_0/d	0.06839	0.08851	0.07100	0.06103	0.05626	0.07280	0.05842	0.05021
	δ_1/d	0.06732	0.04540	0.06986	0.04663	0.05539	0.03735	0.05748	0.03836
	δ_2/d	0.06324	0.07862	0.07223	0.04458	0.05202	0.06468	0.05943	0.03668
	δ_3/d	0.02765	—	0.05473	0.03444	0.02275	—	0.04503	0.02833
Wire rope with steel core		—	—	—	0.05579	—	—	—	0.04590
Fill factor	f	0.627	0.616	0.623	0.628	0.631	0.620	0.625	0.630
Centre strand	δ_0/d	0.05403	0.05555	0.05403	0.05337	0.06627	0.06682	0.06627	0.06559
	δ_1/d	0.05134	0.05277	0.05134	0.0507	0.06291	0.06342	0.06291	0.06234
1. strand layer	δ_0/d	0.04311	0.04344	0.04311	0.04301	0.05551	0.05585	0.05551	0.05563
	δ_1/d	0.04092	0.04126	0.04092	0.0408	0.0527	0.05302	0.05270	0.05283
2. strand layer	δ_0/d	0.07025	0.09092	0.07294	0.06263	0.05786	0.07496	0.06010	0.05160
	δ_1/d	0.06916	0.04664	0.07176	0.04789	0.05696	0.03833	0.05909	0.03940
	δ_2/d	0.06496	0.08076	0.07420	0.04579	0.05349	0.06670	0.06111	0.03772
	δ_3/d	0.02840	—	0.05622	0.03535	0.02344	—	0.04631	0.02910
Strand lay angle $\beta = 20^\circ$, outer wire lay angle $\alpha = 15^\circ$, strand clearance $(s_s/d_s)_{FC} = 0.05$, $(s_s/d_s)_{WRC} = 0.01$, clearance outer wire layer $sw/\delta_1 = 0.02$, clearance first wire layer $sw = 0$		—	—	—	0.05724	—	—	—	0.04712

Table 1.9. Factors for stranded ropes, EN 12385

Rope	construction type	wire rope with fibre core				wire rope with steel core			
		length mass factor	metallic cross- sectional area factor	minimum breaking force factor	K_1	length mass factor	metallic cross- sectional area factor	minimum breaking force factor	
		W_1	C_1			W_2	C_2	K_2	K_3
Single- layer rope	6×7	0.345	0.369	0.332	0.332	0.384	0.432	0.359	0.388
	8×7	0.327	0.335	0.291	0.291	0.391	0.439	0.359	—
	6×19	0.359	0.384	0.330	0.330	0.400	0.449	0.356	—
	8×19	0.340	0.349	0.293	0.293	0.407	0.457	0.356	—
	6×36	0.367	0.393	0.330	0.330	0.409	0.460	0.356	—
	8×36	0.348	0.357	0.293	0.293	0.417	0.468	0.356	—
	6×35 N	0.352	0.377	0.317	0.317	0.392	0.441	0.345	—
	6×19 M	0.344	0.357	0.307	0.307	0.372	0.418	0.332	0.362
	6×37 M	0.334	0.357	0.295	0.295	0.372	0.418	0.319	0.346
	18×7	0.382	0.408	0.328	0.328	0.401	0.433	0.328	—
Multi- strand rope	34×7	0.390	0.416	0.318	0.318	0.401	0.428	0.318	—
subscript 1 for fibre core		rope class $6 \times 19 = 6$ -strand				N for compound lay strands			
subscript 2 for steel rope core		Seale, Warrington or Filler				M for cross-lay strands			
subscript 3 for steel strand core		rope class $8 \times 19 = 8$ -strand							
		Seale, Warrington or Filler							
		rope class 6 or $8 \times 36 = 6$ or 8 -strand Warrington-Seale							

The measured breaking force F_m should not be allowed to fall below the minimum breaking force $F_{\min}$

$$F_m \geq F_{\min}.$$

F_C is the calculated breaking force in kN obtained by calculation from the product of the square of the nominal rope diameter d in mm, the rope grade R_o in N/mm² and the nominal metallic cross-sectional area factor C

$$F_c = \frac{d^2 \cdot R_o \cdot C}{1000}. \quad (1.6b)$$

The constants C , W and K are listed in Table 1.9.

The wire rope tensile force S is very often related to the square of the nominal diameter, to the so-called specific tensile force S/d^2 . The specific minimum breaking forces $F_{\min}/d^2$ based on EN12 185 are listed in Table 1.10 for a simple comparison of the tensile force and the minimum breaking force.

All the values for wire ropes with steel rope cores in Table 1.9 and 1.10 are given for IWRC. These values are also true for wire rope cores enveloped with solid polymer ESWRC. For parallel-closed ropes (parallel steel core with outer strands) PWRC the values are 6% greater. Wire ropes with fibre-enveloped cores EFWRC have values between those of fibre cores and steel cores, but these deviate greatly.

Ratios of the measured and the nominal values evaluated on 49 parallel round strand ropes are listed in Table 1.11. This table shows that the mean measured breaking force is 15.6% greater than the minimum breaking force, Feyrer (1992). This result is confirmed by the measurements made by Chaplin

Table 1.10. Specific minimum breaking force of wire ropes $F_{\min}/d^2$ in N/mm²

Rope	rope class	wire rope with fibre core			wire rope with steel core		
		1,570 N/mm ²	1,770 N/mm ²	1,960 N/mm ²	1,570 N/mm ²	1,770 N/mm ²	1,960 N/mm ²
single-layer rope	7 × 7				609	687	760
	6 × 7	521	588	651			
	8 × 7	457	515	570	564	635	704
	6 × 19	518	584	647	559	630	698
	8 × 19	460	519	574	559	630	698
	6 × 36	518	584	647	559	630	698
	8 × 36	460	519	574	559	630	698
	6 × 35 N	498	561	621	542	611	676
	6 × 19 M	482	543	602	521	588	651
	6 × 37 M	463	522	578	501	565	625
multi-strand rope	18 × 7	515	581	643	515	581	643
	34 × 7	499	563	623	499	563	623

rope class 6 × 19 = 6-strand Seale, Warrington or Filler.

Rope class 8 × 19 = 8-strand Seale, Warrington or Filler.

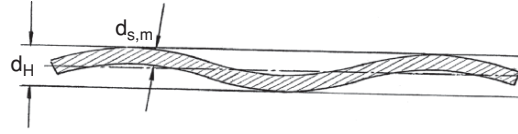
Rope class 6 or 8 × 36 = 6 or 8-strand Warrington-Seale

N for compound lay strands

M for cross-lay strands

Table 1.11. Ratio of measured and nominal values of parallel lay strand ropes

		$\frac{\text{measured}}{\text{nominal}}$	deviation
Wire rope diameter	d_m/d	1.025	0.018
Metallic cross-section	A_m/A	1.049	0.049
Wire strength	R_m/R_o	1.054	0.036
Rope breaking force	$F_m/F_{\min}$	1.156	0.054

**Fig. 1.37.** Strand helix outside the rope

and Potts (1991). They found that the measured breaking force is between 5 and 29% greater than the minimum breaking force. The factor K in the norm has apparently been carefully chosen. Hankus (1983) and Apel (1986) have reported on the measurements and calculations of the spinning loss factor.

Strand and wire forming. Normally, strands are either pre- or post-formed so that they will retain their position in the rope structure when the wire rope is cut. The grade of the strand forming is defined (proposal for the definition) by the strand forming grade

$$\Phi_S = \frac{d_H - d_{S,m}}{d_m - d_{S,m}}. \quad (1.6c)$$

Here d_H is the outside diameter of the loose strand helix taken from the rope, Fig. 1.37, $d_{S,m}$ is the measured strand diameter and d_m is the measured rope diameter. The wire forming grade is defined analogous to

$$\Phi_W = \frac{d_{S,H} - \delta}{d_{S,m} - \delta}. \quad (1.6d)$$

$d_{S,H}$ is the outside diameter of the loose wire helix taken from the strand or the spiral rope, $d_{S,m}$ is the measured strand diameter and δ is the wire diameter.

1.6 The Geometry of Wire Ropes

The geometry of wire rope can be demonstrated in principle by the geometry of a strand. This is true in particular for spiral ropes. For stranded ropes, a strand is to be considered as a wire in a strand. However for stranded ropes, the core, and here especially the fibre core, gives additional problems.

1.6.1 Round Strand with Round Wires

The geometry of both the strand and the wire rope has a great effect on their properties. In the strand, the clearance between the wires of a wire layer should not be too large, but on the other hand there should be no overlapping (or negative clearance). If the clearance is too large, the position of the wires is undefined. This is especially true if the clearance in the inner wire layer of a parallel lay rope is too large as that leads to an irregular structure of the strand with unequal stresses of the outer wires. In some cases, as time passes wire loops may even occur on the outside. In the opposite case where wires overlap (negative clearance), an arching of the wire layer occurs with high secondary tensile stresses when the strand or the rope is bent. Therefore, great care must be taken to dimension wire ropes as well as possible.

There is a great deal of literature already available on strand geometry. All the calculations presented here – with the exception of winding a tape by bending only – have the same following presuppositions

- The cross-section of the wire remains unchanged and perpendicular to the helix for the wire turning point (normally the centre of the wire cross-section) when the straight wire is wound to the helix of the wire.
- The wire is bent with the binormal helix as a neutral axis (this is only of importance for wires which are not round) and twisted in such a way that the neutral axis is located stationary to the wire.

Hruska (1953) calculated the cross-section of round wire strands using the simplification of taking ellipses as the section contour of the round wires. Using the same simplification, Shitkow and Pospechow (German translation, 1957) gave a detailed presentation of this method of calculation. For a long time, their book was a guide for the practical calculation of rope geometry. Jenner (1992) established that the results calculated with the ellipse simplification are accurate enough for the lay angles normally used.

Groß (1954) calculated the first realistic contour (based on the two given presuppositions) for round wires in the cross-section of a strand. Shitkow (1957) and Wiek (1985) came to the same result as Groß. Wolf (1984) and Wang and McKewan (2001) presented the geometry of round strand ropes in vector form. The use of computers for these methods in practice brought a great progress for the rope quality, Wiek (1977), Fuchs (1984), Voigt (1985) etc.

One method of calculating a realistic cross-section of a round wire in a straight strand is presented in a clear parameter form. Using φ for the wire winding angle φ_W , the wire winding radius r_W and the lay angle α , the equations for the wire axis in a strand are

$$\begin{aligned} x_M &= r_W \cdot \sin \varphi \\ y_M &= r_W \cdot \cos \varphi \\ z_M &= r_W \cdot \varphi \cot \alpha. \end{aligned} \tag{1.7}$$

(These and the following equations are also valid for round strands in straight ropes with $\varphi = \varphi_S$ for the winding strand angle, with r_S for the winding strand radius and with the strand lay angle β instead of α .)

The surface of a wire with the diameter δ in a strand is defined by the equations

$$\begin{aligned} x &= r_W \cdot \sin \varphi + \frac{\delta}{2} \cdot \sin \varphi \cdot \cos(\varphi_0 - \varphi) + \frac{\delta}{2} \cdot \cos \varphi \cdot \sin(\varphi_0 - \varphi) \cdot \cos \alpha \\ y &= r_W \cdot \cos \varphi + \frac{\delta}{2} \cdot \cos \varphi \cdot \cos(\varphi_0 - \varphi) - \frac{\delta}{2} \cdot \sin \varphi \cdot \sin(\varphi_0 - \varphi) \cdot \cos \alpha \quad (1.8) \\ z &= r_W \cdot \varphi \cdot \cot \alpha - \frac{\delta}{2} \cdot \sin(\varphi_0 - \varphi) \cdot \sin \alpha. \end{aligned}$$

These equations are to be found in a slightly different version in Andorfer (1983) and Schiffner (1986). There, the cross-section is to be found, for example, with $z = 0$. From that the last equation (1.8) is derived

$$\sin(\varphi_0 - \varphi) = \frac{2 \cdot r_W \cdot \varphi \cdot \cot \alpha}{\delta \cdot \sin \alpha}.$$

From this equation, φ can be calculated for a given φ_0 by iteration. Then the coordinates x and y for a wire in the cross-section of a strand can be calculated using the other two equations.

1.6.2 Round Strand with Any Kind of Profiled Wires

General equations. One method of calculating the strand cross-section when the strand contains any profiled wires has been given in Feyrer and Jenner (1987). According to this method, the coordinates x and y (from the strand axis as origin of the coordinates) for a special profiled wire in the cross-section of a round strand can be calculated with the equations

$$\begin{aligned} x &= \sqrt{(r_W + b)^2 + a^2 \cdot \cos^2 \alpha} \cdot \sin \left(\frac{a \cdot \sin^2 \alpha}{r_W \cdot \cos \alpha} + \arctan \frac{a \cdot \cos \alpha}{r_W + b} \right) \\ y &= \sqrt{(r_W + b)^2 + a^2 \cdot \cos^2 \alpha} \cdot \cos \left(\frac{a \cdot \sin^2 \alpha}{r_W \cdot \cos \alpha} + \arctan \frac{a \cdot \cos \alpha}{r_W + b} \right). \end{aligned} \quad (1.9)$$

Or in polar coordinates with

$$\tan \varphi = \frac{x}{y} = \frac{\sin}{\cos} \left(\frac{a \cdot \sin^2 \alpha}{r_W \cdot \cos \alpha} + \arctan \frac{a \cdot \cos \alpha}{r_W + b} \right).$$

the polar coordinate φ is

$$\varphi = \frac{a \cdot \sin^2 \alpha}{r_W \cdot \cos \alpha} + \arctan \frac{a \cdot \cos \alpha}{r_W + b}. \quad (1.9a)$$

and with

$$r = \sqrt{x^2 + y^2}$$

the polar coordinate r is

$$r = \sqrt{(r_W + b)^2 + a^2 \cdot \cos^2 \alpha}. \quad (1.9b)$$

In these equations, r_W is the wire winding radius and α is the lay angle for the helix of the wire turning point M (the wire centre for symmetric cross-sections). a and b are the coordinates for the cross-section contour of the straight lay wire with the turning point M as origin of the coordinates. The coordinates y and b have the same direction as the radius r between the centre of the strand and the wire turning point M .

Figure 1.38 shows the cross-section of a Z-wire as an example. In the strand (spiral rope) the point B_2 is lying on the point B_1 of the neighbouring wire. The turning point M therefore has to lie on the mid-perpendicular between B_1 and B_2 . The turning point M and the centre of gravity S are not identical in this case.

In Fig. 1.39, the Z-wire is shown both in its ground-plan and elevation. The point P is calculated with the (1.9) or (1.9a) and (1.9b) in a section perpendicular to the strand axis. In the same way, the contour of the wire cross-section perpendicular to the axis of the strand is found by calculating point-by-point.

Equations for round wires. Equations (1.9) or (1.9a) and (1.9b) can of course be used to calculate the cross-section perpendicular to the strand axis of a round lay wire. With the wire diameter δ , the coordinates for the contour of the round wire cross-section are

$$\begin{aligned} a &= (\delta/2) \cdot \sin \psi \\ b &= (\delta/2) \cdot \cos \psi. \end{aligned} \quad (1.10)$$

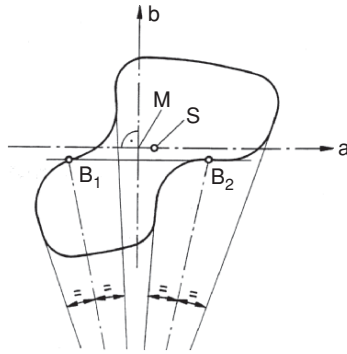


Fig. 1.38. Cross-section of a Z-wire

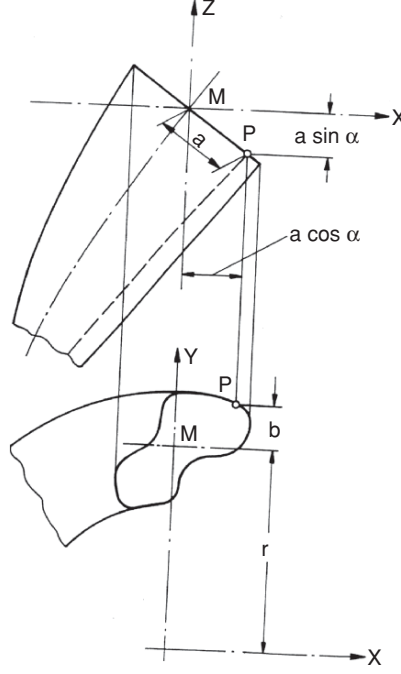


Fig. 1.39. Wire helix in ground-plan and elevation

With these introduced into (1.9), the coordinates for the contour of the round wire cross-section perpendicular to the strand axis is

$$\begin{aligned}
 x &= \sqrt{\left(r_W + \frac{\delta}{2} \cdot \cos \psi\right)^2 + \left(\frac{\delta}{2} \cdot \sin \psi \cdot \cos \alpha\right)^2} \\
 &\quad \cdot \sin \left(\frac{\delta \cdot \sin \psi \cdot \sin^2 \alpha}{2 \cdot r_W \cdot \cos \alpha} + \arctan \frac{\frac{\delta}{2} \cdot \sin \psi \cdot \cos \alpha}{r_W + \frac{\delta}{2} \cdot \cos \psi} \right) \\
 y &= \sqrt{\left(r_W + \frac{\delta}{2} \cdot \cos \psi\right)^2 + \left(\frac{\delta}{2} \cdot \sin \psi \cdot \cos \alpha\right)^2} \\
 &\quad \cdot \cos \left(\frac{\delta \cdot \sin \psi \cdot \sin^2 \alpha}{2 \cdot r_W \cdot \cos \alpha} + \arctan \frac{\frac{\delta}{2} \cdot \sin \psi \cdot \cos \alpha}{r_W + \frac{\delta}{2} \cdot \cos \psi} \right).
 \end{aligned} \tag{1.11}$$

The polar coordinates are

$$\varphi = \frac{\delta \cdot \sin \psi \cdot \sin^2 \alpha}{2 \cdot r_W \cdot \cos \alpha} + \arctan \frac{\frac{\delta}{2} \cdot \sin \psi \cdot \cos \alpha}{r_W + \frac{\delta}{2} \cdot \cos \psi}. \tag{1.11a}$$

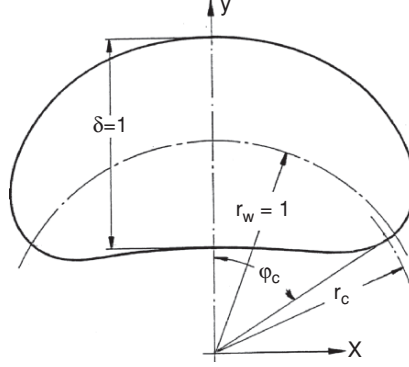


Fig. 1.40. Contour of a round lay wire in the section perpendicular to the strand axis

and

$$r = \sqrt{\left(r_w + \frac{\delta}{2} \cdot \cos \psi\right)^2 + \left(\frac{\delta}{2}\right)^2 \cdot \sin^2 \psi \cdot \cos^2 \alpha}. \quad (1.11b)$$

The result of the calculations with both of the earlier equations (1.11) is of course the same as that of (1.8) but gained in a simpler way with no iteration. Fig. 1.40 shows the contour of a round strand wire in the cross-section perpendicular to the strand axis drawn for $r_w = 1$ and $\delta = 1$. In this example, the lay angle has been given the value $\alpha = 60^\circ$ which is much higher than that used in practice to show clearly the characteristic difference to the ellipse often used for simplification.

Clearance between the round lay wires. The most important aim of geometry calculation is to determine the clearance between two neighbouring wires. An accurate calculation of the clearance based on (1.8) is to be found in Jenner's paper (1992). Griffioen and Wiek (1992) also accurately calculated the distance between two neighbouring wires by using a clearance angle with the vector method. This clearance angle is the angle between the two straight lines coming from the centre of the strand and they only touch the cross-section contour of two neighbouring wires at one point each. Griffioen and Wiek introduced with this clearance angle a new point of view.

The clearance angle for round wires can also be calculated accurately based on the contact angle φ_c as the boundary angle of the wire contour. To do this, the contact angle φ_c for a straight line from the strand centre has to be used as a tangent on the contour of the wire cross-section. The contact angle φ_c will finally be found by iteration of the angle ψ in (1.11a) as the maximum angle $\varphi_{\max}$

$$\varphi_c = \varphi_{\max}(\psi \rightarrow \psi_c)$$

with the angle ψ_c . Then the contact radius r_c up to the contact point can also be derived from (1.11b) with the angle $\psi = \psi_c$.

Using the contact angle φ_c from (1.11a) and the number z_W of wires in the wire layer being considered, the clearance angle between the two neighbouring wires in the cross-section is

$$\Delta\varphi = 2 \cdot \left(\frac{\pi}{z_W} - \varphi_c \right). \quad (1.11c)$$

The clearance between two neighbouring wires is then

$$s_W = 2 \cdot r_c \cdot \sin \left(\frac{\pi}{z_W} - \varphi_c \right) \cdot \cos \alpha. \quad (1.11d)$$

Clearance between the Round Lay Wires, Approximation

Shitkow (1957) (in an other form Costello (1997)) provided the following relatively simple equation for the clearance in the cross-section perpendicular to the strand axis with z_W for the number of wires in the layer being considered

$$s_{WQ} = 2 \cdot r \cdot \sin \frac{\pi}{z_W} - \frac{\delta}{\cos \frac{\pi}{z_W} \cdot \sqrt{\cos^2 \alpha + \tan^2 \frac{\pi}{z_W}}}. \quad (1.12)$$

Then the real clearance between the wires is

$$s_W = s_{WQ} \cdot \cos \alpha. \quad (1.12a)$$

Jenner (1992) found that these equations are sufficiently accurate for the lay angles used in practice.

Example 1.2: Cross-section of a strand wire

Data:

Strand 1 + 6,

Wire diameter $\delta = 1$ mm; winding radius $r = 1.055$ mm; lay angle $\alpha = 20^\circ$.

Results:

For an angle $\psi = 30^\circ$ the coordinates of a point of the cross-section contour are according to (1.11)

$x = 0.2787$ mm and $y = 1.4804$ mm.

The contact angle is (1.11a)

$\varphi_c = 0.5209$ rad resp. $\varphi_c = 29.85^\circ$

with the parameter $\psi_c = 2.01383$ rad

The contact radius is (1.11b)

$r_c = 0.9418$ mm.

The clearance angle is (1.11c)

$\Delta\varphi = 0.0054$ rad $\Delta\varphi = 0.31^\circ$.

The clearance is (1.11d)

$s_W = 0.0048$ mm,

clearance, approximation (1.12a) is

$s_W = 0.0075$ mm.

1.6.3 Fibre Core

Experience has shown that the fibre core should normally be dimensioned to allow sufficient clearance between the strands so that they do not press on each other during the normal life of the rope. A very high strand clearance is required according to rope way rules and, in relation to the strand diameter, this is about $s_S/d_S = 9\%$ for a Seale rope 6×19-FNC and about $s_S/d_S = 12\%$ for a Seale rope 8×19-FNC.

Jenner (1992) evaluated the required fibre core mass of stranded ropes in relation to the clearance between the strands. For his investigations, he considered the cross-section up to the smallest distance between the strands of the fibre core cross-section available, as well as 80% of the wire gussets area in the contact zone to the core. With the same intention, Sivatz (1975) recommended that the number of outer strand wires be taken into consideration.

The density ρ of fibre cores under the specific tensile forces $S/d^2 = 0$ and $S/d^2 = 117 \text{ N/mm}^2$, evaluated by Jenner (1992), is listed in Table 1.12. The length-related mass of a fibre core is in g/m

$$m_F = \rho \cdot A_F, \quad (1.13)$$

with A_F as the cross-section of the fibre core in mm^2 and ρ in g/cm^3 as the density of the fibre core, Table 1.12. The cross-section for the fibre core in the rope is thus as in Fig. 1.41

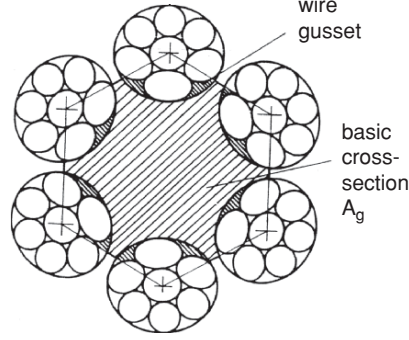
$$A_F = A_g + \Delta A. \quad (1.14)$$

According to Jenner (1992), for round strands with unstructured surface (number of outer strand wires $z_W = \infty$), with the actual rope diameter d_m , the strand diameter d_S , the number of strands z_S and the strand lay angle β , the cross-section for the fibre core without the wire gussets is

$$A_g = z_S \cdot \left[\left(\frac{d_m - d_S}{2} \right)^2 \cdot \sin \frac{\pi}{z_S} \cdot \cos \frac{\pi}{z_S} - \frac{d_S^2}{4 \cdot \cos \beta} \cdot \arcsin \frac{\cos \beta}{\sqrt{\cos^2 \beta + \tan^2 \frac{\pi}{z_S}}} \right]. \quad (1.15)$$

Table 1.12. Density of the fibre core, Jenner (1992)

core	$S/d^2 = 0 \text{ N/mm}^2$		$S/d^2 = 117 \text{ N/mm}^2$	
	density	standard deviation	density	standard deviation
	ρ g/cm ³	s g/cm ³	ρ g/cm ³	s g/cm ³
SFC (polyprop.)	0.749	0.045	0.874	0.044
NFC (sisal)	1.053	0.106	1.233	0.0622

**Fig. 1.41.** Fibre core cross-section, Jenner (1992)**Table 1.13.** Factor q for the wire gusset area, Jenner (1992)

number of outer strand wires	factor q	
	number of strands	
z_W	$z_S = 6$	$z_S = 8$
6	0.222	0.333
9	0.153	0.230
12	0.118	0.177
14	0.102	0.154

The area of the wire gussets is

$$\Delta A = \frac{q \cdot d_S^2}{\cos \beta}. \quad (1.16)$$

The factor q for 80% of the real area of the wire gussets and the usual number of outer strand wires is listed in Table 1.13 and this is valid for layer angles of the outer strand wires $\alpha = 12\text{--}18^\circ$. For more information see Jenner (1992).

The clearance between the strands in the cross-section can be expressed precisely enough in a simple form according to Shitkow (1957) as

$$s_{SQ} = (d_m - d_S) \cdot \sin \frac{\pi}{z_S} - \frac{d_S}{\cos \frac{\pi}{z_S} \cdot \sqrt{\cos^2 \beta + \tan^2 \frac{\pi}{z_S}}} \quad (1.17)$$

and the real clearance between the strands is

$$s_S = s_{SQ} \cdot \cos \beta. \quad (1.18)$$

The relation between the cross-section of the fibre core in a rope with the actual diameter A_F/d_m^2 and the clearance between the strands s_S/d_S has been calculated with the equations given earlier and drawn in Fig. 1.42. This figure is valid for strand lay angle $\beta = 20^\circ$ and the usual wire lay angles α .

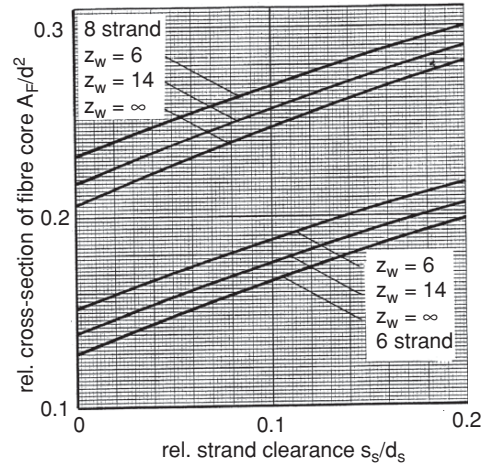


Fig. 1.42. Relative cross-section of the fibre core for ropes with strand lay angle $\beta = 20^\circ$, Jenner (1992)

Table 1.14. Variables and constants for the calculation of the actual rope diameter, Jenner (1992)

Core	IWRC	WSC	PWRC	ESWRC	under the specific tensile force S/d^2
y	$\frac{d_m}{d_{cal}}$	$\frac{d_m}{d_{cal}}$	$\frac{d_m}{d_{cal}}$	$\frac{d_m}{d_{S2}}$	
x_1	$\frac{s_{S2}}{d_{S2}}$	$\frac{s_{S1}}{d_{S1}}$	$\frac{s_{S2}}{d_{S2}}$	$\frac{s_{S2}}{d_{S2}}$	
x_2	$\frac{\delta_2}{\delta_1}$	$\frac{\delta_1}{\delta_0}$	$\frac{\delta_2}{\delta_1}$	$\frac{A}{d_{S2}^2}$	
a_0	0.9924	0.7855	1.026	3.2146	
a_1	-0.1206	0.1587	-0.2375	-0.2216	$S/d^2 = 0$
a_2	-0.0156	0.2095	-0.0226	0.0921	
a_0	0.9759	0.8748	1.0069	2.5867	
a_1	-0.1555	-0.1116	-0.0392	-0.1354	$S/d^2 = 117$
a_2	-0.0117	0.1115	-0.0210	0.1781	N/mm^2

Symbols: d_m : actual rope diameter; d_{cal} : rope diameter calculated with incompressible round strands; d_{S1} , d_{S2} : diameter of the strands 1, 2; A: metallic cross-section of the wire rope

δ_0 , δ_1 , δ_2 : outer wire diameter of the strands 0, 1 and 2

s_{S1} , s_{S2} : clearance between the strands 1 and 2

1.6.4 Steel Core

Unlike in wire ropes with fibre cores, wire ropes with steel cores especially with the usual independent wire rope core (IWRC) and the wire strand core (WSC) should not have high clearance between the strands.

When dimensioning the independent wire rope core, it has to be considered that the wires of the core and the strands comb each other in a complicated way. Jenner (1992) found the results given by a geometrically based calculation with the necessary assumptions are no better than those from a regression calculation using rope measurements. The regression equation for the ropes with different steel cores is

$$y = a_0 + a_1 \cdot x_1 + a_2 \cdot x_2. \quad (1.19)$$

The meanings of the variables y , x_1 and x_2 differ for the different types of steel cores. These meanings and the constants a_i are listed in Table 1.14. To evaluate the constants, Jenner carried out a great number of measurements on wire ropes with steel cores. Using the regression calculation, he found a standard deviation for the actual rope diameter of about 1.5% for wire ropes under no tensile force and of about 1% under the specific tensile force $S/d^2 = 117 \text{ N/mm}^2$.

References

- Albert, W. A. J.: On the manufacture of whim ropes from iron wires. The Mining Journal and Commercial Gazette, Suppl. XII, Feb. 25, 1837 (extracts from Foreign Scientific Works V), pp. 47–48
- Andorfer, K.: Die Zugkraftverteilung in schwingend beanspruchten geraden Drahtseilen. Diss Techn. Universität Graz 1983
- Apel, G. and Nünninghoff, R.: Einflüsse des Zinkschichtaufbaus auf das Ziehergebnis beim Naßziehen dünner, hochfester Stahldrähte. Stahl und Eisen 99 (1979) 25/26, pp. 1482–1486
- Apel, G. and Nünninghoff, R.: Einfluß der Werkstoffalterung auf die Eigenschaften hochfester dünner Stahldrähte. Stahl und Eisen 103 (1983) 24, pp. 1275–1281
- Apel, G.: The stranding factor. WIRE 36 (1986) 3, pp. 137–140 and 7, pp. 279–282
- Becker, K.: On the fatigue strength of wire ropes. OIPEEC Round Table 1977, Luxembourg, Chaps. 1–3
- Benoit, G.: Zum Gedächtnis von W. A. J. Albert und die Erfindung des Drahtseile. Berlin: VDI verlag, 1935
- Birkenmaier, M.: Fatigue resistant tendons for cable-stayed constructions. IABSE-Periodica 2 (1980), pp. 65–79. Hersg. ETH Zürich
- Blanpain, J.: Einfluß der Hartzinkschicht auf die mechanischen Eigenschaften feuerverzinkter Drähte. Stahl und Eisen 84 (1964) 24, pp. 1576–1585

- Briem, U.: Umlaufbiegewechselzahl von Seildrähten. DRAHT 51 (2000) 3, pp. 73–76
- Buchholz, H.: Die Beeinflussung der Dauerfestigkeit der Stähle durch werkstoffliche und technologische Faktoren. IFL-Mitteilungen 4 (1965) 1, pp. 3–6
- Chaplin, C. R. and Potts, A. E.: Wire Rope Offshore – A Critical Review of Wire Rope Endurance Research Affecting Offshore Applications HSE. Publication OTH 91 341, HMSO London, June 1991
- Costello, G. A.: Theory of Wire Rope, Second Edition, Springer, Berlin Heidelberg New York, 1997 ISBN 0–387–98202–7
- Dillmann, U. and Gabriel, K.: Die Streuung von Werkstoffkennwerten–Hochfester Stahldraht Archiv für Eisenhüttenwesen 53 (1982) 5, pp. 181–188
- Donandt, H.: Über denStand unserer Kentnisse in der Frage der Grenzschmierung. Z. VDI 80 (1936) 27, pp. 821–829
- Donandt, H.: Zur Dauerfestigkeit von Seildraht und Drahtseil. Archiv für das Eisenhüttenwesen 21 (1950) 9/10, pp. 283–292
- Dopler, T. M., Nistelberger, F., Jeglitsch, G., and Hampejs: Hydrogen-induced fracturing of patented steel wire during hot-dip galvanizing. WIRE 46 (1996) 1, pp. 15–21
- Erlinger, E.: Umlaufbiegemaschine für Drahtproben. Fördertechnik 5/6 (1942) 3, pp. 43–45
- Faulhaber, R.: Über den Einfluß des Probestabdurchmessers auf die Biegeschwingfestigkeit von Stahl. Mitt. Forschungsinst. Verein. Stahlwerke AG 3 (1932/1933), pp. 153–172
- Feyrer, K.: Effect of bending length on endurance of wire ropes. Wire World 23 (1981), pp. 115–119
- Feyrer, K. and Jenner, T.: Der Querschnitt eines Spiralseiles mit beliebig profilierten Drähten. DRAHT 38 (1987) 12, pp. 939–941
- Feyrer, K.: Reference values for the evaluation of wire rope tests. OIPEEC Bulletin 63. Reading, May 1992. ISSN 1018–8819. Copy: Wire Industry 55 (1992) August, pp. 593–594
- Feyrer, K.: Nachschmierung von laufenden Drahtseilen. DRAHT 49 (1998) 1, pp. 40–46
- Fuchs, D.: Die Verbesserung der Qualität von Förderseilen durch Optimierung des Litzenaufbaus und des Seilaufbaus. Bergbau 35 (1984) 2, 48–51
- Gräbner, P. and Gwenetadse, M.: New research results of lubrication on steel wire ropes. OIPEEC Round Table Delft, 1993
- Greis, P.: Untersuchung über die Lebensdauer von Drähten und Seilen von Krananlagen. Stahl und Eisen 99 (1979) 10, pp. 518–524
- Griffioen, W. U. L. and Wiek: Eine exakte Methode zur Lösung des Berührungsproblems in Stahldrahtseilen in einer Verseillage, DRAHT 43 (1992) 3, pp. 236–239
- Groß, S.: Ein Beitrag zur Geometrie des Drahtseiles. DRAHT 5 (1954) 5, pp. 173–176

- Häberle, B.: Pressung zwischen Drahtseil und Seilscheibe. Diss. Universität Stuttgart, 1995
- Haid, K.-D.: Determination of forces in strand wires. WIRE 33 (1983) 1
- Hankus, J.: The actual breaking force of steel wire ropes. OIPEEC Bulletin 45 Torino, 1983, pp. 101–120
- Hempel, M.: Stand der Erkenntnisse über den Einfluß der Proben-größe auf die Dauerfestigkeit. DRAHT 8 (1957) 9, pp. 385–394
- Hruska, F.: Geometrie im Drahtseil. DRAHT 4 (1953) 5, pp. 173–176
- Jehmlich, G.: Einfluß der Korrosion auf die Wöhlerkuurve von Seildrähten. Bergakademie 21 (1969) 1, pp. 45–47
- Jenner, T.: Ein Beitrag zur Geometrie der Drahtseile. Diss. Universität Stuttgart, 1992
- Kieselstein, S. and Wißuwa, E.: Effective processing of wire rod through draw-peeling. WIRE (2005) 2, pp. 34–36
- Luo, B.: Überprüfung und Weiterentwicklung der Zuverlässigkeits-modelle im Maschinenbau mittels Mono-Bauteil-Systemen. Diss. Universität Stuttgart, 2002
- Lutz, D.: Entwicklung eines Dauerprüfverfahrens zur Ermittlung der Verwendbarkeit von Seildraht. Diss. TH Aachen, 1972
- Marcol, J. and Miculec, Z.: Problematik des Patentierens von Stahldraht. Draht-Welt 72 (1986) 1/2, pp. 3–7
- Matsukawa, A. U. A.: Fatigue resistance analysis of parallel wire strand cables—part 2. Stahlbau 57 (1988) 7, pp. 205–210
- Meebold, R.: Die Drahtseile in der Praxis. Springer, Berlin Heidelberg New York, 1959
- Molkow, M.: Die Treibfähigkeit von gehärteten Treibscheiben mit Keilrillen. Diss. Universität Stuttgart, 1982. Kurzfassung dhf 29 (1983) 7/8, pp. 209–217
- Müller, H.: The properties of wire rope under alternating stresses. Wire World 3 (1961) 5, pp. 249–258
- Müller, H.: Drahtseile im Kranbau, Auswahl und Betriebsverhalten. VDI-Berichte No. 98 und dhf 12 (1966) 11, pp. 714–716 and 12, pp. 766–773
- Müller, H.: Vortrag bei der Jahreshauptversammlung der Drahtseil-Vereinigung am, 1977
- Naumann, B. and Gedecke, G.: Einfluss von Seilschmierstoffe auf Synthese-Fasereinlagen für Drahtseile. DRAHT 22 (1971) 8, pp. 542–545
- NN: A new form of fatigue testing machine for wire. Engineering (1933), p. 567
- Nünninghoff, R. U. K. and Sczepanski: Galfan – an improved corrosion protection for steel wire. WIRE 37 (1987) 3, pp. 240–243 and 4, pp. 321–324
- Nünninghoff, R.: Langzeiterfahrung mit Galfan. DRAHT 54 (2003) 2, pp. 37–39
- Oplatka, G.: Nachschmierung von Drahtseilen. Int. Kolloquium, “150 Jahre Drahtseil” Techn. Akademie Esslingen 13th and 14th Sep. 1984, pp. 1.11–1.16

- Oplatka, G. and Vaclavik, P.: Relubrication of moving stranded wire ropes. WIRE 46 (1996) 2, pp. 132–136
- Pantucek, P.: Pressung von Seildraht unter statischer und dynamischer Beanspruchung. Diss. Universität Karlsruhe, 1977
- Pfister, H. R.: Dauerprüfung von Seildrähten. Dr.-Ing. Diss. TH Stuttgart, 1964
- Pomp, A. and Hempel, M.: Dauerprüfung von Stahldrähten unter wechselnder Zugbelastung. Mitt. Kaiser-Wilh.-Institut für Eisenforschung. Abh. 334 (1937) and Abh. 340 (1938) Düsseldorf
- Reemsnyder, H.: Homer Research Laboratories Reports–August 14, 1969, pp. 1414–1424, 1505–1566
- Reuleaux, F.: Der Konstrukteur. 1. Aufl., Braunschweig: Vieweg 1861
- Sayenga, D.: John Roebling’s initial studies of wire rope endurance and the creation of Three-Size (Warrington) construction. OIPEEC Round Table Reading, 1997, pp. 1–16
- Sayenga, D.: The advent of wire rope, “Constructions”, 1800–1850. OIPEEC Bulletin 86, Reading, Dec. 2003, pp. 21–41
- Schiffner, G.: Spannungen in laufenden Drahtseilen. Diss. Universität Stuttgart, 1986
- Schmidt, K.: Die sekundäre Zugbeanspruchung der Drahtseile aus der Biegung. Diss. TH Karlsruhe, 1964
- Schmidt, W. and Dietrich, H.: Mechanische Eigenschaften kaltgezogener Drähte verschiedener rostfreier Stähle. DRAHT 33 (1982) 3, pp. 111–115 and 4, pp. 166–169
- Schneider, F. and Lang, G.: Stahldraht. Leipzig: VEB Verlag für Grundstoffindustrie, 1973
- Shitkow, D. G. and Pospechow, I. T.: Drahtseile. VEB Verlag Technik, Berlin, 1957
- Siebel, E.: Die Prüfung der metallischen Werkstoffe. 2. Aufl. Springer, Berlin, Heidelberg New York, 1959
- Singenstoth, F.: Das Herz des Drahtseiles – eine Beurteilung der Einlagen. Int. Kolloquium, “150 Jahre Drahtseil” Techn. Akademie Esslingen 13th and 14th, Sep. 1984
- Sivatz, F.: Einflüsse der Seilkonstruktion auf die Einlagenbemessung. Seilbahnbuch 1975, pp. 63–66. Beilage zur ISR
- Unterberg, H.-W.: Die Dauerfestigkeit von Seildrähten bei Biegung und Zug. Diss. TH Karlsruhe, 1967
- Verreet, R.: 100 years of equal-laid wire rope. WIRE 38 (1988) 2, pp. 223–224
- Verreet, R.: Die Geschichte des Drahtseiles. Drahtwelt 75 (1989) 6, pp. 100–106
- Voigt, P. G.: Technical manufacturing of ropes with large diameters and long lengths. OIPEEC Round Table, East Kilbride Glasgow, June 1985
- Votta, F. A.: New wire fatigue testing method. The IRON AGE (1948), pp. 78–81

- Wang, R. C. and McKewan, W. M.: A model for the structure of round-strand wire rope. OIPEEC Bulletin No. 81. Reading, June 2001, pp. 15–42, ISSN 1018 8819
- Wehking, K.-H.: Zukunftsausrichtung des IFT im Bereich der Seiltechnik. 2. Internationaler Stuttgarter Seiltag. 17.a.18. Feb. 2005, pp. 1–9
- Wiek, L.: Computerized rope design and endurance. OIPEEC Round Table, Oct. 1977
- Wiek, L.: Beziehung zwischen Theorie, Versuch, Anwendung und Herstellung von Drahtseilen. DRAHT 36 (1985) 1, pp. 12–15
- Winkler, L.: Beitrag zum Fragenkomplex Schmierung von Drahtseilen. Diss. TH Karl-Marx-Stadt, 1971
- Woernle, R.: Ein Beitrag zur Klärung der Drahtseilfrage. Z. VDI 73 (1929) 13, pp. 417–426
- Wolf, R.: Zur Beschreibung der vollständigen Seilkinematik. Forschung Ing.-Wesen 50 (1984) 8, pp. 81–86
- Wolf, E.: Seilbedingte Einflüsse auf die Lebensdauer laufender Drahtseile. Dr.-Ing. Diss. Universität Stuttgart 1987. Kurzfassung: DRAHT 39 (1987) 11, pp. 1088–1093
- Wyss, Th.: Die Stahldrahtseile. ...Zürich: Schweizer Druck- und Verlagshaus, 1956
- Ziegler, S., Vogel, W. and Wehking, K.-H.: Influence of wire fatigue strength on rope lifetime. WIRE (2005) 3, pp. 44–48

Wire Ropes under Tensile Load

2.1 Stresses in Straight Wire Ropes

The wires in straight wire ropes under tensile load are mainly strained by tensile stresses. The real tensile stress in the wires will not be considered in most cases. Instead of this the stress condition will be normally characterised globally by the rope diameter related tensile force S/d^2 or by the rope tensile stress (nominal tensile stress). This global rope tensile stress is

$$\sigma_z = \frac{S}{A}.$$

In this, S is the rope tensile force and A is the wire rope cross-section, that means the sum of the cross sections of all wires in the rope with the diameters δ_i is

$$A = \frac{\pi}{4} \sum \delta_i^2.$$

The real stress in wires of the layer k is named σ_{tk} in opposition to the (global) rope tensile stress σ_z . The real wire tensile stress σ_{tk} is bigger than the rope tensile stress σ_z . In addition to the tensile stresses, the wires in ropes under tensile force are strained by bending and torsion stresses and normally slightly by pressure. The stresses in all the individual wires are different:

- Systematically according to the different lay angles of the wire and the strand layers and
- Unsystematically because wires or strands very often are lying loosely on their base and therefore do not start to take up the load from the beginning by increasing the tensile force of the rope.

The unsystematic working stresses may be bigger in some cases than the systematic ones. Of course, they cannot be calculated but their influence can always be observed especially in the rope endurance under fluctuating tensile forces.

The systematic working stresses will be determined in the following chapter. Thereby, it will be presupposed that all wires are without self-contained stresses and no wires or strands are loose, so that all wires start to bear when the wire rope will be under a slight tensile force. Furthermore, it will be presupposed that all stresses remain in the elastic region.

The self-contained stresses of the wires resulting from their manufacture will not be considered. These unknown stresses have no importance in the case of static loads. In case of fluctuating loading, they influence the endurance like an increasing or a decreasing of the middle stress.

2.1.1 Basic Relation for the Wire Tensile Force in a Strand

A tensile force loading a strand induces a torque because of the helix form of the wires. Therefore the strand will be turning if the strand ends are not secured against this. In practical usage, the turning of strands and ropes must be prevented because otherwise the strand loosens its structure and because of this very unequal stresses would be induced in the wires. For normal ropes, the turning can be only prevented securing the rope ends. In so-called non-rotating ropes, the turning is more or less prevented because the torque of different right or left wound wire layers or strand layers compensate each other. In the following it will be presupposed that the turning of the strands and ropes are prevented.

For one wire, the portion of the tensile strand force S_i in strand axis direction and the corresponding portion of the circumference force U_i out of torque act as outer forces on the single wire in the wire layer i of a strand. The division of the strand tensile force and torque in the wire forces S_i and U_i will be described later on. For the present, S_i and U_i will be presupposed as known.

Both outer forces S_i and U_i on a wire must be in balance with the inner forces, the wire tensile force F_i and the wire shear force Q_i . The forces on a wire of a wire layer i are shown in Fig. 2.1. From these, using α_i for the lay angle, both of the following equations can be derived

$$F_i = \frac{S_i - Q_i \cdot \sin \alpha_i}{\cos \alpha_i} \quad (2.1)$$

and

$$U_i = F_i \cdot \sin \alpha_i - Q_i \cdot \cos \alpha_i. \quad (2.2)$$

The shear force Q_i of a wire of layer i is caused by the bending and torsion of this wire, of course geometrically limited by the rope extension. As was first presented by Berg (1907), the shear force of a wire in layer i is

$$Q_i = \frac{\sin \alpha_i}{r_i} \cdot (M_{b,i} \cdot \cos \alpha_i - M_{tor,i} \cdot \sin \alpha_i) \quad (2.3)$$

with the wire winding radius $r_{W,i} = r_i$, the bending moment $M_{b,i}$ around the binormal and the torque $M_{tor,i}$ around the wire axis. With this the tensile

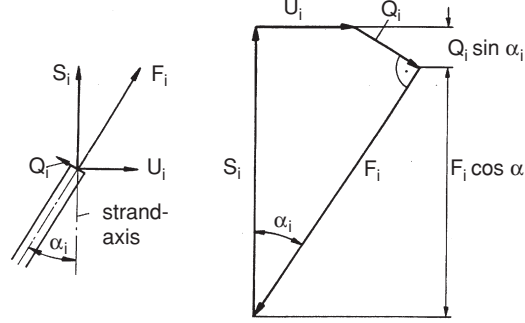


Fig. 2.1. Forces on the wire of a strand

force in a wire of the wire layer i is

$$F_i = \frac{S_i}{\cos \alpha_i} - \frac{\sin^2 \alpha_i}{r_i \cdot \cos \alpha_i} \cdot (M_{b,i} \cdot \cos \alpha_i - M_{\text{tor},i} \cdot \sin \alpha_i). \quad (2.4)$$

According to Berg (1907), the portion of the strand torsion moment caused by a wire of the wire layer i is

$$M_i = F_i \cdot r_i \cdot \sin \alpha_i - Q_i \cdot r_i \cdot \cos \alpha_i + M_{b,i} \cdot \sin \alpha_i + M_{\text{tor},i} \cdot \cos \alpha_i. \quad (2.5)$$

The most recent equations for bending and torsion moment for a wire of the wire layer i were developed by Czitary (1952) as follows

$$M_{b,i} = E_i \cdot J_i \cdot \left(\frac{\sin^2 \alpha_i}{r_i} - \frac{\sin^2 \alpha_{0i}}{r_{0i}} \right) \quad (2.6)$$

and

$$M_{\text{tor},i} = G_i \cdot J_{pi} \cdot \left(\frac{\sin \alpha_i \cdot \cos \alpha_i}{r_i} - \frac{\sin \alpha_{0i} \cdot \cos \alpha_{0i}}{r_{0i}} \right). \quad (2.7)$$

In addition to the known symbols, there is E_i the elasticity module, G_i the shear module, J_i the equatorial and J_{pi} the polar moments of inertia of a wire in the wire layer i . The index 0 means the state before loading by a tensile force. As before, the parameters without the index 0 show the loaded state.

The portion of the strand torque for one wire of the wire layer i can be calculated from (2.3), (2.5)–(2.7)

$$M_i = F_i \cdot r_i \cdot \sin \alpha_i - M_{b,i} \cdot \sin \alpha_i \cdot (1 + \cos^2 \alpha_i) + M_{\text{tor},i} \cdot \cos^3 \alpha_i. \quad (2.8)$$

Both of the moments M_{bi} and M_{ti} are very small, because the lay angle and the winding radius alter only slightly under the tensile load. Therefore, the shear force Q_i is also very slight. As demonstrated by Czitary (1952), both moments and the shear force can be neglected for the calculation of the wire tensile force F_i . This neglect only results in a very minimal deviation. With

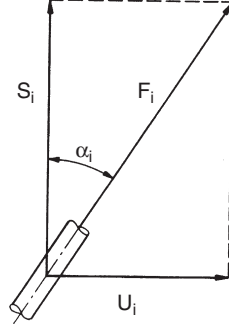


Fig. 2.2. Tensile force of a strand wire neglecting the small shear force

this, out of (2.1) the simple relation for the tensile force in a wire in the layer i depicted in Fig. 2.2 is

$$F_i = \frac{S_i}{\cos \alpha_i} \quad (2.9)$$

and the circumference force out of (2.2) is

$$U_i = F_i \cdot \sin \alpha_i$$

or

$$U_i = S_i \cdot \tan \alpha_i. \quad (2.10)$$

According to (2.4), the portion of the strand torque for a wire in the layer i is now

$$M_i = F_i \cdot r_i \cdot \sin \alpha_i$$

or

$$M_i = S_i \cdot r_i \cdot \tan \alpha_i. \quad (2.11)$$

These equations from Berg (1907) have since been used by nearly all researchers, as for example Heinrich (1937), Costello (1977); Costello and Sinha (1977b). Only Dreher (1933), who first did extensive investigations into wire rope torsion has introduced a basic equation deviating from (2.9). But Dreher's equation is of no value for use with real wire ropes as Heinrich (1942) has already shown. Dreher's equation is only true for a simple wire helix not supported by a strand centre.

A length-related radial force exists between the wire helix and the centre wire or a wire layer (or between a helix strand and the core) in a wire rope under a tensile force. The length-related radial force (when neglecting the bending moment and torque) is

$$q_i = \frac{F_i}{\rho_i} = \frac{F_i \cdot \sin^2 \alpha_i}{r_i}. \quad (2.12)$$

2.1.2 Wire Tensile Stress in the Strand or Wire Rope

Wire Tensile Stress in Strand or Spiral Rope

The first to work out a partition of the wire rope tensile force in wire tensile forces was Benndorf (1904). The following determination of the tensile stress follows his work. Out of the last chapter with (2.9), the wire tensile force component in strand axis direction (neglecting the small shear force) is

$$S_i = F_i \cdot \cos \alpha_i.$$

The strand tensile force is the sum of all wire tensile force components

$$S = \sum_{i=0}^n z_i \cdot S_i = \sum_{i=0}^n z_i \cdot F_i \cdot \cos \alpha_i \quad (2.13)$$

In addition to the known symbols, n is the number of wire layers counted from the inside with $n = 0$ for the centre wire and z_i is the number of wires in the wire layer i .

For the following, it will be presupposed that the strand cross-section rests plane if the strand with the length l_S is elongated with Δl_S by a tensile force. The elongation can now be calculated and, from this, the tensile force of all the wires. The tensile force of a wire in a wire layer i is

$$F_i = \frac{\Delta l_i}{l_i} \cdot E_i \cdot A_i. \quad (2.14)$$

l_i is the wire length, Δl_i the wire elongation, E_i the elasticity module and A_i the cross-section of a wire in the wire layer i . The extension of that wire is

$$\varepsilon_i = \frac{\Delta l_i}{l_i}. \quad (2.15)$$

With l_S for the length of the strand, the length of the wire is

$$l_i = \frac{l_S}{\cos \alpha_i}. \quad (2.16)$$

In Fig. 2.3, the unwound wire about the strand axis is shown before and after the strand elongation. Therefore, when the failures of higher classification are neglected, then the wire elongation is

$$\Delta l_i = (\Delta l_S - \Delta u_i \cdot \tan \alpha_i) \cdot \cos \alpha_i$$

or

$$\Delta l_i = \Delta l_S \cdot \cos \alpha_i - \Delta u_i \cdot \sin \alpha_i. \quad (2.17)$$

The contraction of the winding radius respectively the circumference in relation to the wire extension – that transverse contraction ratio can also be

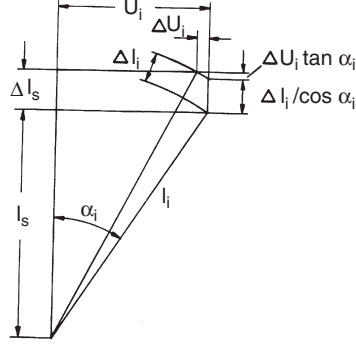


Fig. 2.3. Elongation of a strand wire

designated as “Poisson’s ratio” of the wire helix – is

$$\nu_i = \frac{\Delta u_i / u_i}{\Delta l_i / l_i}.$$

Here u_i is the winding circumference and Δu_i its contraction. Using this and (2.15), the contraction of the winding circumference is

$$\Delta u_i = \varepsilon_i \cdot \nu_i \cdot u_i$$

and with $u_i = l_i \cdot \sin \alpha_i$ is

$$\Delta u_i = \varepsilon_i \cdot \nu_i \cdot l_i \cdot \sin \alpha_i. \quad (2.18)$$

Using (2.17) and (2.18), the wire elongation is then

$$\Delta l_i = \Delta l_S \cdot \cos \alpha_i - \varepsilon_i \cdot \nu_i \cdot l_i \cdot \sin^2 \alpha_i$$

or

$$\Delta l_i + \Delta l_i \cdot \nu_i \cdot \sin^2 \alpha_i = \Delta l_S \cdot \cos \alpha_i.$$

Following this, the elongation of a wire in the wire layer i is

$$\Delta l_i = \frac{\Delta l_S \cdot \cos \alpha_i}{1 + \nu_i \cdot \sin^2 \alpha_i}. \quad (2.19)$$

This equation together with (2.14) and (2.16) supplies the tensile force of a wire in the wire layer i as a function of the strand elongation

$$F_i = \frac{\Delta l_S \cdot \cos^2 \alpha_i}{l_S \cdot (1 + \nu_i \cdot \sin^2 \alpha_i)} \cdot E_i \cdot A_i \quad (2.20)$$

or its component in the direction of the strand

$$S_i = \frac{\Delta l_S \cdot \cos^3 \alpha_i}{l_S \cdot (1 + \nu_i \cdot \sin^2 \alpha_i)} \cdot E_i \cdot A_i. \quad (2.20a)$$

Using (2.13) the strand tensile force is

$$S = \frac{\Delta l_S}{l_S} \cdot \sum_{i=0}^n \left(\frac{z_i \cdot \cos^3 \alpha_i}{1 + \nu_i \cdot \sin^2 \alpha_i} \cdot E_i \cdot A_i \right). \quad (2.21)$$

The tensile force in a wire of a specific wire layer k is found by combining (2.20) and (2.21) with the elimination of $\Delta l_S/l_S$

$$F_k = \frac{\frac{\cos^2 \alpha_k}{1 + \nu_k \cdot \sin^2 \alpha_k} \cdot E_k \cdot A_k}{\sum_{i=0}^n \left(\frac{z_i \cdot \cos^3 \alpha_i}{1 + \nu_i \cdot \sin^2 \alpha_i} \cdot E_i \cdot A_i \right)} \cdot S. \quad (2.22)$$

The tensile stress in this wire is

$$\sigma_{tk} = \frac{F_k}{A_k} = \frac{\frac{\cos^2 \alpha_k}{1 + \nu_k \cdot \sin^2 \alpha_k} \cdot E_k}{\sum_{i=0}^n \left(\frac{z_i \cdot \cos^3 \alpha_i}{1 + \nu_i \cdot \sin^2 \alpha_i} \cdot E_i \cdot A_i \right)} \cdot S. \quad (2.23)$$

Wire Tensile Stress in Stranded Ropes

As before, the same derivation can be used for the stranded rope by now observing a strand as a wire. The wire layers keep the counting index i and a certain wire layer the index k , whereas the strand has the respective indices j and l . The total number of wire layers in a strand is n_W and the total number of strand layers is n_S .

The wire rope tensile force is according to (2.13)

$$S = \sum_{j=0}^{n_S} F_j \cdot z_j \cdot \cos \beta_j$$

and with the strand tensile force

$$F_j = \sum_{i=0}^{n_{Wj}} F_{ij} \cdot z_{ij} \cdot \cos \alpha_{ij}$$

the wire rope tensile force is

$$S = \sum_{j=0}^{n_S} z_j \cdot \cos \beta_j \cdot \sum_{i=0}^{n_{Wj}} F_{ij} \cdot z_{ij} \cdot \cos \alpha_{ij}. \quad (2.24)$$

According to (2.20), the wire tensile force in the wire layer i of the strand j is

$$F_{ij} = \frac{\Delta l_j}{l_j} \cdot \frac{\cos^2 \alpha_{ij}}{1 + \nu_{ij} \cdot \sin^2 \alpha_{ij}} \cdot E_{ij} \cdot A_{ij} \quad (2.25)$$

and according to (2.19) and the wire rope length $L = l_j \cdot \cos \beta_j$

$$\frac{\Delta l_j}{l_j} = \frac{\Delta L}{L} \cdot \frac{\cos^2 \beta_j}{1 + \nu_j \cdot \sin^2 \beta_j}. \quad (2.26)$$

Then, using (2.25) and (2.26), the tensile force of a wire ij is

$$F_{ij} = \frac{\Delta L}{L} \cdot \frac{\cos^2 \beta_j}{1 + \nu_j \cdot \sin^2 \beta_j} \cdot \frac{\cos^2 \alpha_{ij}}{1 + \nu_{ij} \cdot \sin^2 \alpha_{ij}} \cdot E_{ij} \cdot A_{ij}. \quad (2.27)$$

Using (2.27) and (2.24), the wire rope tensile force is

$$S = \frac{\Delta L}{L} \cdot \sum_{j=0}^{n_s} \left(z_j \cdot \frac{\cos^3 \beta_j}{1 + \nu_j \cdot \sin^2 \beta_j} \cdot \sum_{i=0}^{n_{wj}} z_{ij} \cdot \frac{\cos^3 \alpha_{ij}}{1 + \nu_{ij} \cdot \sin^2 \alpha_{ij}} \cdot E_{ij} \cdot A_{ij} \right). \quad (2.28)$$

Combining (2.27) and (2.28) by eliminating $\Delta L/L$, the tensile force in the certain wire k in the strand l is

$$F_{kl} = \frac{\frac{\cos^2 \beta_l}{1 + \nu_l \cdot \sin^2 \beta_l} \cdot \frac{\cos^2 \alpha_{kl}}{1 + \nu_{kl} \cdot \sin^2 \alpha_{kl}} \cdot E_{kl} \cdot A_{kl}}{\sum_{j=0}^{n_s} \left(z_j \cdot \frac{\cos^3 \beta_j}{1 + \nu_j \cdot \sin^2 \beta_j} \cdot \sum_{i=0}^{n_{wj}} z_{ij} \cdot \frac{\cos^2 \alpha_{ij}}{1 + \nu_{ij} \cdot \sin^2 \alpha_{ij}} \cdot E_{ij} \cdot A_{ij} \right)} \quad (2.29)$$

and the tensile stress in that wire is

$$\sigma_{tkl} = \frac{F_{kl}}{A_{kl}}. \quad (2.30)$$

Influence of the Poisson Ratio

The Poisson ratio (transverse contraction ratio) for steel $\nu_i = 0.3$ can also be used for the steel wire helix in the strands. Because the length-related radial force between the wires is very small, the reduction of the wire diameter and winding radius or circumferences in the strands, in the spiral ropes and in the strands of the stranded ropes are practically only caused by the elongation of the wires. This is especially true for the most frequently used parallel lay ropes.

The transverse contraction ratio of the strand helix ν_j in stranded ropes is difficult to estimate. Especially in wire ropes with a fibre core, this “Poisson ratio” is very large. In any case of fluctuating force, there is a great part of the rope contraction and the rope elongation remaining.

The influence of the Poisson ratio of the wires and the “Poisson ratio” of the winding radius or circumferences of wires on the calculated distribution of the wire tensile forces is normally not very large. For strands, the influence

reduces with the increasing number of wires. For a parallel wire strand with 19 wires, the calculated stress of the outer wires is at the most 2% more and that of the centre wire 3% less if the Poisson ratios are neglected.

The influence of the Poisson ratios of wires and of winding circumferences of wires and strands on the wire tensile stress is also small for the stranded ropes. This is true for ropes with steel cores because the “Poisson ratio” ν_j is also small. For ropes with fibre cores, the contraction can be quite large. However, the influence on the distribution of tensile forces of the strands is small.

However, unlike with the calculation of the wire tensile stresses, the “Poisson ratio” ν_j must be used as precisely as possible if the equations given here are to be used later on to calculate the additional stresses, the rope elongation or the rope elasticity module. The Poisson ratio $\nu = 0.3$ can continue to be used for the strands and spiral wire ropes. But that is not valid for the strand helix (strand axis) in stranded ropes.

The cross-section of fibre cores and their effective diameter are very greatly reduced under the effect of the length-related radial force of the bearing strands. This is also true to a lesser extent for wire ropes with steel cores especially for wire ropes with several strand layers especially if the strand layers lie parallel. The “Poisson ratio” of the strand winding radius of these wire ropes is not constant as it depends on the wire rope stress. The “Poisson ratio” of the stranded wire ropes can generally only be evaluated by measurement and not by calculation.

Wire Tensile Stress Neglecting the Poisson Ratios

If the tensile force of the wires in strands or wire ropes is calculated by neglecting the “Poisson ratios”, the equations are much simpler. The tensile force in a strand is in this case

$$S = \frac{\Delta l_S}{l_S} \cdot \sum_{i=0}^{n_W} z_i \cdot \cos^3 \alpha_i \cdot E_i \cdot A_i \quad (2.21a)$$

and the tensile force in the wire k is

$$F_k = \frac{\cos^2 \alpha_k \cdot E_k \cdot A_k}{\sum_{i=0}^{n_W} z_i \cdot \cos^3 \alpha_i \cdot E_i \cdot A_i} \cdot S. \quad (2.22a)$$

By neglecting the contraction, the tensile force of the stranded rope is

$$S = \frac{\Delta L}{L} \cdot \sum_{j=0}^{n_S} \left(z_j \cdot \cos^3 \beta_j \cdot \sum_{i=0}^{n_{Wj}} z_{ij} \cdot \cos^3 \alpha_{ij} \cdot E_{ij} \cdot A_{ij} \right) \quad (2.28a)$$

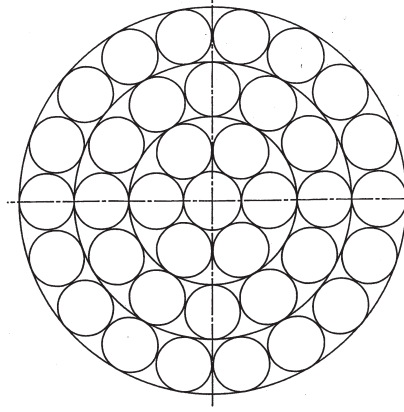


Fig. 2.4. Cross-section of a spiral rope 1×37 , wire diameters $\delta_0 = 1.35$ mm, $\delta_1 = \delta_2 = \delta_3 = 1.25$ mm; wire cross sections $A_0 = 1.431$ mm², $A_1 = A_2 = A_3 = 1.227$ mm²; lay angles $\alpha_0 = 0^\circ$, $\alpha_1 = 14^\circ$, $\alpha_2 = -14^\circ$, $\alpha_3 = 14^\circ$

and the tensile force in the wire k of the strand l is

$$F_{kl} = \frac{\cos^2 \beta_l \cdot \cos^2 \alpha_{kl} \cdot E_{kl} \cdot A_{kl}}{\sum_{j=0}^{n_S} \left(z_j \cdot \cos^3 \beta_j \cdot \sum_{i=0}^{n_{Wj}} z_{ij} \cdot \cos^3 \alpha_{ij} \cdot E_{ij} \cdot A_{ij} \right)} \cdot S. \quad (2.29a)$$

The wire tensile stresses in the spiral rope and in the stranded rope are

$$\sigma_{tk} = \frac{F_k}{A_k} \quad \text{and} \quad \sigma_{tkl} = \frac{F_{kl}}{A_{kl}}.$$

All wires have nearly the same tensile stress if a wire rope has a fibre core and the same lay angle for all wire layers (except, of course, the centre wires in the strands which have a higher stress than the other wires). This common tensile stress is

$$\sigma_t = \frac{S}{A \cdot \cos \alpha \cdot \cos \beta}. \quad (2.31)$$

This equation was previously given by Wiek (1980). In the outer wires, the tensile stress is a little smaller than as calculated in (2.31).

Example 2.1: Wire tensile stress in spiral wire ropes

Calculation of the tensile stress in the wires of the open spiral wire rope 1×37 according to Fig. 2.4 with the global wire rope tensile stress $\sigma_z = 300$ N/mm².

The tensile force of the spiral rope is

$$S = A_m \cdot \sigma_z = (1.431 + (6 + 12 + 18) \cdot 1.227) \cdot \sigma_z = 45.61 \cdot 300 = 13,680 \text{ N}.$$

Using (2.23), the tensile stress in the centre wire is

$$\sigma_{t0} = \frac{45.61 \cdot \sigma_z}{1.431 + \frac{0.9703^3}{1 + 0.3 \cdot 0.2419^2} \cdot (6 + 12 + 18) \cdot 1.227} = \frac{45.61}{41.09} \cdot \sigma_z$$

$$\sigma_{t0} = 1.110 \cdot \sigma_z = 333 \text{ N/mm}^2.$$

With the same equation (2.23), the tensile stress in the wires of the layers 1, 2 and 3 with the same lay angle $\alpha = 14^\circ$ is

$$\sigma_{t,1,2,3} = \frac{\frac{0.9703^2}{1 + 0.3 \cdot 0.2419^2} \cdot 45.61 \cdot \sigma_z}{41.09}$$

$$\sigma_{t,1,2,3} = 1.027 \cdot \sigma_z = 308 \text{ N/mm}^2.$$

2.1.3 Additional Wire Stresses in the Straight Spiral Rope

A straight spiral rope respectively a straight strand becomes longer and thinner under a tensile force. The wire helix will be deformed and – beside the tensile stress – there exist bending stresses, torsion stresses and radial pressures from the small length-related radial force of the wires. The bending and torsion stresses have to be calculated from the alteration in the space curve of the wire.

The space curve of a wire in a straight strand is in parameter form

$$\begin{aligned} x &= -r \cdot \sin \varphi \\ y &= r \cdot \cos \varphi \\ z &= \frac{r}{\tan \alpha} \cdot \varphi. \end{aligned} \tag{2.32}$$

φ is the angle of rotation (running angle), α the lay angle and $r_W = r$ the wire winding radius, Fig. 2.5. The lay length is

$$h_W = \frac{2 \cdot \pi \cdot r}{\tan \alpha}.$$

Although the moments M_b and M_{tor} out of (2.6) and (2.7) can be neglected in calculating the wire tensile force, the bending and torsion stresses resulting from these moments can be considerable. The stresses come from the change of the curvature K and the winding T .

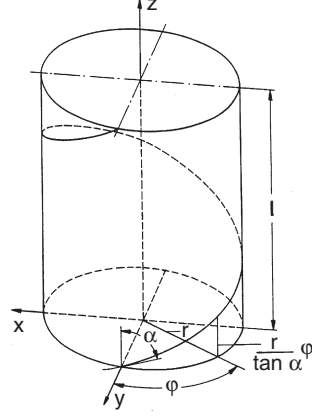


Fig. 2.5. Wire space curve in a straight spiral rope

The curvature K of a space curve is in parameter form according to (2.32) with the curvature radius ρ

$$K = \frac{1}{\rho}$$

$$K = \sqrt{\frac{(x'^2 + y'^2 + z'^2) \cdot (x''^2 + y''^2 + z''^2) - (x' \cdot x'' + y' \cdot y'' + z' \cdot z'')}{(x'^2 + y'^2 + z'^2)^3}}. \quad (2.33)$$

The winding T shows how strongly the space curve differs from the osculating plane in the neighbourhood of a point. The winding is

$$T = \rho^2 \cdot \frac{\begin{vmatrix} x' & y' & z' \\ x'' & y'' & z'' \\ x''' & y''' & z''' \end{vmatrix}}{(x'^2 + y'^2 + z'^2)^3}. \quad (2.34)$$

For the simple case of a wire in a straight strand or spiral rope with the wire winding radius r , the curvature radius ρ is

$$\rho = \frac{r}{\sin^2 \alpha} \quad (2.35)$$

and the winding

$$T = \frac{\sin \alpha \cdot \cos \alpha}{r}. \quad (2.36)$$

The stresses in the wires induced by the alteration of the wire curvature are of special interest. Together with the tensile stresses, they determine the endurance of the strand or spiral rope in the case of fluctuating tensile force.

The bending stress is

$$\sigma_b = \left(\frac{1}{\rho} - \frac{1}{\rho_0} \right) \frac{\delta}{2} E \quad (2.37)$$

or with (2.35)

$$\sigma_b = \left(\frac{\sin^2 \alpha}{r} - \frac{\sin^2 \alpha_0}{r_0} \right) \frac{\delta}{2} E. \quad (2.37a)$$

The torsion stress is

$$\tau = (T - T_0) \frac{\delta}{2} G \quad (2.38)$$

and with (2.36)

$$\tau = \left(\frac{\sin \alpha \cos \alpha}{r} - \frac{\sin \alpha_0 \cos \alpha_0}{r_0} \right) \frac{\delta}{2} G. \quad (2.38a)$$

In addition to the symbols already known, δ is the wire diameter. The index 0 is again of value for the initial state and the symbols without indices designate the state under the effect of tensile force. E is again the elasticity module and G is the shear module. The bending and torsion stresses were first calculated by Schiffner (1986).

Example 2.2: Additional stresses in a spiral rope

Calculation of the bending and torsion stresses in the wires of an open spiral rope according Fig.2.4 with the global wire rope stress $\sigma_z = 300 \text{ N/mm}^2$ (neglecting the influence of the point pressure between the crossing wires).

The winding radius under the effect of the tensile force is (neglecting the small higher tensile stress in the centre wire) with $\sigma_t = \sigma_{t1,2,3}$

$$r_i = r_{0i} \left(1 - \nu \frac{\sigma_t}{E} \right) = r_{0i} \left(1 - 0.3 \frac{308}{196,000} \right) = 0.99953 r_{0i}$$

and the lay angle

$$\sin \alpha = \sin \alpha_0 \cdot \frac{1 - \nu \cdot \frac{\sigma_t}{E}}{1 + \frac{\sigma_t}{E}} = \sin \alpha_0 \cdot \frac{1 - 0.3 \cdot \frac{308}{196,000}}{1 + \frac{308}{196,000}}.$$

$$\alpha_0 = 14^\circ; \quad \sin \alpha_0 = 0.24192; \quad \cos \alpha_0 = 0.97030$$

$$\sin \alpha = 0.9980 \cdot \sin \alpha_0 = 0.9980 \cdot 0.24192 = 0.24144$$

$$\cos \alpha = 0.97042.$$

According to (2.37a), the bending stress in the wires of the different wire layers is

$$\sigma_{bi} = \left(\frac{0.9980^2 \cdot \sin^2 \alpha_0}{0.99953 \cdot r_{0i}} - \frac{\sin^2 \alpha_0}{r_{0i}} \right) \cdot \frac{\delta}{2} \cdot E = 0.00354 \cdot \frac{0.2419^2}{r_{0i}} \cdot \frac{1.25}{2} \cdot 196,000$$

$$\sigma_{bi} = \frac{25.4}{r_{0i}}.$$

According to (2.38a), the torsion stress is

$$\tau_i = \left(\frac{0.23430}{0.99953 \cdot r_{0i}} - \frac{0.23473}{r_{0i}} \right) \cdot \frac{1.25}{2} \cdot 76,000 = \frac{15.2}{r_{0i}}.$$

Then with $r_{01} = 1.3$ mm, $r_{02} = 2.55$ mm, $r_{03} = 3.8$ mm, the bending stresses are

$$\sigma_{b1} = 19.4 \text{ N/mm}^2; \quad \sigma_{b2} = 9.9 \text{ N/mm}^2; \quad \sigma_{b3} = 6.7 \text{ N/mm}^2$$

and the torsion stresses are

$$\tau_1 = 11.7 \text{ N/mm}^2; \quad \tau_2 = 6.0 \text{ N/mm}^2; \quad \tau_3 = 4.0 \text{ N/mm}^2.$$

As shown in the example, the additional wire stresses in spiral ropes are not very large.

2.1.4 Additional Wire Stresses in Straight Stranded Ropes

The wires of straight stranded wire ropes under tensile force are loaded like the wires in spiral ropes by bending and torsion stresses. Besides that, they are loaded with a second tensile stress caused by friction between the wires in the bent strands, Schmidt (1965). The additional stresses will be evaluated using the space curves of the strands and the wires.

According to (2.32), the equations for the space curve of the strand axis in a straight stranded rope are

$$\begin{aligned} x_S &= -r_S \cdot \sin \varphi_S \\ y_S &= r_S \cdot \cos \varphi_S \\ z_S &= \frac{r_S}{\tan \beta} \varphi_S \end{aligned} \tag{2.39}$$

with r_S for the strand winding radius, β for the strand lay angle and φ_S for the angle of rotation of the strand helix. The strand lay length is

$$h_S = \frac{2 \cdot \pi \cdot r_S}{\tan \beta}.$$

Andorfer (1983) derived analytically the equations for the space curve of the double helix of the wire in the straight stranded rope as done before by Bock (1909) using a kinematic method and later on by Wolf (1984) using a vectorial method. The wire winding radius r stands perpendicular on the strand axis helix and the ratio between the wire winding angle φ_W and the strand winding angle φ_S is constant, $\varphi_W/\varphi_S = \text{const}$. Schiffner (1986) pointed out that this constant ratio practically always occurs if the clearance between the wires is – as usual – very small.

The constant ratio between both winding angles φ_W and φ_S is only valid if they both start from $\varphi_W = \varphi_S = 0$. The constant ratio of the winding angles is therefore better described by

$$m^* = \frac{h_S}{h_W \cdot \cos \beta}.$$

That means in any one strand lay length h_S there are m^* wire lay lengths h_W . With $m = m^* \pm 1$ is

$$\varphi_W \pm \varphi_S = m \cdot \varphi_S = \Phi.$$

Bock (1909) nominated Φ as normative phase angle. After $\Phi = 2\pi$, a wire element has the same position as for $\Phi = 0$, Fig. 2.6. The positive sign has to be set for ordinary lay ropes and the negative for lang lay ropes.

To include the case of any phase of φ_W and φ_S a constant winding angle of the wire helix φ_{W0} or shorter φ_0 will be added. Then it is

$$\varphi_W \pm \varphi_S + \varphi_0 = m \cdot \varphi_S + \varphi_0.$$

With this, for the space curve for a wire in a straight stranded wire rope, the equations of Andorfer (1983) are in parameter form

$$\begin{aligned} x &= -r_S \cdot \sin \varphi_S \\ &\quad - r_W \cdot [\cos(\varphi_0 + m \cdot \varphi_S) \cdot \sin \varphi_S + \sin(\varphi_0 + m \cdot \varphi_S) \cdot \cos \beta \cdot \cos \varphi_S] \\ y &= r_S \cdot \cos \varphi_S \\ &\quad + r_W \cdot [\cos(\varphi_0 + m \cdot \varphi_S) \cdot \cos \varphi_S - \sin(\varphi_0 + m \cdot \varphi_S) \cdot \cos \beta \cdot \sin \varphi_S] \\ z &= \frac{h_S}{2 \cdot \pi} \cdot \varphi_S - r_W \cdot \sin(\varphi_0 + m \cdot \varphi_S) \cdot \sin \beta. \end{aligned} \quad (2.40)$$

The equation (2.37) for the bending stress can only be used for the strand center wires of stranded ropes. For the lay wires in the strands this simple equation is not valid, because the curvature plane turns around the wire axis against the wire. Determinant for the change of bending stress is therefore not only the change of the curvature radius ρ but also the turning angle γ_k so that the maximum bending stress occurs in another fibre of the wire. Leider (1977) presented firstly this fact in case of bending a strand. Schiffner (1986) – respecting this – calculated the wire bending and torsion stresses by changing the space curve in a stranded rope under the action of the wire rope tensile force. Depending on the small rope elongation and diameter reduction under rope tensile forces these stresses are also small.

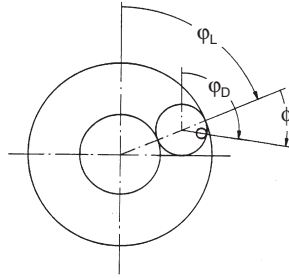


Fig. 2.6. Winding angle of a wire in the wire rope cross section, normative phase angle Φ

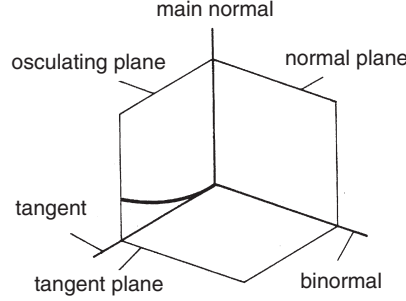


Fig. 2.7. Moving trihedron (tangent, main normal and binormal) of a space curve

The effect of the turning angle γ_k on the wire bending stress can be demonstrated for the case when a strand is bent over a sheave. For the straight strand the curvature radius of a lay wire is $\rho_0 = r_W / \sin^2 \alpha$. For the bent strand Wiek (1973) and with a small correction Leider (1977) have derived the curvature radius ρ of lay wires for the different position of the wire element in relation to the sheave axis. As turning angle Leider (1977) has used the angle between the main normals but Schiffner (1986) found, that the angle between the osculating plane before changing and the main normal after changing is correct for the turning angle

$$\gamma_k = \arcsin \frac{A \cdot a + B \cdot b + C \cdot c}{\sqrt{(A^2 + B^2 + C^2) \cdot (a^2 + b^2 + c^2)}} \quad (2.41)$$

with

$$A = y'_0 \cdot z''_0 - z'_0 \cdot y''_0$$

$$B = z'_0 \cdot x''_0 - x'_0 \cdot z''_0$$

$$C = x'_0 \cdot y''_0 - y'_0 \cdot x''_0$$

and

$$a = y' \cdot (x' \cdot y'' - y' \cdot x'') - z' \cdot (z' \cdot x'' - x' \cdot z'')$$

$$b = z' \cdot (y' \cdot z'' - z' \cdot y'') - x' \cdot (x' \cdot y'' - y' \cdot x'')$$

$$c = x' \cdot (z' \cdot x'' - x' \cdot z'') - y' \cdot (y' \cdot z'' - z' \cdot y'').$$

The equation for the osculating plane is

$$A \cdot (X - x_0) + B \cdot (Y - y_0) + C \cdot (Z - z_0) = 0$$

and for the main normal

$$\frac{X - x}{a} = \frac{Y - y}{b} = \frac{Z - z}{c}.$$

X , Y and Z are the coordinates of the centre of the moving trihedron for the space curve for which the bending stress is considered. The parameter equations x_0 , y_0 and z_0 present the space curve before changing, and x , y and z afterwards.

The maximum change of bending stress resulting from the space curve change is

$$\sigma_b = \frac{\delta}{2} \cdot E \cdot \left(\frac{1}{\rho} \cdot \cos(\psi_{\max} - \gamma_k) - \frac{1}{\rho_0} \cdot \cos \psi_{\max} \right). \quad (2.42)$$

The turning angle $\psi_{\max}$ for the virtual fibre with the maximum stress change is determined by

$$\psi_{\max} = \arctan \left(\frac{\sin \gamma_k}{\cos \gamma_k - \frac{\rho}{\rho_0}} \right). \quad (2.43)$$

Following Schiffner (1986), the calculation of the torsion stress has to be adjusted on the space curve with the winding

$$T^* = \frac{d\varphi_W \cdot \cos \alpha}{ds} = \frac{h_S \cdot \cos \alpha}{h_W \cdot \cos \beta} \frac{1}{\sqrt{x'^2 + y'^2 + z'^2}}.$$

Then according to (2.38), the torsion stress from the winding change is

$$\tau = (T^* - T_0^*) \cdot \frac{\delta}{2} G.$$

When loaded by a tensile force, the wires elongate and contract. The strands will be bent up like the wires in the straight strand under a tensile force. The wires displace each other under the strand bending in core direction. The friction between the wires induces a secondary tensile stress in the wires. Andorfer (1983) calculates this secondary tensile stress to be as Schmidt (1965) first indicated.

When the rope tensile force increases, the secondary tensile force increases in the strand wire of the wire rope from the outside to the inside in the opposite direction to the displacement. The displacement is restricted to the half lay length of the strand wires. The resulting wire tensile force is bigger than the mean tensile force in the wire sections lying directly on the core and smaller than that of the outer wire sections. Contrary to the statement of Andorfer (1983), this is also valid for ordinary lay ropes as well. The force induced by friction will be called secondary tensile force although the force can be either tensile or compression.

On the other hand, the secondary tensile force reverses its direction when the rope tensile force decreases so that the resulting tensile force in the inner wire sections is smaller and in the outer wire sections bigger than the mean wire tensile force. The rope force reversal increases the wire stress amplitude in the case of wire ropes loaded with fluctuating tensile forces.

The secondary tensile stress in a straight stranded rope can reach a considerable size. This stress is especially responsible for the fact that well-lubricated stranded wire ropes have a longer endurance under fluctuating tensile force than unlubricated ones. The lubrication reduces the friction and because of that the secondary tensile stress.

Supplementary to the fluctuating tests, Wang (1989) calculated the stresses in a simple stranded rope ordinary lay FC-6 $\times$ 7-sZ with the diameter 12.2 mm. At about the half endurance for the lower wire rope stress $\sigma_{zunt} = 100 \text{ N/mm}^2$ (with the indices of Wang) the rope extension is $\varepsilon_{sunt} = 1.5\%$ and the lateral contraction $\varepsilon_{qunt} = 5.2\%$ and for the upper wire rope stress $\sigma_{zoben} = 675 \text{ N/mm}^2$ the rope extension is $\varepsilon_{soben} = 5.8\%$ and the lateral contraction $\varepsilon_{qoben} = 9.8\%$. For this rope with fibre core between the lower and the upper rope tensile force the transverse contraction ratio is $\nu = 1.69$ for the rope diameter and $\nu = 1.88$ for the winding radius of the strand axis.

Wang (1989) presented the results of his calculation, done with the relatively high friction coefficient $\mu = 0.25$, in Fig. 2.8. The (global) rope tensile stress range is $2\sigma_{za} = 575 \text{ N/mm}^2$ between 100 and 675 N/mm^2 . After Fig. 2.8 the maximum range of longitudinal stresses in the fibres of the lay wires is $2\sigma_{1.1a \max} = 674 \text{ N/mm}^2$. That is 17.2% more than the range of the rope tensile stress $2\sigma_{za}$.

In addition to the longitudinal stresses the wires will be stressed by torsion, pressure and to a small extent by wear and corrosion. Supplementary to this, secondary bending stresses occur in wire ropes with crossing wire layers or crossing strand layers. All these stresses are systematically unavoidable.

In any case, higher stresses occur unsystematically in some wires because of the unevenly distributed wire tensile forces. This uneven distribution coming

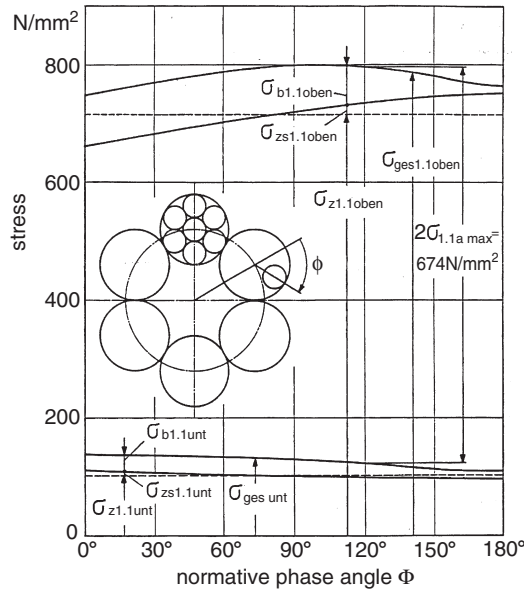


Fig. 2.8. Longitudinal stresses in the lay wires of a wire rope FC + 6 $\times$ 7 sZ under fluctuating forces, Wang (1989)

from the fabrication and the handling of the wire ropes cannot be totally avoided. The calculated stresses compared with the strength show “what is possible in the ideal case and gives the limit that a rope construction can reach but never exceed,” Donandt (1950).

Jiang et al. (1997) and Wehking and Ziegler (2004) recently calculated the stresses in a tensile loaded strand 1 + 6 by the finite element method. In contrast to the analytic method presented here, this method includes the pressure between the centre wire and the lay wires. The maximum stress in the lay wires has nearly the same size as in the analytical calculation but is a little further away from the analytical maximum, the inner wire edge.

2.2 Wire Rope Elasticity Module

2.2.1 Definition

The elongation behaviour of materials under the effect of mechanical stresses is described by elasticity modules. The elongation of a wire rope depends, of course, on elasticity module for wire materials, but the wire rope elasticity module describing wire rope elongation differs from the wire elasticity module. The rope stress-extension curve is not linear. Therefore, for a certain wire rope, the wire rope elasticity module is not constant but depends on the tensile stresses.

As far as strands and spiral ropes are concerned, there is only minimal non-linearity and this can be neglected in most cases. The wire rope elasticity module for these ropes can be calculated approximately using analytical methods (see Sect. 2.2.2), but this is not true for stranded ropes as their rope elasticity modules can only be evaluated by measurements, and – because of the non-linear stress-extension curve – the wire rope elasticity module resulting from these measurements can only be given with a correct definition of the loading.

The main rope elasticity modules which are of importance for practical usage are:

- $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$ as secant between both of the wire rope stresses with a load reverse at the beginning stress (this is especially the case for fluctuating tensile stresses) and, as a special case of this,
- $E_S(0, \sigma_{\text{upper}})$ with the lower stress 0.

Here, and in the following, the stresses refer to the wire rope stresses $\sigma_z = S/A$ with the rope tensile force S and the metallic cross-section A of all rope wires. The index z normally used for the global wire rope stress is left out here for simplification ($\sigma_{z,\text{lower}} = \sigma_{\text{lower}}$ and $\sigma_{z,\text{upper}} = \sigma_{\text{upper}}$).

The rope elasticity modules defined in this way are always meant if they have not been described expressly in a different way. Here it is important that

a stress reverse takes place at the starting stress. The rope elasticity module as secant between two points on a stress–elongation curve (without stress reverse at the beginning) is of no practical importance.

The tangent elasticity module defined by a tangent on the stress-extension curve will be only used in special cases. But later on this tangent elasticity module E_t will be used as an assisting parameter for evaluating the stress-extension curve to find out the rope elasticity module $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$, see Sect. 2.2.3. The measurements of the stress-extension curves for this have always been taken between the lower stress $\sigma_{\text{lower}} = 0$ and the upper rope stress $\sigma_{\text{upper}} = 800 \text{ N/mm}^2$. Because of this, it is not necessary to show either of these end stresses in the symbol of the tangent elasticity module. The tangent elasticity module (as assisting parameter) on the stress-extension curves between the rope tensile stresses 0 and 800 N/mm^2 in the up-and-down direction are therefore given by the symbols where only the rope tensile stress in the tangent point is nominated:

- $E_{t,\text{up}}(\sigma_z)$ rope elasticity module as tangent on the stress-extension curve in the up direction at the rope tensile stress σ_z
- $E_{t,\text{down}}(\sigma_z)$ rope elasticity module as tangent on the stress-extension curve in the down direction at the rope tensile stress σ_z

The rope elasticity modules in the different definitions used here are shown in Fig. 2.9.

2.2.2 Rope Elasticity Module of Strands and Spiral Ropes, Calculation

As already mentioned, the non-linearity of the stress-extension curve is relatively small for strands and spiral ropes. There is also only a small increase of the rope elasticity module with the number of loadings. The smaller the

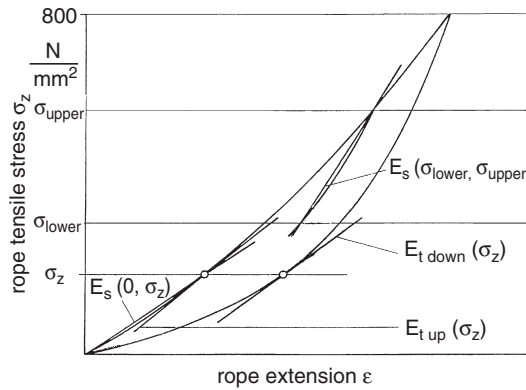


Fig. 2.9. Definitions of the wire rope elasticity modules used

number of wires in the rope, the more likely this is to be true. Buchholz and Eichmüller (1988) found that there was only the very small difference of $\Delta E = 600 \text{ N/mm}^2$ between the first, second and third measurements with an almost constant rope elasticity module $E_S = 198,000 \text{ N/mm}^2$. Taking all these observations into consideration, it is possible to make reliable calculations for the rope elasticity module for strands and spiral ropes with a small number of wires. A method of calculation was first devised for this by Hudler (1937).

The calculation can be done with the help of the equations from Sect. 2.1. The rope elasticity module is by definition

$$E_S = \frac{\sigma_z}{\varepsilon}.$$

With (2.21) for the tensile stress and the definition of the strand extension

$$\varepsilon = \frac{\Delta l_S}{l_S},$$

the rope elasticity module for strands and spiral ropes is

$$E_S = \frac{1}{A} \sum_{i=0}^n \frac{z_i \cdot \cos^3 \alpha_i}{1 + \nu_i \cdot \sin^2 \alpha_i} \cdot E_i \cdot A_i. \quad (2.44)$$

Poisson's ratio can be set $\nu = \nu_i = 0.3$ for all wire diameters and winding radii in steel spiral ropes because the length-related force between the wire layers is small and the lateral contraction is almost only caused by the tensile stress in the wires.

Example 2.3: Elasticity module of an open spiral rope according to Fig. 2.4

According to (2.44) the rope elasticity module is

$$E_S = \frac{196,000}{45.61} \cdot \left(1.431 + \frac{(6 + 12 + 18) \cdot 0.9703^3 \cdot 1.227}{1 + 0.3 \cdot 0.2419^2} \right)$$

$$E_S = 177,000 \text{ N/mm}^2.$$

The rope elasticity module for strands and spiral ropes calculated by (2.44) is independent from the rope tensile stress. But in reality this rope elasticity module always depends slightly on the stress level and it is always a little smaller than the one calculated. This means, the smaller the stress level and the higher the number of wires in the rope, the bigger the difference. The calculated rope elasticity module can only be reached approximately with a strong pre-stressing.

A reference value for the elasticity modules of closed spiral ropes for bridges which have not been pre-stressed is given in Fig. 2.10 by DIN 18809. This shows elasticity modules with different definitions:

- E_g rope elasticity module for the first loading up to the permanent load
- E_p rope elasticity module for the traffic load

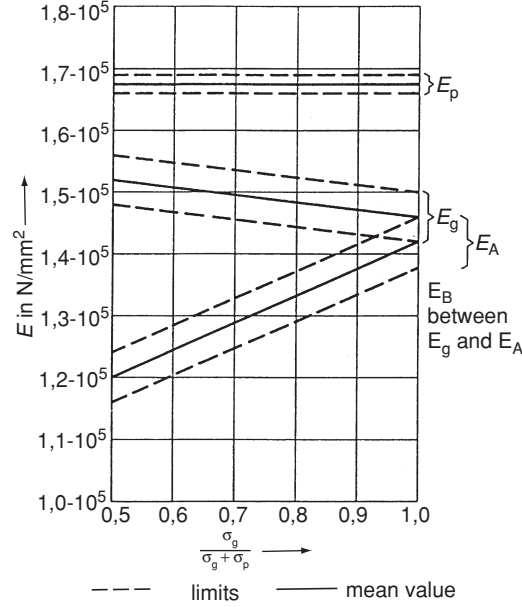


Fig. 2.10. Reference value for rope elasticity modules of locked coil ropes, DIN18809

- E_A rope elasticity module for defining the rope length
- E_B rope elasticity module during bridge erection

2.2.3 Rope Elasticity Module of Stranded Wire Ropes

Because of its lateral contraction, the rope elasticity module of stranded ropes cannot be calculated in the same way as that of strands or spiral ropes. The lateral contraction of the stranded ropes depends on a large unknown quantity at the tensile stress level. Therefore, the elasticity module of stranded ropes can only be evaluated by taking measurements.

Wyss (1957) and Jehmlich (1985) have made series of measurements. They distinguished between a total rope elasticity module starting from the stress 0 (first loading) and the rope elasticity module between two stresses after a longer rope working time. An important contribution to what is known about the rope elasticity module was made by Hankus (1976, 1978 and 1989). He measured the elongation of many ropes of different constructions with fibre and steel cores with the first loading as well as after loading repeatedly in an up-and-down direction. He used these measurements to evaluate the rope elasticity module as secant starting from the stress $\sigma_z = 0$ with multi-dimensional linear regression calculations. He also evaluated the rope elongation after it had been loaded for a long time.

The following remarks about the rope elasticity module relate mostly to the Stuttgart tests conducted by Feyrer and Jahne (1990). These tests were

done with nearly all types of construction for round stranded wire ropes. A lot of the tests were carried out by the students listed in the previous article.

Stress-Extension Curves

The measurements of stress-extension curves – which form the basis of the evaluation for rope elasticity modules – have always been taken in the same manner. The rope elongation ΔL is measured for a rope length of $L = 2,000$ mm with two inductive elongation meters on the right and left of the rope as seen in Fig. 2.11. The results of these measurements are recorded for the first loading cycle up to rope tensile stress $\sigma_z = 800$ N/mm² and after nine loadings between $\sigma_z = 0$ and $\sigma_z = 800$ N/mm² for the tenth loading and unloading.

The nine loading cycles should give nearly the same compression of the rope structure as is found in practice after some time under working conditions (of course with smaller tensile stresses and a greater number of loading cycles). It will be anticipated here that after ten loading cycles the mean residual extension is 4% with a large deviation. A residual extension of approximately the same size was found for wire ropes running over sheaves after 2% of their life time as Woernle (1929) already noticed. However a residual extension of 3% was measured again for elevator ropes FC-8 $\times$ 19 after a long period of operation under these ten loading cycles.

In Fig. 2.12, the stress-extension curves are presented for a wire rope with a fibre core under the first and the tenth loading and unloading. This figure shows the typical progressive increase of tensile stress arising as the rope extends. Especially for wire ropes with a fibre core, a large progressive increase and hysteresis for loading and unloading occurs. The progressive form of the rope stress-extension curve has its origin in the lateral contraction of the stranded ropes. In ropes with fibre cores, this is especially large and non-linear.

The stress-extension curves of wire ropes with steel cores are given as an example in Fig. 2.13. This also shows the progressive increase of the stress

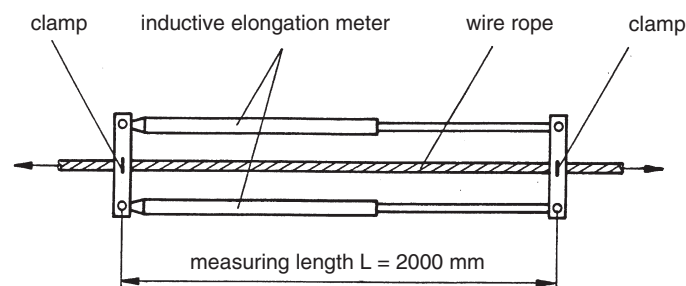


Fig. 2.11. Arrangement for measuring the wire rope elongation, Feyrer and Jahne (1990)

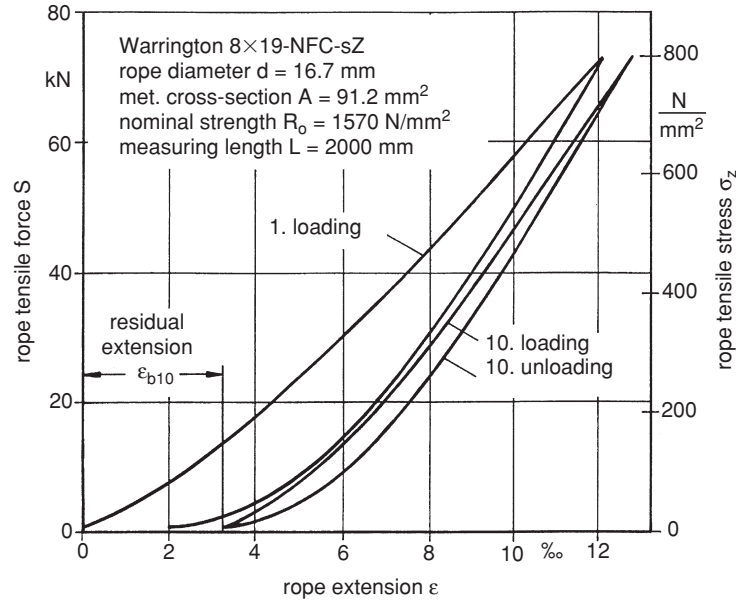


Fig. 2.12. Stress-extension curves for a stranded wire rope with fibre core, Feyrer and Jahne (1990)

when a rope with a steel core becomes extended. Normally, this is not as large as in the case of wire ropes with fibre cores. However, in this special case, the residual extension is greater. The stress-extension curve is always different for loading and unloading. The enclosed area in the hysteresis loop is a mark of the inner frictional work of the wire rope.

Figure 2.14 shows the stress-extension curves for the loading and unloading of the wire rope from Fig. 2.12 after the tenth loading cycle. Between the rope tensile stresses 0 and 800 N/mm^2 , the tensile stress changes in small steps. In loop A, the tensile stress increases starting from $\sigma_z = 0$ in steps of $\Delta\sigma_z = 100 \text{ N/mm}^2$ and reduces the stress at every level reached in a small stress loop $\sigma_{\text{upper}} - \sigma_{\text{lower}} = \Delta\sigma_z = 100 \text{ N/mm}^2$. The two lowest partial loops still show a clear hysteresis, but the others do not.

Loop B is again loaded in stress steps of $\Delta\sigma_z = 100 \text{ N/mm}^2$ but now starting from $\sigma_z = 800 \text{ N/mm}^2$ in a “down” direction. The two lowest partial loops show a clear hysteresis as in loop A. The partial loops for the same stresses σ_{lower} and σ_{upper} in the loops A and B are practically parallel. They represent the rope elasticity modules $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$.

In loop C some partial loops of stress-extension curves are shown, starting from $\sigma_z = 0$ to the upper stresses $\sigma_{\text{upper}} = 200, 400$ and 600 N/mm^2 . The loading curves are the same for all upper stresses. The unloading curves from these upper stresses can be taken approximately as a part of the entire unloading curve from the upper stress $800 - 0 \text{ N/mm}^2$, turned around the point for $\sigma_z = 0$.

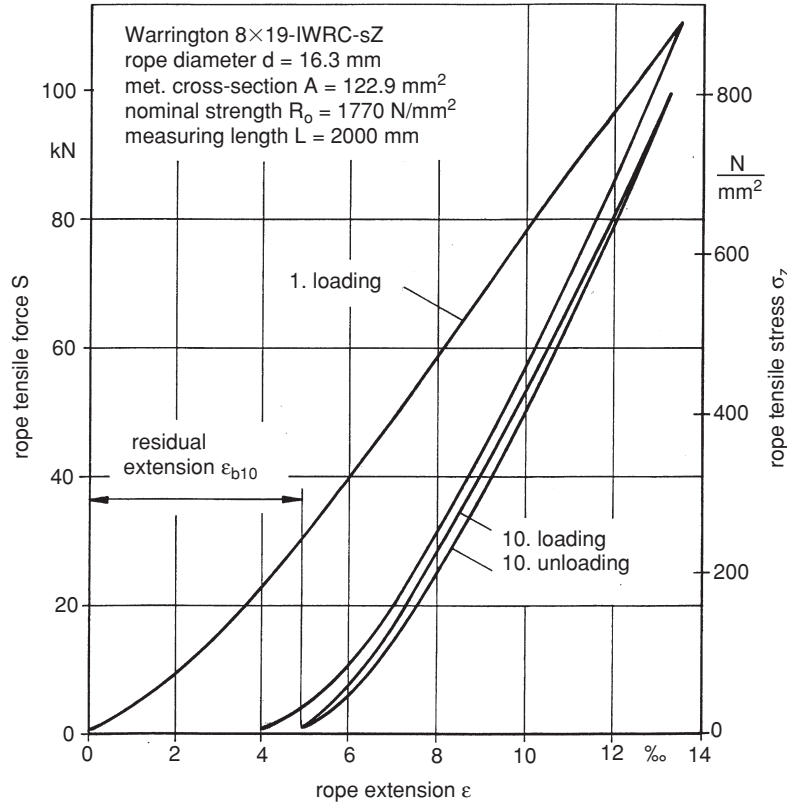


Fig. 2.13. Stress-extension curves for a stranded wire rope with steel core, Feyrer and Jahne (1990)

Assistant Parameter: Tangent Elasticity Module

The stress-extension curves of the different wire ropes measured between the rope tensile stresses 0 and 800 N/mm^2 – as seen in Figs. 2.12 and 2.13 – will be used to evaluate the rope elasticity module $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$. The calculation based on the rope tangent module has the advantage of being very precise. The tangent module has been taken point-for-point from the stress-extension curves. Figure 2.15 gives an example of the tangent module based on the diagram in Fig. 2.12 after the tenth loading and unloading. It should be realised here that the tangent module depends strongly on the rope tensile stress and the direction of the loading or unloading.

Common linear regression calculation was used to work out the rope tangent module from numerous wire ropes. After a number of trials, the best regression equation was found to be

$$E_t(\sigma_z) = C_0 + \frac{C_1}{\sigma_z + A} + \sum_{i=2}^n C_i \cdot x_i. \quad (2.45)$$

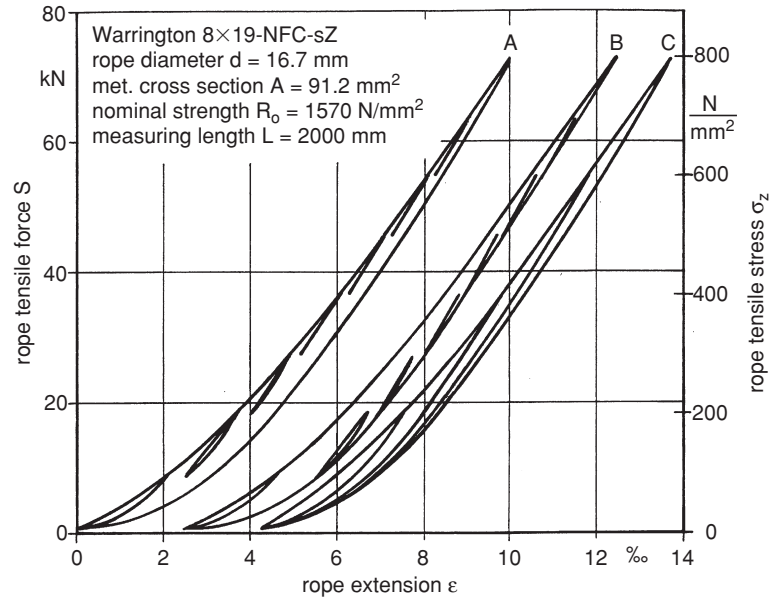


Fig. 2.14. Stress-extension curves with loading between different stresses, Feyrer and Jahne (1990)

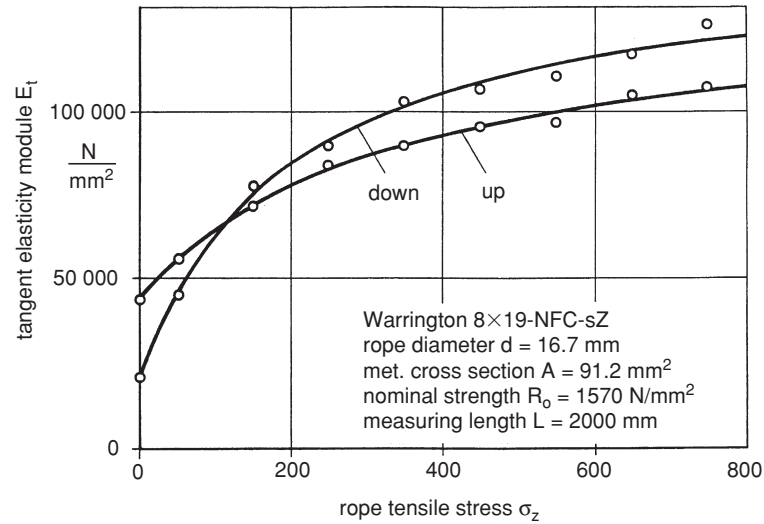


Fig. 2.15. Assistant parameter: tangent elasticity module E_t , Feyrer and Jahne (1990)

The constant A in the equation has to be worked out by iteration, but this does not cause a problem when using computers. The wire rope construction is characterised by the variables x_i . For example $x_2 = 0$ is set for 6-strand ropes and $x_2 = 1$ is set for 8-strand ropes. Separate regression calculations have been done for ropes with fibre cores, ropes with steel cores and for spiral round strand ropes, and also, of course, both for loading and unloading.

With a common constant

$$B = C_0 + \sum_{i=2}^n C_i \cdot x_i \quad \text{and with} \quad C = C_1,$$

the tangent elasticity module for the rope tensile stress σ_z on the stress-extension curve between the tensile stresses $\sigma_z = 0$ and 800 N/mm^2 in an “up-and-down” direction is

$$E_t(\sigma_z) = B + \frac{C}{\sigma_z + A}. \quad (2.46)$$

The constants A , B and C are listed in Tables 2.1 and 2.2. The constant B for wire ropes with fibre-covered steel cores in Table 2.1 has been changed unlike constant B in Feyrer and Jahne (1990).

Rope Elasticity Module with the Lower Tensile Stress $\sigma_z = 0$

The extension ε of a wire rope is

$$\varepsilon = \int \frac{1}{E_t(\sigma_z)} \cdot d\sigma_z.$$

Using (2.46), the extension of a wire rope between the two stresses σ_{lower} and σ_{upper} – in the stress-extension curve coming from $\sigma_z = 0$ – in the “up” direction – is

$$\varepsilon = \int_{\sigma_{\text{lower}}}^{\sigma_{\text{upper}}} \frac{1}{B_{\text{up}} + \frac{C_{\text{up}}}{\sigma_z + A_{\text{up}}}} \cdot d\sigma_z = \int_{\sigma_{\text{lower}}}^{\sigma_{\text{upper}}} \frac{\sigma_z + A_{\text{up}}}{B_{\text{up}} \cdot \sigma_z + A_{\text{up}} \cdot B_{\text{up}} + C_{\text{up}}} \cdot d\sigma_z$$

and after integration

$$\varepsilon = \frac{\sigma_{\text{upper}} - \sigma_{\text{lower}}}{B_{\text{up}}} - \frac{C_{\text{up}}}{B_{\text{up}}^2} \cdot \ln \frac{\sigma_{\text{upper}} + A_{\text{up}} + C_{\text{up}}/B_{\text{up}}}{\sigma_{\text{lower}} + A_{\text{up}} + C_{\text{up}}/B_{\text{up}}}. \quad (2.47)$$

According to (2.47), the important rope elasticity module with the lower stress $\sigma_{\text{lower}} = 0$ and the upper stress σ_{upper} is

$$E_S(\sigma_z) = E_S(0, \sigma_z) = \frac{\sigma_z}{\varepsilon}$$

or

$$E_S(\sigma_z) = \frac{\sigma_z}{\frac{\sigma_z}{B_{\text{up}}} - \frac{C_{\text{up}}}{B_{\text{up}}^2} \cdot \ln \left(1 + \frac{\sigma_z}{A_{\text{up}} + C_{\text{up}}/B_{\text{up}}} \right)}. \quad (2.48)$$

This rope elasticity module is especially important for the first loading when it is installed. The constants A , B , and C are listed in Tables 2.1 and 2.2.

Table 2.1. Constants A , B and C for calculating the elasticity module in N/mm^2 of round stranded ropes, Feyrer and Jahne (1990)

rope condi- tion	core	coefficient of determina- tion standard deviation	constant A	constant C	constant B					
					6-strand wire layers			8-strand wire layers		
					1	2	3	1	2	3
new up	NFC	64%	161	-10,700,000	118,000	102,000	101,000	116,000	100,000	99,000
	SFC	14,000			112,000	96,000	95,000	110,000	94,000	93,000
	IWRC	75%	81	-5,140,000	120,000	105,000	104,000	103,000	88,000	87,000
	PWRC	13,000			137,000	122,000	121,000	120,000	105,000	104,000
	ESWRC				126,000	111,000	110,000	109,000	94,000	93,000
ten times loaded up	EFWRC				108,000	93,000	92,000	91,000	76,000	75,000
	NFC	82%	161	-14,400,000	152,000	141,000	138,000	149,000	138,000	135,000
	SFC	11,000			141,000	130,000	127,000	138,000	127,000	124,000
	IWRC	89%	131	-12,500,000	160,000	149,000	147,000	149,000	138,000	136,000
	PWRC	10,000			163,000	152,000	150,000	152,000	141,000	139,000
ten times loaded down	ESWRC				156,000	145,000	143,000	145,000	134,000	132,000
	EFWRC				145,000	134,000	132,000	134,000	123,000	121,000
	NFC	89%	192	-25,500,000	177,000	166,000	163,000	174,000	163,000	160,000
	SFC	11,000			166,000	155,000	152,000	163,000	152,000	149,000
	IWRC	91%	167	-20,500,000	177,000	166,000	164,000	166,000	155,000	153,000
	PWRC	10,000			180,000	169,000	167,000	169,000	158,000	156,000
	ESWRC				173,000	162,000	160,000	162,000	151,000	149,000
	EFWRC				162,000	151,000	149,000	151,000	140,000	138,000

Table 2.2. Constants A , B and C for calculating the elasticity module of spiral round strand ropes, Feyrer and Jahne (1990)

rope condition	new	ten times loaded	
load direction	up	up	down
constant A	35	149	229
constant B			
two strand layers	90,000	123,000	151,500
three strand layers	89,000	121,000	149,500
constant C	-1,700,000	-11,200,000	-26,700,000
coefficient of determination	62%	75%	86%
standard deviation s	12,000	11,000	11,000

Rope Elasticity Module E_S between two Stresses

The rope elasticity module $E_S(\sigma_{\text{lower}}; \sigma_{\text{upper}})$ between the two stresses σ_{lower} and σ_{upper} is defined by the secant between these two stresses of the stress-extension curve with a load reverse at the beginning stress, Figs. 2.9 and 2.14. The loading direction changes in most practical applications at the beginning of the considered loading. This is especially true in cases with a fluctuating load. $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$ is therefore the rope elasticity module normally used.

A very good approximation of this rope elasticity module can be obtained by quasi-turning the stress-extension curve (between $\sigma_z = 0$ and $\sigma_z = 800 \text{ N/mm}^2$) in the “down” direction around the origin of coordinate ($\sigma_z = 0$) so far until its extension at the upper stress is the same as that of the “up” direction (between $\sigma_z = 0$ and $\sigma_z = 800 \text{ N/mm}^2$)

$$\varepsilon_{\text{up}}(\sigma_{\text{upper}}) = \varepsilon_{\text{down}}(\sigma_{\text{upper}}). \quad (2.49)$$

The rope elasticity module $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$ can be taken from the “down” direction stress-extension curve turned as described. The turning will be brought about by exchanging the constant B_{down} to $B_{\text{down,upper}}$ for $\varepsilon_{\text{down}}(\sigma_{\text{upper}})$. Equation (2.48) set in (2.49) gives (with this new constant for the “down” direction curve) the equation for calculating the new constant $B_{\text{down,upper}}$

$$\begin{aligned} & \frac{\sigma_{\text{upper}}}{B_{\text{up}}} - \frac{C_{\text{up}}}{B_{\text{up}}^2} \cdot \ln \left(1 + \frac{\sigma_{\text{upper}}}{A_{\text{up}} + C_{\text{up}}/B_{\text{up}}} \right) \\ &= \frac{\sigma_{\text{upper}}}{B_{\text{down,upper}}} - \frac{C_{\text{down}}}{B_{\text{down,upper}}^2} \cdot \ln \left(1 + \frac{\sigma_{\text{upper}}}{A_{\text{down}} + C_{\text{down}}/B_{\text{down,upper}}} \right). \end{aligned} \quad (2.50)$$

The constant B_{downup} has to be calculated by iteration using (2.50). The rope extension $\varepsilon_{\text{lower,upper}}$ can be calculated with (2.47), the constant $B_{\text{down,upper}}$, and the constants A_{down} and C_{down} using Tables 2.1 and 2.2. Then the rope

elasticity module is

$$E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}}) = \frac{\sigma_{\text{upper}} - \sigma_{\text{lower}}}{\varepsilon_{\text{lower, upper}}} \quad (2.51)$$

or

$$E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}}) = \frac{\sigma_{\text{upper}} - \sigma_{\text{lower}}}{\frac{\sigma_{\text{upper}} - \sigma_{\text{lower}}}{B_{\text{down, upper}}} - \frac{C_{\text{down}}}{B_{\text{down, upper}}^2} \cdot \ln \frac{\sigma_{\text{upper}} + A_{\text{down}} + C_{\text{down}}/B_{\text{down, upper}}}{\sigma_{\text{lower}} + A_{\text{down}} + C_{\text{down}}/B_{\text{down, upper}}}} \quad (2.52)$$

Calculating the rope elasticity module without the aid of a computer involves a certain amount of effort. For some chosen rope stresses σ_{lower} and σ_{upper} , the rope elasticity module $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$ is listed in tables. Table 2.3 shows the rope elasticity module for 6-strand ropes with two wire layers and for spiral round strand ropes. In case of rope oscillations with the middle stress σ_{m} and small amplitude stress σ_{a} , the elasticity module required is $E_S(\sigma_{\text{m}} \pm 0)$. This rope elasticity module is listed for some middle stresses in Table 2.3 as

$$E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}}) = E_S(\sigma_{\text{m}}; \sigma_{\text{m}}).$$

For example, for a rope 6×19 – IWRC with $\sigma_{\text{m}} = 200 \text{ N/mm}^2$

$$E_S(200 \pm 0) = E_S(200; 200) = 117 \text{ kN/mm}^2.$$

Table 2.4 gives correction constants ΔE for 8-strand ropes and for one and three wire layers. With this, the rope elasticity module $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$ is

$$E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}}) = E_S(\text{Table 2.3}) + \Delta E. \quad (2.53)$$

The standard deviation can be taken from the Tables 2.1 and 2.2.

Example 2.4: Wire rope elasticity module

Data:

wire rope IWRC + 8×19

rope tensile stresses between $\sigma_z = 100$ and $\sigma_z = 220 \text{ N/mm}^2$.

Results:

From (2.50) and (2.52)

$$E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}}) = E_S(100; 220) = 98 \text{ kN/mm}^2$$

Alternative from tables:

From Table 2.3 the rope elasticity module for a rope IWRC + 6×19 is $E_S(100; 200) = 107 \text{ kN/mm}^2$ and $E_S(100; 300) = 113 \text{ kN/mm}^2$ and as a middle value $E_S(100; 220) = 108 \text{ kN/mm}^2$.

Table 2.3. Rope elasticity module as secant between the lower and upper rope tensile stress, 6-strand ropes with two wire layers and spiral round strand ropes

rope-condition	rope-construct.	rope-core	Rope elasticity module E in kN/mm^2						
			lower tensile stress in N/mm^2						
			0	upper tensile stress in N/mm^2					
			40	100	200	300	400	600	800
new	6-strands two wire layers	NFC	42	49	57	62	66	71	75
		SFC	36	43	51	56	60	65	69
		IWRC	53	62	71	76	80	85	88
		PWRC	70	79	88	93	97	102	105
		ESWRC	59	68	72	84	86	91	94
		EFWRC	41	50	59	64	68	73	76
	spiral-round strand rope	two strand layer	57	66	72	76	78	81	82
		three strand layer	55	65	71	75	77	80	81
rope-condition	rope-construct.	rope-core	Rope elasticity module E in kN/mm^2						
			lower tensile stress in N/mm^2						
			0	upper tensile stress in N/mm^2					
			40	100	200	300	400	600	800
ten times loaded	6-strands two wire layers	NFC	61	70	80	87	93	100	105
		SFC	50	59	69	76	82	89	94
		IWRC	65	76	88	96	101	109	114
		PWRC	70	79	91	99	104	112	117
		ESWRC	61	72	84	92	97	105	110
		EFWRC	50	61	73	81	86	94	99
	spiral-round strand rope	two strand layer	57	67	73	79	84	90	94
		three strand layer	55	65	71	77	82	88	92

Table 2.3. *Continued*

rope-condition	rope-construct.	rope-core	Rope elasticity module E in kN/mm^2							
			lower tensile stress in N/mm^2							
			upper tensile stress in N/mm^2							
ten times loaded	6-strands two wire layers	NFC	72	81	91	97	102	109	110	110
		SFC	61	70	80	86	91	98	101	101
		IWRC	77	88	98	105	110	117	121	121
		PWRC	80	91	101	108	113	120	124	124
		ESWRC	73	84	94	101	106	113	117	117
		EFWRC	62	73	83	90	95	102	106	106
	spiral-round strand rope	two strand layer	65	73	81	87	91	96	100	100
		three strand layer	63	71	79	85	89	94	98	98
rope-condition	rope-construct.	rope-core	Rope elasticity module E in kN/mm^2							
			lower tensile stress in N/mm^2							
			upper tensile stress in N/mm^2							
new	6-strand two wire layers	NFC	92	101	107	111	116	119	123	127
		SFC	81	90	96	100	105	108	112	116
		IWRC	98	107	113	118	123	127	130	134
		PWRC	101	110	116	121	126	130	134	138
		ESWRC	94	103	109	114	119	123	127	131
		EFWRC	83	92	98	103	108	112	117	121
	spiral-round strand rope	two strand layer	81	89	94	98	103	106	110	114
		three strand layer	79	87	92	96	101	104	108	112

Table 2.4. Correction constants ΔE for round strand ropes with 6- and 8-strands of one, two and three wire layers

rope condition	rope core	correction constant					
		6-strands			8-strands		
		wire layers			wire layers		
		1	2	3	1	2	3
new	fibre core	16	0	-1	14	-2	-3
	steel core	15	0	-1	-2	-17	-18
ten times loaded	fibre core	11	0	-3	8	-3	-6
	steel core	11	0	-2	0	-11	-13

Fibre core = NFC, SFC; steel core = IWRC, PWRC, ESWRC, EFWRC

From Table 2.4 the correction constant for 8-strand ropes is $\Delta E = -11 \text{ kN/mm}^2$.

This means that with (2.53), the rope elasticity module for the wire rope IWRC + 8 $\times$ 19 is $E_S(100; 250) = 110 - 11 = 97 \text{ kN/mm}^2$. Nearly the same as 98 kN/mm^2 .

According to Table 2.1, the standard deviation is $s = 10 \text{ kN/mm}^2$.

2.2.4 Waves and Vibrations

Longitudinal Waves

If a long wire rope receives a shock load, a tensile force wave (strain wave) moves along the wire rope starting from the initial point of impact. The velocity of the wave is

$$c = \sqrt{\frac{E}{\rho}} \quad (2.54)$$

with E for the elasticity module and ρ for the mass density. For a single wire with, for example, $E = 196,000 \text{ N/mm}^2 = 196,000 \times 10^6 \text{ N/m}^2$ and $\rho = 7,800 \text{ kg/m}^3 = 7,800 \text{ N s}^2/\text{m}^4$ the velocity of the wave is

$$c = \sqrt{\frac{196,000 \cdot 10^6}{7,800}} = 5010 \text{ m/s}.$$

The wave velocity is of some importance for the understanding of accidents related to wire rope installations. The tensile stress of a wave will be practically doubled when it is reflected from the termination of the rope and it is possible that the wire rope will break if the velocity v of the impact is big enough. For example, the shock load can be effected on the hanging rope by a falling weight with the striking velocity v . According to Irvine's fundamental theory (1981), the tensile rope force F produced by the shock load is

$$F = m_T \cdot c \cdot v \cdot e^{(-m_T \cdot c \cdot t)/M}. \quad (2.55)$$

In this equation, m_T is the length-related rope mass, c the wave velocity, v the striking velocity, M the falling mass and t the time. For $t = 0$, the wire rope shock force is $F_0 = m_T cv$ and this fades away in time if the tensile shock force is not great enough to break the rope. The size of the mass hitting the wire rope has no influence on the tensile shock force but only on its fading. (If the falling mass M is very large, the wire rope can of course break even if the velocity v is small. This can be the case if the weight force Mg is greater than the rope breaking force or if the falling energy is greater than the stress-extension energy of the rope.)

Irvine's theory can be used to explain the terrible accident with an aerial rope way at Cavalese on 3rd February, 1998, when an aircraft with a relatively fragile structure severed a solid track rope and the haulage rope and was still able to fly afterwards. The velocity of the aircraft was 241 m/s, the length of the tears in the aircraft wings caused by the track rope were about 1 m and those caused by the haulage rope, 0.5 m, Oplatka and Volmer (1998). They pointed out that the aircraft wings would have been totally torn off if the aircraft velocity had been lower than the limit velocity. Spontaneous wire rope breakages caused by aircraft impacts also occurred prior to Cavalese, Lombard (1998a).

The wire rope breakage caused by the impact of an aircraft hitting the wire occurs if its velocity v is big enough. According to Irvine, the minimum velocity is

$$v = c \cdot \sqrt{\varepsilon^2 + 2\varepsilon \cdot \sqrt{\varepsilon}}. \quad (2.56)$$

In this equation, ε is the breaking extension and c is once more the wave velocity. With the breaking extension of rope wires $\varepsilon = 0.007$ with a safety margin, Irvine (1981) calculated a minimum velocity $v = 150$ m/s for the aircraft. Lombard (1998a, b) calculated a minimum velocity $v = 156$ m/s with one-third of the wire elasticity module and a more realistic breaking extension $\varepsilon = 0.018$. He used his own extended theory for this calculation.

Longitudinal Vibrations

A mass hanging on a wire rope can be made to vibrate along the axis of the rope. Without taking the damping into consideration, the angular frequency is

$$\omega_0 = \sqrt{\frac{c_S}{M}}$$

and the frequency

$$f_0 = \frac{\omega_0}{2 \cdot \pi} = \frac{1}{2 \cdot \pi} \cdot \sqrt{\frac{c_S}{M}}. \quad (2.57)$$

Here it is presupposed that the rope mass is much smaller than the hanging mass M and can be neglected. The wire rope as a spring has the spring constant

$$c_S = \frac{E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}}) \cdot A}{L} \quad (2.58)$$

with the rope elasticity module $E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}})$, the metallic rope cross-section A and the rope length L . When the stress amplitude changes, the rope elasticity module will be nearly constant if the middle stress remains the same. The rope elasticity module

$$E_S(\sigma_{\text{lower}}, \sigma_{\text{upper}}) = E_S(\sigma_m \pm \sigma_a) \quad (2.58a)$$

with the amplitude σ_a and with the middle rope tensile stress

$$\sigma_m = \frac{M \cdot g}{A} \quad (2.58b)$$

can be evaluated using (2.50) and (2.52) or Tables 2.3 and 2.4.

In addition to the frequency, the damping of the longitudinal vibrations is of interest. Wehking et al. (1999) have made some decay tests. Figure 2.16 shows the test situation. A main mass M and a dropping mass M_A hang on a wire rope with the diameter $d = 10$ mm and the length $l = 12$ m. After cutting the thin rope between the main mass M and the dropping mass M_A the main mass swings with decreasing amplitude.

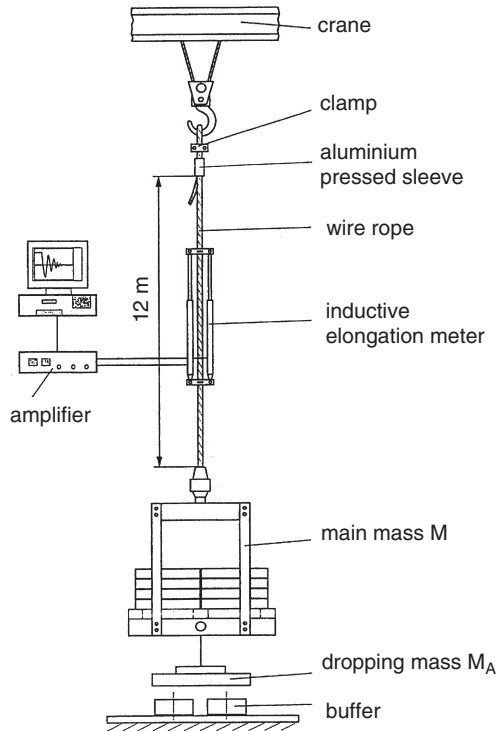


Fig. 2.16. Test situation for the measurement of rope damping, Wehking et al. (1999)

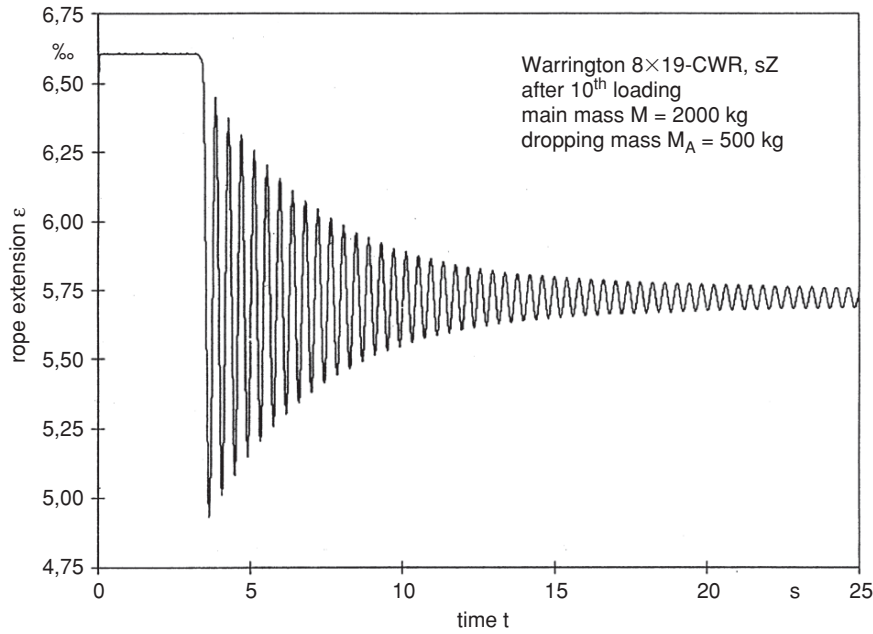


Fig. 2.17. Decay behaviour of a mass hanging on a wire rope, Wehking et al. (1999)

Figure 2.17 shows the typical behaviour of a decay test. Only the tests with wire ropes which were ten times loaded before will be considered here. With the metallic cross section of the wire rope $A = 45.1 \text{ mm}^2$, the middle rope tensile stresses for both of the main masses $M = 400$ and $2,000 \text{ kg}$ are

$$\sigma_m = \frac{400 \cdot 9.81}{45.1} = 87 \text{ N/mm}^2 \quad \text{and} \quad \sigma_m = \frac{2000 \cdot 9.81}{45.1} = 435 \text{ N/mm}^2.$$

Using (2.50) and (2.52), the rope elasticity module of the Warrington rope is

$$E_S(87 \pm \sigma_a) = 83,000 \text{ N/mm}^2 \quad \text{and} \quad E_S(435 \pm \sigma_a) = 125,000 \text{ N/mm}^2.$$

Or alternative, by interpolation for the 6-strand rope from Table 2.3 and ΔE from Table 2.4, the rope elasticity module E_S is approximately

$$E_S(87 \pm \sigma_a) = E_S(40; 40) + (E_S(100; 100) - E_S(40; 40)) \cdot \frac{87 - 40}{100 - 40} + \Delta E$$

$$E_S(87 \pm \sigma_a) = 77 + (98 - 77) \cdot \frac{47}{60} - 11 = 82 \text{ kN/mm}^2$$

and

$$E_S(435 \pm \sigma_a) = E_S(400; 400) + (E_S(600; 600) - E_S(400; 400)) \cdot \frac{435 - 400}{600 - 400} + \Delta E$$

$$E_S(435 \pm \sigma_a) = 134 + (141 - 134) \cdot \frac{35}{200} - 11 = 124 \text{ kN/mm}^2.$$

For the rope elasticity modules – evaluated using (2.50) and (2.52) – the spring constants according to (2.58) are

$$c_{S87} = \frac{83,000 \cdot 45.1}{12,000} = 312 \text{ N/mm} \rightarrow 3,12,000 \text{ N/m} \quad \text{and}$$

$$c_{S435} = \frac{125,000 \cdot 45.1}{12,000} = 470 \text{ N/mm} \rightarrow 470,000 \text{ N/m}.$$

Without taking the damping into consideration, according to (2.57) the frequency is then

$$f_{0;87} = 4.45 \text{ Hz} \quad \text{and} \quad f_{0;435} = 2.44 \text{ Hz}.$$

Under the influence of the damping, the amplitude (stress or extension) is continuously reduced and the frequency is somewhat less. With the small damping of the inner rope friction, the amplitude is

$$x = x_0^{-\delta t} \cos \omega t. \quad (2.59)$$

In this, δ is the decay coefficient, ω the angular frequency of the damped vibration and t the time. The decay coefficient is

$$\delta = \frac{\Lambda}{T} = \Lambda \cdot f = \frac{\Lambda \cdot \omega}{2 \cdot \pi}.$$

The logarithmic decrement Λ is the natural logarithm of the ratio of two consecutive maximum amplitudes

$$\Lambda = \ln \frac{\hat{x}_i}{\hat{x}_{i+1}}.$$

The frequency of the poorly damped vibration is

$$\omega = \sqrt{\omega_0^2 - \delta^2}. \quad (2.60)$$

The results of the decay tests conducted by Wehking et al. (1999) are presented in Table 2.5. Because there is only a very small decay coefficient, the frequency is hardly reduced by the damping. The frequencies which were measured and calculated are compared in Table 2.5. For the small middle rope tensile stress 87 N/mm^2 , the difference between the measured and the calculated frequency is 13%. That is probably caused by the big deviation occurring in the measured elasticity module for small stress levels. For the big middle rope tensile stress 435 N/mm^2 there is practically no difference between the measured and the calculated frequencies.

As expected, the damping of wire ropes with longitudinal vibrations is much greater for the small mean stress than for the big one. This behaviour is caused by the inner rope friction, Andorfer (1983). The hysteresis area, enclosed by the loading and unloading loop, shows the damping energy. In Fig. 2.14, it can be clearly seen that the higher the stress level is, the smaller the enclosed area. Certainly, as far as the wire rope with fibre core in Fig. 2.14 is concerned, the damping is greater than that found for a rope with a steel core. No explanation was found for the smaller logarithm decrement Λ which was measured for the smaller dropping mass.

Table 2.5. Results from decay tests, Wehking et al. (1999)

main mass M kg	middle stress σ_z N/mm ²	dropping mass M_A kg	measured frequency f_{mes} 1/s	calculated frequency f_{cal} 1/s	logarithm decrement Λ —	decay coefficient δ 1/s
400	87	134	5.03	4.45	0.125	0.629
2,000	435	134	2.48	2.44	0.046	0.115
2,000	435	500	2.41	2.44	0.089	0.215

Example 2.5: Frequency of a mass hanging on a wire rope**Data:**

Filler $6 \times 19\text{-IWRC-}sZ$ (ten times loaded)

Mass $M = 1,000$ kg

Rope diameter $d = 10$ mm

Rope length $L = 50$ m

Results:

With the wire rope-cross section $A = C_2 d^2 = 45.7 \text{ mm}^2$ with $C_2 = 0.457$ accordingly Table 1.9, the rope tensile stress is

$$\sigma_m = \frac{M \cdot g}{A} = \frac{M \cdot g}{C_2 \cdot d^2} = \frac{1,000 \cdot 9.81}{0.457 \cdot 10^2} = 218 \text{ N/mm}^2.$$

According to (2.50) and (2.52), the wire rope elasticity module is

$$E_S = 119,300 \text{ N/mm}^2.$$

From that, according to (2.58), the spring constant is

$$c_S = \frac{119,300 \cdot 44.9}{50} = 107,130 \text{ N/m}$$

and the frequency according to (2.57) is

$$f_0 = \frac{1}{2 \cdot \pi} \sqrt{\frac{107,130}{1,000}} = 1.651 \text{ 1/s}.$$

Transverse Waves

A short-time local (lateral) deflection moves as a wave along the wire rope. Czitary (1931) investigated these waves theoretically and he pointed out that the tensile force of a wire rope can be calculated by measuring the wave running time. According to Zweifel (1961) the velocity of a transverse wave is

$$v = \sqrt{\frac{g \cdot S}{q} \left[1 + \frac{EI}{S} \left(\frac{2 \cdot \pi}{\lambda} \right)^2 \right]}. \quad (2.61)$$

In this S is the rope tensile force in N;

g , the acceleration due to gravity in m/s²;

q , the length-related rope weight force in N/m;

E , the rope elasticity module in N/m²;

I , the equatorial moment of inertia in m^4 ;
and λ is the wave length in m.

If a wire rope is knocked with a lead hammer, transverse waves of different wave lengths will be initiated and run along the rope. As (2.61) shows, the velocity of the wave increases with the decreasing length of the wave. The velocity of a wave package with different wave lengths is therefore inhomogeneous with a scatter which increases with time. The lead hammer method is therefore unsuitable for evaluating the tensile force of a rope. Instead of this, Zweifel (1961) recommended using the kind of impulse which produces a wave length which is as large and homogeneous as possible so that the bending stiffness of the wire rope can be neglected and (2.61) can be simplified to

$$v = \sqrt{\frac{g \cdot S}{q}}.$$

With the length-related rope mass $m_r = q/g$, the wave velocity is

$$v = \sqrt{\frac{S}{m_r}}. \quad (2.62)$$

The failure for the rope tensile force calculated is smaller than 1% if the wave length is at least $\lambda > 250; 300$ and $450d$ for the tensile rope stress $\sigma_z = 600; 400$ and 200 N/mm^2 , Zweifel (1961). The desired wave length λ will be obtained if the impulse brought on the rope is not too sharp and $\lambda/4$ away from the end of the rope. Zweifel recommended winding a fibre rope around the wire rope and pulling on that shock-wise by hand. He supposed that the force of one hand was sufficient for a wire rope of 20 mm diameter. For thicker wire ropes, it would be necessary to have several persons pulling (increasing in number with the rope diameter squared). After pulling, the fibre rope should be slightly stressed by hand, so that the waves coming back can be sensed. For n cycles in the measured time t and for the rope length L , the length of the rope from one end to the other, the wave velocity is

$$v = \frac{2 \cdot n \cdot L}{t}. \quad (2.63)$$

According to that and (2.62), the rope tensile force in the middle of the rope field is

$$S = m_r \cdot \left(\frac{2 \cdot n \cdot L}{t} \right)^2. \quad (2.64)$$

For very large rope fields, Zweifel presented equations to calculate the rope tensile force considering the chain line.

Example 2.6: Rope tensile force from the running time of the transverse wave

Data:

Seale rope $6 \times 19\text{-NFC-zZ}$
rope diameter $d = 20 \text{ mm}$

distance between rope terminations $L = 250$ m

number of cycles $n = 12$

running time t for n cycles $t = 40$ s

Results:

According to equation (1.5b) and Table 1.9, the length-related mass m_r of the rope is

$$m_r = \frac{1}{100} \cdot W_1 \cdot d^2 = \frac{1}{100} \cdot 0.359 \cdot 20^2 = 1.436 \text{ kg/m}$$

Then according to (2.64), the rope tensile force in the middle of the rope field is

$$S = m_r \cdot \left(\frac{2 \cdot n \cdot L}{t} \right)^2 = 1.436 \cdot \left(\frac{2 \cdot 12 \cdot 250}{40} \right)^2 = 32,300 \text{ N}.$$

Transverse Vibrations

Transverse vibrations are to be understood as standing waves. The equations for the velocity of the waves can be used to calculate the frequency. Because the wave length is large, the influence of the bending stiffness is very small and can be neglected. So (2.62) can be used and the running time of the wave, there and back, is

$$t_L = \frac{2 \cdot L}{v} = 2 \cdot L \sqrt{\frac{m_r}{S}}. \quad (2.65)$$

In this, L is once again the rope length (or the distance between the ends of the rope for a small curvature). The period T of a standing wave is

$$T = \frac{t_L}{i}$$

and with (2.65)

$$T = \frac{2 \cdot L}{i} \sqrt{\frac{m_r}{S}}. \quad (2.66)$$

In this i is the number of the antinodes of vibration on the rope length. The frequency is

$$f = \frac{1}{T} = \frac{i}{2 \cdot L} \sqrt{\frac{S}{m_r}}. \quad (2.67)$$

In rope fields which are not too long i.e. about 100 m, it is possible to make the rope vibrate, Zweifel (1961). Using the frequency f observed here, the rope tensile force S can be calculated according to the converted (2.67)

$$S = m_r \left(\frac{2 \cdot f \cdot L}{i} \right)^2. \quad (2.68)$$

There are strong variations of the rope tensile force in rope-ways due to braking. The movements of the ropes and their connected masses in such systems can only be calculated with large-scale methods. Such methods are presented by Czitary (1975), Engel (1977), Schlauderer (1990) and Beha (1994). For transverse vibrations, the damping depends on the rope construction, the rope tensile force and the amplitude. Raoof and Huang (1993) reported investigating into the damping of spiral ropes.

Example 2.7: Frequency of transverse vibration (string)

Data from example 2.5.

According to Table 1.9, the length-related rope mass is

$$m_r = \frac{1}{100} \cdot W_2 \cdot d^2 = \frac{1}{100} \cdot 0.4 \cdot 10^2 = 0.40 \text{ kg/m}$$

The frequency of the transverse vibration with one antinode of vibration $i = 1$ is

$$f = \frac{i}{2 \cdot L} \sqrt{\frac{S}{m_r}} = \frac{1}{2 \cdot 50} \sqrt{\frac{1,000 \cdot 9.81}{0.40}} = 1.5661 \text{ 1/s.}$$

2.3 Reduction of the Rope Diameter due to Rope Tensile Force

The reduction of the rope diameter due to the rope tensile force is caused by the lateral contraction of the wires, the strands and, in particular, the cores. The lateral contraction of the wires caused by its tensile stress is small. Even for a tensile stress of 670 N/mm^2 the wire contraction is only one per mil of the diameter of the wire. In comparison, the effect of the relatively low length-related compressive force of the wires and in particular of the strands on the core is much greater.

The length-related compressive force first results in resetting any loose wires and strands and then in deforming the rope in a different way. There is also some minor deformation due to the pressure between wires crossing. As far as fibre-core wire ropes are concerned, a large diameter reduction occurs and this is mainly due to the compression of the core. The diameter reduction of steel-core wire ropes, on the other hand, is normally less than that found with fibre-core wire ropes and this is mainly caused by the wires of the strands and the core becoming adjusted to one another.

Measurements were taken of the diameters of a great number of wire ropes affected by different tensile forces. Figures 2.18 and 2.19 show the diameter reduction measured as the relative rope diameter d_s/d_0 for the first loading. The diameter for the loaded wire rope is d_s and the actual diameter for the not loaded wire rope is d_0 . Figure 2.18 presents the relative rope diameter of

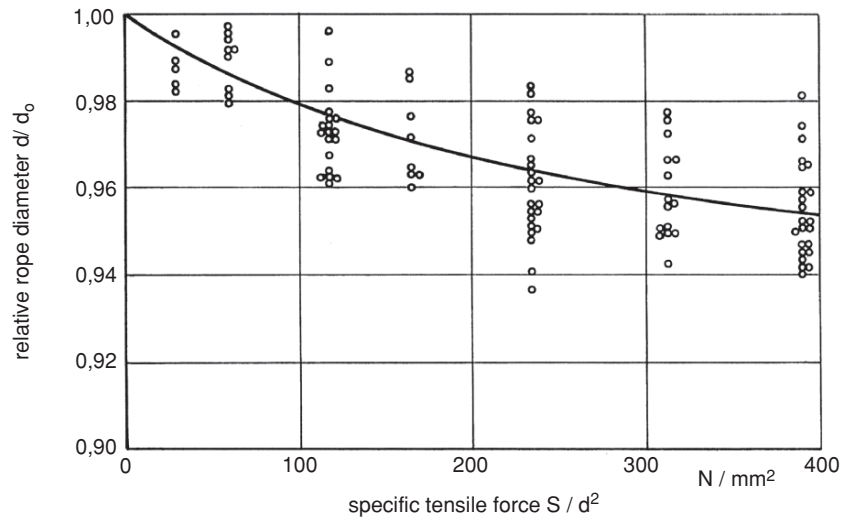


Fig. 2.18. Relative rope diameter d/d_0 of 8-strand ropes with fibre core

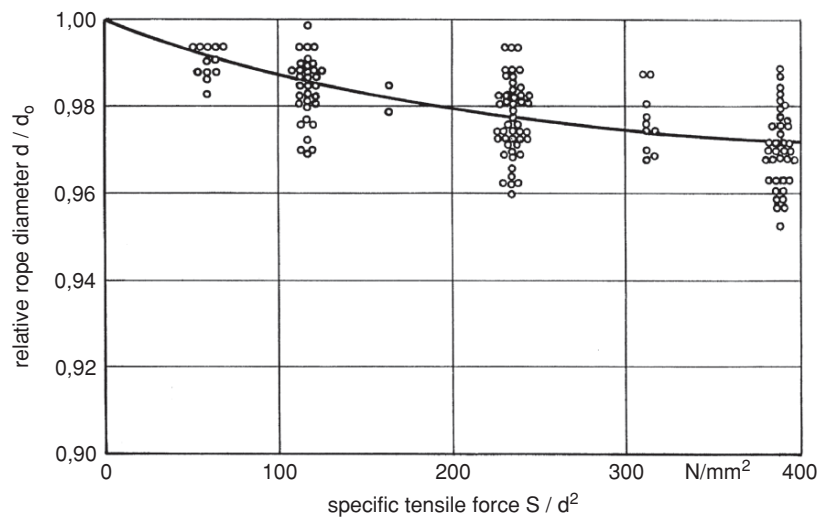


Fig. 2.19. Relative rope diameter d/d_0 of 8-strand ropes with steel core

8-strand fibre-core wire ropes and Fig. 2.19 those with steel cores. The nominal rope diameter – which is normally smaller than the actual diameter – of all these ropes is 16 mm.

Relative rope diameters d/d_0 deviate to a great extent. The most important influences are due to wires and strands loosening and to variations in

core density. An unexpectedly small diameter reduction can result from the strands arching. Such arched strands reduce the working life of running ropes and should therefore be avoided. For the fibre-core wire ropes normally used for rope ways, the regulations therefore recommend that up to half of the wire rope breaking force the rope diameter should be at least 3.1 times the strand diameter for 6-strand ropes and 3.8 times the strand diameter for 8-strand ropes.

2.4 Torque and Torsional Stiffness

2.4.1 Rope Torque from Geometric Data

If a wire rope is loaded by a tensile force, a rope torque will occur due to the helix structure of the rope. The torque can be calculated if the geometric data of the wire rope and the rope tensile force are known. Heinrich (1942) was the first to investigate the torque of a strand by consistently taking any changes in the strand diameter and the lay length into account. Costello and others (1977, 1979) have also arrived at this derivation. In contrast, most authors such as Dreher (1933), Hruska (1953), Unterberg (1972) and Haid (1983) made use of a practical calculation which neglected minor influences. Engel (1957, 1958) calculated the torque and the torsional stiffness as well.

By neglecting the same minor influences, it is possible to come up with a calculation method for the torque using the equations from Sect. 2.1. According to (2.11) the torque of a not twisted strand or spiral rope is (with the wire winding radius $r_W = r$)

$$M = \sum_{i=1}^n S_i \cdot r_i \cdot z_i \cdot \tan \alpha_i. \quad (2.69)$$

The symbols are the same as in Sect. 2.1. The lay angle α for a different lay direction is used with a different sign. In neglecting the contraction (Poisson ratio $\nu = 0$) and with the same elasticity module E for all wires, according to (2.20a) the torque is

$$M = \frac{\Delta l_S}{l_S} \cdot E \cdot \sum_{i=1}^n z_i \cdot r_i \cdot A_i \cdot \cos^2 \alpha_i \cdot \sin \alpha_i. \quad (2.70)$$

By neglecting the same influences, according to (2.21) the rope tensile force is

$$S = \frac{\Delta l_S}{l_S} \cdot E \cdot \sum_{i=0}^n z_i \cdot A_i \cdot \cos^3 \alpha_i. \quad (2.71)$$

By eliminating $E \cdot \Delta l_S / l_S$ from (2.70) and (2.71), the torque of a not twisted strand or spiral rope is

$$M = \frac{S \cdot \sum_{i=1}^n z_i \cdot r_i \cdot A_i \cdot \cos^2 \alpha_i \cdot \sin \alpha_i}{\sum_{i=0}^n z_i \cdot A_i \cdot \cos^3 \alpha_i}. \quad (2.72)$$

It is advisable to introduce a torque constant c_{1S} . With that constant, the torque for a strand or a spiral rope is

$$M = c_{1S} \cdot d_S \cdot S. \quad (2.73)$$

The torque constant c_{1S} depends only on the rope geometry. Out of (2.72) and (2.73), the torque constant for strands or spiral ropes is

$$c_{1S} = \frac{\sum_{i=1}^n z_i \cdot r_i \cdot A_i \cdot \cos^2 \alpha_i \cdot \sin \alpha_i}{d_S \cdot \sum_{i=0}^n z_i \cdot A_i \cdot \cos^3 \alpha_i}. \quad (2.74)$$

Analogous to (2.72), the torque for a stranded rope can be expressed with a torque constant c_1

$$M = c_1 \cdot d \cdot S. \quad (2.72a)$$

The torque constant c_1 for a stranded rope with some round strand layers (strand lay angle β_j) with fibre or steel core is

$$c_1 = \frac{\sum_{j=1}^n z_j \cdot A_j \cdot r_{Sj} \cdot \cos^2 \beta_j \cdot \sin \beta_j + \sum_{j=0}^n z_j \cdot A_j \cdot d_{Sj} \cdot c_{1Sj} \cdot \cos^3 \beta_j}{d \cdot \sum_{j=0}^n z_j \cdot A_j \cdot \cos^3 \beta_j}. \quad (2.75)$$

For a one-layer round strand rope with fibre core the equation for the torque constant can be simplified enormously to

$$c_1 = \frac{r_S \cdot \tan \beta + d_S \cdot c_{1S}}{d}.$$

The symbols here are the same as in Fig. 2.20. For spiral round strand ropes with the same strands in all strand layers the torque constant is a simplification of (2.75)

$$c_1 = \frac{\sum_{j=1}^n z_j \cdot r_{Sj} \cdot \cos^2 \beta_j \cdot \sin \beta_j + c_{1S} \cdot d_S \cdot \sum_{j=0}^n z_j \cdot \cos^3 \beta_j}{d \cdot \sum_{j=0}^n z_j \cdot \cos^3 \beta_j}. \quad (2.76)$$

It is possible to calculate the torque of round strands and round strand ropes to a satisfactory degree of accuracy with the equations presented here provided

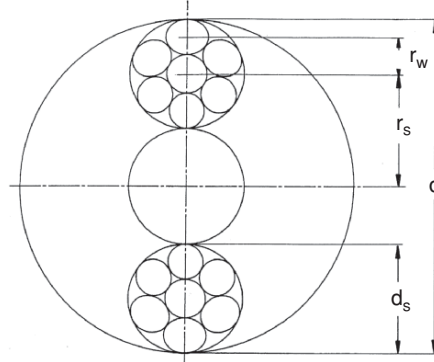


Fig. 2.20. Wire rope with one strand layer

that there is sufficient known geometric data for the rope. These methods of calculation are of particular use to rope manufacturers when designing new ropes, especially for so-called non-rotating ropes. Such ropes have to be designed in such a way that the resulting rope torque is as close to zero as possible.

Calculating the rope torque with the equations presented here is only possible for ropes which are not twisted. For twisted ropes, the torque is strongly influenced by the torsional stiffness.

Example 2.8: Torque constant c of the open spiral rope in Fig. 2.4

According to (2.74), the torque constant is

$$c_{1S} = \frac{(6 \cdot 1.3 - 12 \cdot 2.55 + 18 \cdot 3.8) \cdot 1.227 \cdot 0.970^2 \cdot 0.242}{8.85 \cdot [1 \cdot 1.431 + (6 + 12 + 18) \cdot 1.227 \cdot 0.970^3]}$$

$$c_{1S} = \frac{12.74}{369.45} = 0.0345.$$

2.4.2 Torque of Twisted Round Strand Ropes

Measurements

The torque of twisted and not twisted round strand ropes has been investigated by measurements. With the results of these measurements a simple calculation method will be derived with that the customer can calculate the torque for a given wire rope. This method will have the advantage that it can be used without knowing the precise geometrical data of the rope.

Torque measurements with different tensile forces and different twist angles are carried out with a series of ropes. The equipment of Feyrer and Schiffner (1986) comprises a torque meter and a rope twisting device, mounted in a

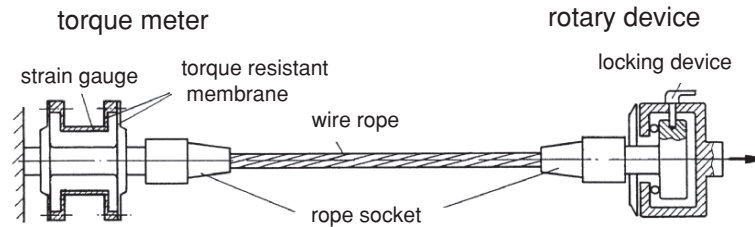


Fig. 2.21. Equipment for measuring the torque and the rotary angle, Feyrer and Schiffner (1986)

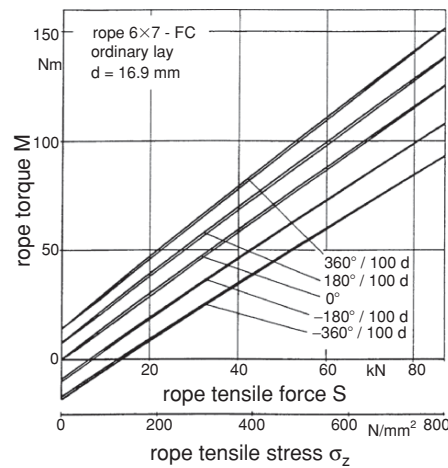


Fig. 2.22. Torque of a wire rope $6 \times 7\text{-FC-sZ}$, Feyrer and Schiffner (1986)

tensile testing machine for carrying out the measurements. The torque meter, which measures the torque by means of strain gauges, is because of the installed membranes nearly not influenced by the tensile force. The entire equipment is shown in Fig. 2.21. Rebel and Chandler (1996) presented a measuring equipment with the opportunity to measure in addition the rope elongation and the rope diameter reduction.

The torques measured with different twisted wire ropes (positive sign for turn off) are shown in Figs. 2.22 and 2.23 for ordinary lay ropes $6 \times 7\text{-FC}$ and Warrington $8 \times 19\text{-FC}$. As for all wire ropes with fibre core the torque increases nearly linear with the tensile force. The distance of the lines for the different twist angles is for the wire rope $6 \times 7\text{-FC}$ bigger than for the Warrington rope $8 \times 19\text{-FC}$. That means the wire rope $6 \times 7\text{-FC}$ with 42 wires is more torsion rigid than the Warrington rope with 152 wires. The Warrington rope has been measured lubricated and not lubricated with practically no different torque.

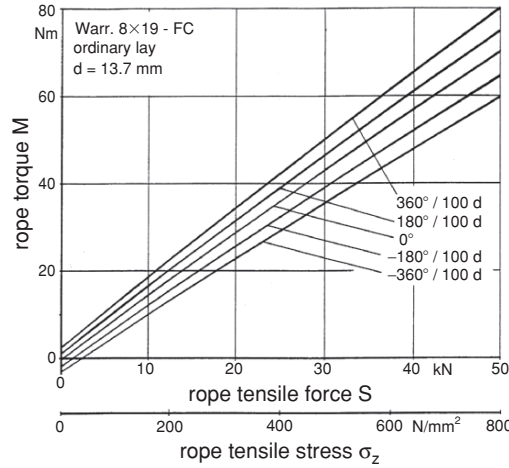


Fig. 2.23. Torque of a Warrington rope 8 × 19-FC-sZ, Feyrer and Schiffner (1986)

The torque for the increasing and decreasing tensile force shows nearly no hysteresis. In the following diagrams only the lines for the increasing tensile force are shown.

For wire ropes with independent made steel wire rope core IWRC the torque also increases nearly linear with the tensile force. To demonstrate this, in addition to the measured torque lines, straight lines are sketched in Fig. 2.24 for a Filler rope 8 × (19 + 6F)-IWRC-sZ.

Double parallel wire ropes (PWRC) have only a nearly linear relation between torque and tensile force for small twist angles. Figure 2.25 shows the torque of a Seale rope 8 × 19-PWRC-zZ. In the twisted state the strands and the core are loaded very differently. Therefore in the 360° untwisted wire rope on the rope length $L = 100d$, the core breaks very soon at the relative small rope tensile stress $\sigma_z = 640 \text{ N/mm}^2$, as to be seen in Fig. 2.25.

In addition to the described investigation (1986) a lot of torque measurements with spiral strand ropes have been done from the Institut für Fördertechnik der Universität Stuttgart in a great part by students. This work was sponsored from AVIF and the Drahtseilvereinigung. The results of this investigation are presented by Feyrer (1997), in which the work of the students are listed.

The torque of a spiral strand rope with two strand layers is shown in Fig. 2.26 and that with three strand layers in Fig. 2.27. The torque-tensile-force lines are all buckled even for the not twisted rope. The reason for this buckling of the not twisted rope is that the different strands are not loaded from the load beginning. The relative big distance between the torque lines shows that the spiral strand ropes are twist rigid.

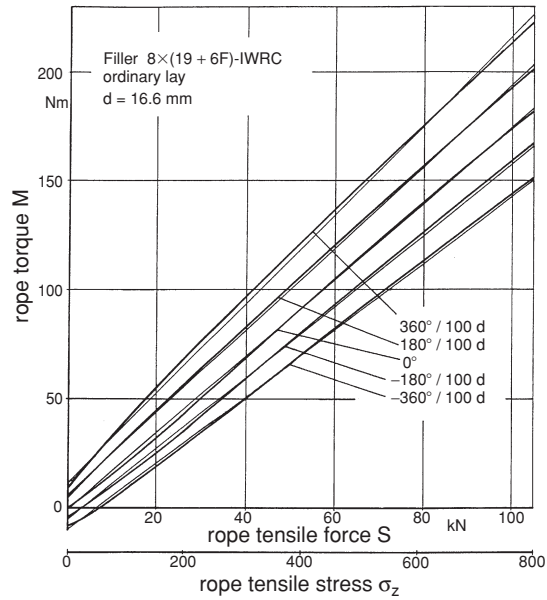


Fig. 2.24. Torque of a Filler rope $8 \times (19 + 6F)$ -IWRC-sZ, Feyrer and Schiffner (1986)

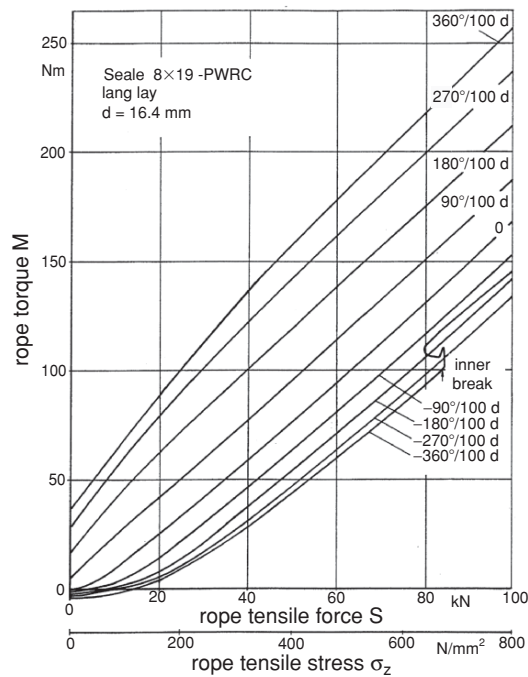


Fig. 2.25. Torque of a Seale rope 8×19 -PWRC-zZ, Feyrer and Schiffner (1986)

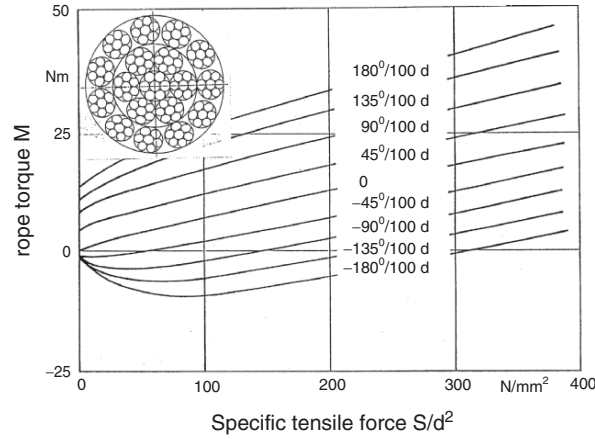


Fig. 2.26. Torque of a spiral round strand rope with two strand layers, Feyrer (1997)

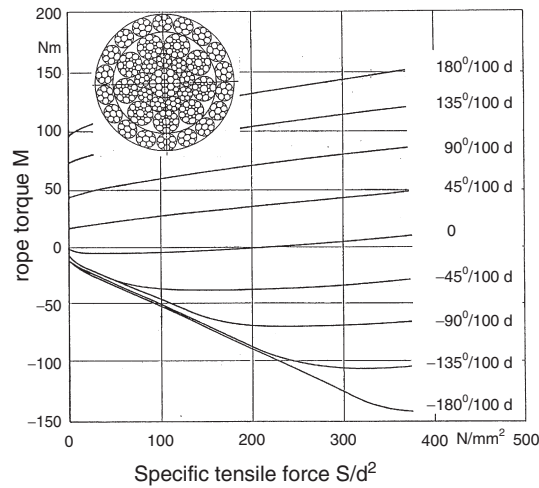


Fig. 2.27. Torque of a spiral round strand rope with three strand layers, Feyrer (1997)

Calculation of the Torque for Wire Ropes

The results of the torque measurements with the round strand wire ropes with one strand layer can be very good evaluated by a regression calculation. Kollros (1974, 1976) evaluated first his torque measurements with such a regression. Based on theoretical considerations he creates an equation with two constants for the regression. Forerunner of these constants are the torque constant $\mu = M/S = c_1 d$ and the torsional stiffness $D = M/\omega$ from Engel (1957, 1958 and 1966).

The torque measurements with many wire ropes by Feyrer and Schiffner (1986) show that two constants are not enough to describe the results with good precision. Therefore the regression for the results of these measurements has been made practically with the equation of Kollros but with three constants. The torque is then

$$M = c_1 \cdot d \cdot S + c_2 \cdot d^2 \cdot S \cdot \omega + c_3 \cdot G \cdot d^4 \cdot \omega. \quad (2.77)$$

Therein M is the torque;
 φ , the rotary angle in rad;
 d , the rope diameter;
 $\omega = \varphi/L$, the twist angle;
 S , the tensile force;
 L , the rope length;
 G , the shear module;
and c_1, c_2, c_3 , are constants.

The twist angle ω has to set positive for turning off the rope and negative for turning on the rope. The constants c and their standard deviation are listed in Table 2.6. These constants have been found by regression of Feyrer and Schiffner (1986) with their own test results, with many test results of students and with the test results of Kollros (1974) and Unterberg (1972). As limit for the use of (2.77) with the constants c , the maximum allowed twist angle $\omega_{\max} = \varphi_{\max}/100d$ (angle for a rope length of 100 times rope diameter) is also given in Table 2.6.

By measurements with wire ropes of diameters 55.6 and 76 mm Kraincanic and Hobbs (1997) evaluated torque constants c_1 that corresponds respecting the standard deviation with those in Table 2.6. Cantin et al. (1993) found in measurements with a 6-strand rope constants c_1 and c_2 comparable with that of Table 2.6 but the constant c_3 deviates more than 30%. For lang's lay triangular strand ropes Rebel (1997) found that (2.77) cannot describe satisfactory the measured torques. Therefore Rebel established an equation with nine constants what he evaluated out of his measurements.

Example 2.9: Wire rope torque

Data:

Filler rope $6 \times (19 + 6F) - \text{NFC-sZ}$
rope diameter $d = 16 \text{ mm}$
rope length $L = 5,000 \text{ mm}$
shear module $G = 76,000 \text{ N/mm}^2$
tensile force $S = 40,000 \text{ N}$
angle of turn on $\varphi = -600^\circ$
 $\varphi = -2 \cdot \pi \cdot 600/360 = -10.47 \text{ rad}$
twist angle $\omega = -10.47/5000 = -0.002094 \text{ rad/mm}$

Table 2.6. Constants c_1 , c_2 , c_3 to the torque (2.77), Feyrer and Schiffner (1986)

rope construction			6-strand				8-strand			
core	layer	strands number of wires	c_1	c_2	$c_3 \cdot 10^3$	$\pm\varphi_{\max}$ for $100d$	c_1	c_2	$c_3 \cdot 10^3$	$\pm\varphi_{\max}$ for $100d$
FC	sZ	7	0.100	0.157	0.765	360°	0.106	0.166	0.658	360°
		19 Seale	0.109	0.207	0.400	360°	0.115	0.216	0.293	360°
		19 Filler ^a ,	0.102	0.212	0.376	360°	0.108	0.222	0.268	360°
		19 Warr.	0.102	0.212	0.376	360°	0.108	0.222	0.268	360°
		36 Warr.- Seale	0.105	0.212	0.376	360°	0.111	0.222	0.268	360°
FC	zZ	7	0.123	0.127	0.732	360°	0.129	0.137	0.624	360°
		19 Seale	0.132	0.177	0.367	360°	0.138	0.186	0.259	360°
		19 Filler ^a ,	0.126	0.183	0.342	360°	0.131	0.194	0.234	360°
		19 Warr.	0.126	0.183	0.342	360°	0.131	0.194	0.234	360°
		36 Warr.- Seale	0.128	0.183	0.342	360°	0.134	0.194	0.234	360°
IWRC	sZ	7	0.080	0.131	0.921	180°	0.086	0.141	0.813	180°
		19 Seale	0.089	0.181	0.556	180°	0.095	0.190	0.448	180°
		19 Filler ^a ,	0.082	0.187	0.531	180°	0.088	0.196	0.424	180°
		19 Warr.	0.082	0.187	0.531	180°	0.088	0.196	0.424	180°
		36 Warr.- Seale	0.085	0.187	0.531	180°	0.091	0.196	0.424	180°
IWRC	zZ	7	0.103	0.101	0.888	180°	0.109	0.112	0.779	180°
		19 Seale	0.112	0.151	0.523	180°	0.118	0.160	0.414	180°
		19 Filler ^a ,	0.105	0.158	0.497	180°	0.111	0.168	0.390	180°
		19 Warr.	0.105	0.158	0.497	180°	0.111	0.168	0.390	180°
		36 Warr.- Seale	0.108	0.158	0.497	180°	0.114	0.168	0.390	180°
standard deviation			0.012	0.028	0.080		0.012	0.028	0.080	

^aFiller 19 = Filler 19 + 6F

The constants out of Table 2.6 are

$$c_1 = 0.102; \quad c_2 = 0.212; \quad c_3 = 0.376 \times 10^{-3}$$

Results:

According (2.77) the rope torque is

$$M = 0.102 \cdot 16 \cdot 40,000 - 0.212 \cdot 16^2 \cdot 40,000 \cdot 0.002094 \\ - 0.376 \cdot 10^{-3} \cdot 16^4 \cdot 0.002094 \cdot 76,000$$

$$M = 65,800 - 4,540 - 3,920$$

$$M = 56,800 \text{ N mm} = 56.8 \text{ N m.}$$

Definition of Non-Rotating Rope

The spiral strand ropes are designated for supporting loads without turning protection. Therefore they should be rotation-resistant to a great extent. This will be succeeded only approximately. Really non-rotating spiral strand ropes do not exist. But it is useful to define the limit up to this a wire rope can be declared as a non-rotating one.

A proposal for the definition of a non-rotating wire rope is:

A wire rope counts as non-rotating if the twist angle rests smaller than

$$\frac{\varphi}{L} \leq \frac{360^\circ}{1,000 \cdot d}$$

during the tensile loading between

$$\frac{S}{d^2} = 0 \quad \text{to} \quad \frac{S}{d^2} = 150 \text{ N/mm}^2.$$

Spiral Round Strand Ropes

From 48 spiral round strand ropes with three strand layers (with between 14 and 20 outer strands) seven ropes are not non-rotating for the given definition. On the other hand from the 25 tested spiral round strand ropes with two strand layers (with between 10 and 12 outer strands) six ropes are still non-rotating.

The non-rotating spiral strand ropes show – if not twisted – torque-tensile-force lines with a small buckling and a mean constant $c_1 = 0.026$ with the standard deviation $s = 0.012$. For all these ropes the torque constant c_1 , calculated with (2.76) on the base of geometrical data, has been very well confirmed by the measurements. The deviation from the torque zero is to lead back on the rope geometry not optimal chosen. Under the specific tensile force $S/d^2 = 0\text{--}150 \text{ N/mm}^2$ the “non-rotating ropes” show the mean twist $\varphi/1,000d = -40^\circ/1,000d$ in turning on direction with the standard deviation $s = 140^\circ/1,000d$.

The low-rotating spiral round strand ropes with two strand layers show – if not twisted – a nearly straight torque-tensile-force line with $c_1 = 0.058$. For a small twisting up to $90^\circ/100d$ a nearly straight torque-tensile-force line is only to expect for specific tensile forces above $S/d^2 > 70 \text{ N/mm}^2$, as can be seen in Fig. 2.26. For that the constants are

$$c_1 = 0.058$$

$$c_2 = 0.269$$

$$c_3 = 0.00853.$$

2.4.3 Rotating of the Bottom Sheave

As an important possible use of the evaluated torque equation, the height between the bottom sheave and the rope termination respectively the drum can be calculated, at which the bottom sheave turns too much so that the bearing ropes overlap each other. Unterberg (1972) investigated first this problem. Out of the energy W for lifting the bottom sheave and the load when the bottom sheave turns, he found for the reverse moment

$$M_{\text{rev}} = \frac{dW}{d\varphi} = \frac{r_1 \cdot r_2 \cdot \sin \varphi}{\sqrt{h_0^2 - 2 \cdot r_1 \cdot r_2 \cdot (1 - \cos \varphi)}} Q. \quad (2.78)$$

Q is the force from the mass of the load and the bottom sheave. The meaning of the other symbols can be taken out of Fig. 2.28.

In normal cases the height h_0 is much bigger than r_1 and r_2 . Then the reverse moment is with no big error

$$M_{\text{rev}} = \frac{r_1 \cdot r_2 \cdot \sin \varphi}{h_0} \cdot Q$$

and the maximum reverse moment is for $\varphi = 90^\circ$

$$M_{\text{rev,max}} = \frac{r_1 \cdot r_2}{h_0} \cdot Q. \quad (2.79)$$

To prevent the rope overlapping, this maximum reverse moment must be at least greater than the torque of the bearing ropes

$$M_{\text{rope}} < M_{\text{rev,max}}. \quad (2.80)$$

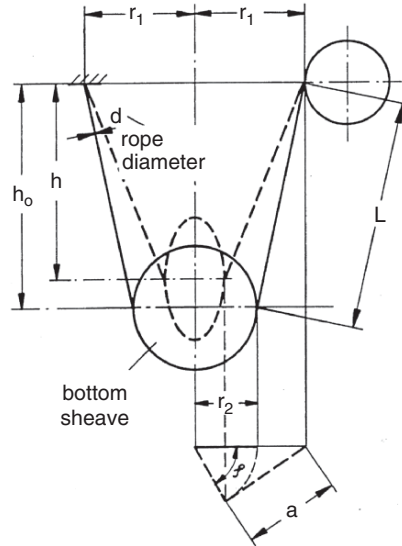


Fig. 2.28. Rotation of a bottom sheave

For the normal height h_0 the angle φ is small. The torque for one rope is then according (2.77)

$$M = c_1 \cdot d \cdot S. \quad (2.81)$$

In addition a torque M_0 can occur by a wrong rope mounting. This torque will be at rough guess

$$M_0 = 0.0002 \cdot d \cdot F_{\min} \quad (2.82)$$

with $F_{\min}$ for the minimum breaking force of the rope. The torque for the z bearing ropes is then

$$M_{\text{rope}} = (M + M_0) \cdot z. \quad (2.83)$$

An overlapping of the bearing ropes will be prevented if the condition (2.80) is provided. Because of the estimated constant trouble-moment M_0 the critical case is that for the force Q out of the bottom sheave mass only.

Example 2.10: Rotating of the bottom sheave

Data:

force out of the bottom sheave mass $Q = 600 \text{ N}$
distance $r_1 = 200 \text{ mm}$
bottom sheave radius $r_2 = 160 \text{ mm}$
height $h_0 = 8,000 \text{ mm}$
Warr.-Seale $6 \times 36\text{-IWRC-}sZ$ diameter $d = 16 \text{ mm}$
min. breaking force $F_{\min} = 160,000 \text{ N}$
number of bearing ropes $z = 2$

Results:

The maximum reverse moment is according (2.79)

$$M_{\text{rev,max}} = \frac{r_1 \cdot r_2}{h_0} \cdot Q = \frac{160 \cdot 200 \cdot 600}{8,000} = 2,400 \text{ N mm.}$$

The rope torque from rope tensile force $S = Q/2$ is according (2.81)

$$M = c_1 \cdot d \cdot S = 0.085 \cdot 16 \cdot 300 = 408 \text{ N mm.}$$

The torque M_0 according (2.82) is

$$M_0 = 0.0002 \cdot d \cdot F_{\min} = 0.0002 \cdot 16 \cdot 160,000 = 512 \text{ N mm.}$$

The torque for the z bearing ropes is then

$$M_{\text{rope}} = (M + M_0) \cdot z = (408 + 512) \cdot 2 = 1,840 \text{ N mm.}$$

Because $M_{\text{rope}} < M_{\text{rev,max}}$ an overlapping of the bearing ropes is prevented.

2.4.4 Rope Twist Caused by the Height-Stress

Wire Rope Supported Non-Rotated at both Ends

Because of the rope weight the tensile force in a suspending rope has on the upper end a bigger tensile force than on the lower end. The rope stress increasing with the height of the suspending rope is called height-stress. Because the rope torque along the rope length must be constant, the wire rope supported non-rotated on the upper and the lower end will twist between the both ends. The rotary angle of a vertical hanging wire rope is demonstrated in Fig. 2.29. The rope turns on in the upper field and off in the lower field.

Engel (1957) and little later Hermes and Bruuens (1957) derived at first the rotary angle caused by the height-stress, see also Gibson (1980). Engel (1959) calculated the twist angle for haul and traction ropes of rope ways. Rebel (1997) calculated with his own equation the rotation of triangular strand ropes in deep shafts.

In the following the rotary angle will be derived on the base of (2.77). That has the advantage that the constants for the different wire ropes from Table 2.6 can be used. Transforming (2.77) the twist angle is

$$\omega = \frac{d\varphi}{dx} = \frac{M - c_1 \cdot d \cdot S}{c_2 \cdot d^2 \cdot S + c_3 \cdot G \cdot d^4}. \quad (2.84)$$

The rope tensile force increases from the lower end with the rope length x and the angle β_F between the horizontal and the secant of the small rope bow ($S \geq m \cdot g \cdot L \cdot \cos \beta_F$ as normal in practice) approximately

$$S \approx S_0 + m \cdot g \cdot x \cdot \sin \beta_F$$

with the tensile force S_0 on the lower end and the length-related rope mass m (exactly for $\beta_F = 90^\circ$). Equation (2.84) is with that

$$d\varphi = \frac{M - c_1 \cdot d \cdot (S_0 + m \cdot g \cdot x \cdot \sin \beta_F)}{c_2 \cdot d^2 \cdot (S_0 + m \cdot g \cdot x \cdot \sin \beta_F) + c_3 \cdot G \cdot d^4} \cdot dx. \quad (2.85)$$

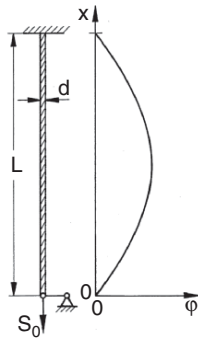


Fig. 2.29. Rotation of a vertical hanging wire rope supported non-rotated at both ends

By integration the rotary angle φ is

$$\begin{aligned} \varphi = & \frac{c_1 \cdot x}{c_2 \cdot d} - \left(\frac{M}{c_2 \cdot d^2 \cdot m \cdot g \cdot \sin \beta_F} + \frac{c_1 \cdot c_3 \cdot d \cdot G}{c_2^2 \cdot m \cdot g \cdot \sin \beta_F} \right) \\ & \times \ln[c_2 \cdot d^2 \cdot (S_0 + m \cdot g \cdot x \cdot \sin \beta_F) + c_3 \cdot G \cdot d^4] + B. \end{aligned} \quad (2.86)$$

As preproposed the rotary angle φ is $\varphi = 0$ for $x = 0$ and $\varphi = 0$ for $x = L$.

From this and (2.86) the torque M and the constant B can be derived. The torque is

$$\begin{aligned} M = & -\frac{c_1 \cdot c_3 \cdot G \cdot d^3}{c_2} \\ & - \frac{c_1 \cdot d \cdot m \cdot g \cdot L \cdot \sin \beta_F}{\ln \left(\frac{c_2 \cdot S_0 + c_3 \cdot G \cdot d^2}{c_2 \cdot S_0 + c_2 \cdot m \cdot g \cdot L \cdot \sin \beta_F + c_3 \cdot G \cdot d^2} \right)}. \end{aligned} \quad (2.87)$$

Then with (2.86) the rotary angle is

$$\varphi = \frac{c_1 \cdot x}{c_2 \cdot d} - \frac{c_1 \cdot L}{c_2 \cdot d} \cdot \frac{\ln \left(\frac{c_2 \cdot m \cdot g \cdot x \cdot \sin \beta_F}{c_2 \cdot S_0 + c_3 \cdot G \cdot d^2} + 1 \right)}{\ln \left(\frac{c_2 \cdot m \cdot g \cdot L \cdot \sin \beta_F}{c_2 \cdot S_0 + c_3 \cdot G \cdot d^2} + 1 \right)}. \quad (2.88)$$

The maximum rotary angle occurs for the rope length

$$x(\varphi_{\max}) = -\frac{c_2 \cdot S_0 + c_3 \cdot G \cdot d^2}{c_2 \cdot m \cdot g \cdot \sin \beta_F} + \frac{L}{\ln \left(\frac{c_2 \cdot m \cdot g \cdot L \cdot \sin \beta_F}{c_2 \cdot S_0 + c_3 \cdot G \cdot d^2} + 1 \right)}. \quad (2.89)$$

The maximum rotary angle is given with $x = x(\varphi_{\max})$ in (2.88). The maximum twist angle $\omega_{\max}$ occurs on the lower rope end, $x = 0$. It will be calculated with (2.85) for $x = 0$ and the torque out of (2.87).

Wire Rope Supported Non-Rotated at Both Ends, Simplified Calculation

The torque M in the wire rope supported non-rotated at both ends can be set simplified with only a small failure

$$M = c_1 \cdot d \cdot (S_0 + m \cdot g \cdot L/2 \cdot \sin \beta_F). \quad (2.90)$$

Then with (2.85) the twist angle is

$$\omega = \frac{c_1 \cdot m \cdot g \cdot (L/2 - x) \cdot \sin \beta_F}{c_2 \cdot d \cdot (S_0 + m \cdot g \cdot x \cdot \sin \beta_F) + c_3 \cdot G \cdot d^3}. \quad (2.91)$$

With that on the upper rope end the twist angle is

$$\omega_{\text{upper}} = -\frac{c_1 \cdot m \cdot g \cdot L/2 \cdot \sin \beta_F}{c_2 \cdot d \cdot (S_0 + m \cdot g \cdot L \cdot \sin \beta_F) + c_3 \cdot G \cdot d^3} \quad (2.91a)$$

and on the lower rope end

$$\omega_{\text{lower}} = \frac{c_1 \cdot m \cdot g \cdot L/2 \cdot \sin \beta_F}{c_2 \cdot d \cdot S_0 + c_3 \cdot G \cdot d^3}. \quad (2.91b)$$

For the integration to evaluate the rotary angle φ , the denominator of (2.91) can be further simplified with $x = L/2$. The failure for that is very small if the rope weight force $mgL \sin \beta_F$ is smaller than the rope tensile force S_0 . Then the twist angle is

$$\omega = \frac{c_1 \cdot m \cdot g \cdot (L/2 - x) \cdot \sin \beta_F}{c_2 \cdot d \cdot (S_0 + m \cdot g \cdot L/2 \cdot \sin \beta_F) + c_3 \cdot G \cdot d^3} \quad (2.92)$$

and after integration the rotary angle φ is

$$\varphi = \frac{-c_1 \cdot m \cdot g \sin \beta_F \cdot (L - x) \cdot x/2}{c_2 \cdot d \cdot (S_0 + m \cdot g \cdot \sin \beta_F \cdot L/2) + c_3 \cdot G \cdot d^3}. \quad (2.93)$$

The maximum rotary angle (for $x = L/2$) is

$$\varphi_{\text{max}} = \frac{1}{8} \cdot \frac{-c_1 \cdot m \cdot g \cdot \sin \beta_F \cdot L^2}{c_2 \cdot d \cdot (S_0 + m \cdot g \cdot \sin \beta_F \cdot L/2) + c_3 \cdot G \cdot d^3}. \quad (2.94)$$

Example 2.11: Wire rope supported non-rotated at both ends

Data:

Warr. $8 \times 19\text{-NFC-}sZ$,
 rope diameter $d = 16 \text{ mm}$ or $d = 0.016 \text{ m}$
 rope length related mass $m = 0.89 \text{ kg/m}$
 shear module $G = 76,000 \text{ N/mm}^2$ or $G = 76 \times 10^9 \text{ N/m}^2$
 rope length $L = 500 \text{ m}$
 lower tensile load $S_0 = 10,000 \text{ N}$
 angle $\beta_F = 90^\circ$
 constants (Table 2.6) $c_1 = 0.108$; $c_2 = 0.222$; $c_3 = 0.000268$

Results:

According to (2.87) the torque is

$$M = -40.58 + 61.56 = 20.98 \text{ N m.}$$

According (2.89) the maximum angle occurs at the rope length

$$x(\varphi_{\text{max}}) = 4080.8 - 3835.8 = 245 \text{ m}$$

The maximum rotary angle – (2.88) – is

$$\varphi_{\text{max}} = -232.8 \text{ rad.}$$

The maximum number of rope turns is then

$$n_{\text{max}} = \frac{\varphi_{\text{max}}}{2 \cdot \pi} = -37.$$

According to (2.85) and (2.87), the maximum twist angle is on the lower rope end ($x = 0$)

$$\omega_{\text{lower}} = \omega(x = 0) = 1.94 \text{ rad/m} = 111^\circ/\text{m} = 178^\circ/100d.$$

The twist angle on the upper rope end is

$$\omega_{\text{upper}} = \omega(x = L) = -1.79 \text{ rad/m} = -103^\circ/\text{m} = -164^\circ/100d.$$

The calculated twist angles are smaller than the allowed limit $360^\circ/100d$ (Table 2.6).

Simplified calculation:

With the simplified calculation the maximum rotary angle is

$$\varphi_{\text{max}} = -232.6 \text{ rad}$$

and the twist angles are

$$\omega_{\text{lower}} = \omega(x = 0) = 1.98 \text{ rad/m} \quad \text{and} \quad \omega_{\text{upper}} = \omega(x = L) = -1.75 \text{ rad/m}.$$

Suspended Wire Rope Without Rotation Protection at the Lower End

At the lower end the wire rope has no rotation protection. However this rope end is like the upper rope end fixed in a termination so that the relative motion of wires and the strands are prevented. (This is the condition for the validity of the constants c of the Table 2.6 and all the equations based on equation (2.77)).

With the torque $M = 0$ the twist angle is (again for $S_0 \geq m \cdot g \cdot L \cdot \cos \beta_F$) according to (2.85)

$$\omega = \frac{-c_1 \cdot d \cdot (S_0 + m \cdot g \cdot x \cdot \sin \beta_F)}{c_2 \cdot d \cdot (S_0 + m \cdot g \cdot x \cdot \sin \beta_F) + c_3 \cdot G \cdot d^4}. \quad (2.85a)$$

$$\begin{aligned} \varphi = & -\frac{c_1 \cdot (L - x)}{c_2 \cdot d} - \frac{c_1 \cdot c_3 \cdot d \cdot G}{c_2^2 \cdot m \cdot g \cdot \sin \beta_F} \\ & \times \cdot \ln \frac{c_2 \cdot d^2 (m \cdot g \cdot x \cdot \sin \beta_F + S_0) + c_3 \cdot G \cdot d^4}{c_2 \cdot d^2 (m \cdot g \cdot L \cdot \sin \beta_F + S_0) + c_3 \cdot G \cdot d^4}. \end{aligned} \quad (2.86a)$$

The maximum rotary angle φ_{max} occurs at the lower end of the rope, that means for $x = 0$. The most interesting maximum twist angle ω_{max} occurs at the upper rope end, for $x = L$.

According to (2.85), the maximum twist angle is

$$\omega_{\text{max}} = \frac{-c_1 \cdot d \cdot (S_0 + m \cdot g \cdot L \cdot \sin \beta_F)}{c_2 \cdot d^2 \cdot (S_0 + m \cdot g \cdot L \cdot \sin \beta_F) + c_3 \cdot G \cdot d^4}. \quad (2.85b)$$

Example 2.12: Suspended wire rope without rotation protection at the lower end**Data:**

The same data will be used as in example 2.11, but the tensile force at the lower rope end is $S_0 = 0$.

Results:

According to (2.93), the maximum rotary angle at the lower rope end is

$$\varphi_{\max} = -15,202 - 81,795 \cdot \ln 0.8433 = -15,292 + 13,943$$

$$\varphi_{\max} = -1,259 \text{ rad.}$$

With that, the number of rope turns at the lower end is

$$n_{\max} = \frac{\varphi_{\max}}{2\pi} = \frac{-1,259}{2\pi} = -200.4.$$

According to (2.94), the maximum twist angle (rotary angle per length unit at the upper rope end) is

$$\omega_{\max} = -4.766 \text{ rad/m} \rightarrow \omega_{\max} = -273^\circ/\text{m} \rightarrow \omega_{\max} = -437^\circ/100d$$

The numbers are given with four or more digits to make it easier to follow the calculation. But, of course, the results are only valid in a scattering following the standard deviation of the constants c from Table 2.6. Above that, the maximum twist angle $437^\circ/100d$ exceeds the limit $360^\circ/100d$ for the validity of the constants. But for the rope considered here according to Fig. 2.23, there is practically no change of the constants c to be expected.

2.4.5 Change of the Rope Length by Twisting the Rope

By twisting a wire rope, the rope length and the lay length will be increased in the “on” rotary direction and decreased in the “off” rotary direction. For this problem, Hankus (1997) remembered the equations of Glushko (1996). He measured and calculated the rotary angles of wire ropes in mining shafts, Hankus (1993, 1997).

In the following, the change of rope lengths will be calculated using geometric data for wire ropes with one strand layer and a fibre core. It can be presupposed that the strand length l and the strand winding radius r will remain constant. On the base of Fig. 2.30 the change of the rope length is given by the equation

$$L + \Delta L_D = \sqrt{l^2 - (u - \Delta u)^2}. \quad (2.95)$$

In this L is the rope length and ΔL_D is the rope elongation when the rope is twisting.

Δu is the change of the circle bow length for the strand helix. With the strand winding radius $r_S = r = \text{const.}$ and the rotary angle φ , the circle bow and the change of the circle bow length are

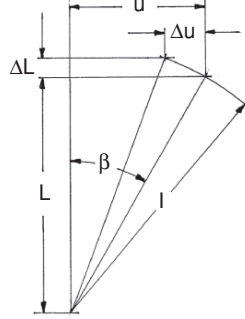


Fig. 2.30. Change of rope length by twisting the rope

$$u = r \cdot \varphi \quad \text{and} \quad \Delta u = r \cdot \Delta \varphi.$$

Then, from (2.95), the rope elongation by twisting the rope is

$$\Delta L_D = \sqrt{l^2 - r^2 \cdot (\varphi - \Delta \varphi)^2} - L. \quad (2.96)$$

With the strand lay angle β it is

$$l = L / \cos \beta, \quad r \cdot \varphi = L \cdot \tan \beta \quad \text{and} \quad \Delta \varphi / L = -\omega.$$

Using that, the rope elongation (+) or shortening (−) is

$$\Delta L_D = \sqrt{\frac{L^2}{\cos^2 \beta} - L^2 \cdot \tan^2 \beta - 2 \cdot r \cdot L^2 \cdot \omega \cdot \tan \beta - r^2 \cdot \omega^2 \cdot L^2} - L.$$

Divided by L , the rope extension by twisting the rope is

$$\varepsilon_D = \frac{\Delta L_D}{L} = \sqrt{1 - 2 \cdot r \cdot \omega \cdot \tan \beta - r^2 \cdot \omega^2} - 1. \quad (2.97)$$

For a constant rope twisting over the rope length L , the change of the rope length is

$$\Delta L_D = \varepsilon_D \cdot L$$

and if the twisting over the rope length is not constant, the change of the rope length is

$$\Delta L_D = \int_{x=0}^L \varepsilon_D \cdot dx = \int_{x=0}^L \left(\sqrt{1 - 2 \cdot r \cdot \omega \cdot \tan \beta - r^2 \cdot \omega^2} - 1 \right) \cdot dx \quad (2.97a)$$

and with ω from (3.85) the change of the rope length is

$$\begin{aligned} \Delta L_D &= \int_0^L \left(\sqrt{1 - \frac{(M - c_1 \cdot d \cdot (S_0 + m \cdot g \cdot x)) \cdot 2 \cdot r \cdot \tan \beta}{c_2 \cdot d^2 \cdot (S_0 + m \cdot g \cdot x) + c_3 \cdot G \cdot d^4} - r^2 \cdot \left(\frac{M - c_1 \cdot d \cdot (S_0 + m \cdot g \cdot x)}{c_2 \cdot d^2 \cdot (S_0 + m \cdot g \cdot x) + c_3 \cdot G \cdot d^4} \right)^2} - 1 \right) dx \\ &\quad (2.98) \end{aligned}$$

The equation has to be calculated numerically.

The lay angle β' of the twisted wire rope is

$$\tan \beta' = \frac{u - \Delta u}{L + \Delta L_D} = \frac{L \cdot \tan \beta - L \cdot r \cdot \omega}{L + \Delta L_D}.$$

With $L + \Delta L_D = L \cdot (1 + \varepsilon_D)$, the lay angle of the twisted wire rope is then

$$\beta' = \arctan \frac{\tan \beta - r \cdot \omega}{1 + \varepsilon_D}. \quad (2.99)$$

and with the elastic rope extension ε_E the lay angle is

$$\beta'' = \arctan \frac{\tan \beta - r \cdot \omega}{1 + \varepsilon_D + \varepsilon_E}. \quad (2.99a)$$

The lay length of the twisted wire rope is

$$h'_S = h_S \frac{\tan \beta}{\tan \beta'}. \quad (2.100)$$

Example 2.13: Change of length of a twisted wire rope

To demonstrate the rope length calculations, examples 2.9, 2.11 and 2.12 will be continued. The lay angle is $\beta = 20^\circ$.

Example 2.13a: Constant twist angle over the entire rope length, continuation of example 2.9

According to (2.97), the rope elongation by twisting the rope is

$$\Delta L_D = (\sqrt{1 + 2 \cdot 5.4 \cdot 0.002094 \cdot 0.364 - 5.4^2 \cdot 0.002094^2} - 1) \cdot 5,000$$

$$\Delta L_D = (\sqrt{1.008103} - 1) \cdot 5,000 = 20.2 \text{ mm}$$

With the metallic rope cross-section $A = 100.5 \text{ mm}^2$ and the rope elasticity module $E_S = 93,000 \text{ N/mm}^2$ from Table 2.3 for the pre-loaded rope between $\sigma_z = 0$ and 400 N/mm^2 , the elastic rope elongation is

$$\Delta L_E = \frac{S \cdot L}{A \cdot E_S} = \frac{40,000 \cdot 5,000}{100.5 \cdot 93,000} = 21.4 \text{ mm}.$$

The overall elongation is

$$\Delta L = \Delta L_D + \Delta L_E = 20.2 + 21.4 = 41.6 \text{ mm}.$$

The small neglected part of the elongation ΔL_T from twisting the strands, see example 2.14a.

Example 2.13b: Wire rope supported non-rotated at both ends, continuation of example 2.11

Integrating (2.98) leads to a small reduction of the rope length for the twisted wire rope

$$\Delta L_D = -0.01138 \approx -0.011 \text{ m}$$

From the length $x = 245$ m, the lower rope section turns off and the upper rope section turns on with the rotary angle $\varphi_{\max}$. The rope elongation from twisting on the larger upper rope section $L_u = 255$ m is 0.494 m shorter than the shortening by 0.505 m of the relatively small lower rope section $L_l = 245$. The difference is therefore $\Delta L_D = -0.011$ m.

The elastic rope elongation is

$$\Delta L_E = \varepsilon_E \cdot L = \frac{\sigma_z}{E_S} \cdot L = \frac{S_0 + m \cdot g \cdot L/2}{A \cdot E_S} \cdot L.$$

$$\Delta L_E = \frac{10,000 + 2,180}{87.5 \times 71,000} \cdot 500 = 0.979 \text{ m.}$$

The shortening by twisting of the rope supported non-rotated at both ends is much less than the elastic rope elongation. In comparable cases, this rope shortening can always be neglected.

According to (2.99) and with extension $\varepsilon_D(x = 500) = 0.00378$ according to (2.97a), the lay angle of the twisted rope is on the upper end

$$\beta'(x = 500) = \arctan \frac{\tan \beta - r \cdot \omega}{1 + \varepsilon_D} = \arctan \frac{0.364 - 0.0059 \cdot 1.789}{1 + 0.00378} = 19.40^\circ$$

and with the elastic rope extension $\varepsilon_E(x = 500) = 0.00231$ the lay angle is

$$\beta''(x = 500) = \arctan \frac{\tan \beta - r \cdot \omega}{1 + \varepsilon_D + \varepsilon_E} = 19.35^\circ.$$

Example 2.13c: Suspended wire rope without rotation protection at the lower end, continuation of example 2.12

For the twisted wire rope, the integration of (2.98) leads to the rope elongation

$$\Delta L_D = 2.62 \text{ m.}$$

The elastic rope elongation is

$$\Delta L_E = \varepsilon_E \cdot L = \frac{\sigma_{zm}}{E_S} \cdot L = \frac{m \cdot g \cdot L/2}{A \cdot E_S} \cdot L.$$

$$\Delta L_E = \frac{2180}{87.5 \times 59,000} \cdot 500 = 0.000422 \cdot 500 = 0.211 \text{ m.}$$

According to (2.99), the lay angle of the twisted rope on the upper end is

$$\beta'(x = 500) = \arctan \frac{\tan \beta - r \cdot \omega}{1 + \varepsilon_D} = \arctan \frac{0.364 - 0.0059 \cdot 4.765}{1 + 0.00979} = 18.40^\circ$$

and, together with the elastic rope extension on the upper rope end $\varepsilon_E(500) = 0.000844$, the lay angle is

$$\beta''(x = 500) = \arctan \frac{\tan \beta - r \cdot \omega}{1 + \varepsilon_D + \varepsilon_E} = \arctan \frac{0.364 - 0.0059 \cdot 4.765}{1 + 0.00979 + 0.000844} = 18.38^\circ.$$

On the upper end, the lay length of the twisted and elastic elongated rope is

$$h_S''(x = 500) = \frac{\tan \beta}{\tan \beta''} \cdot h_S = \frac{0.364}{0.332} \cdot 0.1019 = 1.095 \cdot 0.1019 = 0.1116 \text{ m.}$$

The big increase of the lay length on the upper rope end can lead to a lasting change of the rope structure. When a wire rope is let down, the lower rope end should always be protected against turning. This rule has to be followed for wire ropes with steel cores even if there are only small differences in height.

2.4.6 Wire Stresses Caused by Twisting the Rope

When twisting the wire rope, the wires will be stressed by torsion. Furthermore the wires of the different wire layers and the strands of the different strand layers will be elongated or shortened differently. However, these different elongations will be prevented because the cross-sections of the strands and of the rope must remain plane. Therefore a common elongation or shortening of the twisted strands or ropes is forced by inducing longitudinal stresses in the wires and the strands.

The influence of the rope twisting on the rope stresses is so great that parts of the rope can be broken far below the normal wire rope breaking force. For example, in Fig. 2.25 the effect of the breakage of the rope core can be seen. For an untwist angle $\omega = -360^\circ/100d$ of the rope, the rope core has been broken at about 40% of the normal rope breaking force. In the diagram the core breakage is shown by an abrupt increase of the rope torque.

Torsional Stress

When a round strand wire rope is twisted by rope twist angle ω (+ for turn off), the strand twist angle of the strand layer j is

$$\omega_j = \pm\omega \cdot \cos \beta_j$$

(+ for lang lay rope) and the twist angle ω_{ij} of the wire i in the strand j of the round strand rope is

$$\omega_{ij} = \omega \cdot \cos \alpha_{ij} \cdot \cos \beta_j.$$

The torsional stress for a wire with the diameter δ_{ij} and the shear module G is

$$\tau_{ij} = \omega_{ij} \cdot G \cdot \frac{\delta_{ij}}{2}. \quad (2.38b)$$

Unimpeded Change of Lengths

The unimpeded change of lengths of the different wire and strand helixes means that the wires and strands can move against each other and a cross-section of the strands and the rope will not remain plane. For a spiral rope twisted with the rope twist angle ω , the unimpeded change in the length of a wire helix from the wire layer i in the direction of the spiral rope axis (or a

strand axis) is according to (2.97) for the winding radius r_i (presupposed as constant) and the lay angle α_i of a wire i

$$\frac{\Delta l_i}{l} = \sqrt{1 - 2 \cdot r_i \cdot \omega \cdot \tan \alpha_i - r_i^2 \cdot \omega^2} - 1. \quad (2.97b)$$

In a twisted stranded wire rope with fibre core and one strand layer $j = 1$, the unimpeded change of a wire helix of the wire layer $i,1$ in the direction of the strand axis of the strand layer 1 is

$$\frac{\Delta l_{i,1}}{l_1} = \sqrt{1 - 2 \cdot r_{i,1} \cdot \omega_1 \cdot \tan \alpha_{i,1} - r_{i,1}^2 \cdot \omega_1^2} - 1. \quad (2.97c)$$

In this equation the strand twist angle is

for lang lay ropes $\omega_1 = \omega \cdot \cos \beta_1$ and

for ordinary lay ropes $\omega_1 = -\omega \cdot \cos \beta_1$.

$r_{i,1}$ is the winding radius of a wire i in the strand 1 and

$\alpha_{i,1}$, is the lay angle of a wire i in the strand 1.

In a wire rope with steel core or in a multi-strand layer rope, the unimpeded length change of a strand helix of the strand layer j in the direction of the rope axis is

$$\frac{\Delta L_j}{L} = \sqrt{1 - 2 \cdot r_j \cdot \omega \cdot \tan \beta_j - r_j^2 \cdot \omega^2} - 1. \quad (2.97d)$$

r_j is winding radius of a strand j and

β_j is lay angle of a strand j .

The length changes of the wire helix and the strand helix can be calculated independently from each other.

Longitudinal Stress

With practically all rope terminations the relative motion of the wires and strands in straight ropes are prevented and the cross-sections remain plane when the rope is twisted. By preventing the relative motions, longitudinal forces are induced by extensions ε of the wires and strands. The sum of the components in the rope axis direction of all these forces is

$$\sum S_{i,j} = 0. \quad (2.97e)$$

The stress in a wire i of a wire rope with fibre core and only one strand layer will be looked at a great detail as an example. The unimpeded wire elongation $\Delta l_{i,1}$ is transformed to the common strand elongation Δl_1 as a real elongation. Then the component in the strand axis direction from the necessary longitudinal force of the wire i is with the same elasticity module E for all wires

$$S_{i,1} = \frac{\Delta l_1}{l_1} \cdot A_{i,1} \cdot E - \frac{\Delta l_{i,1}}{l_1} \cdot A_{i,1} \cdot E. \quad (2.97f)$$

According to (2.97e) with A_1 for the sum of the cross-sections of all strand wires is

$$0 = \sum_{i=0}^{n_s} \frac{\Delta l_{i,1}}{l_1} \cdot A_{i,1} - \frac{\Delta l_1}{l_1} \cdot A_1.$$

From that the common extension of the strand is

$$\varepsilon_T = \frac{\Delta l_1}{l_1} = \frac{1}{A_1} \cdot \sum_{i=0}^{n_s} \frac{\Delta l_{i,1}}{l_1} \cdot A_{i,1}. \quad (2.97g)$$

Introduced in (2.97f) the component of the longitudinal force of the wire i in the strand axis direction – induced by the rope twisting – is

$$S_{i,1} = \frac{A_{i,1}}{A_1} \cdot \sum_{i=0}^{n_s} \frac{\Delta l_{i,1}}{l_1} \cdot A_{i,1} \cdot E - \frac{\Delta l_{i,1}}{l_1} \cdot A_{i,1} \cdot E.$$

and the enforced extension of the wire i in strand axis direction is

$$\varepsilon_i = \frac{1}{A_1} \cdot \sum_{i=0}^{n_s} \frac{\Delta l_{i,1}}{l_1} \cdot A_{i,1} - \frac{\Delta l_{i,1}}{l_1}.$$

With this equation and the relation for the parallel lay strands

$$\tan \alpha_i = \frac{r_i}{r_n} \cdot \tan \alpha_n = \frac{r_i}{r_n} \cdot \tan \alpha$$

the enforced extension of the wire i in the strand axis direction is

$$\begin{aligned} \varepsilon_i = \sum_{i=0}^{n_s} \left(\frac{A_{i,1}}{A_1} \cdot \left(\sqrt{1 - 2 \cdot r_{i,1} \cdot \omega_1 \cdot \frac{r_{i,1}}{r_{n,1}} \cdot \tan \alpha - r_{i,1}^2 \cdot \omega_1^2 - 1} \right) \right. \\ \left. - \sqrt{1 - 2 \cdot r_{i,1} \cdot \omega_1 \cdot \frac{r_{i,1}}{r_{n,1}} \cdot \tan \alpha - r_{i,1}^2 \cdot \omega_1^2 - 1} \right). \end{aligned} \quad (2.97h)$$

The longitudinal stress of a wire i according to (2.20) – neglecting the strand contraction and left out the index 1 as there is only one strand layer – is

$$\sigma_{\text{long},i} = \varepsilon_i \cdot E.$$

In addition to that the tensile stress from an outer rope tensile force is according to (2.31)

$$\sigma_{t,i} = \frac{S}{A \cdot \cos \alpha_i \cdot \cos \beta}.$$

The both stresses can be added to a resulting tensile stress

$$\sigma_{\text{res},i} = \sigma_{\text{long},i} + \sigma_{t,i}.$$

Example 2.14: Wire stresses caused by twisting the rope**Example 2.14a: Constant twist angle over the entire rope length, continuation of example 2.9**

The data used for example 2.9 is again valid. Further data for the Filler rope being considered has been taken from Tables 1.8, 1.9 and 2.6. The wire and strand lay angles are

$$\alpha = \alpha_{3,1} = 15^\circ \quad \text{and} \quad \beta = \beta_1 = 20^\circ.$$

Torsional stress:

The twist angle of the most interesting outer wires (wire layer 3) is

$$\omega_{3,1} = -\omega \cdot \cos \alpha_{3,1} \cdot \cos \beta_1$$

$$\omega_{3,1} = 0.02094 \cdot \cos 15^\circ \cdot \cos 20^\circ = 0.00190 \text{ rad/mm}$$

and – according to (2.38b) – their torsional stress is

$$\tau_{3,1} = \omega_{3,1} \cdot G \cdot \delta_{3,1}/2 = 0.00190 \cdot 76,000 \cdot 1.01/2$$

$$\tau_{3,1} = 73 \text{ N/mm}^2.$$

Longitudinal stresses:

The strand twist angle of the ordinary rope is

$$\omega_j = -\omega \cdot \cos \beta_1 = 0.002094 \cdot \cos 20^\circ = 0.00197 \text{ rad/mm}$$

The longitudinal stresses of the wires caused by the rope twisting have been calculated with (2.97f) and (2.97g):

wire layer	rel. cross-sect. $A_{i1\text{layer}}/A_{\text{strand}}$	rel. winding radius r_{i1}/d_{strand}	longitud.-stress $\sigma_{\text{long},i}$
0	0.0554	0	-154 N/mm^2
1	0.3220	0.2114	-94 N/mm^2
2 (F)	0.0543	0.3593	22 N/mm^2
3	0.5683	0.4015	68 N/mm^2

By untwisting the ordinary lay rope, the strands are twisted off. The longitudinal stresses of the wires from the rope tensile force will be reduced for the centre wire and the wires of the first wire layer by $\sigma_{0,1}$ respectively $\sigma_{1,1}$ and increased for the filler wires and the outer wires by $\sigma_{2,1}$ respectively $\sigma_{3,1}$.

The change of the rope length from twisting off the strands is according to equation (2.97f)

$$\Delta L_T = \varepsilon_T \cdot L = -0,00077 \cdot 5000 = -3.85 \text{ mm}.$$

Stresses in Wire Ropes Supported Non-Rotated at Both Ends

For the wire rope supported non-rotated at both ends, (2.38b) is again valid for the torsional stress and (2.97f) or similar equations for the longitudinal stresses. The twist angles ω to be set in these equations have been derived in Sect. 2.4.4. In the present case, the twist angle from the simplified (2.91) is precise enough. The maximum stresses on both of the rope ends can then be calculated with the twist angles from (2.91a) and (2.91b).

By analysing the equations, it will be found that the stresses are independent from the rope diameter and depend only on:

- The constants involved (Tables 1.8, 1.9 and 2.6) the rope construction being considered
- The wire rope length L and
- The specific tensile force S_0/d^2 on the lower rope end.

Example 2.14b: Wire rope supported non-rotated at both ends, continuation of example 2.11

Data:

The data of example 2.11 is again valid. Further data of the Warrington rope being considered has been taken from Tables 1.8, 1.9 and 2.6. The rope lay angles are

$$\alpha = \alpha_{3,1} = 15^\circ \quad \text{and} \quad \beta = \beta_1 = 20^\circ.$$

According to (2.91a) and (2.91b), the rope twist angles are at the lower rope end $\omega_{\text{lower}} = 0.00194 \text{ rad/mm}^2$ and at the upper rope end $\omega_{\text{upper}} = -0.00172 \text{ rad/mm}^2$.

Torsional stress:

The twist angle of the most interesting thicker outer wires (wire layer 2) for the ordinary rope is

$$\omega_{2,1} = -\omega \cdot \cos \alpha_{2,1} \cdot \cos \beta_1$$

$$\omega_{2,1-\text{lower}} = -0.00177 \text{ rad/mm} \quad \text{and} \quad \omega_{2,1-\text{upper}} = 0.00157 \text{ rad/mm}$$

and – according to (2.38b) – their torsional stress is

$$\tau_{2,1} = \omega_{2,1} \cdot G \cdot \delta_{2,1}/2 = \omega_{2,1} \cdot 76,000 \cdot 0.92/2$$

$$\tau_{2,1-\text{lower}} = -64 \text{ N/mm}^2 \quad \text{and} \quad \tau_{2,1-\text{upper}} = 57 \text{ N/mm}^2.$$

Longitudinal stresses:

The strand twist angle of the ordinary rope is

$$\omega_1 = -\omega \cdot \cos \beta$$

$$\omega_{1-\text{lower}} = -0.00182 \text{ rad/mm} \quad \text{and} \quad \omega_{1-\text{upper}} = 0.00162 \text{ rad/mm}$$

In this, the longitudinal stresses of the wires from the rope twisting have been calculated with (2.97f) and (2.97g):

wire-layer	relative cross-section $A_{i1\text{layer}}/A_{\text{strand}}$	rel. winding- radius r_{i1}/d_{strand}	longitudinal-stress	
			lower $\sigma_{\text{long},i}$	upper end $\sigma_{\text{long},i}$
0	0.0603	0	108 N/mm ²	-97 N/mm ²
1	0.3503	0.2200	61 N/mm ²	-54 N/mm ²
2	0.3744	0.3842	-39 N/mm ²	35 N/mm ²
3	0.2150	0.4145	-64 N/mm ²	57 N/mm ²

For all wires and especially for the outer wires, the resulting tensile stresses from the rope rotation and from the rope tensile force are positive. Therefore the outside wires will not be loose.

Steel Core

If instead of a rope with fibre core a rope with steel core were to be used, then the longitudinal and the torsional stresses in the wires would be a little smaller than those in the wire rope with fibre core. However, a tensile strand stress has to be added to these tensile stresses and a large tensile stress range will occur in the steel core.

At the upper rope end the steel core can – depending on the core construction – even be loaded by the whole rope tensile force with totally unloaded strands. And at the lower rope end a large compressive stress of the core exists because normally the core cannot escape laterally.

2.4.7 Rope Endurance Under Fluctuating Twist

Test Machines

The first people to test how the endurance of wire ropes is affected by fluctuating twist and tension were Oplatka and Roth (1996). In their test machine which they designed themselves, the wire rope is stressed by a fluctuating tensile force and a fluctuating twist. The twist angle range is – for a constant middle tensile force S_m – approximately proportional to the tensile force range $2S_a$. They have carried out fatigue tests with a stress level where a high rope endurance has to be expected if there would be no fluctuating twist. Together with the fluctuating twist the numbers of load cycles are only about $N = 50,000$ for ropes with cast sockets as terminations. With Oplatka's clamp-sockets which allow slight movements of the wires and thus reduce the longitudinal stresses from the rope twist, they get more than ten times the number of load cycles for relatively short ropes.

Chaplin (2002) started his investigations in this field by defining the demand for a special testing machine which would enable rope endurance to be evaluated when the rope is stressed by constant or fluctuating twist in combination with constant or fluctuating tensile stress. Now Chaplin (2005) has reported that the new testing machine functions. He has presented first results in a diagram with the axis not scaled, because the results – belonging to a sponsor – are still confidential.

Stationary Wire Ropes

A wire rope supported non-rotated at the upper and the lower ends rotates with a rotary angle φ as can be seen in Fig. 2.29. The twist angle on the upper and the lower rope ends are expressed accurately enough by the simplified equations (2.91a) and (2.91b). The fluctuating twist angle depends on the sum of the constant force from the rope mass and the fluctuating force S_0 . In most cases these fluctuating twist angles and the fluctuating stresses from that are relative small.

Running Ropes

Fluctuating twist angles occur in running ropes of elevators, mine hoistings and rope ways. Between the guided car and the drum or traction sheave, the wire rope is twisted as a hanging stationary rope. When the car mounts, the twist angles – caused by the rope weight – in the remaining rope length will be continually reduced. In addition to the variable twist angle, there is also a constant twist angle ω_{con} – constant over the rope length – which usually arises from the installation itself and its loading history. A third twist angle ω_{side} can be produced, when the wire rope is wound in the groove of the sheave or drum. This occurs especially if the wire rope moves under side deflection sliding and rolling over the groove flank in the groove, Neumann (1987) and Schönherr (2005).

The maximum fluctuating twist angle and therefore the maximum fluctuating stress occurs in the rope piece above the car or the counter weight. The twist angles in that rope piece can be calculated under the supposition that no twist angle exists from installing the rope and its loading history and that no further twist angle is introduced by winding the wire rope in the sheave or drum groove with side deflection or other influences. For the lowest car position, the twist angle is given by (2.91b) and for the highest car position the twist angle is about zero. After passing a traction sheave, that piece of rope is twisted in the opposite direction according (2.91a) and this causes a great range of stresses, as in the Oplatka and Roth (1996) tests described.

The real twist angles should be investigated under different influences by computer simulation and measuring in installations. Furthermore a method should be found by which the influence of fluctuating torsion and longitudinal stresses from rope twisting can be introduced in the endurance calculation of wire ropes.

For installations not covering too great a height difference, wire ropes with fibre core can be used as they have been up to now. Because of the great fluctuating stresses especially in the steel core which even result in the total loosening of the strands at the upper end of the rope, wire ropes with steel core in normal construction with relative large torque should only be used to cover relatively small differences in height. For installations with a very height difference rotation-resistant ropes should be used.

However wire ropes with special steel core can be used for longer differences of height, for that they are qualified by good experience or by calculated relative small stresses.

2.5 Wire Rope Breaking Force

Measured Breaking Force

The breaking force F_m of the wire ropes has of course to be evaluated by measuring it. In the pieces of rope to be tested, it should be ensured that

there are no visible loose strands or wires. To compensate for any unavoidable minor loosening, the rope length between the terminations should be at least longer than 30 times the diameter of the rope. For the standard tension test, metal sockets are used as terminations. If the wire rope breaks in or near a termination, the measured breaking force of the wire rope may not be really obtained and the test should be repeated.

The tension test can of course be done with every kind of rope termination. But the breaking force so determined is not the breaking force of the wire rope. It is normally smaller. However, the tension test with resin sockets is an exception. In most cases the breaking force with these resin sockets is a little higher than with metal sockets and can therefore be taken too as the measured breaking force. But normally the measured breaking force will be evaluated by the standard tension test with metal sockets.

Minimum Breaking Force

For standardised wire ropes, the measured breaking force F_m is mostly greater than the minimum breaking force $F_{\min}$ given in the norm. From tests with 49 round strand ropes, about half with fibre and half with steel cores, the mean ratio of the measured breaking force (metal sockets) to the standardised minimum breaking force is

$$\left(\frac{F_m}{F_{\min}} \right)_m = 1.156$$

with the standard deviation $s = 0.054$.

In accordance with that, Chaplin and Potts (1991) found that the measured breaking force is 5–20% bigger than the minimum breaking force. The reason for this difference is that the measured strength is normally greater than the nominal strength and that the minimum breaking force in the norm is carefully chosen.

Wire Rope Breaking Force with Different Terminations

The wire rope breaking force is valid for wire ropes terminated with either resin or metal sockets. For wire ropes with other terminations, the wire rope breaking force (more or less reduced) can be estimated with the breaking force factor f_F . The breaking force factor is the ratio of the rope breaking force with a certain termination F_{mT} and the measured rope breaking force terminated with metal sockets F_m .

The required minimum breaking force of the wire rope with the terminations T is then

$$F_{\min T} = f_F F_{\min} \geq \nu \cdot S.$$

The breaking force factor is listed in Table 2.7.

Table 2.7. Breaking force factor f_F for rope terminations related to metal sockets

rope termination	breaking force factor f_F
splice eye	0.50–0.80
cylindric aluminum ferrule eye	0.85–1.00
Flemic eye with steel clamp	0.90–1.00
press bolt	0.90–1.00
wedge socket (rope lock)	0.80–0.95
U-bolt clamp DIN 1142	0.85–0.95

2.6 Wire Ropes Under Fluctuating Tension

2.6.1 Conditions of Tension-Tension Tests

A wire rope can only be loaded in connection with rope terminations. The wire rope and the terminations form one unit and the results from tension-tension tests refer only to this unit. In order to determine the actual characteristics of the wire ropes themselves fairly accurately, it is therefore necessary to use terminations which exert only minimal influence. In any case, it is difficult to avoid the effect of the terminations completely. Even a wire rope breakage occurring in the free length of the rope is no certain indication that there is no influence from the termination.

Of all known terminations, resin sockets exert the least influence on the endurance of wire ropes. Just how small this influence is can be seen by the small deviation in the numbers of load cycles reached in repeated tension-tension tests with specimens of the same rope, see the following Figs. 2.35 and 2.36. Results gained using metal sockets have much greater deviation with smaller numbers of load cycles.

The tension-tension tests for determining rope characteristics as described here, in particular those used to determine rope endurance, are conducted using resin sockets. Normally the ropes are lubricated. The rope ends, which were degreased before fitting the resin sockets, were lubricated again on the outside of the sockets after fitting.

The temperature needs to be kept low during tension-tension fatigue tests to ensure that the lubrication remains fairly effective. A top limit can be set at about 50°C or for a lubricant with very high viscosity at the most 60°C. The temperature increases greatly with the diameter of the rope. A certain limit for the frequency of testing in relation to the rope diameter cannot be given because the maximal possible frequency also depends on the extension hysteresis occurring during the load cycles. As Fig. 2.14 (Sect. 2.2.3) shows, the hysteresis effect increases the greater the stress range is and the smaller the lower stress. Ventilation can help to reduce the temperature.

The tension-tension fatigue tests normally end with rope breakage. Strand breakage or rope deformations count as rope breakage too if they result in the tests being discontinued. The results of rare tests where rope breakage occurs

near the terminations (about two times rope diameter) are to be disregarded as untrue for the rope itself. Of 49 tests, three resulted in breakage occurring at a distance up to two rope diameters and two others at a distance up to 2.5 rope diameters. Four of these five reached a number of load cycles higher than the mean number.

Tension-tension fatigue tests with wire ropes are much rarer than bending tests. Where the results of tension-tension fatigue tests have been published, it is often not possible to evaluate them in common with other tests. As Chaplin and Potts (1991) pointed out, one problem is that there are no precise specifications laid down for the wire ropes to be tested or for the test conditions. One other problem at that time was that there was no convincing regression formula. To overcome the first problem OIPEEC (1991) passed OIPEEC-Recommendation No. 7 laying down that the specifications for wire ropes and test conditions should at least be described. The solution for the other problem will be described in the next chapter.

2.6.2 Evaluating Methods

Goodman Line

Wire rope endurance under fluctuating tension depends on the amplitude force S_a and the middle force S_m or the lower force S_{lower} . These forces are defined for a sinus course in Fig. 2.31. For the evaluation of a group of wire ropes with varying rope diameters, all these forces have to be replaced by the wire rope stresses or by the specific forces S/d^2 .

The first proposals to evaluate the results of tension-tension fatigue tests with wire ropes came from Yeung and Walton (1985) and at the same time from Matsukawa and others (1985). For spiral ropes, they proposed to combine the force range $2S_a$ and the middle force S_m to produce an equivalent force S_q on the basis of the Goodman line. According to their proposal, the equivalent force is

$$S_q = \frac{F}{F + S_a - S_m} \cdot 2 \cdot S_a \quad \text{or} \quad S_q = \frac{F}{F - S_{lower}} \cdot 2 \cdot S_a \quad (2.101)$$

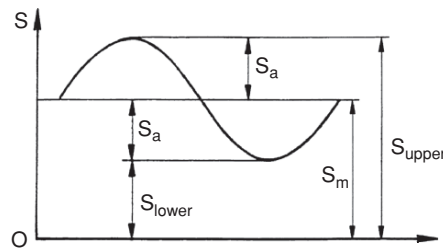


Fig. 2.31. Fluctuating tensile force

F is the wire rope breaking force, for this Yeung and Walton, and Matsukawa and others differ in their definitions. The lower force is $S_{\text{lower}} = S_m - S_a$.

The use of the equivalent force seems very attractive, because this means that the number of variables is reduced. The endurance of a wire rope can be described with the single variable S_q by the very simple equation for the number of the load cycles

$$N = a \cdot S_q^b.$$

This equation has been used to evaluate the results of different tension-tension fatigue tests and it discloses a profound difficulty. For the same equivalent force, the number of load cycles is much smaller with a small lower force than with large lower force. Therefore, an evaluation using this equation can only be done separately for different lower force segments. This means that this method is unsatisfactory.

Haigh diagrams have been designed for a bigger number of test results. As an example, the best one – that is, the diagram where the test results follow the Goodman line at least in part – is shown in Fig. 2.32. In this figure, the force range $2S_a/d^2$ has been drawn for the number of load cycles $N = 100,000$ of the wire rope C , in Table 2.10. Because the number of load cycles cannot be gained using direct testing, the drawn force ranges $2S_a$ in Fig. 2.32 are evaluated by interpolation. For smaller middle forces, the force range is drawn using points. For larger middle forces, the force range follows the Goodman line which is also drawn in Fig. 2.32. A limit line starting from the origin of coordinates has been introduced for the lower force $S_{\text{lower}} = 0$, because the wire rope cannot transfer a compressive force.

The force range $2S_a$ is small for small middle forces S_m . It increases at first and then reduces with the growing middle force S_m along the Goodman line. Where this reduction begins, the upper force $S_{\text{upper}} = S_m + S_a$ (of the oscillating force) reaches 75% of the calculated breaking force of the rope.

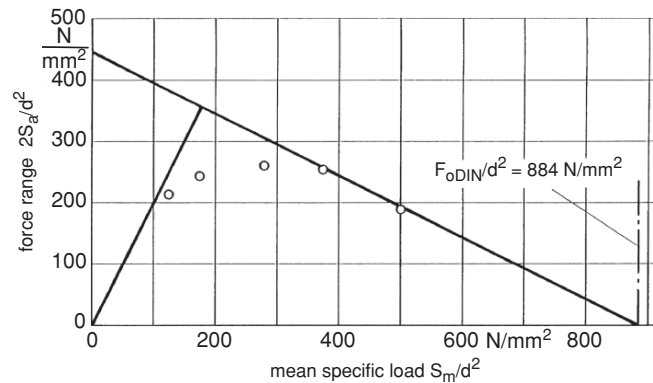


Fig. 2.32. Haigh diagram for wire rope C , resin socket, number of load cycles $N = 100,000$

Therefore the force range $2S_a$ resulting from the tests with a stranded wire rope is not represented at all by a Goodman line in the region of practical usage.

The reduced force range $2S_a$ for small middle forces is caused by the additional stresses arising from the bigger fluctuating contraction of the wire rope in this force region, see Sect. 2.1. The wire rope is not a piece of material for which the Goodman line is valid. The additional stresses distinguish the wire rope as a machine element rather than as a piece of material.

Endurance Formula

Numerous tension-tension fatigue tests have been carried out on three resin-socketed round strand wire ropes A, B and C (as listed in Table 2.10) by systematically varying the forces in order to determine a better method for evaluating the test results. After a number of trials, the best regression equation for the number of load cycles N up to the wire rope breakage has been found to be, Feyrer (1995)

$$\lg N = a_0 + a_1 \cdot \lg \frac{2 \cdot S_a \cdot d_e^2}{d^2 \cdot S_e} + a_2 \cdot \frac{S_{\text{lower}} \cdot d_e^2}{d^2 \cdot S_e} + a_3 \cdot \left(\frac{S_{\text{lower}} \cdot d_e^2}{d^2 \cdot S_e} \right)^2 + a_4 \cdot \lg \frac{d}{d_e}. \quad (2.102)$$

To make them dimensionless, $S_e = 1 \text{ N}$ and $d_e = 1 \text{ mm}$ are introduced into the equation. The other symbols are known already. For using this equation for practical purposes, the forces are divided by the rope diameter square as so-called specific forces. The rope diameter d is the nominal rope diameter as used for the regression calculation. This has the advantage that the deviation of the rope diameter is included in the standard deviation for the calculated number of load cycles.

In Table 2.9, the coefficient of determination B and the standard deviation $\lg s$ for the three stranded ropes A, B and C – calculated with this regression – are listed as well as the constants a_i . The great coefficients of determination $B = 0.916\text{--}0.941$ show that the test results are well expressed with this equation. Wehking and Klöpfer (1999) found that (2.102) was equally valid both for spiral wire ropes and for Warrington–Seale ropes. For the regression calculation, the numbers of load cycles $N > 10^6$, or in other cases $N > 1.75 \times 10^6$, are not taken into account as they are considered to be outside the sphere of finite life strength.

In Fig. 2.33, the lines for the calculated number of load cycles N with (2.102) as well as the test results for the wire rope C are drawn as an example. It is to be seen that the lines and the test results up to $N = 10^6$ are close together. The number of load cycles N increases at first with increasing lower tensile force S_{lower} . As additional information, Fig. 2.33 also includes lines of the upper force with 50% (surely the maximum allowed upper force in all cases) and 70% of the calculated rope breaking force.

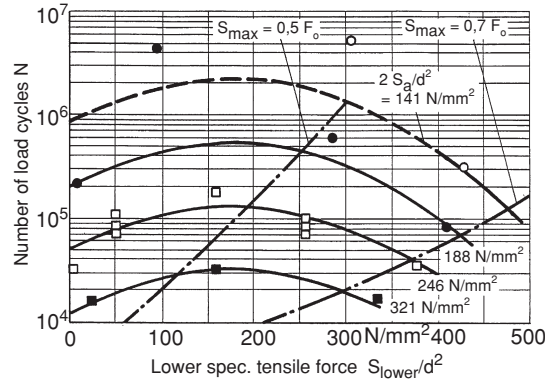


Fig. 2.33. Number of load cycles N for wire rope C, Warr. 8×19 -IWRC-zZ, resin socket

The mean number of load cycles $\bar{N}$ is calculated with (2.102) and the constants. The number of load cycles where with a certainty of 95% the highest quantile $\gamma\%$ (for example, 10 or 1%) of the wire ropes are broken can be calculated with

$$\lg N_\gamma = \log \bar{N} - k_{T\gamma} \cdot \lg s. \quad (2.103)$$

The standard deviation $\lg s$ is determined with the regression calculation. The constant $k_{T\gamma}$ has to be calculated as a mean value for the region of the wire rope forces being considered, Stange (1971).

In contrast to the rope bending fatigue tests, all the known tension fatigue tests have been carried out up to wire rope breakage mostly without detecting any outside wire breaks or other discard criteria. Magnetic inspection to detect inner wire breaks during the fatigue tests has not been used up to now, Feyrer and Wehking (2006). For practical purposes in connection with safety requirements, for the time being it seems reasonable to evaluate the number of load cycles N_1 as that which – with a certainty of 95% – not more than 1% of the wire ropes under consideration are broken. It can be expected that up to this limit possible rope defects will be detected and show that the rope has to be discarded. For wire ropes without safety requirements, the number of load cycles N_{10} may be used as having – with a certainty of 95% – not more than 10% of the ropes broken.

Woehler Diagram

With the help of (2.102) a Woehler diagram can be drawn for the sphere of finite life strength. The test results let us see that the sphere of finite life fatigue strength ends for not much more than $N = 1,000,000$. There are only a small number of test results available above this number of load cycles and

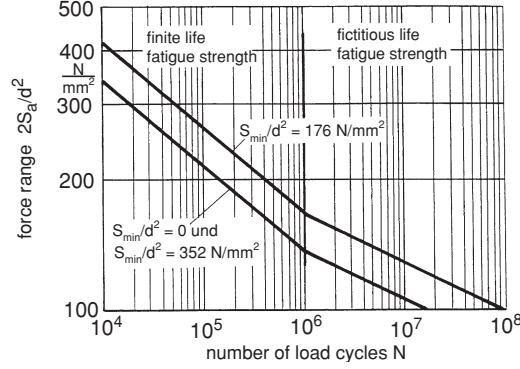


Fig. 2.34. Woehler diagram, wire rope C, Warr. 8×19 -IWRC-zZ, resin socket

from these results it is not possible to derive the relation between the acting forces and the number of load cycles. Supposing a more or less constant fatigue strength does not exist, a fictitious continuation of the fatigue strength line according to Haibach (1989) can be drawn as a conservative form, Sonsino (2005). To be on the safe side the fictitious continuation may start at the limiting load cycles $N_D = 2,000,000$. The number of load cycles for this fictitious continuation is

$$N = N_D \cdot \left(\frac{2 \cdot S_a/d^2}{2 \cdot S_{aD}/d^2} \right)^{2 \cdot a_1 + 1}. \quad (2.104)$$

In this equation, $2 \cdot S_{aD}/d^2$ is the force range at the number of load cycles $N_D = 2,000,000$. The Woehler diagram in Fig. 2.34 is still drawn for wire rope C in two lines for $N_D = 1,000,000$ (as found from Fig. 2.33) with the help of (2.102) and (2.104). The first line has the value $S_{\text{lower}}/d^2 = 0$ for the lower specific force and at the same time for $S_{\text{lower}}/d^2 = 352 \text{ N/mm}^2$. The second line with the maximum possible mean number of load cycles has the value $S_{\text{lower}}/d^2 = 176 \text{ N/mm}^2$ for the lower specific force. A Woehler line can be calculated and drawn between these lines for other lower specific forces.

Distribution of the Number of Load Cycles

As can be seen from the form of (2.102), the described regression is based on the logarithm normal distribution. This is justified because it was found, for example, that the logarithm normal distribution provided a very good degree of conformity for the number of load cycles of the specimens from wire ropes A and C which were each tested under nominally identical conditions as shown in Figs. 2.35 and 2.36.

Raoof and Hobbs (1994) found on the contrary that it was preferable to use the Gumbel distribution for the number of load cycles in the tension fatigue tests on stranded ropes tested repeatedly under the same conditions.

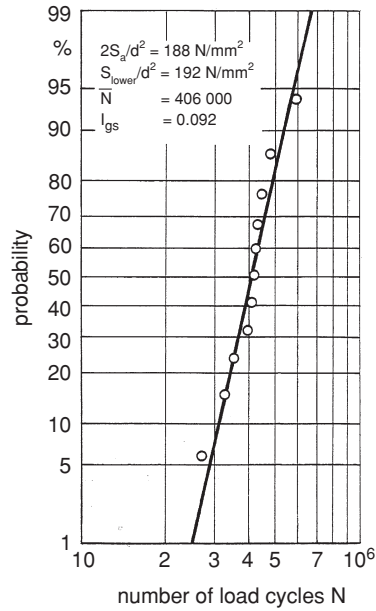


Fig. 2.35. Number of load cycles N , wire rope A, Warr. 8×19 -SFC- sZ , resin socket

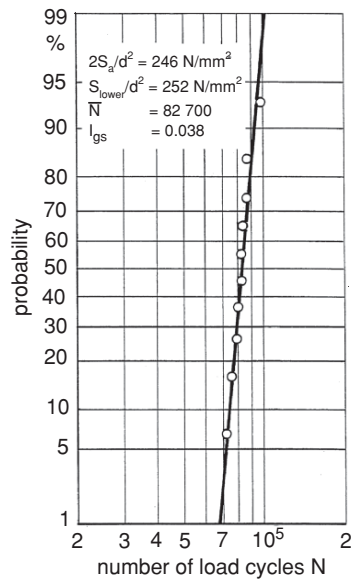


Fig. 2.36. Number of load cycles N , wire rope C, Warr. 8×19 -IWRC- zZ , resin socket

Unfortunately, the numbers of load cycles they counted are in the region of $N = 355,000 - 1,636,000$ which is where finite life fatigue strength ends. Also, with its difficult relation to the regression, the Gumbel distribution does not describe their test results better than the logarithm normal distribution would have done.

Castillo and others (1990) proposed using the Weibull distribution with three parameters to describe the number of load cycles for repeated tension tests with the same conditions. This distribution has the disadvantage that many more tests would be needed to evaluate the three parameters and, above all, these parameters cannot be combined simply with a regression calculation.

2.6.3 Results of Tension Fatigue Test-Series

Spiral Wire Ropes with Resin Sockets

Wehking and Klöpfer (2000) in Stuttgart, and Casey (1993) and Paton and others (2001) in East Kilbride, Glasgow have completed extensive tension fatigue investigations with open spiral ropes. In all cases the wire ropes were fastened in resin sockets. The results of these tests have been evaluated by regression with (2.102). The constants a_i and the rope and test data are listed in Table 2.8.

Wehking and Klöpfer (2000) and Klöpfer (2002) tested open spiral ropes with round wires $1 + 6 + 12 + 18$ (short 1×37) with different diameters. The free length between the sockets was uniformly $L = 40d$. The wire ropes had zinc coated wires and were lubricated. The numbers of load cycles up to $N = 1.75 \times 10^6$ are included in the regression calculation. For every wire rope, the coefficient of determination is high but the standard deviation varies between $\lg s = 0.094$ and 0.236 . The constants a_i in Table 2.8 determined by Wehking and Klöpfer (2000) was corrected slightly by Klöpfer (2002) by neglecting the results with rope breakages near the terminations. This then reduces the standard deviation to $\lg s = 0.227$.

The maximum number of load cycles has been reached for the mean lower specific force $S_{\text{lower}}/d^2 = 140 \text{ N/mm}^2$ with a relatively large deviation. The numbers of load cycles for a spiral rope with the diameter $d = 16 \text{ mm}$ are presented in Fig. 2.37 as an example. The line for the upper force as a half rope breaking force has been included in the figure to show the maximum usable region. It can be seen in Fig. 2.37 that the endurance curves are relatively flat. In accordance to that Alani and Raoof (1997) found that under fluctuating tensile forces the endurance of spiral ropes has been nearly independent from the lower respectively the middle stress.

Two of the seven wire ropes tested had the unusual wire lay direction SSZ. In comparison with the normal lay direction ZSZ, the wire lay direction SSZ has lower endurance.

Table 2.8. Constants for the number of load cycles of open spiral wire ropes for (2.102), resin sockets

Reference	Wire rope	Nominal rope diameter d mm	Nominal strength R_0 N/mm ²	Test frequency Hz	Free rope length L/d	Number of tests n	Constants					Standard deviation $\lg s$	$\bar{N}$ for $d = 30$ mm, $S_u/d^2 = 60$, $2S_u/d^2 = 300$ N/mm ²
							a_0	a_1	a_2	a_3	a_4	a_5	
Wehking, Klöpfer (2000) and Klöpfer (2002)	open spiral ropes	4	1,370			12							
	$1 \times 37zn$, lubricated	4	1,770			4							
		5	1,370	1.5		12							
		10	1,770	bis	40	11							
		16	1,770	8.0 ^b		19	15.90	-3.862	0.0009	-0.0000030	-0.779	-	168,000
		16	1,770			11							
Cassey (1993) and Paton et al. (2001) ^a		24	1,770			16							
	open spiral	40	1,570	≥ 0.3	100	4							
	$1 \times 292zn$												
	1×135	66.6				4							
	1×147	70				2	20.587	-5.420	-0.00019	-0.000024	-1.040	-	337,000
Wehking and Klöpfer (2000), Cassey (1993) and Paton et al. (2001)		1×127				2							
	$1 \times 292zn$	127	1,570		55	4							
	open spiral ropes												
	1×37	4	1,370										for $z = 37$ 169,000
	to 1×292	127	to 1,770		40 100	101	15.401	-3.910	0.00118	-0.0000037	-0.793	0.399	0.214 for $z = 199$ 331,000

^aLower specific tensile force $S_u/d^2 = 84$ N/mm²^bPartly with ventilator for cooling

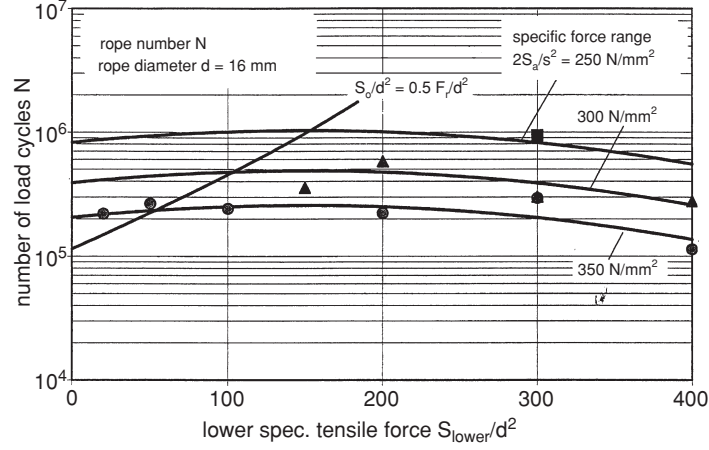


Fig. 2.37. Number of load cycles of an open spiral rope 1×37 , Wehking and Klöpfer (2000)

Casey (1993), and Paton and others (2001), National Engineering Laboratory (NEL) East Kilbride, Glasgow, have done numerous tension-tension fatigue tests with open spiral ropes having larger diameters. The wire ropes have different numbers of wires. The wire ropes with the diameters $d = 40$ mm and 127 mm have the biggest number of wires with the construction $1 + 7 + 7/7 + 14 + 19 + 25 + 31 + 42 + 48 + 49 = 292$, Casey (1993).

The constants a_i and other results of the regression calculation for these spiral ropes are listed in Table 2.8. The results of Casey, and Paton and others are used for the regression again up to the number of load cycles $N \leq 1.75 \times 10^6$. The lower specific force is at maximum $S_{\text{lower}}/d^2 = 84 \text{ N/mm}^2$. Therefore, with the constants a_i of these wire ropes, (2.102) is only valid up to this lower specific force.

A common regression calculation has been carried out using the results of Wehking and Klöpfer, Casey, and Paton and others. Because of the very different numbers of wires z , the regression equation has been – compared with Feyrer (2003) – complemented here by the number of wires z .

$$\begin{aligned} \lg N = & a_0 + a_1 \cdot \lg \frac{2 \cdot S_a \cdot d_e^2}{d^2 \cdot S_e} + a_2 \cdot \frac{S_{\text{lower}} \cdot d_e^2}{d^2 \cdot S_e} + a_3 \cdot \left(\frac{S_{\text{lower}} \cdot d_e^2}{d^2 \cdot S_e} \right)^2 \\ & + a_4 \cdot \lg \frac{d}{d_e} + a_5 \cdot \lg z. \end{aligned} \quad (2.102a)$$

The constants a_i from the common regression are also listed in Table 2.8. According to the ropes used, the mean number of load cycles is given with these constants and (2.102a) for open spiral ropes with the diameter $d = 4 - 127$ mm and with the number of wires $z = 37 - 292$. The standard deviation is $\lg s = 0.214$. Using the constants $k_{T10} = 1.69$ respectively $k_{T1} = 2.93$, with

a certainty of 95% at the most 10% respectively 1% of the wire ropes are broken at the number of load cycles

$$N_{10} = 0.435 \cdot \bar{N} \quad \text{respectively} \quad N_1 = 0.236 \cdot \bar{N}.$$

For open spiral ropes, the relation between the wire rope stress and the specific force is about

$$\sigma_z = 1.70 \cdot \frac{S}{d^2}.$$

With an increasing rope diameter, the endurance decreases. On the other hand the endurance increases with the number of wires. Thereby the influence of the rope diameter predominates. In the range tested, the number of load cycles decreases with the rope diameter exponent $a_4 = -0.793$ and increases with the number of wires exponent $a_5 = 0.399$. All the spiral ropes tested were zinc coated and lubricated. It is not possible to evaluate the influence of the nominal strength on the rope endurance from the existing database. The constants in Table 2.8 are therefore valid for wire ropes with a nominal strength between 1,370 and 1,770 N/mm².

In the last column in Table 2.8, the mean numbers of load cycles are registered, calculated with (2.102a) for a rope diameter $d = 30$ mm. The difference between the numbers of load cycles from regressions of both diameter ranges tested with the mean numbers of wires $z = 37$ respectively 199 and those of the common regression is very small. Therefore, the common regression is legitimated.

Spiral Wire Ropes with Metal Sockets

Hugo Müller has carried out tension-tension fatigue tests (unpublished) with a locked coil spiral rope with metal sockets. The rope with a diameter $d = 28$ mm has round wires 1+6+12+18 and 19Z-wires outside. His results ($a_0 = 12.528$; $a_1 = -2.960$) showed about 55% of the endurance found for a comparable open spiral rope terminated with resin sockets. Yeung and Walton (1985) have done numerous tension-tension fatigue tests with spiral ropes. They did not use constant forces but a light force collective. This means that the results cannot be compared with those using constant forces.

Round Strand Wire Ropes with Resin Sockets

Tension-tension fatigue tests have been carried out with various round strand wire ropes using resin sockets as terminations. The wire ropes tested are listed in Table 2.9. The results of the tests have been used in regression calculations based on (2.102). The constants thus determined are also included in Table 2.9.

A very large number of tension-tension fatigue tests have been carried out using Warrington–Seale 6-strand ropes in ordinary lay with steel core. Wehking and Klöpfer (2000) tested ropes with diameters $d = 8 - 36$ mm and

Casey (1993) ropes with diameters $d = 38 - 127$ mm. For the ropes with smaller diameters of up to 40 mm, the strands had 36 wires, the wire rope with a diameter of 70 mm had 41 wires and the 127 mm wire rope had 49 wires.

Of the Warrington–Seale ropes tested by Wehking and Klöpfer, seven ropes were zinc coated and five bright. The wire ropes Casey tested were all zinc coated. All the wire ropes were lubricated. From the existing database, it is not possible to evaluate whether either the zinc coating or the nominal strength influence the endurance of the rope. Therefore the constants for the Warrington–Seale ropes in Table 2.9 are valid for ropes, whether zinc coated or bright, with a nominal strength between 1,570 and 1,960 N/mm².

The results of the regression calculations based on the data from the tests on the Warrington–Seale ropes are also listed in Table 2.9. The common regression calculation done on the basis of the data from Wehking and Klöpfer and Casey is very well-founded as the numbers of load cycles for both test series calculated with their constants come very close to that for a rope diameter 38 mm. As the ropes tested varied greatly in quality, the coefficient of determination is only $B = 0.68$ and the standard deviation is $\lg s = 0.266$. Then, using the constants $k_{T10} = 1.575$ respectively $k_{T1} = 2.76$, with a certainty of 95% at the most 10% respectively 1% of the wire ropes are broken at the number of load cycles

$$N_{10} = 0.38 \cdot \bar{N} \quad \text{respectively} \quad N_1 = 0.184 \cdot \bar{N}.$$

For the Warrington–Seale ropes with 6-strands and steel core IWRC, the relation between the wire rope stress and the specific force is

$$\sigma_z = 2.195 \cdot \frac{S}{d^2}.$$

Of the wire ropes with steel core, the 6-strand Warrington–Seale ropes reach a much higher number of load cycles than both of the 8-strand Warrington ropes compared in the last column of Table 2.9. Even for the same specific forces, the 8-strand Warrington rope with fibre core has shown a higher endurance than both of those with steel cores. For the same wire rope stress, Reemsnyder (1972) also found that wire ropes with fibre cores had an advantage as far as endurance is concerned.

Round Strand Wire Ropes with Metal Sockets

The results of tension-tension fatigue tests on wire ropes with metal sockets are listed in the Table 2.10. These results are presented because wire ropes with metal sockets are frequently used and because it is very informative to see the endurance results with metal sockets under different conditions. Most of the results come from Müller (1962, 1963 and 1966 as well as other unpublished results). For all his tests, the lower specific force was about $S_{\text{lower}}/d^2 = 20$ N/mm². He found that the parallel lay wire ropes always have

Table 2.9. *Continued*

Reference	Wire rope	Nominal diameter d mm	Nominal strength R_0 N/mm ²	Test frequency Hz	Free rope length L/d	Number of tests n	Constants					$\bar{N}$ for $d = 16$ mm, $S_u/d^2 = 20$, $2S_u/d^2 = 200$ N/mm ²
							a_0	a_1	a_2	a_3	a_4	Standard deviation lg s
Wehking and Klöpfer (2000) and Casey (1993)	WS-IWRC	8	1,570		40		16.302	-3.939	0.00326	-0.000012	-1.180	0.266
	6 × 36 - sZ to bright or zinc lubricated	127	1,960		100	217						755,000
	spiral- ^b multistrand rop.	40			100	6						
Casey (1993)	(1 + 5 + 11 + 17 + 23) × 7 = 57 × 7zn lubricated	127	1,570	≥ 0.3	55	4	17.669	-4.736	0.00110	-0.000016	0.849	0.072
												-
Feyrer (1995)	A: War 8 × 19 FC - sZ	16	1,570			22	19.66	-6.201	0.00382	-0.0000185	-	0.132
	B: War 8 × 19 IRWC - sZ	16	1,960	3.3	87	19	14.40	-4.078	0.00246	-0.0000066	-	0.148
	C: War- 8 × 19 IWRC - zZ	16	1,960	- ^b		16	16.74	-5.033	0.00493	-0.0000140	-	0.131
	lubricated											
Ridge (1993)	Filler 6 × 19 ^a IWRC - sZ	13	1,770	3.0	50	9	25.108	-8.565	0.00288	0.000041	-	0.141
	lubricated											

^aLower specific tensile force $S_u/d^2 = 20$ N/mm²^bPartly with ventilator for cooling

Table 2.10. Constants a_i for the number of load cycles of stranded wire ropes for (2.102), metal sockets

Reference	Wire ropes	Nominal diameter d mm	Nominal strength R_0 N/mm ²	Test frequency Hz	Free rope length L/d	Number of tests n	Constants					Standard deviation $\lg s$	$\bar{N}$ for $d = 16$ mm, $S_u/d^2 = 20$, $2S_u/d^2 = 200$ N/mm ²
							a_0	a_1	a_2	a_3	a_4		
H. Müller (1962, 1963, 1966)	cross lay ropes $6 \times 19 - \text{NFC} - sZ$	2.5	2,160	4.7	320	1/1							
		3.2	1,570	4.7	280	5/3			lubricated				
		5.0	1,770	4.2	190	4/5	14.03	-3.461	-	-	-1.021	0.393	68,200
		8.5	1,570	4.2	110	6/7			unlubricated				
H. Müller (1962, 1963, 1966)	cross lay ropes $6 \times 19 - \text{NFC} - zZ$	16	1,570	3.0	65	7/6	12.82	-3.215	-	-	-0.718	0.385	35,800
		28	1,570	3.0 ^b	27	6/6 ^c							
		2.5	2,160	4.7	320	1/1			lubricated				
		3.2	1,570	4.7	280	3/3	12.74	-3.205	-	-	-0.535	0.178	52,700
H. Müller (1962, 1963, 1966)	Filler ^a $6 \times 19 - \text{NFC} - zZ$	5.0	1,770	4.2	190	5/5			unlubricated				
		8.5	1,570	4.2	110	5/6							
		16	1,570	3.0	40	4/6	13.08	-3.245	-	-	-0.821	0.341	42,200
		28	1,570	3.0 ^b	27	7/6 ^c			lubricated				
H. Müller (1962, 1963, 1966)	Seale ^a $6 \times 19 - \text{NFC} - sZ$	16	1,570	3.0	50	2/2	18.740	-5.729	-	-	-	-	381,000
									unlubricated				
H. Müller (1962, 1963, 1966)	Filler ^a $8 \times 19 - \text{NFC} - sZ$	16	1,570	3.0	50	5	15.324	-4.470	-	-	-	-	109,000
							16.055	-4.680	-	-	-	0.021	193,000
H. Müller (1962, 1963, 1966)	Filler ^a $8 \times 19 - \text{NFC} - sZ$												
H. Müller (1962, 1963, 1966)	lubricated												
H. Müller (1962, 1963, 1966)	Filler ^a $8 \times 19 - \text{NFC} - sZ$	16	1,570	3.0	50	5	16.891	-5.476	-	-	-	0.055	136,000
	lubricated												

Table 2.10. *Continued*

Reference	Wire ropes	Nominal diameter d mm	Nominal strength R_0 N/mm ²	Test frequency Hz	Free rope length L/d	Number of tests n	Constants					Standard deviation $\bar{N}$ for $d = 16$ mm, $S_u/d^2 = 20$, $2S_u/d^2 = 200$ N/mm ²	
							a_0	a_1	a_2	a_3	a_4		lg s
H. Müller (1962, 1963, 1966)	Warrington ^a $8 \times 19 - \text{NFC} - sZ$ lubricated	16	1,570	3.0	50	6	18.025	-5.109	-	-	-	0.194	266,000
Suh and Chang (2000)	Warrington $6 \times 19 - \text{IWRC} - sZ$ lubricated	12.5	1,570		70 140 210	8 6 4	10.555	-2.137	0.000459	-0.0000015	-	0.137	
H. Müller (1962, 1963, 1966)	cross lay ropes lubricated ^a $6 \times 7 - \text{NFC} - sZ$ $6 \times 19 - \text{NFC} - sZ$ $6 \times 37 - \text{NFC} - sZ$ $6 \times 61 - \text{NFC} - sZ$	16 16 16 16	1,570 1,570 1,570 1,570	3.0	50 50 50 50	5 10 7 7	12.219	-3.175	-	-	-	0.035 0.143 0.080 0.112	81,900 58,400 63,400 65,200

^a Lower specific tensile force $S_u/d^2 = 20$ N/mm²^b Partly with ventilator for cooling^c lubricated/unlubricated

much higher endurance than cross lay ropes although the cross lay ropes with the same wire lay angle in all wire layers have the advantage of having theoretically the same tension in the different wire layers. The reason for the smaller endurance of the cross lay ropes may just be due to the pressure between the crossing wires.

There is not much variation in the number of load cycles of the cross lay ropes FNC + 6 × 19, 6 × 37 and 6 × 61 whereby the higher number of wires tends to show an advantage. The simple wire rope FNC + 6 × 7 has a slightly higher endurance. Here again, the reason may be that there are no crossing wires. This may overcome the disadvantage of the thicker wires. In all cases, he found that the lubrication gave higher endurance. This result comes from the smaller second tensile stress, see Sect. 2.1.4.

2.6.4 Further Results of Tension Fatigue Tests

Number of Load Cycles for Wire and Wire Rope

Setzer (1976) did tension fatigue tests with a Warrington–Seale rope and compared them to the strands and wires of this rope before being manufactured into a rope. He presented the result of these tests as a Smith diagram shown here in Fig. 2.38. The diagram is based on a number of load cycles $N = 2 \times 10^6$. For the middle stress $\sigma_m = 500 \text{ N/mm}^2$ Setzer found a stress range $2\sigma_a = 550 - 600 \text{ N/mm}^2$ for the wires and only $2\sigma_a = 140 \text{ N/mm}^2$ for the wire rope.

A further comparison of the stress range for the wires and the wire rope is shown on the basis of the data of Table 2.9 for the Warrington–Seale ropes. Figure 2.39 shows the stress range in the outside wires for a rope with the diameter $d = 16 \text{ mm}$ where the wire rope breaks at the number of load cycles $N = 10^6$, with a probability of 1, 10 or 50%. In order to take the additional stresses into consideration, the stress range for the wire rope (better for the outside wires of the rope) has been drawn 20% above the global wire rope stress $\sigma_z = S/A_m$ calculated using (2.102). In comparison, the strength range for straight wires with the same diameter as the outside rope wires is drawn in Fig. 2.39. This strength range has been calculated with (1.3) and (1.3b) for a breakage probability of 50%.

Even for a high quality wire rope (failure probability 1%), the stress range for the breakage at the number of load cycles $N = 10^6$ is clearly smaller than the mean stress range of the straight wires. The remaining difference can be declared by the unsystematic increased stresses of individual wires or strands due to the loosening of the others, Evans and others (2001). Furthermore, the pressure between the wires has not been included in the stress calculation.

Size Effect Wire Rope Diameter

Müller (1966) was the first to investigate the effect of the size of the rope diameter on cross lay wire ropes 6 × 19 - FNC - sZ. Figure 2.40 shows his

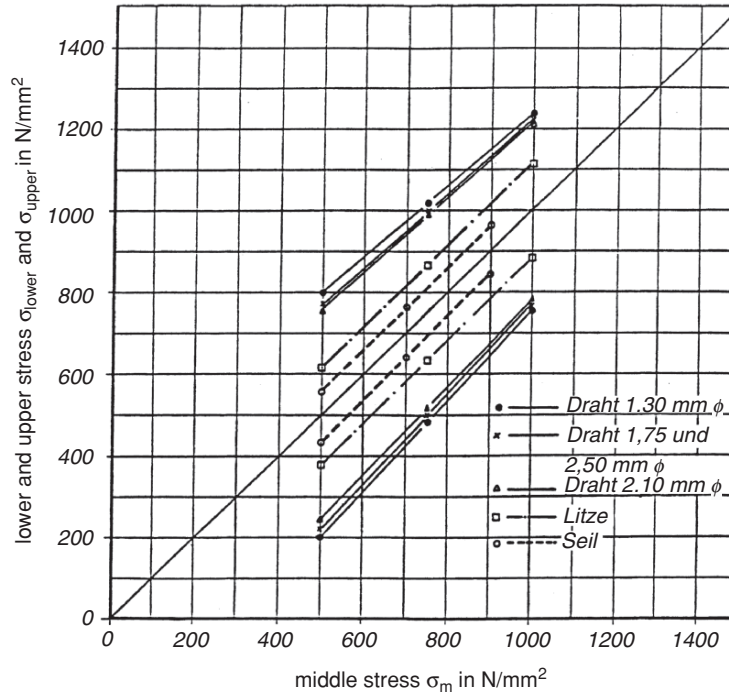


Fig. 2.38. Smith diagram for a Warrington-Seale rope $6 \times 36\text{-FC-s-Z}$ in comparison with the strands and wires, Setzer (1978)

results. The mean ratio of the number of load cycles N_1/N_2 of two wire ropes with the diameters d_1 and d_2 is

$$\frac{N_1}{N_2} = \left(\frac{d_1}{d_2} \right)^{a_4}. \quad (2.105)$$

For the lubricated cross lay ropes, Müller found exponents $a_4 = -1.021$ and 0.535 . For different test series the constants a_4 are listed in the Tables 2.8, 2.9 and 2.10. For the whole diameter sphere of the open spiral ropes 4–127 mm, the constant is $a_4 = -0.793$. For the diameter sphere 8–127 mm of the Warrington-Seale ropes, the constant is $a_4 = -1.180$. The influence of the rope diameter is much higher in the case of tension-tension fatigue than in the case of bending fatigue with the exponent -0.32 .

There is no explanation for this big difference between the exponents for tension and bending. It could have been expected that the size of the diameter had a greater influence on bending due to the stress gradient effect. In any case, the results emphasize Unterberg's statement (1967) that a stress gradient effect does not exist for rope wires.

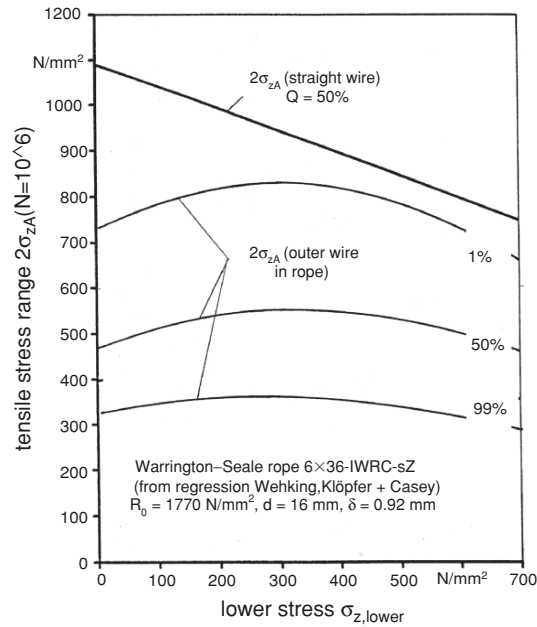


Fig. 2.39. Stress range in outside wires of a Warrington-Seale rope in comparison to straight wires of the same diameter at the number of load cycles $N = 10^6$

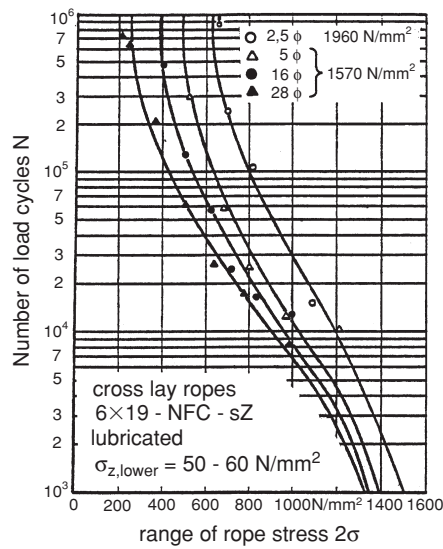


Fig. 2.40. Number of rope cycles N for cross lay ropes 6×19 -FNC - sZ with different diameters, Müller (1966)

Size Effect Wire Rope Length

Suh and Chang (2000) carried out tension-tension fatigue tests on a Warrington rope with rope lengths of 10, 20 and 30 lay lengths. To their surprise, they found that the number of load cycles increased slightly with the length of the rope. They want to do further tests to help to understand this unexpected result. It is to be supposed that the reason for the greater endurance of the longer rope pieces lies in the fact that the loosening of the rope structure, especially in the neighbourhood of the sockets, could be compensated better in longer pieces of rope.

Esslinger (1992) carried out tension-tension fatigue tests with a simple strand 1×7 with 0.6 in. and rope lengths $L = 1,040$, $2,030$ and $19,430$ mm. Figure 2.41 shows his results. Contrary to the findings of Suh and Chang, he discovered that, as expected, the mean number of cycles decreases with the rope length. For the simple strand Esslinger tested there can only be minimal possible loosening of the rope structure.

Therefore, the loosening of the structure and the sockets can be considered as not affecting the influence of the length on the strand endurance.

The number of load cycles can be calculated with the methods of the reliability theory. Without any explanation, Gabriel (1979) first presented this method in a diagram for wire ropes of different lengths. The survival probability of the rope with the length L as a serial grouping of the pieces with the length L_0 is (while neglecting the influence of the sockets)

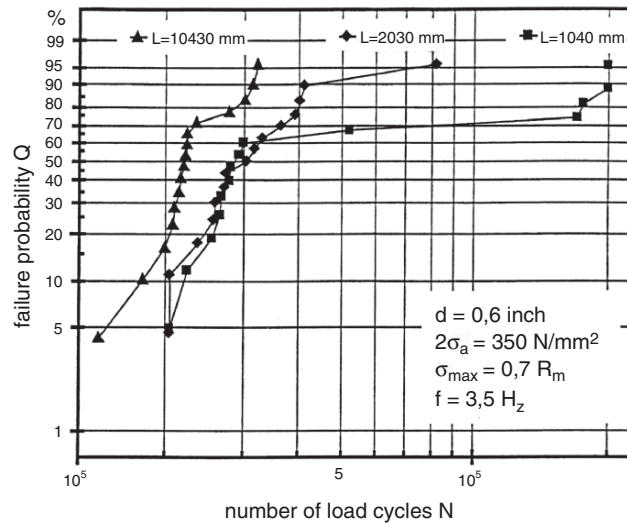


Fig. 2.41. Number of load cycles for a strand 1×7 for different lengths L , Esslinger (1992)

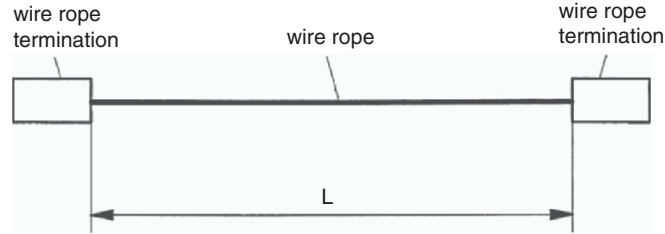


Fig. 2.42. Test sample, wire rope with termination

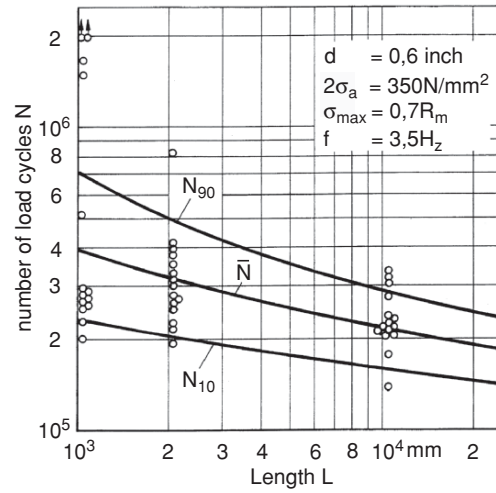


Fig. 2.43. Number of load cycles of a strand 1×7 for different lengths using Esslinger's results in Fig. 2.41

$$R = R_0^{L/L_0}. \quad (2.106)$$

Figure 2.42 shows the rope with sockets and defines the rope lengths. Once more, the logarithm normal distribution has been used to evaluate Esslinger's results. In Fig. 2.43 the numbers of load cycles found by Esslinger are introduced from Fig. 2.41 and in addition the lines calculated are drawn for the mean number of load cycles $\bar{N}$ and for the number of load cycles N_{10} and N_{90} , at which point, at the most 10% respectively 90% of the ropes will be broken. The numbers of load cycles N for the rope length $L_0 = 2,030$ mm have been taken as the basis for the calculation because there is only one extreme number of load cycles. For that distribution, the mean number of load cycles is $\bar{N}_0 = 318,000$ and the standard deviation is $\lg s = 0.148$. The test results and the calculated lines harmonise quite well.

From the results of (2.106), an equation can be derived for the load cycles ratio of the rope lengths L and L_0 [(2.107)]. With this, the endurance (2.102) can be corrected for different rope lengths. Equation (2.102) and their constants in Table 2.8, 2.9 and 2.10 are related on a mean rope length of about

$L_0 = 60d$ of the test rope lengths $40d$, $55d$ and $100d$. Based on this rope length, the ratio of the numbers of load cycles is

$$\lg \frac{N_L}{N_{60}} = \lg N_L - \lg N_{60} = \frac{1}{a_{6\gamma} + \lg(L/d)} - \frac{1}{a_{6\gamma} + \lg 60}. \quad (2.107)$$

The results of (2.106) and (2.107) depend on the standard deviation of the number of load cycles. The standard deviation, known until now for two Warrington ropes with a the length $87d$ is $\lg s = 0.038$ (Fig. 2.36, $\bar{N} = 82700$) and $\lg s = 0.092$ (Fig. 2.35, $\bar{N} = 406,000$) and for the strand 1×7 with a length $133d$, is $\lg s = 0.148$ (Fig. 2.41, $\bar{N} = 318,000$). The standard deviation is non-uniform and probably increases as for materials normally found with the number of load cycles.

As for the rope bending, a mean standard deviation is set at $\lg s = 0.05$ for the rope length $L/d = 60$. The standard deviation for fluctuating tension is probably greater. On the other hand, the rope endurance will possibly at first not decrease with the rope length as shown in the findings of Suh and Chang (2000). With this standard deviation, the decrease of the number of load cycles with the rope length has been at least partly taken into consideration. In (2.107), the constant $a_{6\gamma}$ for this together with the constants from Tables 2.8 and 2.9 are listed in Table 2.11.

Palmgren-Miner Rule (Damage Accumulation Hypothesis)

For roller bearings loaded by a series of load cycles with different loads, Palmgren (1924) stated the hypothesis that the sum of ratios n_i/N_i (called damage sum) will be 1. That means

$$\sum_{i=1}^m \frac{n_i}{N_i} = 1. \quad (2.108)$$

In this, n_i is the number of load cycles under the load i (load defined by i) and N_i is the endurance under the load i . Miner (1945) found that this rule is also valid for other elements and special kinds of loads.

According to (2.108a) the endurance Z of an element under a series of different loads i will be

$$Z = \frac{1}{\sum_{i=1}^m \frac{w_i}{N_i}}. \quad (2.108a)$$

Here $w_i = n_i/Z$ is the portion of the number of load cycles n_i under the load i .

However, this rule is only a hypothesis and it must be checked to see whether it can be used for wire ropes under fluctuating tension. From the results of tension-tension tests in four series of block loads, Chaplin (1988) found damage sums between 0.897 and 1.109, and Rossetti and Maradei (1992) found damage sums between 1.24 and 1.28. From similar tests with Warrington-Seale ropes, Casey (1993) got damage sums between 0.6022 and 1.2584.

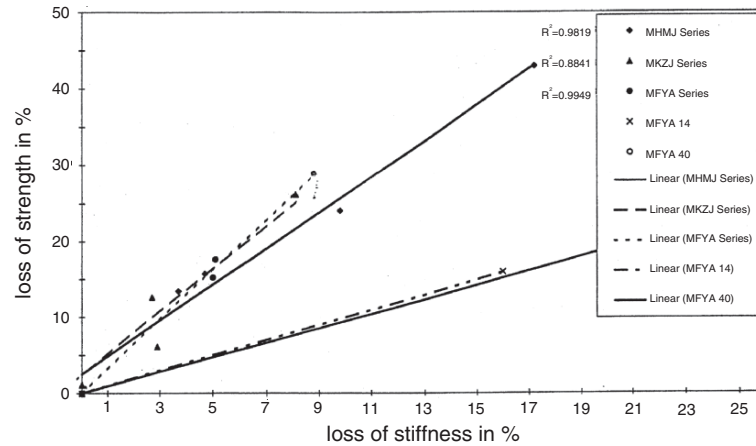


Fig. 2.44. Relation between loss of strength and loss of stiffness, Paton and others (2001)

All these results show that the Palmgren-Miner rule can be used for wire ropes under fluctuating tension.

Discard Criteria

The amplitude stresses for the inner wires are normally greater than in the outer wires, Chaps. 2.12–2.14. Therefore, outside wire breaks cannot be detected in most cases during tension-tension tests, Wehking and Klöpfer (2000). That means, the point at which the wire rope requires replacement is not defined by the number of outer wire breaks. Wehking and Klöpfer therefore recommend inspecting wire ropes under fluctuating tension by means of a magnet inductive test. Because a relation between rope endurance and the number of (inner) wire breaks is still unknown, the wire rope should be designed for the number of load cycles where at most 1% of the wire ropes are broken.

Paton and others (2001) tested the residual rope breaking force after having different numbers of load cycles. They found a relation between the loss of rope breaking force and the loss of length stiffness $S/\Delta L$. In Fig. 2.44 this relation is shown for 6-strand Warrington–Seale ropes with steel cores of 40 and 70 mm diameters. They recommend using a discarding criterion of a loss of 10% of the wire rope breaking force measured.

2.6.5 Calculation of the Number of Load Cycles

Resin Sockets

With the test results and the related equations, the number of load cycles prior to rope breakage can be calculated for open spiral ropes of nominal

strength 1,370–1,770 N/mm², zinc coated and for Warrington–Seale ropes 6 × 36 to 6 × 49-IWRC-*sZ* of the nominal strength 1,570–1,960 N/mm², bright or zinc coated, lubricated. For these, the regression (2.102) respectively (2.102a), (2.103) for the varying quantile γ and (2.107) for the influence of the rope length will be combined. To give a better overview, the unit factors $S_e = 1$ N and $d_e = 1$ mm (to make the ratios dimensionless) will be removed. Then, for a rope with the length L , the number of load cycles – where with a certainty of 95% at most a quantile γ of the wire ropes has been broken – is

$$\lg N_\gamma = a_0 + a_1 \cdot \lg \frac{2S_a}{d^2} + a_2 \cdot \frac{S_{\text{lower}}}{d^2} + a_3 \cdot \left(\frac{S_{\text{lower}}}{d^2} \right)^2 + a_4 \cdot \lg d + a_5 \cdot \lg z \\ + \frac{1}{a_{6\gamma} + \lg(L/d)} - \frac{1}{a_{6\gamma} + \lg 60} - k_{T\gamma} \cdot \lg s.$$

This equation is also valid for the Warrington–Seale ropes if here the constant a_5 is set $a_5 = 0$. For the failure quantiles of 50, 10 and 1%, the constant parts can be summarised to

$$a_{G\gamma} = a_0 - k_{T\gamma} \cdot \lg s - \frac{1}{a_{6\gamma} + \lg 60}. \quad (2.109)$$

With this constant $a_{G\gamma}$ the number of load cycles N_γ – where with a certainty of 95% at most a quantile γ of wire ropes has been broken – is

$$\lg N_\gamma = a_{G\gamma} + a_1 \cdot \lg \frac{2S_a}{d^2} + a_2 \cdot \frac{S_{\text{lower}}}{d^2} + a_3 \cdot \left(\frac{S_{\text{lower}}}{d^2} \right)^2 + a_4 \cdot \lg d \\ + a_5 \cdot \lg z + \frac{1}{a_{6\gamma} + \lg(L/d)}. \quad (2.110)$$

In Table 2.11, the constants a_i for (2.110) are listed Feyrer u. Wehking (2006). The constants a_1 to a_5 have been taken from Table 2.8 for the open spiral ropes and from Table 2.9 for the Warrington–Seale ropes. The constant $a_{G\gamma}$ – listed in Table 2.11 – has been calculated with (2.109) for the different quantiles.

Equation (2.110) and the constants of Table 2.11 are valid up to the limiting number of load cycles N_D . With the reduced gradient of Haibach (1989), the number of load cycles above $N_D = 2 \times 10^6$ is (as explained under Woehler Diagram)

Table 2.11. Constants for calculating the number of loading cycles, (2.110)

wire ropes	$\gamma(\%)$	$a_{G\gamma}$	a_1	a_2	a_3	a_4	a_5	$a_{6\gamma}$
open spiral ropes	50	15.065						1.2
	10	14.767	−3.910	0.00118	−0.0000037	−0.793	0.399	1.9
	1	14.540						2.5
Warr.-Seale ropes IWRC – <i>sZ</i>	50	15.966						1.2
	10	15.611	−3.939	0.00326	−0.000012	−1.180	0	1.9
	1	15.334						2.5

$$N_k = N_D \left(\frac{2S_a/d^2}{2S_{aD}/d^2} \right)^{2a_1+1}. \quad (2.104)$$

S_{aD} is the amplitude of the tensile force for which the limiting number of load cycles $N_D = 2 \times 10^6$ has to be expected. This limiting amplitude of tensile force can be calculated with the following equation (inverted from equation (2.110)).

$$\begin{aligned} \lg \frac{2S_{a\gamma}}{d^2} = \frac{\lg N_\gamma}{a_1} - \frac{1}{a_1} \cdot \left(a_{G\gamma} + a_2 \cdot \frac{S_{\text{lower}}}{d^2} + a_3 \cdot \left(\frac{S_{\text{lower}}}{d^2} \right)^2 + a_4 \cdot \lg d \right. \\ \left. + a_5 \cdot \lg z + \frac{1}{a_{6\gamma} + \lg \frac{L}{d}} \right). \end{aligned} \quad (2.111)$$

with $N_j = N_D = 2 \cdot 10^6$.

With (2.110), the number of load cycles will be calculated, directly valid up to 2×10^6 for all quantiles γ . Numbers of load cycles above that should be corrected with (2.104). By using the limiting number of load cycles 2×10^6 for all quantiles, the standard deviation will be – as in reality – strongly extended in the region above the limiting number of load cycles.

Example 2.15: Number of load cycles

Data:

Warrington–Seale rope $6 \times 36 - \text{IWRC} - sZ$

Wire rope diameter $d = 20 \text{ mm}$, nominal strength $R_0 = 1,770 \text{ N/mm}^2$, lubricated

Rope length $L = 120 \text{ m}$, terminated with resin sockets

The fluctuating tensile forces are

Lower tensile force $S_{\text{lower}} = 30 \text{ kN}$, $S_{\text{lower}}/d^2 = 75 \text{ N/mm}^2$

Upper tensile force $S_{\text{upper}} = 80 \text{ kN}$

The range of the specific force is

$$2S_a/d^2 = \frac{S_{\text{upper}} - S_{\text{lower}}}{d^2} = 125 \text{ N/mm}^2.$$

Results:

Using (2.110) and the constants from Table 2.11, the numbers of load cycles are

$$N_{50} = 3,540,000 \quad N_{10} = 1,480,000 \quad N_1 = 750,000$$

From these numbers only

$$N_1 = 750,000 \quad \text{and} \quad N_{10} = 1,480,000$$

are directly valid. The mean number of loading cycles – greater than 2×10^6 – has to be corrected. For that, using (2.111), the limit range of the specific tensile force is

$$2S_{aD50}/d^2 = 144.5 \text{ N/mm}^2.$$

Then, with (2.104) the mean number of load cycles is

$$N_{50k} = 5,420,000.$$

Example 2.16: Number of load cycles Z , load collective

Data:

The data from example 2.15 are valid again. The lower force remains constant $S_{\text{lower}} = 30$ kN. The load collective for the force range is given by

part of the number of cycles	w_i	0.2	0.3	0.5
relative force range	q_i	1	0.8	0.6

Results:

The three specific force ranges $q_i * 2Sa/d^2 = q_i * 125$ are

125	100	75
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and according to equation (2.110) the numbers of load cycles N_1 are

750000	1810000	5610000.
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The last number of load cycles – greater than $2 \cdot 10^6$ – has to be corrected. For that, using the equation (2.111), the limit range of the specific tensile force is

$$2S_{aD1}/d^2 = 97.5 \text{ N/mm}^2.$$

Then, according to equation (2.104) the corrected number of load cycles is

$$N_{1K} = 12\,100\,000.$$

The common number of load cycles Z_1 , at which with a certainty of 95% at most 1% of the wire ropes are broken, is according to equation (2.106a)

$$Z_1 = \frac{1}{\frac{0.2}{750000} + \frac{0.3}{1810000} + \frac{0.5}{12100000}} = 2\,110\,000.$$

Rope Terminations

The number of load cycles will be reduced if rope terminations other than resin sockets are used. For ropes with these terminations, the number of load cycles will be calculated again with (2.110). Here, all the constants of Table 2.11 can be used except $a_{G\gamma}$. This constant is now

$$a_{G\gamma} = a_0 + \lg f_V - k_{T\gamma} \cdot \lg s_V - \frac{1}{a_{6\gamma} + \lg 60}. \quad (2.109a)$$

In contrast to (2.109a), the logarithm of the endurance factor f_V for the termination taken from Table 2.12 (still from a very small database) has been added. Furthermore, the standard deviation is replaced by those of rope pieces ($L_0 = 60d$) with the considered termination $\lg s_V$. However, up to now, this standard deviation is still unknown for all other terminations than for resin

Table 2.12. Endurance factor $f_V = N_V/N_w$, for rope with terminations related to resin sockets

Rope termination	Number of ropes z	Number of tests n_V/n_w	Logarithm of the endurance factor $\lg f_V$	Standard deviation $\lg s_{f_V}$	Endurance factor $f_V = N_V/N_w$
Metal socket ^{a,d}	Cross-NFC - 6×37 Warr.-SFC - 8×19 Warr.-IWRC - 8×19	2/2	-0.544	0.367	0.286
		12/19			
		5/26			
Splice eye ^b	Warr.-SFC - 8×19	2/19	-1.008	0.161	0.098
Cylindric aluminum ferrule eye termination ^{c,d,e,f,g,h}	Cross-NFC - 6×37	1/2	-0.463	0.368	0.345
	Warr.-NFC - 8×19	15/19			
	Warr.-IWRC - 8×19	5/26			
	WS-IWRC - 6×36	8/26			
	Cross-NFC - 6×37	9/2			
Flemic eye with steel clamp ^{i,d}	Warr.-SFC - 8×19	9/19	-0.428	0.314	0.373
Press bolt ^j	Spiral 37×1	7	-0.700	0.263	0.20
Wegde socket (rope lock) ^{d,k,n}	Warr.-SFC - 8×19	36/19	$= -1.347 + 0.0071\alpha$ $+ 0.145 l_K/d$	0.306	$\alpha = 14^\circ$ 28° 14° 28°
	Warr.-IWRC - 8×19	46/26			$l_K/d = 4$ 4 6 6
					$f_V = 0.216$ 0.271 0.420 0.529
U-bolt clamp DIN 1142 ^{d,l,m}	Warr.-SFC - 8×19	5/19	-0.871	0.152	0.108

^aMüller (1971); ^bMüller (1976); ^cMüller (1966); ^dFeyrer (1995); ^eFeyrer et al. (1987); ^fHemminger (1989); ^gWehking et al. (2000); ^hKlöpper (2002); ⁱSchneidersmann et al. (1980); ^jVogel (2005); ^kFeyrer (1984); ^lUlrich (1973); ^mMüller (1975); ⁿfor clamping angle $\alpha = 14^\circ$ to 30° and clamping length $l_K/d = 3.5$ to 6 .

sockets and cannot be derived from the standard deviation $\lg s_{fV}$ of the factor f_V in Table 2.12. It is therefore not possible at the time being to use this replacement and it will only be of interest for the future. Now with $\lg s_V = \lg s$, the constant $a_{G\gamma}$ can be derived simply from the constant in Table 2.11

$$a_{G\gamma} = a_{G\gamma\text{Tab}} + \lg f_V. \quad (2.109b)$$

2.7 Dimensioning Stay Wire Ropes

Stay wire ropes have to be dimensioned in such a way that they can stand up to extreme forces which only occur rarely, be sufficiently durable in case of fluctuating forces and have safe discard criteria. These safety limits are characterised by:

- Extreme forces
- Fluctuating forces
- Discard criteria

Stationary wire ropes have to meet all these requirements independent of each other.

2.7.1 Extreme Forces

To prevent a wire rope breaking due to an extreme force which occurs only rarely, technical regulations normally require that the minimum breaking force $F_{\min}$ is several times higher than the nominal rope tensile force S

$$F_{\min} \geq \nu \cdot S. \quad (2.112)$$

The so-called safety factor ν takes the increase of the tensile force due to possible overloading into consideration as well as the weakening of the wire rope breaking force due to fatigue occurring over time in the case of fluctuating forces or by corrosion. Paton and others (2001) found a reasonable weakening of the wire rope breaking force occurs long before the rope breaks under the fluctuating tensile force. One of their results shows that the breaking force for the spiral ropes tested is reduced by 15% at between about 20 and 70% of their endurance. The 15% loss of breaking force occurs late if the endurance is low ($N \approx 50,000$) and earlier if it is high ($N \approx 5 \times 10^6$).

In technical regulations, experts have defined the reference values for the safety factors based on their own experience combined with theoretical considerations. Of course, for each individual technical field, the safety factor varies according to whatever extreme forces may occur there. For example the safety factor for stay ropes for cranes is about $\nu = 3.2$. For steel constructions and bridges, the safety factor is smaller being about $\nu = 2.2$ as the greater part of the forces comes from their own constant weight.

Table 2.13. Range of specific force of open spiral ropes for certain numbers of load cycles (full load) N_1 , at which with a certainty of 95% at most 1% of the wire ropes are broken

Rope-diameter d (mm)	Number of wires z	range of specific force $2S_{a1}/d^2$ in N/mm^2							
		$N_1 = 20,000$	50,000	125,000	320,000	800,000	2,000,000	10,000,000	
10	37	400	326	258	203	161	127	100	
12.5	37	395	312	247	194	154	122	96	
16	61	396	313	248	195	154	122	96	
20	61	379	300	237	186	147	117	92	
25	85	375	297	235	184	146	115	91	
32	85	357	283	224	176	139	110	87	
40	100	348	275	218	171	135	107	85	
50	125	341	269	213	168	133	105	83	
63	160	334	264	209	164	130	103	81	
80	200	326	258	204	160	127	100	79	
100	250	319	253	200	157	124	98	78	
125	292	311	246	194	153	121	96	76	

Rope tensile stress $\sigma_z = 1.70 S/d^2$

The wire rope breaking force is valid for wire ropes terminated with either resin or metal sockets. For wire ropes with other terminations, the wire rope breaking force (more or less reduced) can be calculated with the breaking force factor f_F from Table 2.7.

The required minimum breaking force of the wire rope with the terminations T is then

$$F_{\min T} = f_F \cdot F_{\min} \geq \nu \cdot S. \quad (2.113)$$

2.7.2 Fluctuating Forces

Technical Rules – Test Results

In the existing technical regulations, the larger the lower stress is, the smaller the stress range that is allowed. This restriction of the stress range for large lower stresses cannot be explained at all by the test results. On the contrary, the test results show that for a certain number of load cycles the stress range tends to increase with the lower stress up to the allowed maximum stress. As an example, Fig. 2.45 from Wehking and Klöpfer (2000) shows the range of the specific rope force allowed after DIN 15018 compared with the test results. The comparison was made for Warrington-Seale ropes 6×36 -IWRC-*sZ* with rope diameters 30 and 40 mm. The specific force range allowed in the DIN loading group B6 for a strong load collective is compared with test results where the ropes are loaded with the full force range for all load cycles (full load cycles) and where only 10% of the ropes break before 2×10^6 load cycles have been reached.

In future technical regulations, the range of the specific force found for the lower force 0 can simply be allowed for all lower forces, or, taking the

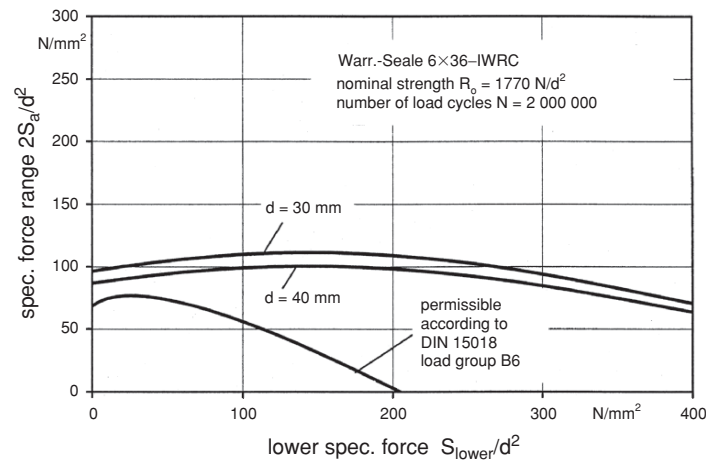


Fig. 2.45. Allowed range of specific force of Warrington-Seale ropes after DIN 15018 and from tests, Wehking and Klöpfer (2000)

influence of the lower forces into consideration, the allowed range of specific forces (stress) can be calculated for a given lower specific force and for a given number of load cycles. That is possible at this time for open spiral ropes zinc coated and lubricated and for ordinary lay Warrington–Seale ropes, bright or zinc coated and lubricated.

Force Range

As has been repeatedly pointed out, the range of specific force should be calculated in such a way that with a certainty of 95% at most 1% of the ropes is broken for a required endurance. If the required number of load cycles (full load cycles) is smaller than 2×10^6 , the allowed specific force range can be calculated directly with (2.111) and the constants in Table 2.11.

If the required number of load cycles is bigger than 2×10^6 , first the specific force range $2S_{aD\gamma}/d^2$ has to be calculated with (2.111) for $N_\gamma = 2 \times 10^6$. Then for the required number of load cycles bigger than 2×10^6 , according to the inverted (2.104) the range of the specific force is

$$2S_{a\gamma}/d^2 = 2S_{aD\gamma}/d^2 \cdot \left(\frac{N_\gamma}{N_D} \right)^{1/(2a_1+1)}. \quad (2.114)$$

For both (2.111) and (2.114) the constants have to be taken from Table 2.11.

A survey of the range of specific forces $2S_{a1}/d^2$ is presented in Table 2.13, with which open spiral ropes with resin sockets can reach a given number of load cycles N_1 . These numbers N_1 mean the number of full load cycles at which with a certainty of 95% at most 1% of the wire ropes are broken. The lower specific force is $S_{\text{lower}}/d^2 = 0$. With increasing rope diameters d , an increased number of wires z in the wire rope have been inserted as is usual in practice. The rope length is $L = 100$ m.

As Table 2.13 shows, the allowed range of specific force is strongly reduced with an increasing rope diameter. For the smallest rope diameter $d = 10$ mm and the smallest required number of load cycles $N_1 = 20,000$, the force range is restricted by the maximum allowed specific force. This maximum allowed specific force for spiral ropes in steel constructions is about $S_{\text{max}}/d^2 = 400$ N/mm². For open spiral ropes (in contrast to Warrington–Seale ropes) the influence of the lower force is relatively low.

Example 2.17: Allowed specific force range

Data:

Required number of load cycles $N_1 = 5,000,000$ at which with a certainty of 95% at most 1% of the wire ropes are broken.

Spiral rope 1×61 , lubricated

$d = 20$ mm, strength $R_0 = 1,770$ N/mm²

Rope length $L = 120$ m, $L/d = 6,000$

Lower rope force $S_{\text{lower}} = 30$ kN, $S_{\text{lower}}/d^2 = 75$ N/mm²

Resin sockets

Results:

According to (2.111), the specific force range for the limiting number of load cycles $N_{D1} = 2 \times 10^6$ is

$$\lg \frac{2S_{aD1}}{d^2} = \frac{\lg 2,000,000}{-3.910} - \frac{1}{-3.910} \cdot \left(14.540 + 0.00118 \times 75 - 0.0000037 \times 75^2 - 0.793 \times \lg 20 + 0.399 \times \lg 61 + \frac{1}{2.5 + \lg 6,000} \right)$$

$$2S_{aD1}/d^2 = 121.4 \text{ N/mm}^2.$$

For the required number of load cycles $N_1 = 5,000,000$, the specific force range according to (2.114) is

$$2S_{a1}/d^2 = 121.4 \left(\frac{5,000,000}{2,000,000} \right)^{1/(-2 \times 3.910 + 1)}$$

$$2S_{a1}/d^2 = 106 \text{ N/mm}^2$$

and the force range and the stress range are

$$2S_{a1} = 42.4 \text{ kN} \quad \text{and} \quad 2\sigma_{a1} = 180 \text{ N/mm}^2.$$

Rope Termination

The calculation of the allowed range of specific force for wire ropes under fluctuating forces described here is based on test results relating to wire ropes terminating in resin sockets. For ropes with other terminations, the allowed range of specific forces can be calculated in the same way. In this case the constant a_G has to be estimated with (2.109a) or (2.109b).

2.7.3 Discard Criteria

Wire ropes always have a limited working life. Prior to rope breakage, the rope has to be discarded and replaced. It is necessary to have safety inspections to ascertain the state of the wire rope, i.e. the state at which the wire rope should be discarded.

The discarding state of stay wire ropes will be indicated by damage near the terminations as well as wire breaks or corrosion on the free rope length. The inner wires of ropes with tensile loads are always stressed to a greater degree than the outer wires. This means that in wire ropes suffering under fluctuating tension, it is the wires in the inner rope in particular which break. Therefore the wire rope has to be inspected by magnetic methods Feyrer u. Wehking (2006).

In any case, wire breaks in or close to the sockets are promoted by transverse vibrations of the ropes, Hobbs and Smith (1983), Oplatka and Roth (1991, 1993), Brevet and Siegert (1996), Siegert and others (1997), Gourmelon (2002) and Siegert and Brevet (2005). Gabriel and Nürnberger

(1992) pointed out that in the most cases, the stay wires rope has to be discarded because of damage near the terminations or corrosion but not because of wire breaks on the free length.

The transversal vibration of stay ropes should be minimised by dampers, Gourmelon (2002). However these vibrations that induce wire breaks in the sockets cannot totally avoided. As Oplatka and Roth (2000) stated, there is no method found that can show in field-test the condition of the rope even in resin sockets with sufficient accuracy. Therefore the transversal vibration should be kept away from the sockets. The wire rope should be hold by a fastening in front of the socket on that the transversal vibration ends. This fastening should be removable so that the wire rope can be inspected in this region with magnetic methods.

References

- Alani, M. and Raoof, M.: Effect of mean axial load on axial fatigue life of spiral strands. *Int. J. Fatigue* 19 (1997) 1, 1–11
- Andorfer, K.: Die Zugkraftverteilung in schwingend beanspruchten geraden Drahtseilen. Diss. Techn. Universität Graz 1983
- Becker, K.: On the fatigue strength of wire ropes. OIPEEC Round Table 1977, Luxembourg, Chaps. 1–3
- Beha, R.: Bewegungsverhalten und Kraftwirkungen des Zugseiles und der Fahrzeuge von Zweiseilbahnen zur Berechnung der Dynamik des Gesamtsystems. Diss. Universität Stuttgart 1994. Kurzfassung *ISR* (1995) 1, S11–15
- Benndorf, H.: Beiträge zur Theorie der Drahtseile. *Zeitschr. d. öster-reichischen Ingenieur- u. Architektenvereins* 56 (1904) 30, S433–437 (u. 31, S449–453)
- Berg, F.: Der Spannungszustand einfach geschlungener Drahtseile. Diss. TH Hannover 1907 und *Dinglers Polytech. Journal* 88 (1907) 19, S289–292 (u. 20, S307–311)
- Bock, E.: Die Bruchgefahr der Drahtseile. Diss. TH Hannover 1909
- Brevet, P. and Siegert D.: Fretting fatigue of seven wire strands axially loaded in free bending fatigue tests. *OIPEEC Bull.* 71 (1996)
- Buchholz, G. and Eichmüller, H.: Tätigkeitsbericht 1986–1988. Staatl. Materialprüfungsamt Nordrhein-Westfalen, Dortmund 1988, S58–62
- Cantin, M., Cubat, D. and Nguyen Xuan, T.: Experimental analysis and modelisation of the stiffness in torsion of wire ropes. *OIPEEC Round Table*, Delft Sept. 1993, pp. II.67–II.77
- Casey, N. F.: The fatigue endurance of wire ropes for mooring offshore structures. *OIPEEC Round Table*, Delft Sept. 1993, pp. I.21–I.49
- Casey, N. F. and Waters, D. M.: Condition monitoring for fatigue test assessment and life prediction of six-strand rope. *OIPEEC Round Table*, Zürich Sept. 1989, pp. 7.1–7.20

- Castillo, E. and Fernandez-Canteli, A.: Statistical models for fatigue analysis of long elements. Introductory Lectures of the IABSE-Workshop "Length Effect on the Fatigue of Wires and Strands". Madrid, Sept. 1992
- Castillo, E., Fernandez-Canteli, A., Ruiz-Tolosa and Sarabia, J. M.: Statistical models for analysis of Fatigue life of long elements. Trans. ASCE J. Eng. Mech. 116 (1990) 5, 1036–1049 (paper 24618)
- Chaplin, C. R.: Tension-tension fatigue in mooring offshore structures. OIPEEC Bull. 56 (1988) 9–22
- Chaplin, C. R.: Problems of torque and rotation in wire ropes. First International Stuttgart Rope Day, Institut für Fördertechnik und Logistik, Universität Stuttgart 21. Feb. 2002
- Chaplin, C. R.: Deepwater Moorings: Challenges, Solutions and Torsion. Second International Stuttgart Rope Day, Institut für Fördertechnik und Logistik, Universität Stuttgart 17. and 18. Feb. 2005
- Chaplin, C. R. and Potts, A. E.: Wire rope offshore - a critical review of wire rope endurance research affecting offshore application. HSE publication OHT, 91341, HMSO, London June 1991
- Costello, G. A.: Theory of wire rope, 2nd edn. New York: Springer, 1997, ISBN 0-387-98202-7
- Costello, G. A. and Miller, R. E.: Lay effect of wire rope. J. Eng. Mech. Div. ASCE 105 (1979) No EM5, 597–608
- Costello, G. A. and Sinha, S. K.: Torsional stiffness of twisted wire cables. J. Eng. Mech. Div. ASCE 103 (1977a) No EM5, 766–770
- Costello, G. A. and Sinha, S. K.: Static behaviour of wire rope. J Eng. Mech. Div. ASCE 103 (1977) No EM5, 1011–1022
- Czitary, E.: Spannkraftermittlung in Seilen durch Schwingungs-messung. Wasserwirtschaft (1931) 15/16, S246–249
- Czitary, E.: Seilschwebbahnen. Wien Springer, 1952
- Czitary, E.: Über das Schwingungsverhalten des Trag- und Zugseiles von Seilschwebbahnen. ISR Int. Seilbahn-Rundschau, Seilbahnbuch 1975, S27–34
- Donandt, H.: Zur Dauerfestigkeit von Seildraht und Drahtseil. Archiv für (1950) das Eisenhüttenwesen 21 (1950) 9/10, S283–292
- Dreher, F.: Ein Beitrag zur Theorie der Drehung und Spannungsverteilung bei zugbelasteten Litzen und Seilen. Diss. TH Karlsruhe, 1933
- Engel, E.: Ein Beitrag zur Berechnung der Verdrehungen von Draht-seilen und deren Bedeutung bei Seilbahnen. Diss. TH Wien, 1957
- Engel, E.: Das Drehbestreben der Seile und ihre Drehsteifigkeit. Österreichische Ingenieur-Zeitschrift 1 (1958) 1, S33–39
- Engel, E.: Verdrehungserscheinungen an Seilen bei Seilbahnen. Österreichische Ingenieur-Zeitschrift 2 (1959) 6, S215–220
- Engel, E.: Der Seildrall. Int. Berg- und Seilbahn-Rundschau 9 (1966) 2, S33–35
- Engel, E.: Nichtlineare Seilschwingungen bei Seilbahnen. ISR Int. Seilbahn-Rundschau (1977) 3, S39–40

- Esslinger, V.: Fatigue testing of wires and strands. Introductory. Lectures of the IABSE-Workshop "Length Effect on the Fatigue of Wires and Strands". Madrid, Sept. 1992
- Evans, J. J. and Chaplin, C. R.: The effect of wire breaks and overload on wire strain differences in six strand wire ropes under tensile fatigue. OIPEEC Round Table Reading, Sept. 1997, pp. 45–57
- Evans, J. J., Ridge, I. M. L. and Chaplin, C. R.: Wire strain variations in normal and overloaded ropes in tension-tension fatigue and their effect on endurance. *J. Strain Anal.* 36 (2001) 2, 219–230
- Feyrer, K.: Effect of bending length on endurance of wire ropes. *Wire World Int.* 23 (1981) 115–119
- Feyrer, K.: Das Tragverhalten von Seilklemmen und Seilschlössern. *DRAHT* 35 (1984) 5, S239–245
- Feyrer, K. and Jahne, K.: Seilelastizitätsmodul von Rundlitzenseilen. *DRAHT* 41 (1990) 4, S498–504
- Feyrer, K. and Schiffner, G.: Torque and torsional stiffness of wire ropes. *WIRE* 36 (1986) 8, 318–320 and 37 (1987) 1, 23–27
- Feyrer, K.: Klemmwinkel und Klemmwinkel von Seilschlössern. Schriftenreihe der Bundesanstalt für Arbeitsschutz Fb 622. Bremerhaven: Wirtschaftsverlag NW 1991 Clamping angle and clamping length of rope wedge. . . HSE Translation No 14265 1, Nov. 1991, Health and Safety Executive. Harpur Hill, Buxton Derbyshire SK 179 JN, England
- Feyrer, K.: Reference values for the evaluation of wire rope tests. OIPEEC Bulletin 63 and Wire Industry 55 (1992) August pp. 593–594
- Feyrer, K.: Endurance formula for wire ropes under fluctuating tension. OIPEEC Technical Meeting Stuttgart Sept. 1995, pp. 2.1–2.10
- Feyrer, K.: Torsion of multilayer round strand ropes. *WIRE* (1997) 3, pp. 45–47. Deutsch: *DRAHT* 48 (1997) 2, S 34–36
- Feyrer, K.: Endurance of wire ropes under fluctuating tension. OIPEEC Bulletin 85, Reading June 2003, pp. 19–26
- Feyrer, K. and Wehking, K. H.: Lebensdauer von Drahtseilen unter schwelender Zugkraft - Wissensstand und Ausblick. *Bauingenieur* 81 (2006) Dec. pp. 533–537
- Fuchs, D., Spas, W.: A method of calculating the hoisting cycles of a rope as a function of stress to the point of discarding. OIPEEC Round Table, Delft Sept. 1993, pp. I.91–I.102
- Fuchs, D., Spas, W. and Dürrer, F.: Elastische Dehnung von Förderseilen während des Betriebs. *DRAHT* 47 (1996) 4/5, S281–287
- Gabriel, K.: Anwendungen von statistischen Methoden und Wahrscheinlichkeitsbetrachtungen auf das Verhalten von Bündeln und Seilen aus vielen langen Drähten. Vorbericht zum 2. Int. Symposium des Sonderforschungsbereiches 64, Stuttgart 1979
- Gabriel, K.: Fatigue resistance of locked coil ropes. OIPEEC Round Table, Delft Sept. 1993, pp. Ill.27–Ill.39

- Gabriel, K. and Nürnberger, U.: Failure mechanisms in fatigue. Introductory Lectures of the IABSE-Workshop "Length Effect on the Fatigue of Wires and Strands" Madrid, Sept. 1992
- Gibson, P. T.: Wire rope behaviour in tension and bending. Proceedings of the First Annual Wire Rope Symposium. Denver Colorado. Engineering Extension Service Washington State University 1980, 3–31
- Glushko, M. F.: Steel lifting ropes. Kiev Technica 1996, p. 327
- Gourmelon, J. P.: Fatigue of staying cables, Organisation and results of the research programme. OIPEEC Bulletin 84 Reading Dec. 2002
- Gräbner, P. and Thomasch, A.: Zur Dimensionierung drehungsarmer Seile. Hebezeuge und Fördermittel 23 (1983) 6, S166–169
- Haibach, E.: Betriebsfestigkeit. Düsseldorf VDI Verlag GmbH 1989. ISBN 3–18–400828–2
- Haid, K.-D.: Determination of forces in strand wires. WIRE 33 (1983) 1
- Hankus, J.: Längsverformungen von Förderseilen. Glückauf-Forschungshefte 37 (1976) 2, S19–21
- Hankus, J.: Regressionsmodelle der Längsverformungen und des (1978) Elastizitätsmoduls von Förderseilen. Glückauf-Forschungshefte 39 (1978) 6, S252–256
- Hankus, J.: The actual breaking force of steel wire ropes. OIPEEC Bull. 45 (1983), 101–112
- Hankus, J.: Mechanische Eigenschaften von Drahtseilen. Drahtwelt 75 (1989) 4, S9–17
- Hankus, J.: Non-typical process of the progressive weakening of a mining hoisting rope. OIPEEC Round Table, Delft Sept. 1993. pp. II21–II34
- Hankus, J.: Consideration of mine hoisting rope contraction. OIPEEC Bull. 73 (1997), 9–19
- Heinrich, G.: Zur Statik des Drahtseiles. Wasserwirtschaft und Technik 4 (1937) 30, S267–271
- Heinrich, G.: Über die Verdrehung der zugbelasteten Litzen. Der Stahlbau (Beilage zu Bautechnik). Berlin 15 (1942) 12/13, S41–45
- Hemminger, R.: Drahtseile mit Aluminium-Preßverbindungen und Kauschen. DRAHT 40 (1989) 10, S781–785
- Hermes, J. M. and Bruens, F. P.: The twist variations in a non-spin rope of a hoist installation. Geologie and Mijnbouw (NW. SER.) 19. Jaargang, November 1957, pp. 467–476 (Holländisch)
- Hobbs, R. E. and Ghavami, K.: The fatigue of structural wire strands. Int. J. Fatigue 1982, 69–72
- Hobbs, R. E. and Smith, B. W.: Fatigue performance of socketed terminations of structural strands. Proc. Inst. Civ. Eng. Part 2 75 (1983), 35–48
- Hruska, F. H.: Calculation of stresses in wire ropes. Wire Wire Prod. 26 (1951), 798
- Hruska, F. H.: Radial forces in wire ropes. Wire Wire Prod. 27 (1952) 1, 44
- Hruska, F. H.: Tangential forces in wire ropes. Wire Wire Prod. 28 (1953) 5, 455–460

- Hudler, S.: Der Elastizitätsmodul des Drahtseiles. Wasserwirtschaft und Technik (1937) 28/30, S271–279
- Irvine, H. M.: Cable structures. The Massachusetts Institute of Technology 1981, ISBN 0-262-09023-6, espec. pp. 119–129
- Jehmlich, G.: Anwendung und Überwachung von Drahtseilen. VEB (1985) Verlag Technik, Berlin, 1985
- Jiang, W. G., Yao, M. S. and Walton, J. M.: Modelling of rope strand under axial and torsional loads by Finite Element Method. OIPEEC Round Table Conference, Reading Sept. 1997, pp. 17–35
- Klöpfer, A.: Untersuchung zur Lebensdauer von (2002) zugschwellbeanspruchten Drahtseilen. Diss. Universität Stuttgart Stuttgart, 2002
- Kollros, W.: Der Zusammenhang zwischen Torsionsmoment, Zugkraft und Verdrillung in Seilen. Int. Berg- und Seilbahn-Rundschau 18 (1974) 2, S. 49–58 and DRAHT 26 (1975) 10, S. 475–480
- Kollros, W.: Relationship between torque, tensile force and twist in wire ropes. WIRE 26 (1976) 1, 19–24
- Kraincanic, I. and Hobbs, R. E.: Torque induced by axial load in a 76 mm wire rope. Comparison of experimental results and theoretical prediction. OIPEEC Round Table. Reading Sept. 1997, pp. 173–185
- Leider, M. G.: Krümmung und Biegespannungen von Drähten in gebogenen Drahtseilen. Draht 28 (1977) 1, S1–8
- Lombard, J.: Aeroplane versus rope - what happened? OIPEEC Bull. 76 (1998), 25–29. Deutsch ISR (1998) 5, S8–9
- Lombard, J.: Deformation of a rope during impact from an aeroplane. OIPEEC Bull. 76 (1998), 31–50
- Martin, P. A. and Berger, J. R.: On mechanical waves along aluminium conductor steel reinforced (ACSR) power lines. J. Appl. Mech. 69 (2002), 740–748
- Matsukawa, A., Kamei, M., Fukui, Y. and Sasaki, Y.: Fatigue resistance analysis of parallel wire strand cables based on statistical theory of extreme. Stahlbau 54 (1985) 11, S326–335
- Miner, M. A.: Cumulative damage in fatigue. J. Appl. Mech. Trans. ASNE 67 (1945), 159–164
- Müller, H.: The properties of wire rope under alternating stresses. Wire World 3 (1961) 5, 249–258
- Müller, H.: Beziehungen zwischen Seilbeanspruchung und Seil-konstruktion. Vortrag Drahtseilvereinigung 23. Nov. 1962
- Müller, H.: Fragen der Seilauswahl und der Seilbemessung an (1962) Turmdrehkränen. Technische Überwachung 4 (1963) 2, S62–66
- Müller, H.: Drahtseile im Kranbau. VDI-Bericht Nr. 98 und dhf 12 (1966) 11, S714–716 und 12, S766–773
- Müller, H.: Untersuchung an Seilvergußmetallen. Goldschmidt informiert 3 (1971) 16, S23–38
- Müller, H.: Untersuchungen an Drahtseilklemmen. DRAHT 26 (1975) 8, S371–378

- Müller, H.: Untersuchungen an Augensleißen von Drahtseilen. DRAHT 27 (1976) 6, S264–269
- NEL-Report: The fatigue of 40 mm diameter six strand wire rope in a sea-water environment. National Engineering Laboratory (NEL) Report No. ENER/14 for the UK Department of Energy, March 1984
- Neumann, P.: Untersuchungen zum Einfluß tribologischer Beanspruchung auf die Seilschädigung. Diss. TH Aachen 1987
- OIPEEC Recommendation No 7: Tension-tension fatigue test. OIPEEC Bull. 61 (1991), 50
- Oplatka, G. and Roth, M.: Bending fatigue of locked coil ropes in the neighbourhood of cast sockets. OIPEEC Technical Meeting, Nantes Oct. 1991
- Oplatka, G. and Roth, M.: Bending fatigue of locked coil ropes in the neighbourhood of cast sockets. Part 2. Influence of lubrication. OIPEEC Technical Meeting, Delft Sept. 1993
- Oplatka, G. u. Roth, M.: Endurance of steel wire ropes under fluctuating tension and twist. OIPEEC Bull. 71 (1996), 13–22
- Oplatka, G. and Roth, M.: Non-destructive testing of resin cast sockets - possibilities and limits. OIPEEC Bull. 79, Reading 2000
- Oplatka, G. and Volmer, M.: Wieso bricht das Seil und nicht der (1998) Flügel? ISR Int. Seilbahn-Rundschau (1998) 4, S13–14
- Palmgren, A.: Die Lebensdauer von Kugellagern. Z.VDI 68 (1924), S339–341
- Paton, A. G., Casey, N. F., Fairbairn, J and Banks, W. M.: Advances in the fatigue assessment of wire ropes. Ocean Eng. 28 (2001), 491–518
- Raoof, M. and Hobbs, R. E.: Analysis of axial fatigue data wire ropes. Int. J. Fatigue 16 (1994) 7, 494–501
- Raoof, M. and Huang, Y. P.: Lateral vibrations of steel cables including structural damping. Proc. Inst. Civ. Eng. Struct. Build. 99 (1993), 123–133
- Reemsnyder, H. S.: The mechanical behaviour and fatigue resistance of steel wire, strand and rope. Homer Research Laboratories, Bethlehem Steel Corporation, Bethlehem PA. June 1972
- Rebel, G.: The torsional behaviour of triangular strand ropes for drum winders. OIPEEC Bull. 74 (1997), 29–55
- Rebel, G. and Chandler, H. D.: A machine for the tension-tension testing of steel wire ropes. OIPEEC Bull. 71 (1996), 55–73
- Ridge, I.: Bending-tension fatigue of wire rope. OIPEEC Bull. Nr. 66 Reading Nov. 1993, 31–50
- Rossetti Rossetti, U. and Maradei, F.: Check on the validity of the Miner's hypothesis for tension-tension fatigue. OIPEEC Bull. 64. Reading Nov. 1992, 23–28
- Schiffner, G.: Spannungen in laufenden Drahtseilen. Diss. Universität Stuttgart 1986
- Schlauderer, A.: Untersuchungen zur Zug- und Biegebeanspruchung beweglicher Anschlussleitungen von Leitungswagen- Stromversorgungsanlagen. Fortschritt-Berichte VDI, Reihe 13 Nr 35. Düsseldorf VDI Verlag 1990
- Schmidt, K.: Die sekundäre Zugbeanspruchung der Drahtseile aus der Biegung. Diss. TH Karlsruhe 1965

- Schneidersmann, E. O., Kraft, G. und Domita, E.: Festigkeitsverhalten von Seilendverbindungen. Stahl und Eisen 100 (1980) 14, S770–775
- Schönherr Schönherr, S.: Einfluss der seitlichen Seilablenkung auf die Lebensdauer von Drahtseilen beim Lauf über Seilscheiben. Diss. Universität Stuttgart 2005
- Setzer, M.: Feststellung der an die Dauerfestigkeit von Drähten, Litzen und Seilen für Fördereinrichtungen zu stellenden Anforderungen. Forschungsbericht der Seilprüfstelle Bochum, Nov. 1976
- Siebert, D. and Brevet, P.: Fatigue of stay cables inside end fittings: high frequencies of wind induced vibrations. OIPEEC Bull. 89, Reading 2005
- Siebert, D., Brevet, P. and Royer, J.: Failure mechanisms in spiral strands under cyclic flexural loading close to terminations. OIPEEC Round Table, Reading Sept. 1997, pp. 111–119
- Sonsino, C. M.: Dauerfestigkeit - Eine Fiktion. Konstruktion (2005) 4, S87–92
- Stange, K.: Angewandte Statistik. 2. Teil, Mehrdimensionale Probleme. Springer. Berlin Heidelberg New York, 1971
- Suh, J.-I. and Chang, S. P.: Experimental study on fatigue behavior of wire ropes. Int. J. Fatigue 22 (2000), 339–347
- Ulrich, E.: Untersuchungen über die Tragfähigkeit von Seileinbänden und Seilendverbindungen mit Doppelbackenklemmen. Diss. Aachen, 1973
- Unterberg, H.-W.: Die Dauerfestigkeit von Seildrähten bei Biegung und Zug. Diss. TH Karlsruhe 1967
- Unterberg, H.-W.: Das Verdrillen der Seilstränge bei Kranen mit großen Hakenwegen. Fördern und Heben 29 (1972) 2, S90–92
- Utting, W. S. and Jones N.: The response of wire rope strands to axial tensile loads. Int. J. Mech. Sci. 29 (1987) 9, pp. 605–636
- Verreet, R.: Steel wire ropes with variable lay lengths for mining application. OIPEEC Bull. 81, 63–70. University of Reading, June 2001. ISSN 1018–8819
- Vogel, W.: Prüf-, Überwachungs- und Zertifizierungsstelle IFT... 2nd International Stuttgart Rope Day, Institut für Fördertechnik und Logistik, Universität Stuttgart 17. and 18. Feb. 2005
- Wang, N.: Spannungen in einem geraden Rundlitzenseil. Studienarbeit. Inst. Fördertechnik, Universität Stuttgart, 1989
- Wehking, K.-H., Vogel, W. and Schulz, R.: Dämpfungsverhalten von Drahtseilen. Fördern und Heben 49 (1999) 1/2, S60–61
- Wehking, K.-H. and Klöpfer, A.: Lebensdauer und Ablegereife von Drahtseilen unter Zugschwellbeanspruchung. Abschlussbericht d. Forschungsprojekts AVIF und DRAHT 51 (2000) 2, S138–144
- Wehking, K.-H.: Zukunftsausrichtung des IFT im Bereich der Seiltechnik. 1. Internationaler Stuttgarter Seiltag. 21. Februar 2002, S1–14.
- Wehking, K.-H. and Ziegler, S.: Berechnung eines einfachen Seils mit Hilfe der Finite-Element-Methode. Fördern u. Heben 53 (2003) 12, S753–754 und 54 (2004) 1/2. S58–60
- Wiek, L.: Strain gauge measurements at multistrand non spinning ropes. OIPEEC Bull. 37, 30–53, Torino 1980

- Wiek, L.: Stress deviations in steel wire ropes. OIPEEC Round Table, Cracow 1981
- Wiek, L.: Experiments with shock loads on steel wire ropes. OIPEEC Bull. 70, 75–91, Reading 1995
- Woernle, R.: Ein Beitrag zur Klärung der Drahtseilfrage. Z. VDI 72 (1929) 4, S9–17
- Wolf, R.: Zur Beschreibung der vollständigen Seilkinetik. Forsch. Ing.-Wes. 50 (1984) 3, S81–86
- Wyss, Th.: Stahldrahtseile der Transport- und Förderanlagen. Schweizer Druck- und Verlagshaus AG Zürich 1957
- Yeung, Y. T. and Walton, J. M.: Accelerated block tension fatigue testing of wire ropes for offshore use. OIPEEC Round Table 1985, East Kilbride Glasgow, Scotland
- Zweifel, O.: Zugkraftmessung in Drahtseilen mit Transversalwellen. Schweizerische Bauzeitung 79 (1961) 21

Wire Ropes Under Bending and Tensile Stresses

3.1 Stresses in Running Wire Ropes

3.1.1 Bending and Torsion Stress

Global Stresses

The global tensile stress in wire ropes is the so called rope tensile stress σ_Z as a quotient between the tensile force S and the nominal metallic cross-section A of the rope

$$\sigma_Z = \frac{S}{A}. \quad (3.1a)$$

Another global form describing the tensile stress is the specific tensile force S/d^2 as a quotient between the tensile force S and the square nominal rope diameter d .

The corresponding global bending stress of a wire in a bent wire rope is the bending stress from Reuleaux (1861)

$$\sigma_b = \frac{\delta}{D} \cdot E. \quad (3.1b)$$

In this equation, δ is the wire diameter, D is the middle curvature diameter (diameter of the rope axis bent over a sheave), and E is the elasticity module (of material).

With (3.1b), the bending stress is calculated as if the wire in the rope did not have a helix form. For a long time, there was a dispute about whether this equation was able to supply a result which was more or less true. The main contributions here came from the famous Bach (1881) with his correcting factor $3/8$ and in opposition from Benoit (1915). Now it is clear that the real bending stress of wires in a rope is both partly smaller and greater than – and well represented by – the global bending stress according to Reuleaux.

Bending of a Strand

Important early contributions to our knowledge about bending and torsion stresses in rope wires came from Bock (1909) and Woernle (1913). Paetzel (1969) and Wiek (1973) were the first to establish that the stresses have to be calculated for the difference in the space curve of the wires before and after the rope is bent. For the space curve of the wire in the bent strand, they presented equations with a constant ratio between the winding angle ϑ around the sheave axis and the wire winding angle φ_W around the strand axis.

Without this restriction, the equations are

$$\begin{aligned} x &= -r_W \cdot \sin \varphi_W \\ y &= \frac{D}{2} \cdot \cos(\vartheta - \vartheta_0) + r_W \cdot \cos \varphi_W \cdot \cos(\vartheta - \vartheta_0) \\ z &= \frac{D}{2} \cdot \sin(\vartheta - \vartheta_0) + r_W \cdot \cos \varphi_W \cdot \sin(\vartheta - \vartheta_0). \end{aligned} \quad (3.2)$$

The meaning of the symbols is to be found in Fig. 3.1. The minus in the first equation (3.2) is valid for a right winding wire in a bent strand.

For (3.2), a supplementary relation between the winding angles ϑ and φ_W has to be made. Wiek (1973) and Leider (1977) presupposed a constant ratio for both of the winding angles

$$k = \frac{\vartheta}{\varphi_W} = \frac{h_W}{\pi \cdot D} = \frac{2 \cdot r_W}{D \cdot \tan \alpha}. \quad (3.3)$$

This means that the lay angle α in the bent strand is not constant. According to Czitary (1951), the lay angle is expressed by

$$\tan \alpha = \tan \alpha_0 \cdot \frac{1}{1 + \frac{2 \cdot r_W}{D} \cdot \cos \varphi_W}$$

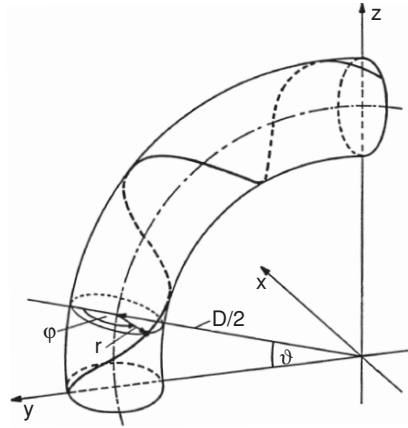


Fig. 3.1. Wire space curve in bent strand, Schiffner (1986)

with α_0 for the lay angle in the straight strand, r_W for the wire winding radius and D for the diameter of the axis in the bent strand.

Schiffner (1986) found it useful to consider the bending of the strand with the supposition of a constant lay angle $\alpha = \text{const}$. In this case the length dl of a wire element in relation to the length dL of the strand element in the bent rope is

$$dl = \frac{dL}{\cos \alpha}. \quad (3.3a)$$

The length dL is

$$dL = \left(\frac{D}{2} + r_W \cdot \cos \varphi_W \right) d\vartheta \quad (3.3b)$$

with the radius $D/2$ of the strand axis and the winding angle ϑ around the sheave axis and the winding radius r_W and the winding angle φ_W around the strand axis (wire helix in the strand). The length dL is therefore

$$dL = \frac{r_W \cdot d\varphi}{\tan \alpha}.$$

According to (3.3b), the winding angle $d\vartheta$ around the sheave axis is

$$d\vartheta = \frac{1}{\tan \alpha \cdot \left(\frac{D}{2 \cdot r_W} + \cos \varphi_W \right)} \cdot d\varphi. \quad (3.3c)$$

By integration, Schiffner (1986) calculated the winding angle ϑ

$$\vartheta = \frac{2}{\tan \alpha \cdot \sqrt{\frac{D^2}{4 \cdot r_W^2} - 1}} \cdot \arctan \frac{\left(\frac{D}{2 \cdot r_W} - 1 \right) \cdot \tan \frac{\varphi_W}{2}}{\sqrt{\frac{D^2}{4 \cdot r_W^2} - 1}}. \quad (3.4)$$

For a constant ratio of the winding angles, the space curve of the wires must then be calculated using (3.2) and (3.3) and for a constant lay angle (3.2) and (3.4) are to be used. The bending stress and the torsion stress can be calculated for the difference in the space curve of the wires before and after the rope is bent by using (2.33), (2.34), and (2.41)–(2.43) given in Sect. 2.1.

Figure 3.2, shows the difference between the bending stresses found in a wire for the bent and the straight strand which some authors have calculated. In the case of a constant ratio between the winding angles, the correct bending stress comes from Leider (1977) and in the case of a constant lay angle, the correct bending stress comes from Schiffner (1986). Costello (1977) describes the bending stress, when the wire helix is not supported by a centre.

Schiffner (1986) shows that the space curve with a constant lay angle has the smallest distance between two points over a bent cylinder, the wire will not be stressed by torsion and the bending stress is smaller than in the case of a constant ratio of the winding angles. However, in contrast to wires with a constant ratio of the winding angles and compared with the straight helix, the bent wire helix with a constant lay angle needs additional lateral space.

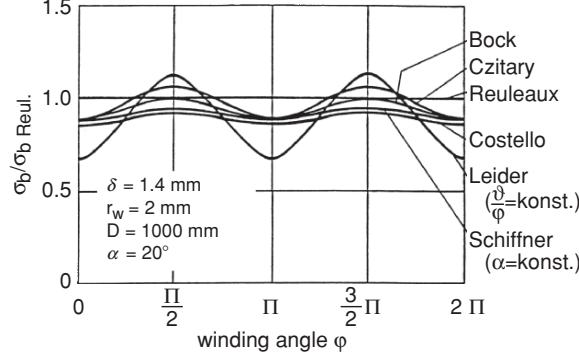


Fig. 3.2. Wire bending stress in a strand between straight and bent, Schiffner (1986)

Therefore a constant ratio of the winding angles occurs in modern strands where there is not much clearance between the neighbouring wires.

Bending of a Stranded Rope

Schiffner (1986), whose work is related to that of Paetzel (1969) and Wiek (1975) – and later on independently Hobbs and Nabijou (1995) – was the first to present the equations for the space curve of the wires in the bent stranded rope. He has also shown that it is not possible for the ratio of the winding angles around the strand axis and the rope axis and, on the other hand, the ratio of the winding angles around the rope axis and the sheave axis to be constant at the same time. He found that it can be either

$$\frac{\Delta\vartheta}{\Delta\varphi_S} = \text{const.} \quad \text{and} \quad \frac{\Delta\varphi_S}{\Delta\varphi_W} \neq \text{const.}$$

or

$$\frac{\Delta\vartheta}{\Delta\varphi_S} \neq \text{const.} \quad \text{and} \quad \frac{\Delta\varphi_S}{\Delta\varphi_W} = \text{const.}$$

With the constant ratio of the winding angles ϑ/φ_S the space curve of a wire in a bent stranded rope is according to Schiffner (1986)

$$\begin{aligned} x &= -r_S \cdot \sin \varphi_S - r_W \cdot [\cos(\varphi_W - \varphi_S) \cdot \sin \varphi_S + \sin(\varphi_W - \varphi_S) \cdot \cos \varphi_S \cdot \cos \beta] \\ y &= \cos\left(\frac{h_S}{\pi \cdot D} \cdot \varphi_S - \vartheta_0\right) \cdot \left(\frac{D}{2} + r_S \cdot \cos \varphi_S + r_W \cdot \begin{bmatrix} \cos(\varphi_W - \varphi_S) \cdot \cos \varphi_S \\ -\sin(\varphi_W - \varphi_S) \cdot \sin \varphi_S \cdot \cos \beta \end{bmatrix}\right) \\ &\quad + \sin\left(\frac{h_S}{\pi \cdot D} \cdot \varphi_S - \vartheta_0\right) \cdot r_W \cdot \sin(\varphi_W - \varphi_S) \cdot \sin \beta \\ z &= \sin\left(\frac{h_S}{\pi \cdot D} \cdot \varphi_S - \vartheta_0\right) \cdot \left(\frac{D}{2} + r_S \cdot \cos \varphi_S + r_W \cdot \begin{bmatrix} \cos(\varphi_W - \varphi_S) \cdot \cos \varphi_S \\ -\sin(\varphi_W - \varphi_S) \cdot \sin \varphi_S \cdot \cos \beta \end{bmatrix}\right) \\ &\quad - \cos\left(\frac{h_S}{\pi \cdot D} \cdot \varphi_S - \vartheta_0\right) \cdot r_W \cdot \sin(\varphi_W - \varphi_S) \cdot \sin \beta. \end{aligned} \quad (3.5)$$

The winding angle of the wire in the strand is

$$\varphi_W = \varphi_{W0} + \frac{h_S}{h_W \cdot \cos \beta_0} \cdot \left(\varphi_S + \sin \varphi_S \cdot \cos^2 \beta_0 \cdot \frac{2 \cdot r_S}{D} \right) \quad (3.6)$$

with the lay angle

$$\beta = \arctan \left(\frac{\tan \beta_0}{1 + \frac{2 \cdot r_S}{D} \cdot \cos \varphi_S} \right). \quad (3.7)$$

And with the constant strand lay angle the space curve of a wire in a bent stranded rope is according to Schiffner (1986)

$$\begin{aligned} x &= -r_S \cdot \sin \varphi_S - r_W \cdot [\cos(\varphi_W - \varphi_S) \cdot \sin \varphi_S + \sin(\varphi_W - \varphi_S) \cdot \cos \varphi_S \cdot \cos \beta] \\ y &= \cos(\vartheta - \vartheta_0) \cdot \left(\frac{D}{2} + r_S \cdot \cos \varphi_S + r_W \cdot \begin{bmatrix} \cos(\varphi_W - \varphi_S) \cdot \cos \varphi_S \\ -\sin(\varphi_W - \varphi_S) \cdot \sin \varphi_S \cdot \cos \beta \end{bmatrix} \right) \\ &\quad + \sin(\vartheta - \vartheta_0) \cdot r_W \cdot \sin(\varphi_W - \varphi_S) \cdot \sin \beta \\ z &= \sin(\vartheta - \vartheta_0) \cdot \left(\frac{D}{2} + r_S \cdot \cos \varphi_S + r_W \cdot \begin{bmatrix} \cos(\varphi_W - \varphi_S) \cdot \cos \varphi_S \\ -\sin(\varphi_W - \varphi_S) \cdot \sin \varphi_S \cdot \cos \beta \end{bmatrix} \right) \\ &\quad - \cos(\vartheta - \vartheta_0) \cdot r_W \cdot \sin(\varphi_W - \varphi_S) \cdot \sin \beta. \end{aligned} \quad (3.8)$$

with

$$\vartheta = \frac{2}{\tan \beta_0 \cdot \sqrt{\frac{D^2}{4 \cdot r_S^2} - 1}} \cdot \arctan \frac{\left(\frac{D}{2 \cdot r_S} - 1 \right) \cdot \tan \frac{\varphi_S}{2}}{\sqrt{\frac{D^2}{4 \cdot r_S^2} - 1}} \quad (3.9)$$

and

$$\varphi_W = \varphi_{W0} + \frac{h_S}{h_W \cdot \cos \beta_0} \cdot \varphi_S. \quad (3.10)$$

For calculating either the curvature, the winding, the bending stress and the torsion stress or the change in these values when the rope is bent, (2.33), (2.34), and (2.41)–(2.43) from Chap. 2 can be used. Using the results of these calculations, the bending stresses of the wires in different positions are drawn in Fig. 3.3 for an ordinary lay rope and in Fig. 3.4 for a lang lay rope. Figure 3.5 shows a very small torsion stress for an ordinary lay rope. The upper number always shows the stress for a constant ratio of the winding angles ϑ/φ_S and the lower number shows the stress for a constant lay angle β .

In the example chosen, the global bending stress according to Reuleaux is 500 N/mm². If the bending stress is calculated more precisely, it is both larger and smaller than that in the various positions. In any case, the maximum bending stress occurs laterally in the neighbourhood of the core. For a constant strand lay angle, this bending stress is just the same as the bending stress according to Reuleaux. For the constant ratio of the winding angles ϑ/φ_S , this bending stress is 23% greater for ordinary lay ropes and 18% greater for lang lay ropes. At the bottom of the groove, the bending stress of the wires

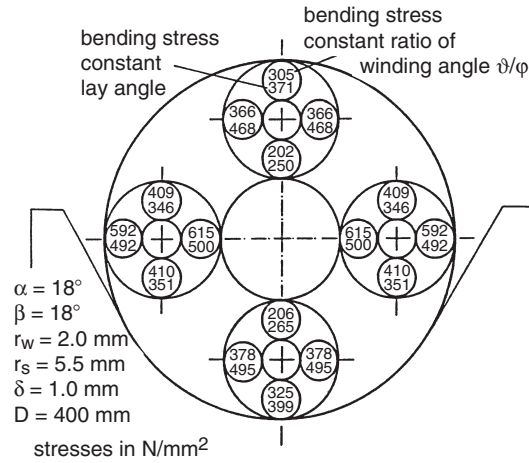


Fig. 3.3. Bending stresses in the wires of an ordinary lay rope between straight and bent, Schiffner (1986)

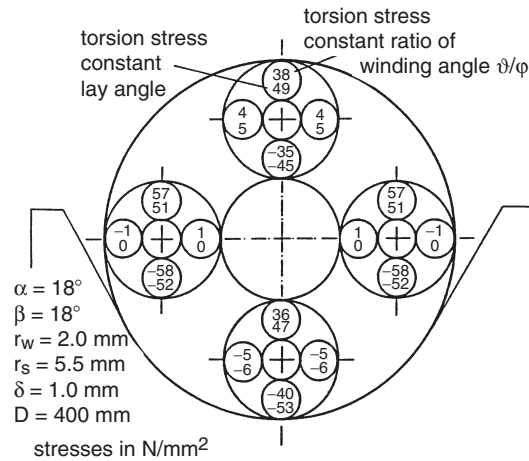


Fig. 3.4. Bending stresses in the wires of a lang lay rope between straight and bent, Schiffner (1986)

is relatively small. For ordinary lay ropes it is 325 resp. 399 N/mm^2 and for lang lay ropes 199 resp. 255 N/mm^2 .

The highest fluctuating bending stress (between straight and bent) occurs in wires inside the rope. It is here, therefore, that the first wire breaks have to be expected if all the other wire stresses are small. This is the case for wire ropes running over sheaves with soft grooves (small elasticity module) which keep the pressure on the wires down. With grooves made of steel, cast iron or other hard material (high elasticity module), the pressure on the wires together with the bending stress can be great enough to produce wire breaks,

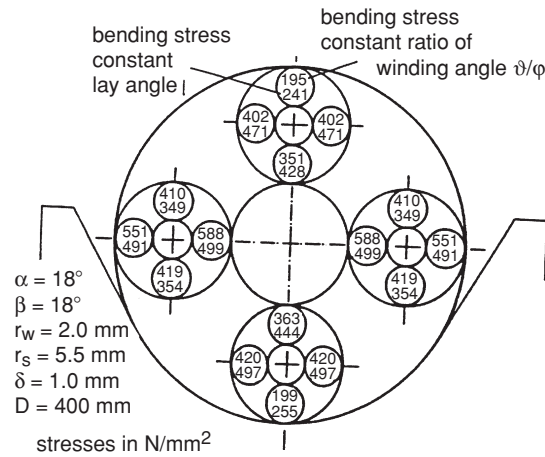


Fig. 3.5. Torsion stresses in the wires of an ordinary lay rope between straight and bent, Schiffner (1986)

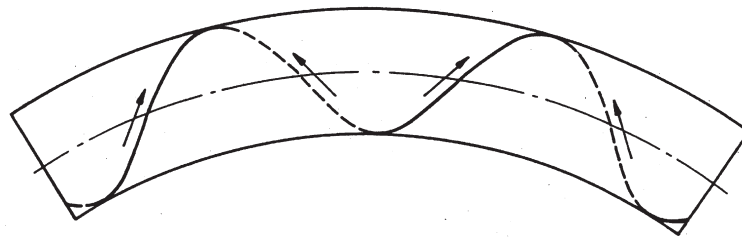


Fig. 3.6. Wire displacement in a uniformly bent strand

first of all in the contact zone with the groove. This is normally the case for ordinary lay ropes but mostly not for lang lay ropes which only have a very low bending stress at this point. This is the reason why – except for special cases – visible wire breaks are only reliable as a wire rope discarding criterion for ordinary lay wire ropes running over sheaves with grooves made of steel or cast iron.

3.1.2 Secondary Tensile Stress

Displacement of Wires and Strands

Strands and wire ropes can only be bent over sheaves because the wires in the strands and the strands in the rope are able to move against each other. When a strand is bent the wires generally move in the direction of the wire axis as shown in Fig. 3.6. For reasons of symmetry, it follows that the inner and the outer wire element of the strand bow lie unchanged in the same position before and after the strand is uniformly bent, Schmidt (1965). By changing the bending state uniformly, the wire elements will be only displaced in the

wire bow between both of these fixed points. The displacement of the wire elements are calculated using the position of the wire before and after the bending of the strand.

a) Constant ratio of the winding angles ϑ/φ

For constant ratio of the winding angles ϑ/φ in a strand (shortened symbols $\varphi = \varphi_W$ and $r = r_W$), (3.3) is valid

$$\frac{\vartheta}{\varphi} = \frac{r}{\frac{D}{2} \cdot \tan \alpha_0}. \quad (3.3)$$

Together with (3.3b) the circle bow length dL around the sheave centre as component of the wire element length dl in the bent strand is

$$dL = \left(\frac{D}{2} + r \cdot \cos \varphi \right) \cdot \frac{r}{\frac{D}{2} \cdot \tan \alpha_0} \cdot d\varphi.$$

The wire element length is

$$dl = \sqrt{dL^2 + (r \cdot d\varphi)^2}$$

or

$$dl = \sqrt{\left(\frac{D + 2 \cdot r \cdot \cos \varphi}{D} \cdot \frac{r}{\tan \alpha_0} \right)^2 + r^2} \cdot d\varphi.$$

The wire length l can be calculated by numerical integration. Then the wire displacement by bending the strand is

$$s = l_{\text{bent}}(\varphi) - l_{\text{straight}}(\varphi)$$

or

$$s = \int_0^\varphi \sqrt{\left(\frac{D + 2 \cdot r \cdot \cos \varphi}{D} \cdot \frac{r}{\tan \alpha_0} \right)^2 + r^2} \cdot d\varphi - \frac{r \cdot \varphi}{\sin \alpha_0}. \quad (3.3d)$$

The maximum wire displacement occurs for $\varphi = \pi/2$.

For the half lay length that means $\varphi = \pi$, the required length for the wire in the bent strand is theoretically a little greater than in the straight one. As, however, for reasons of symmetry no displacement is possible in this position, the wire should theoretically become a little elongated with a very small theoretical tensile stress

$$\sigma_{\text{th}} = \left(\frac{l_{\text{bent}}(\varphi = \pi)}{l_{\text{straight}}(\varphi = \pi)} - 1 \right) \cdot E.$$

For all the wires in the different layers of the strand, nearly the same small theoretical elongation can be calculated in comparison to the centre wire which is not elongated. In reality, the great number of layer wires enforces a small reduction of the centre wire length and a very small elongation of the layer wires as a small part of the theoretical elongation.

b) Constant lay angle α

When there is a constant lay angle α of the wire in the strand, there is no wire displacement in the direction of the wire axis but in the direction of the strand axis with the angle $\Delta\vartheta$ around the sheave centre. Because of the greater space required, a wire helix with constant lay angle can only occur if the clearance between the wires is relatively large. Equation (3.4) is once again valid for the angle ϑ around the sheave centre as a function of the winding angle φ of the wire in the strand

$$\vartheta = \frac{2}{\tan \alpha \sqrt{\left(\frac{D}{2 \cdot r}\right)^2 - 1}} \arctan \frac{\left(\frac{D}{2 \cdot r} - 1\right) \cdot \tan \frac{\varphi}{2}}{\sqrt{\left(\frac{D}{2 \cdot r}\right)^2 - 1}}. \quad (3.4)$$

Using (3.3a) and (3.3c), the length of the wire in the bent strand (lay angle $\alpha = \text{const.}$) is

$$dl = \frac{r}{\sin \alpha} \cdot d\varphi \quad \text{and} \quad l = \frac{r}{\sin \alpha} \varphi,$$

and it has the same length as a wire in a straight strand. For every wire winding angle φ , a wire element will be displaced over a winding angle $\Delta\vartheta$

$$\Delta\vartheta = \vartheta - \frac{\varphi}{\pi} \vartheta(\varphi = \pi). \quad (3.4a)$$

According to (3.4), the winding angle ϑ for $\varphi = 180^\circ$ is

$$\vartheta(\varphi = \pi) = \frac{\pi}{\tan \alpha \sqrt{\left(\frac{D}{2 \cdot r}\right)^2 - 1}}.$$

Using (3.3b) and $\Delta\vartheta$ from (3.4a), the length of the displacement bow is

$$\Delta L = \Delta\vartheta \cdot \left(\frac{D}{2} + r \cdot \cos \varphi \right).$$

The equations for the displacement of the wires in the strand are also valid for the displacement of the strands in the rope. For this, the lay angle α is to be replaced by β and $r = r_S$ and $\varphi = \varphi_S$.

The clearance between the wires is always small and the clearance between the strands is normally small or the strands are more or less fixed laterally by the core. Therefore, at most, the wire and the strand helix show a nearly constant ratio of the winding angles.

Example 3.1: Strand displacement by bending the rope

Data:

Diameter ratio of sheave and rope $D/d = 25$

Lay angle $\beta = 20^\circ$

Winding radius (8 strand rope with steel core) $r/d = 0.363$

Results:

For a rope with a constant ratio of the winding angles, the strand displacement in strand direction for $\varphi = 90^\circ$ is $s/d = 0.0273$

Theoretical ratio of strand length in bent or straight rope $l/l_0 = 1.0000218$
 Theoretical tensile stress in the strand $\sigma_{th} = 4.27 \text{ N/mm}^2$.
 For a rope with a constant lay angle of the strands, the
 displacement angle for $\varphi = 90^\circ$ is $\Delta\vartheta = 0.0023 \text{ rad}$ or $\Delta\vartheta = 0.133^\circ$
 Length of the displacement bow for $\varphi = 90^\circ$ is $\Delta L/d = 0.0290$

Simplified Calculation of the Secondary Tensile Force

When a wire rope is bent, a wire tensile stress will be caused by the friction between the wires and the strands. This stress is called the secondary tensile stress and it increases or reduces the normal tensile stress of the wires differently in different parts of the rope. The sum of all wire tensile forces in a rope cross-section remains unchanged and results in the wire rope tensile force. Like the bending stress and the ovalisation stress in the sheave groove, the secondary tensile force is a fluctuating stress which reduces the endurance of a wire rope.

Already Isaachsen (1907) presented a first equation to calculate the secondary tensile stress. Benoit (1915) and Ernst (1933) improved it. Schmidt (1965) evaluated a correct equation for the secondary stress in the wire of a uniformly bent strand. For reasons of symmetry, he noticed the wire can only be displaced between the inner and the outer point of the rope bow, as shown in Fig. 3.6. In the case considered with uniformly bent strands, according to Schmidt (1965) the secondary tensile stress in an outside wire is

$$\sigma_{ts} = \sigma_t \cdot \left(e^{\mu \cdot \sin \alpha \cdot (\varphi_0 - \varphi)} - 1 \right). \quad (3.11)$$

In this equation, σ_t is the normal tensile stress, μ is the friction coefficient, α is the wire lay angle and φ is the wire winding angle. φ_0 is the winding angle for that the secondary tensile force is zero. This angle is a little greater than $\pi/2$, Fig. 3.7.

On the inner layer wires in parallel lay strands, friction forces work in opposite directions, thus resulting in only a relatively small secondary tensile force, Schmidt (1965) and Leider (1974). In strands with crossing wire layers,

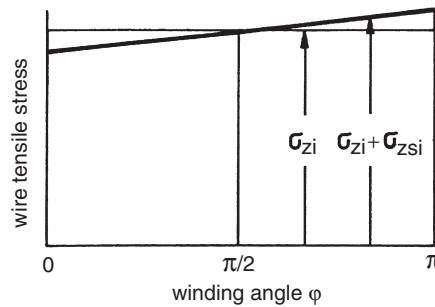


Fig. 3.7. Secondary tensile stress in a half wire winding, uniformly bent strand

however, the secondary tensile stress increases for the inner wires from layer to layer. In two wire layer strands, the secondary tensile stress of the inner layer wires is about three times and in three wire layer strands – with the same lay angle in an alternating direction and the same wire diameter – five times that found in the outer wires, Ernst (1933), Schmidt (1965) and Leider (1974). Cross lay ropes only have a relatively low bending endurance, both for this reason and because of the pressure in the crossing points.

Step-by-Step Calculation of the Secondary Tensile Stress

Equation (3.11) for the calculation of the secondary tensile stress presupposes that the rope will be uniformly bent over the whole of its length. In fact, however, the curvature of the rope changes gradually depending on the distance to the contact point of the sheave. Very early on, Donandt (1934) stated that the displacements of the strand elements can therefore occur over some lay lengths. Schmidt (1965) found that these displacements cannot be calculated using a single equation. Leider (1973, 1975) arrived at the same result. Therefore he calculated the secondary tensile stress using an approximation method which follows the rope elements that move step by step over the sheave. His method has the simplification that the rope running over the sheave remains straight right up to its contact point with the sheave and that the wires and strands in the rope piece lying on the sheave cannot be displaced.

Using a comparable method but without the two simplifications, Schiffner (1986) calculated the secondary tensile stresses in the wires of a stranded rope. The rope as a chain of rope elements runs step by step over the sheave whereby the step width meets the length of the rope element. The calculation starts with as realistic a bending line of the free rope as possible. The different tensile stresses in the rope's cross-section produce a bending moment. With this bending moment and the moment from the outer tensile rope force, a new bending line is calculated for the free rope. Using this new bending line, the calculation is repeated until the bending line calculated for the rope coincides with last bending line of the free rope.

In Fig. 3.8, the secondary tensile stress has been drawn for the wire rope moving over the sheave, together with the bending stress in an outer wire ($\varphi_S = 0$; $\varphi_W = 0$). The calculated stress and the stress course correspond to the one which had been measured by Schmidt (1965), Wiek (1973) and Mancini (1973).

With his method, Schiffner (1986) is not only able to calculate the secondary tensile stress but also the bending line of the free rope both before and after the rope is bent over a sheave. Figure 3.9 shows such bending lines of a rope before and after moving over a sheave with increased scale in the cross direction. As is to be seen in this figure, the wire rope is not only deformed in the sheave plane but also perpendicular to that. The deflection difference of the rope ends in the sheave plane for the rope running on or off is a criterion for the necessary bending work and for the bending stiffness caused by the friction, Schmidt (1965), Schraft (1997).

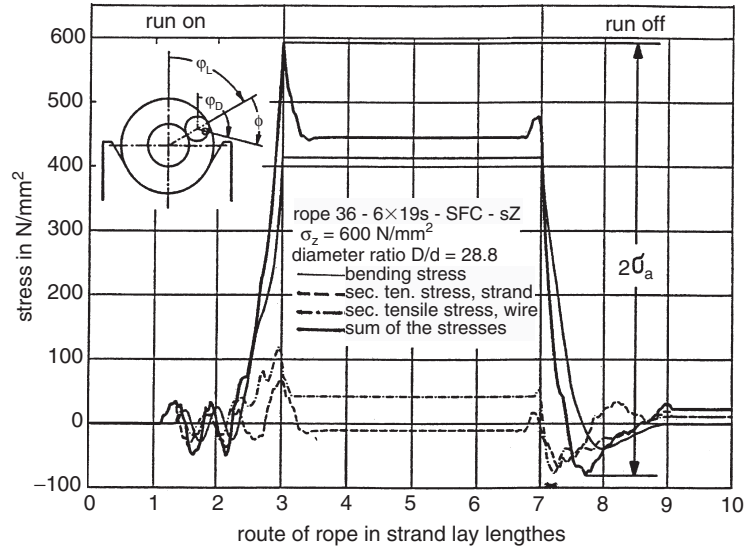


Fig. 3.8. Bending stress and secondary tensile stress in a wire, $\varphi_S = 0$ and $\varphi_W = 0$, Schiffner (1986)

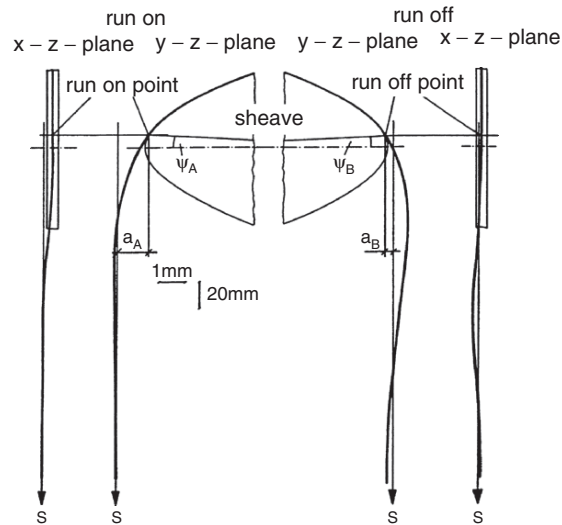


Fig. 3.9. Bending line of a wire rope moving on and off a sheave, Schiffner (1986)

One effect of the secondary tensile stress is, for example, that cross lay ropes 6×37 -FC with thin wires have lower endurance than wire ropes 6×19 -FC with thicker wires even though their bending stress is 40% higher, Woernle (1929). In fibre ropes running over sheaves there is practically no bending stress in the fibres. The endurance of these ropes mainly depends on the secondary tensile stress, Feyrer and Vogel (1992) and Wehking (1997).

3.1.3 Stresses from the Rope Ovalisation

The groove radius of a rope sheave is normally greater than the half rope diameter. When a wire rope loaded by a tensile force moves over the sheave, an ovalisation of the rope arises, Bechtloff (1969). On the bottom of the groove, at least part of the rope takes on the radius of the groove. Outside this contact bow, the cross-section is deformed in unknown way.

Schiffner (1986) calculated the bending and torsion stress due to rope ovalisation. For this, he substituted the round rope cross-section by an ellipse with the same area and then calculated the stresses arising from the curvature changes of the space curves before and after ovalisation. In contrast to the bending stress, the torsion stress arising from ovalisation is small enough to be neglected.

As an example the bending stress at the bottom of the groove can be found simply for the centre wire of a strand. Equation (2.37) from Chap. 2 can be used to calculate the bending stress

$$\sigma_{b,ov} = \left(\frac{\sin^2 \beta_{ov}}{r_{S,ov}} - \frac{\sin^2 \beta}{r_S} \right) \cdot \frac{\delta}{2} \cdot E.$$

The winding radius of the strand with a round form is

$$r_S = \frac{d}{2} - \frac{d_S}{2}.$$

In Fig. 3.10, with r for the groove radius (and supposing that the strand cross-section remains unchanged), the winding radius of the strand in an ovalised rope is

$$r_{S,ov} = r - \frac{d_S}{2}.$$

The lay angle of the strand is hardly changed at all by rope ovalisation and so it can be set as $\beta = \beta_{ov}$. The groove radius is normally $r = 0.53d$. If,

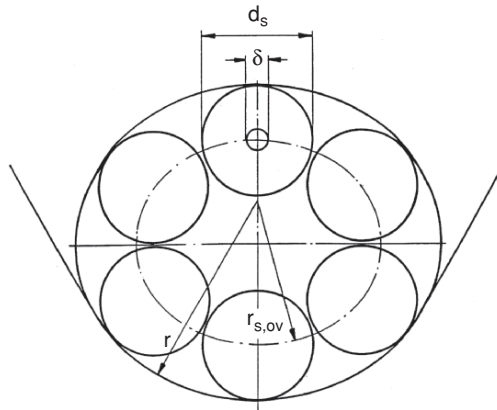


Fig. 3.10. Ovalised rope in a sheave groove

for example, the rope diameter is equal to the nominal rope diameter d , the centre wire diameter is $\delta = d/16$, the strand lay angle is $\beta = 18^\circ$, then the bending stress of the strand centre wire due to ovalisation is

$$\sigma_{b,ov} = 134 \text{ N mm}^{-2}.$$

Part of the rope ovalisation will be permanent and it will increase slowly with the number of times the rope runs over the sheave, Bechtloff (1969) and Dietz (1971). However, of all the bending stress arising due to ovalisation, a great deal will be fluctuating stress. Wire ropes with a fibre core are easily ovalised which means that they will have wide contact with the groove even if the groove has a large radius. The great influence exerted by the groove radius is demonstrated in Fig. 3.45 which shows test results from different authors.

3.1.4 Secondary Bending Stress

In cross lay strands, the outside wires are only supported by inner wires at single points. Due to this, the outside wires are already bent a little, reducing the small wire bow by the rope's tensile force. The bending stress resulting from the compressive forces from the groove is much greater, Fig. 3.11. This bending stress is called secondary bending stress.

The pressure from the compressive forces on the wires depend mainly on the tensile force of the wire rope, the sheave diameter, the groove radius and the form elasticity of the wire rope and the groove. The compressive force on a wire can be estimated using (3.37a). In most cases the secondary bending stress and the pressure will be reduced after the first loading by plastification of small regions of contact. This reduced secondary bending stress and pressure then works as fluctuating stresses in every bending cycle of the wire rope.

Müller (1966) found in bending tests that wire ropes with cross lay strands have only about one third of the endurance of those with parallel lay strands. The reason for this lies mainly in the secondary bending stress. In addition to the cross lay strands, the outside wires are loaded by secondary bending stress in compound strands. Nevertheless, ropes constructed with these strands can be used successfully in mining hoisting installations as, due to the great sheave

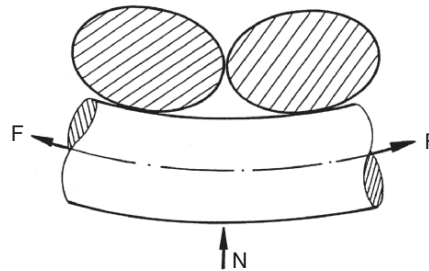


Fig. 3.11. Secondary bending of a wire

diameters and smooth groove material used here, the secondary bending stress is small.

The running rope normally used today is one with parallel lay strands as there are no wires crossing. Nevertheless, in these ropes there is still a form of secondary bending stress, i.e. the bending stress of the wires due to the bending of the strands in wire ropes with independent steel cores. In these ropes, the outer strands are supported by the inner strands at single points as shown for wires in Fig. 3.11. This bending stress arising from the strands bending also could be called tertiary bending stress.

Wolf (1987) found that the endurance of wire ropes increases with the number of supporting points (smaller distance between these points, Apel (1981)) for the strands of the independent steel wire core IWRC. This is beside the pressure due to the tertiary bending stress. Of course, this tertiary bending stress can be avoided by using fibre cores, steel cores with parallel-closed ropes PWRC or steel cores enveloped with solid polymer ESWRC. Running ropes with steel cores such as PWRC or ESWRC have a much higher endurance than wire ropes with independent wire rope cores IWRC.

3.1.5 Sum of the Stresses

The secondary and the tertiary bending stresses can be avoided with right rope construction. From the other stresses the tensile stress, the bending stress, the secondary tensile stress and the rope ovalisation stress sum up to a longitudinal stress in every fibre of the rope wires. Together with the pressure and the small torsion stress this total longitudinal stress has the main influence to the endurance of running wire ropes. It still has to be considered just what the maximum and minimum total longitudinal stress should be over the whole course.

In Fig. 3.12, Wiek (1973) has shown the typical course of the longitudinal stress in a wire of a rope running over a sheave. For the rope's endurance, the important stresses are the stress range

$$2 \cdot \sigma_a = \sigma_2 + \sigma_3 + \sigma_4 \quad (3.12)$$

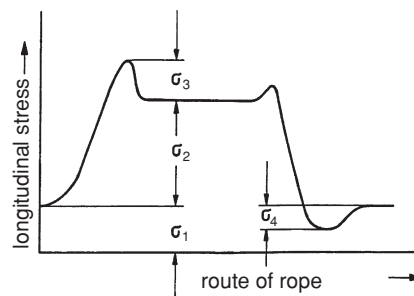
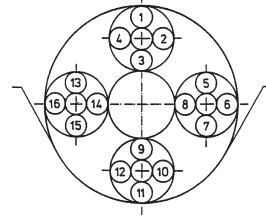


Fig. 3.12. Course of the longitudinal wire stress in rope running over a sheave, Wiek (1973)

Table 3.1. Stress range and middle stress in wires during rope running over a sheave, Schiffner (1986)

wire position	ordinary lay				lang lay			
	tensile side		compressive side		tensile side		compressive side	
	σ_m	$2 \sigma_a$	σ_m	$2 \sigma_a$	σ_m	$2 \sigma_a$	σ_m	$2 \sigma_a$
	N/mm ²		N/mm ²		N/mm ²		N/mm ²	
1	139	467	560	343	128	287	504	513
2, 4	7	452	648	352	137	416	541	572
3	164	316	528	237	101	475	574	540
5, 13	531	776	110	685	591	968	121	642
6, 16	563	812	30	797	599	810	22	781
7, 15	522	690	102	780	552	652	33	923
8, 14	604	857	2	797	590	804	21	926
9	719	524	-115	557	937	798	-262	708
10, 12	1025	726	-446	803	915	914	-327	761
11	864	644	-305	754	779	871	-135	561

Parallel lay rope, six strand with fibre core; lubricated, $\sigma_z = 300 \text{ N/mm}^2$; $D/d = 28.8$; $r = 0.53d$; $\delta = 0.076d$; $d_s = 0.311d$; $h_s = 6.19d$; $h_w = 2.63d$

and the middle stress

$$\sigma_m = \sigma_1 + \frac{\sigma_2 + \sigma_3 + \sigma_4}{2}. \quad (3.13)$$

Schiffner (1986) calculated the range stress and the middle stress on the tensile and the compressive side for wires in different positions. These stresses are listed in Table 3.1 for ordinary and lang lay ropes. The positions of the wires considered here are shown in the sketch accompanying Table 3.1. The calculation has been made for a third of the expected endurance. Where it was supposed that the friction coefficient has been increased up to $\mu = 0.15$.

Table 3.1 shows that the maximum stress range occurs in the strands on the side of the rope. This is true especially for tensile stress. For an ordinary rope, this maximum stress lies in positions 8 and 14 with contact to the core and for the lang lay rope in positions 5 and 13 between the neighbouring strands. The maximum stress range is greater for lang lay ropes than for ordinary ropes. Nevertheless lang lay ropes have higher endurance than ordinary ropes when running over sheaves. This is due to the stress range for ordinary ropes on the bottom of the groove. Combined with the pressure in the groove of steel or cast iron sheaves, this stress is the first cause of the end of the rope's life. For lang lay ropes, the range stress in this position is so small that the wires mostly break first in the inner rope due to high fluctuating stresses.

3.1.6 Force Between Rope and Sheave (Line Pressure)

Apart from all the stresses, wires in ropes are loaded by different pressures, both within the rope as well as between the rope and the sheave. The pressure from the force between the rope and the sheave groove varies along and across the sheave groove. With line pressure, these pressures are considered as concentrated in one line. The definition of line pressure (length related contact force) is contact force/length of contact bow.

Measurements taken by Wiek (1982b) and Molkow (1982) showed that the pressure is much higher at the points where the wire rope meets the sheave than for the remaining contact bow between. These pressure peaks at the contact points of the wire rope and the sheave are caused by the bending stiffness of the wire rope. These pressure peaks will therefore appear in all cases where a stiff tape or wire is stretched over a sheave.

Line Pressure Between Tape and Sheave

Tape force at the contact point.

First of all, the calculation of the pressure between a tape and a sheave will be presented to demonstrate the principle of what occurs as these calculations are much less complicated than doing the same calculations for a wire rope, Feyrer (1986a). Figure 3.13 shows a tape bent over a sheave and stretched by rope tensile force S . The tape has a constant bending stiffness EJ , with E for the elasticity module and J for the equatorial moment of inertia.

First, in Fig. 3.14, that part of the tape will be examined which runs on the sheave. The curvature of the tape increases from the point where the force S is effective to the point where the tape meets the sheave. At this point, the radius of tape curvature is the same as the radius of the sheave. As a simplification, it is supposed that the pressure is constant and does not deform either the tape or the sheave which means the tape and the sheave are

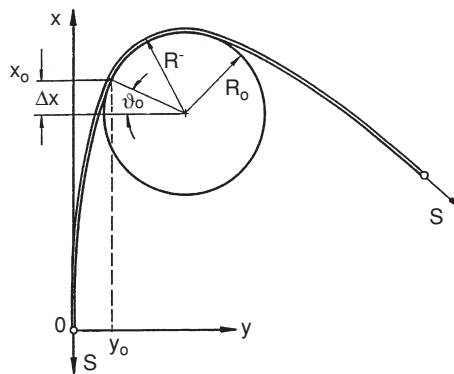


Fig. 3.13. Tape on sheave



Fig. 3.14. Forces on the tape piece running on the sheave

to be considered as being rigid in the crosswise direction. The force F at the boundary angle ϑ_0 is

$$F = S \cdot \cos \vartheta_0. \quad (3.14)$$

Without taking its direction into account, this tensile force F is constant over the contact bow, and a little less than the outer force S . A contact force Q exists between the tape and the sheave

$$Q = S \cdot \sin \vartheta_0. \quad (3.15)$$

According to the assumed simplification, the tape and sheave are rigid in a crosswise direction and the effects of the contact force Q are only to be found in a line parallel to the sheave axis.

Boundary angle ϑ_0 .

In order to calculate the bending curvature of the tape, the method will be used which Isaachsen (1907) already applied for calculating ropes in aerial rope ways. In Fig. 3.13, the bending moment M referred to the point x is,

$$M = S \cdot y. \quad (3.16)$$

From this, the differential equation is

$$\frac{d^2 y}{dx^2} \approx \frac{1}{\rho} = \frac{S \cdot y}{E \cdot J}, \quad (3.17)$$

where ρ stands for the radius of curvature. With the abbreviation

$$\omega = \sqrt{\frac{S}{E \cdot J}}$$

in (3.17), it is

$$y'' = y \cdot \omega^2. \quad (3.18)$$

The trial solution for this differential equation is

$$y = c_1 \cdot e^{\omega \cdot x} + c_2 \cdot e^{-\omega \cdot x} \quad (3.19)$$

with the deviations

$$\begin{aligned} y' &= \omega \cdot (c_1 \cdot e^{\omega \cdot x} - c_2 \cdot e^{-\omega \cdot x}) \\ y'' &= \omega^2 \cdot (c_1 \cdot e^{\omega \cdot x} + c_2 \cdot e^{-\omega \cdot x}). \end{aligned} \quad (3.20)$$

The boundary conditions are

$$\begin{aligned} \text{for } x = 0 \quad y &= 0 \\ \text{for } x = x_0 \quad y'' &= 2/D \end{aligned}$$

because at the point x_0 , the radius of curvature of the tape is equal to the radius $R = D/2$ of the sheave in the middle of the tape. From the boundary conditions we get the constants

$$c_1 = -c_2 = \frac{2}{D \cdot \omega^2 \cdot (e^{\omega \cdot x_0} - e^{-\omega \cdot x_0})}.$$

With these constants and (3.19) and (3.20), the equations of the bending line and its deviations are

$$\begin{aligned} y &= \frac{2}{D \cdot \omega^2} \cdot \frac{\sinh \omega x}{\sinh \omega x_0} \\ y' &= \frac{2}{D \cdot \omega} \cdot \frac{\cosh \omega x}{\sinh \omega x_0} \\ y'' &= \frac{2}{D} \cdot \frac{\sinh \omega x}{\sinh \omega x_0}. \end{aligned} \quad (3.21)$$

From these the bending stress of the tape with the thickness δ is

$$\sigma_b = E \cdot \frac{\delta}{2 \cdot \rho} \approx E \frac{\delta}{2} \cdot y'' = \frac{E \cdot \delta}{D} \cdot \frac{\sinh \omega x}{\sinh \omega x_0}. \quad (3.22)$$

With $x = x_0$ in (3.21), the lever-arm y_0 is

$$y_0 = \frac{2}{D \cdot \omega^2} = \frac{2 \cdot E \cdot J}{D \cdot S}. \quad (3.23)$$

The boundary angle ϑ_0 on the contact point can also be determined with (3.21)

$$\tan \vartheta_0 = y'(x_0) = \frac{2}{D \cdot \omega} \cdot \frac{1}{\tanh \omega x_0}.$$

For $\omega \cdot x_0 \geq 2.5$ is $\tanh \omega x_0 \approx 1$ with a failure smaller than 1%. This condition is fulfilled for all practical applications. With this the boundary angle is

$$\vartheta_0 = \arctan \frac{2}{D \cdot \omega} = \arctan \frac{2}{D \cdot \sqrt{\frac{S}{E \cdot J}}}. \quad (3.24)$$

The contact force Q is given with (3.14) and the boundary angle from (3.24). This angle is very small; so it is nearly $\tan \vartheta_0 \approx \sin \vartheta_0$ and the contact force is

$$Q = \frac{2}{D} \cdot \sqrt{S \cdot E \cdot J}.$$

The contact angle ϑ_c for the contact bow between tape and sheave is

$$\vartheta_c = \vartheta_D - 2 \cdot \vartheta_0$$

with ϑ_D for the tape deflection angle.

Line pressure between tape and sheave.

Presupposing that the tape between both of the contact points is bending limp in the contact bow, the line pressure q (length related contact force) between the tape and the sheave can be derived using the known tensile force F . In Fig. 3.15, the force F works on both cross-sections of the tape element. On the inner side, the force dQ exists. Figure 3.16 shows the diagram of these forces. Therefore the equation for these forces is

$$dQ = 2 \cdot F \cdot \sin \frac{d\vartheta}{2}$$

and for small angles with $\sin \vartheta = \vartheta$ it is

$$dQ = F \cdot d\vartheta. \quad (3.25)$$

On the side it is

$$dQ = q \cdot \frac{D_0}{2} \cdot d\vartheta. \quad (3.26)$$

Derived from (3.25) and (3.26), the global line pressure is then $S \approx F$ and $D_0 \approx D$

$$q_0 = \frac{2 \cdot S}{D}. \quad (3.27)$$

Equation (3.27) shows in a slightly simplified derivation that the line pressure is constant over the contact bow except for the two contact points which are loaded by the single force Q .

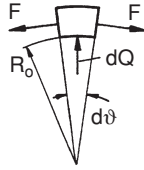


Fig. 3.15. Tape element from the contact bow

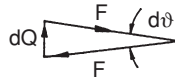


Fig. 3.16. Force plan of the tape element forces

Example 3.2: Tape stretched over sheave, Fig. 3.17**Data:**

Steel tape width	$b = 30 \text{ mm}$
Thickness	$\delta = 2 \text{ mm}$
Sheave diameter	$D_0 = 800 \text{ mm}$
Tensile force	$S = 10 \text{ kN}$

Results:

contact force	$Q = 500 \text{ N}$	(3.15)
boundary angle	$\vartheta_0 = 2.86^\circ$	(3.24)
tape tensile force	$F = 9,988 \text{ N}$	(3.14)
line pressure	$q = 25 \text{ N/mm}$	(3.27)
lever arm	$y_0 = 1.0 \text{ mm}$	(3.23)

Line Pressure Between a Wire Rope and a Sheave

When a wire rope is stretched over a sheave, the contact force Q known from the tape does not only work in a lateral line between rope and sheave. Because of the deformation of the rope, the force in the contact point is more effective in a small area. In any case, the pressure and the line pressure in this area are much higher than in the remaining contact bow. The amount of this pressure depends on the construction of the wire rope, as well as on the material and the shape of the groove. In a steel sheave with a normal groove, Wiek (1982a) found a pressure peak which was 50% higher than in the remaining bow.

Partly due to the measuring devices used, Wiek's measurements (1982a,b) in round grooves and Molkow's (1982) in V-grooves give in their own opinion only first rough results. Häberle (1995) made his measurements to evaluate

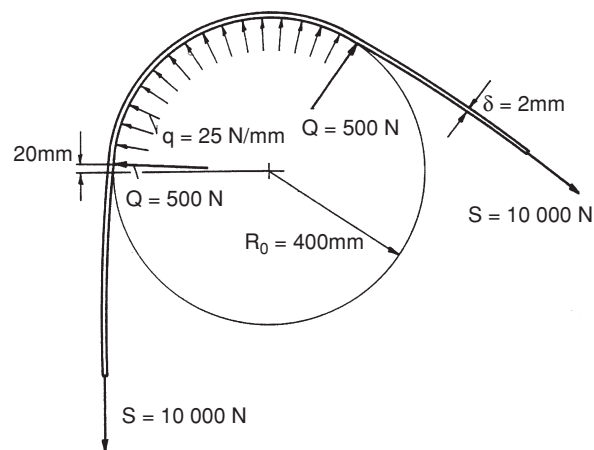


Fig. 3.17. Contact force Q and line pressure q , Example 3.2

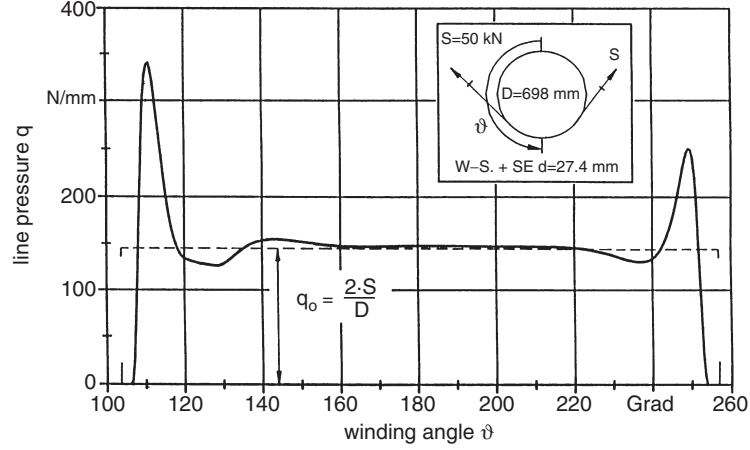


Fig. 3.18. Line pressure q between rope and sheave, Häberle (1995)

the pressure and the line pressure between a wire rope and a longer piece of sheave grooves consisting only out of force measuring elements. Due to these measuring arrangements, the resulting pressure and the line pressure evaluated here are reasonably precise.

Figure 3.18 from Häberle (1995) shows the line pressure measured while a wire rope is running over the sheave. A high peak can be observed in the line pressure at the point where the rope runs on and a lower peak where the rope runs off the sheave. In addition, the global line pressure

$$q_0 = \frac{2 \cdot S}{D} \quad (3.27)$$

is drawn as a dashed curve. The contact angle of the rope is about 3° smaller on both sides than the calculated angle for a limp-bending yarn (that is equal to the deflection angle ϑ_D).

The line pressure measured for a wire rope running over a sheave under different tensile forces is shown in Fig. 3.19. Häberle (1995) evaluated the ratio of the maximum and the global line pressure $q_{\max}/q_0$ from these and other measurements with different ropes. The results are shown in Fig. 3.20. In this figure, there is a list of the wire ropes used and the diameter ratio of sheave and rope was $20 < D/d < 70$. Häberle (1995) used regression calculation to find all the evaluated ratios $q_{\max}/q_0$ for the Warrington-Seale ropes 6×36 sZ with fibre or steel core, lubricated, which were used in the tests

$$\lg \frac{q_{\max}}{q_0} = 1.887 - 0.607 \cdot \lg \frac{S}{d^2} - 0.939 \cdot \lg \frac{D}{d} + 0.316 \cdot \lg \frac{S}{d^2} \cdot \lg \frac{D}{d}. \quad (3.28)$$

The specific tensile force S/d^2 is to be taken in N/mm^2 . The standard deviation is $\lg s = 0.050$.

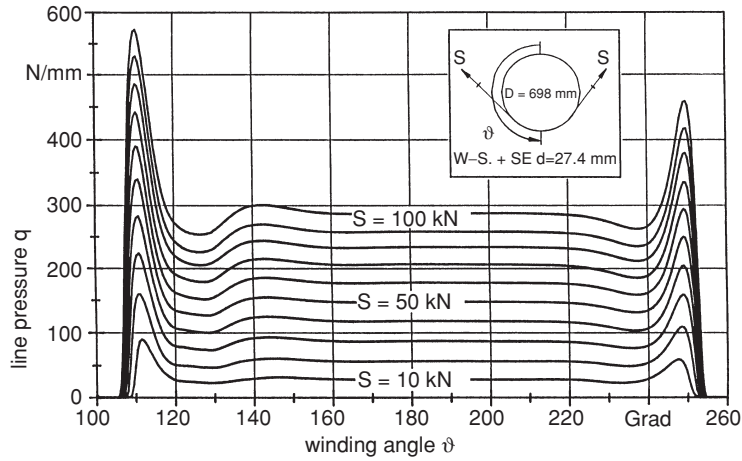


Fig. 3.19. Line pressure q between rope and sheave under different tensile force, Häberle (1995)

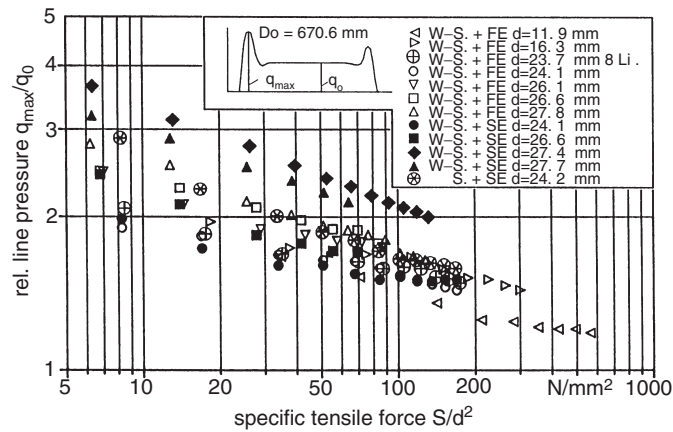


Fig. 3.20. Maximum relative line pressure $q_{\max}/q_0$, Häberle (1995)

Häberle (1995) also evaluated the difference of the sheave winding angle $\Delta\vartheta = \Delta\vartheta_1 + \Delta\vartheta_2$ between the run on and run off angle for limp-bending yarn and the angle of the line pressure peaks for wire ropes ($\Delta\vartheta_1$ and $\Delta\vartheta_2$ correspond with ϑ_0 in case of tape). In Fig. 3.21, the evaluated winding angle difference is only shown for specific tensile force $S/d^2 = 68.3 \text{ N/mm}^2$ (The results with specific tensile forces $S/d^2 = 30$ and 300 N/mm^2 as points have not been shown here). From the regression calculation for the Warrington-Seale ropes $6 \times 36 \text{ sZ}$ with fibre or steel core used for the tests, the winding angle difference is

$$\lg \Delta\vartheta = 2.870 - 0.383 \cdot \lg \frac{S}{d^2} - 1.073 \cdot \lg \frac{D}{d} + 0.171 \cdot \lg \frac{S}{d^2} \cdot \lg \frac{D}{d}. \quad (3.29)$$

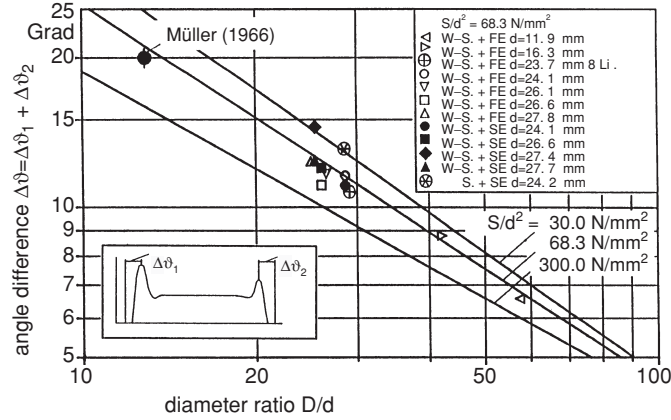


Fig. 3.21. Winding angle difference $\Delta\vartheta$, Häberle (1995)

The winding angle difference $\Delta\vartheta$ is given in degree and the specific tensile force has to be taken in N/mm². The standard deviation is $\lg s = 0.0337$.

The influence of the line pressure peak is shown very impressively by the number of bending cycles which a wire rope attains with different deflection angles. Müller (1961, 1966) carried out such endurance tests. As one of his test results the number of bending cycles of a wire rope as a function of the deflection angle is given in Fig. 3.51, Sect. 3.2.3. The smallest number of bending cycles from this figure exists for the deflection angle $\vartheta_{D,dip} = 20^\circ$.

At this angle, the pressure peaks of the running on and the running off sides of the wire rope work together. Furthermore, in this range the free bending radius of the wire rope bow is the same as that of the sheave radius which means that all the rope bending stresses work together with the pressure to their full extent. This deflection angle $\vartheta_{D,dip} = 20^\circ$ from Müller's diagram, which is given for a specific tensile force $S/d^2 = 62.5 \text{ N/mm}^2$, is drawn in Häberle's Fig. 3.21 and conforms to a great extent with the angle difference $\Delta\vartheta$ for the specific tensile force $S/d^2 = 68.3 \text{ N/mm}^2$. Admittedly, however, the construction of the wire ropes used in Fig. 3.21 and 3.51 is not identical.

Example 3.3: Rope line pressure

Data:

Warrington-Seale rope $6 \times 36 \text{ sZ}$
 Rope diameter $d = 16 \text{ mm}$
 Ratio $r/d_m = 0.54$
 Sheave diameter $D = 400 \text{ mm}$
 Rope tensile force $S = 30 \text{ kN}$

Results:

Global line pressure, (3.27): $q_0 = \frac{2 \cdot 30,000}{400} = 150 \text{ N/mm}$.
 Maximum line pressure, (3.28)

$$\lg \frac{q_{\max}}{q_0} = 1.887 - 0.607 \cdot \lg 117 - 0.939 \cdot \lg 25 + 0.316 \cdot \lg 117 \cdot \lg 25 = 0.2324$$

$$\frac{q_{\max}}{q_0} = 1.71$$

$$q_{\max} = 1.71 \cdot 150 = 256 \text{ N/mm.}$$

Winding angle difference, (3.29)

$$\lg \Delta\vartheta = 2.870 - 0.383 \cdot \lg 117 - 1.073 \cdot \lg 25 + 0.171 \cdot \lg 117 \cdot \lg 25 = 1.0722$$

$$\Delta\vartheta = 11.8^\circ.$$

3.1.7 Pressure Between Rope and Sheave

The rope pressure is the imagined pressure between a rope which is entirely round without any surface structure due to strands and wires and the groove. The line pressure q is distributed lateral to the strands and the wires in contact with the groove. In dimensioning wire ropes for mining hoistings and for elevators simplified forms of rope pressure are used. The real pressure occurs between the most prominent points of the wires and the sheave groove.

Global Rope Pressure

The global rope pressure p_0 is defined as the pressure between a limp-bending tape with the rope diameter d as its width and a cylindrical sheave. The global rope pressure is

$$p_0 = \frac{q_0}{d} = \frac{2 \cdot S}{D \cdot d} \quad (3.27a)$$

with the global line pressure q_0 , the rope tensile force S , the rope diameter d and the sheave diameter D measured from rope centre to rope centre of a rope wound around the sheave. For sheaves with round grooves, the global rope pressure represents all the working pressures between rope and groove. If round grooves made of the same material are always used in combination with the same wire rope surface, the global pressure is a dimensioning criterion. This is the case for mining installations where the sheaves always have round grooves made of soft material (small elasticity module).

Specific Pressure

The specific pressure is a special form of rope pressure for use in elevators. For traction sheaves in elevators, in particular for those with undercut grooves, the specific pressure is used as a criterion for making sure that the rope and the traction sheave are sufficiently durable and for calculating the friction force for driving the car.

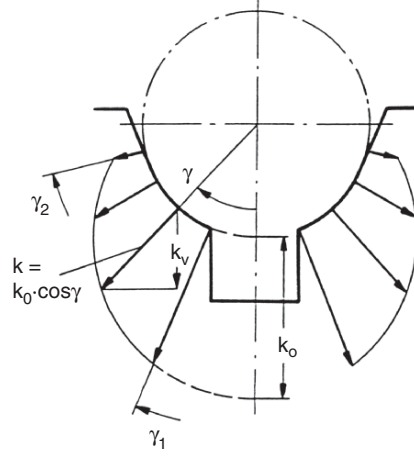


Fig. 3.22. Pressure between rope and undercut groove, $k = k_0 \cos \gamma$

Donandt (1927) and Hyman and Hellborn (1927) were the first to calculate the specific pressure. They supposed that the pressure in the groove has a cosinus-like distribution as shown in Fig. 3.22 for an undercut groove. With this supposition the specific pressure is

$$k = k_0 \cdot \cos \gamma. \quad (3.30)$$

The pressure k_0 is the pressure at the bottom of the groove (groove angle $\gamma = 0$) although this does not exist in undercut grooves. The groove angle is defined in Fig. 3.22. For traction sheaves where the specific pressure has to be considered, the maximum tensile force $S_{\max}$ has to be introduced as the maximum rope tensile force over the whole length of the groove. The part of the pressure in the direction to the sheave axis is

$$k_v = k \cdot \cos \gamma = k_0 \cdot \cos^2 \gamma.$$

The integral of this part of the pressure over the groove angle γ is equal to the line pressure

$$q = 2 \cdot \int_{\gamma_1}^{\gamma_2} k_v \cdot \frac{d}{2} \cdot d\gamma = k_0 \cdot d \cdot \int_{\gamma_1}^{\gamma_2} \cos^2 \gamma \cdot d\gamma$$

or

$$q = k_0 \cdot d \cdot \left(\frac{1}{4} \cdot \sin 2\gamma_2 + \frac{1}{2} \cdot \gamma_2 - \frac{1}{4} \cdot \sin 2\gamma_1 - \frac{1}{2} \cdot \gamma_1 \right). \quad (3.30a)$$

With this and $q = 2S_{\max}/D$ and according to (3.30) for the groove angle γ , the rope pressure is

$$k = \frac{\cos \gamma}{\left(\frac{1}{4} \cdot \sin 2\gamma_2 + \frac{1}{2} \cdot \gamma_2 - \frac{1}{4} \cdot \sin 2\gamma_1 - \frac{1}{2} \cdot \gamma_1 \right)} \cdot \frac{2 \cdot S_{\max}}{d \cdot D}.$$

The maximum specific pressure existing for the groove angle $\gamma = \gamma_1$ is

$$k_{\max} = \frac{\cos \gamma_1}{\left(\frac{1}{4} \cdot \sin 2\gamma_2 + \frac{1}{2} \cdot \gamma_2 - \frac{1}{4} \cdot \sin 2\gamma_1 - \frac{1}{2} \cdot \gamma_1 \right)} \cdot \frac{2 \cdot S_{\max}}{d \cdot D}. \quad (3.31)$$

The friction coefficient of the groove, traditionally written $f(\mu)$, is

$$f(\mu) = \frac{q'}{q} \cdot \mu. \quad (3.31a)$$

In this, the line pressure on both sides of the groove is

$$q' = 2 \cdot \int_{\gamma_1}^{\gamma_2} k \cdot \frac{d}{2} \cdot d\gamma = k_0 \cdot d \cdot \int_{\gamma_1}^{\gamma_2} \cos \gamma \cdot d\gamma$$

or

$$q' = k_0 \cdot d \cdot (\sin \gamma_2 - \sin \gamma_1). \quad (3.31b)$$

From (3.30a), (3.31a) and (3.31b), the friction coefficient for the undercut groove is

$$f(\mu) = \frac{4 \cdot (\sin \gamma_2 - \sin \gamma_1)}{\sin 2\gamma_2 - \sin 2\gamma_1 + 2 \cdot \gamma_2 - 2 \cdot \gamma_1} \cdot \mu. \quad (3.32)$$

For grooves which are not undercut, $\gamma_1 = 0$ and $\gamma_2 = 60^\circ$, the maximum specific pressure (at the bottom of the groove) is $k_{\max} = 1.351(2 \cdot S_{\max})/(dD)$ and the friction coefficient of the groove is $f(\mu) = 1.170\mu$.

Example 3.4: Undercut groove

Data:

Groove angles $\gamma_1 = 47.5^\circ \rightarrow 0.829 \text{ rad}$

$\gamma_2 = 80^\circ \rightarrow 1.396 \text{ rad}$

Results:

The maximum specific pressure according to (3.31) is

$$k_{\max} = \frac{\cos 0.829}{\left(\frac{1}{4} \cdot \sin(2 \cdot 1.396) + \frac{1}{2} \cdot 1.396 - \frac{1}{4} \cdot \sin(2 \cdot 0.829) - \frac{1}{2} \cdot 0.829 \right)} \cdot \frac{2 \cdot S_{\max}}{d \cdot D}.$$

$$k_{\max} = 5.627 \cdot \frac{2 \cdot S_{\max}}{d \cdot D}.$$

The friction coefficient of the undercut groove according to (3.32) is

$$f(\mu) = \frac{4 \cdot (\sin 1.396 - \sin 0.829)}{\sin(2 \cdot 1.396) - \sin(2 \cdot 0.829) + 2 \cdot 1.396 - 2 \cdot 0.829} \cdot \mu$$

$$f(\mu) = 2.062 \cdot \mu.$$

Rope Pressure from Measurements

As described before, Häberle (1995) measured the rope pressure in a test sheave with a longer piece of round groove that only consisted of force measuring elements. In addition to the line pressure, Häberle also measured the pressure over the groove angle γ for some wire ropes with these elements. In Fig. 3.23, the pressure relative to the global pressure k/p_0 over the groove angle γ is given for two wire ropes as an example. This figure shows the relative pressure k/p_0 in the peak of the line pressure $q_{\max}$ when the rope runs onto the sheave. The bottom of the groove is well-suited to the rope ($r/d = 0.504$) with the relative pressure being $k/p_0 = 2.3$ and for the rope's relatively small diameter ($r/d = 0.581$) the relative pressure is $k/p_0 = 4.7$. The maximum pressure in the groove is always a little displaced in the same direction as the lay direction of the rope, Fig. 3.23.

By using regression calculation, Häberle (1995) evaluated the pressures measured on the bottom of the groove for the fibre or steel core Warrington-Seale ropes 6×36 sZ which were used in the tests

$$k = \frac{2 \cdot S}{D \cdot d} \cdot \left(1 + \frac{17.4}{\left(\frac{S}{d^2} \right)^{0.229}} \cdot (1 - e^{-4.52 \cdot (r/d - 0.5)}) \right). \quad (3.33)$$

With the maximum line pressure $q_{\max}/q_0$ from (3.28) and the pressure k_0 from (3.33), the maximum pressure on the bottom of the groove (peak pressure when the rope runs onto the sheave) is

$$k_{\max} = k \cdot \frac{q_{\max}}{q_0}. \quad (3.33a)$$

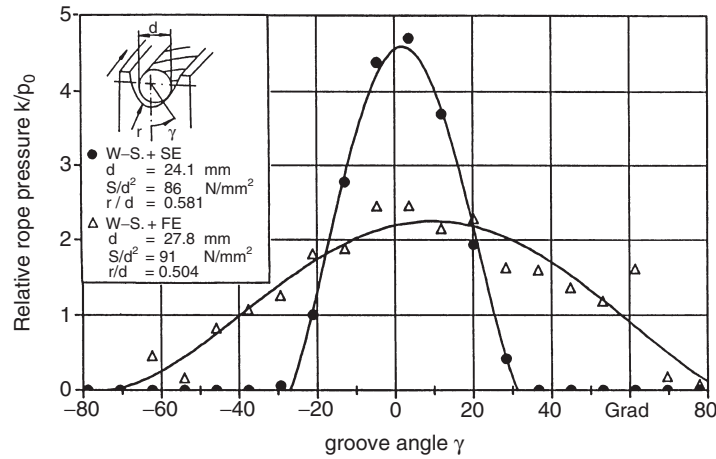


Fig. 3.23. Relative pressure k/p_0 for the line pressure $q_{\max}$, Häberle (1995)

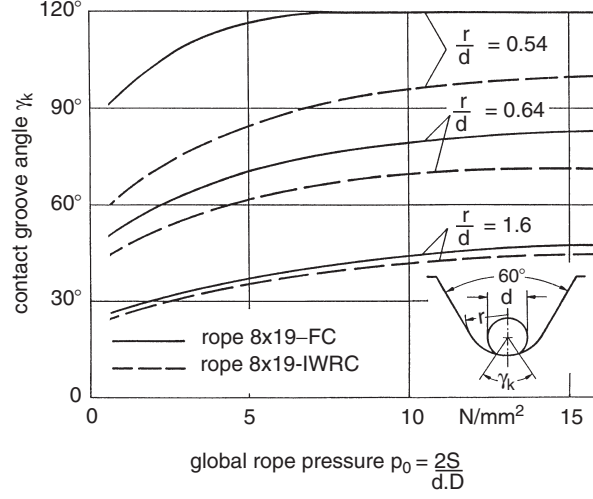


Fig. 3.24. Lateral contact angle γ_k for eight strand ropes in round grooves

The pressure on the bottom of the groove k and $k_{\max}$ increases and the lateral contact angle $\gamma_k = \gamma_1 + \gamma_2$ decreases – see Häberle (1995) – with the ratio of the groove radius and the measured rope diameter r/d_m .

The contact angle γ_k can be evaluated approximately by placing a thin paper under the rope while it is running over the sheave. Figure 3.24 shows the result of such an evaluation with two wire ropes and different ratio of groove radius and rope diameter. Of the two 8×19 ropes, one has a fibre core and the other a steel core. The contact angle for the rope with a fibre core is 5–20% greater than for the rope with a steel core.

As an example the relative rope pressure k/p_0 over the sheave groove (winding angle ϑ and groove angle γ) is drawn in Fig. 3.24a from Häberle.

3.1.8 Force on the Outer Arcs of the Rope Wires

The rope-pressures k and $k_{\max}$ are only comparable values but not really existing pressures. The real wire pressure (the material pressure) can be derived from the contact force between the outer arcs of the rope wires and the groove. In the following this force executed by the outer wire arcs of a rope will be called *wire arc force*. The maximum wire arc force on bottom of the groove can be calculated using the maximum rope pressure $k_{\max}$ from (3.33)

$$F_{Wcal} = k_{\max} \cdot A_f. \quad (3.34)$$

In this, the area A_f related to the arc force of one wire as a part of the rope surface A_{rope} in one lay length h_s is

$$A_f = \frac{A_{\text{rope}}}{z_K}. \quad (3.35)$$

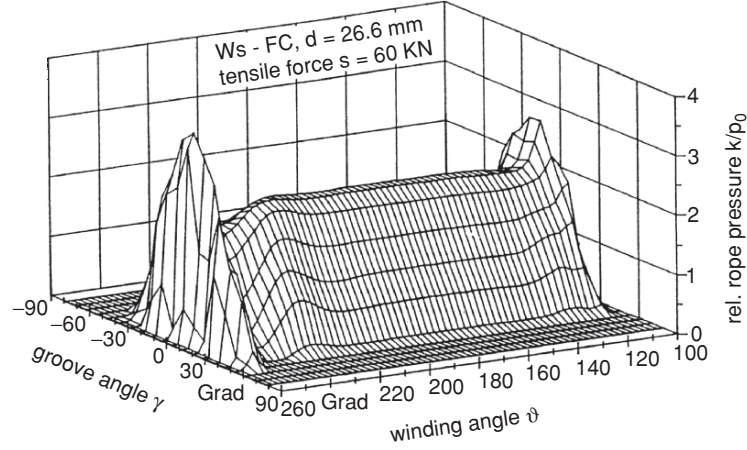


Fig. 3.24a. Relative pressure k/p_0 between wire rope and sheave, Häberle (1995)

According to Recknagel (1972), the number of wire arcs on the rope surface in one lay length h_S is

$$z_K = z_S \cdot z_W \cdot \left(\frac{h_S}{h_W \cdot \cos \beta} \pm 1 \right). \quad (3.36)$$

In this, z_S is the number and h_S the lay length of the outer strands of the rope, z_W is the number of outer wires and h_W the lay length of these outer wires, β is the lay angle of the outer strands. Then, using (3.34)–(3.36), the calculated maximum force on the wire arcs is

$$F_{Wcal} = k_{max} \cdot \frac{\pi \cdot d \cdot h_S}{z_S \cdot z_W \cdot \left(\frac{h_S}{h_W \cdot \cos \beta} \pm 1 \right)}. \quad (3.37)$$

This calculated maximum arc force F_{Wcal} is only true for a perfectly round rope. This is normally not the case. This means that in a real wire rope some of the arcs of the wires bear a very high force and others even do not come into contact with the groove at all. Between these two extremes, there are varying degrees of force existing for the arcs. Häberle (1995) measured the real forces for the arcs of the wires. Due to the dimensions of the force measuring elements, they normally measure the force of two wire arcs at a time. Based on his measurements, Häberle presented the ratio $(F_{W1} + F_{W2})/2F_{Wcal}$ as the ratio between the measured and the calculated forces for the wire arcs as shown in Fig. 3.25 and these results show a wide deviation that decreases with the specific tensile rope force S/d^2 , respectively, with the global rope pressure p_0 . Häberle considered the results as a normal distribution with the mean force ratio $(F_{W1} + F_{W2})/2F_{Wcal} = 1$ and with the standard deviation

$$s_2 = \left(\frac{0.614}{p_0} + 0.146 \right) \cdot \sqrt{2}$$

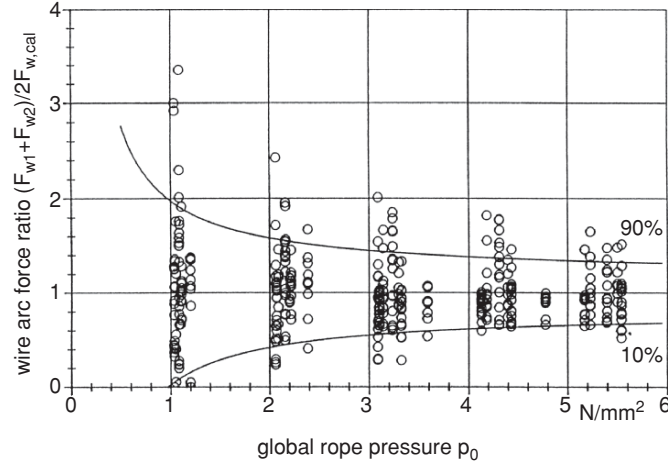


Fig. 3.25. Ratio wire arc forces $(F_{W1} + F_{W2})/2F_{Wcal}$ of Warrington-Seale ropes 6×36 sZ with fibre or steel core, Häberle (1995)

and according to the addition theorem of normal distribution for the arc force of a single wire F_W/F_{Wcal} , the standard deviation is

$$s = \frac{0.614}{p_0} + 0.146.$$

Then for Warrington-Seale ropes 6×36 sZ running on the sheave, 10% of the wire arc forces on the bottom of the groove is higher than

$$F_{W10} = \left[1 + u_{10} \cdot \left(\frac{0.614}{p_0} + 0.146 \right) \right] \cdot F_{Wcal} \quad (3.37a)$$

with forces F_{W10} and F_{Wcal} in N and the global pressure p_0 in N/mm^2 . Over the whole length of the rope, this high arc force F_{W10} is a frequent occurrence.

Wyss (1956) calculated the Hertz pressure between a wire arc and steel or cast iron groove. He found that the outer arcs of the rope wires and the groove material or both yield for relatively small forces. Häberle came to the same result. For example, he found that the yield stress will be exceeded for a global pressure smaller than $p_0 = 2 \text{ N/mm}^2$. Pantucek (1977) analysed the stresses in flattened wire arcs in relation to the different breaking forms of the wires in running ropes.

Example 3.5: Wire arc force

Data:

Data the same as in Example 3.3 and additional data:

Strand lay length $h_S = 6d$

Wire lay length $h_W = 3.1d$

Results:

Global pressure, (3.27a)

$$p_0 = \frac{2 \cdot 30,000}{16 \cdot 400} = 9.38 \text{ N/mm}^2.$$

Pressure at the bottom of the groove, (3.33)

$$k_0 = 150 \cdot \left(1 + \frac{17.4}{1170.229} (1 - e^{-4.52(0.54-0.5)}) \right) = 18.4 \text{ N/mm}^2.$$

Max. pressure at the bottom of the groove, (3.33a)

$$k_{\max} = 18.4 \cdot 1.71 = 31.5 \text{ N/mm}^2.$$

Max. calculated wire arc force, (3.37)

$$F_{\text{Wcal}} = 31.5 \frac{\pi \cdot 16 \cdot 6 \cdot 16}{6 \cdot 18 \cdot \left(\frac{6}{3.1 \cdot 0.94} + 1 \right)} = 501 \text{ N}.$$

The wire arc forces for 10% of the wires on the bottom of the groove is according to (3.37a) greater than

$$F_{\text{W10}} = \left[1 + 1,282 \cdot \left(\frac{0.614}{9.38} + 0.146 \right) \right] \cdot 501 = 637 \text{ N}.$$

3.2 Rope Bending Tests

When wire ropes run over sheaves, the wires are loaded by constant and fluctuating stresses and pressure, Sect. 3.1. However, it is not possible to derive the endurance of a particular wire rope even if all these stresses and the wire endurance under these stresses are known. This is the case partly because the relative motion causes the wires to become worn and also because of irregularities which result in neighbouring wires being loaded by very different tensile stresses. Therefore, wire ropes running over sheaves always have a finite life and the number of bending cycles can be only evaluated by wire rope bending fatigue tests.

3.2.1 Bending-Fatigue-Machines, Test Procedures

Test Principle

The predominant test principle used today for bending-fatigue machines is shown in Fig. 3.26. The wire rope to be tested is reeved in a loop over the traction sheave and the test sheave. The traction sheave that moves the rope has a much bigger diameter than the test sheave so that it is always the rope piece running over the test sheave that will break. Therefore the distance between

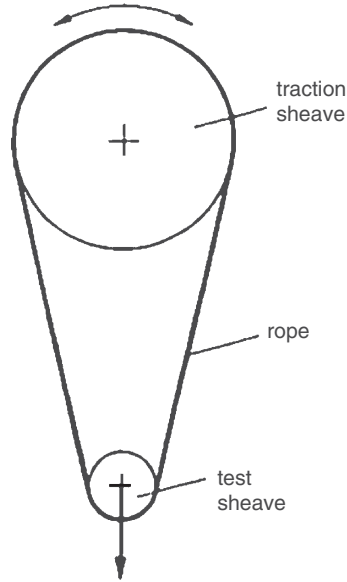


Fig. 3.26. Arrangement of sheaves for testing the number of simple bending cycles

the sheaves is larger than the rope stroke so that the rope test piece does not move over the traction sheave.

The rope can be bent with a simple bending test or (with several test sheaves) as reverse bending. In Fig. 3.27, both simple and reverse bending is defined and the symbols used are taken from the OIPEEC Recommendations, OIPEEC Bulletin 56 (1988). In addition to this, combined fluctuating tension and bending is introduced in Fig. 3.27. Combined fluctuating tension and bending means that the tensile force will be enlarged before the rope is bent and reduced again afterwards. In this case the longitudinally fluctuating wire stresses are composed of those of the rope bending and those of the changing tensile force. These combined fluctuating stresses are much higher than in the case of simple bending under a constant tensile force. Therefore this combined fluctuating bending and tensile force reduces the rope endurance enormously.

In Fig. 3.27 the number of bending cycles are defined by two different indices. The indices *sim*, *rev* and *com* are introduced for the case that the characterising indices are not available for printing.

For testing the number $N_{\angle} = N_{\text{sim}}$ of simple bending cycles, the rope stroke h is of importance. If the stroke h is smaller than the rope contact bow u with the test sheave, then for each machine cycle the rope is bent by one simple rope bending. The number of machine cycles Z is equal to the number of the rope's simple bending cycles $N_{\angle} = N_{\text{sim}}$. The bending length l on two bending zones h is

$$l = 2h \quad \text{and} \quad N_{\angle} = N_{\text{sim}} = Z \quad \text{for } h < u. \quad (3.38)$$

	SIMPLE BENDING	N_{sim} 
	straight – bent – straight or bent – straight – bent	
	REVERSE BENDING	N_{rev} 
	bent – straight – reverse bent	
	FLUCTUATING TENSION AND BENDING	N_{comb} 
	straight, tensile force enlarged – bent – straight, tensile force reduced	

Fig. 3.27. Symbols for standard loading elements and numbers of bending cycles

If the rope stroke h is greater than the rope contact bow u with the test sheave, then for each machine cycle the rope is bent twice. For the number of machine cycles Z , the number of simple rope bending cycles is $N_{\angle} = N_{\text{sim}} = 2Z$. In this case the rope bending length is

$$l = h - u \quad \text{and} \quad N_{\angle} = N_{\text{sim}} = 2Z \quad \text{for } h > u. \quad (3.39)$$

On both sides of this bending length $l = h - u$ with the number of bending cycles $N_{\angle} = N_{\text{sim}} = 2Z$ there are two bending zones of the length u with the number of bending cycles $N_{\angle} = N_{\text{sim}} = 1Z$. Of course these bending zones have no influence on the endurance of the rope.

Reverse bending cycles without additional simple bending cycles on the same bending length are only possible on a relatively small rope bending length. Figure 3.28 makes this clear. If the rope stroke is just $h = u + a$, then on a bending length $l = u$ for the machine cycle Z there are two reverse bending cycles $N_{\text{rev}} = 2Z$. On other zones, the rope is only bent by simple bending cycles.

To evaluate the number of reverse bending cycles for each bending length, it is practical to use a sheave arrangement as shown for example in Fig. 3.29. Then the most stressed rope zone is bent for one machine cycle with several reverse bending cycles and only two simple bending cycles. This can be seen in Fig. 3.29 where the bending sequence for the sheave arrangement is shown. The number of reverse bending cycles can be separated from the machine

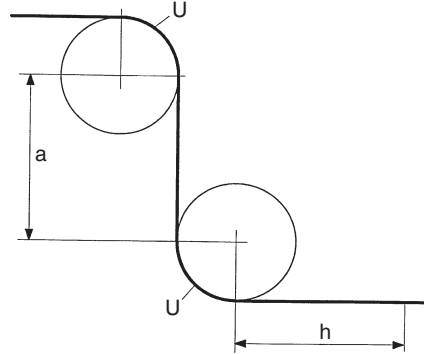


Fig. 3.28. Reverse bending

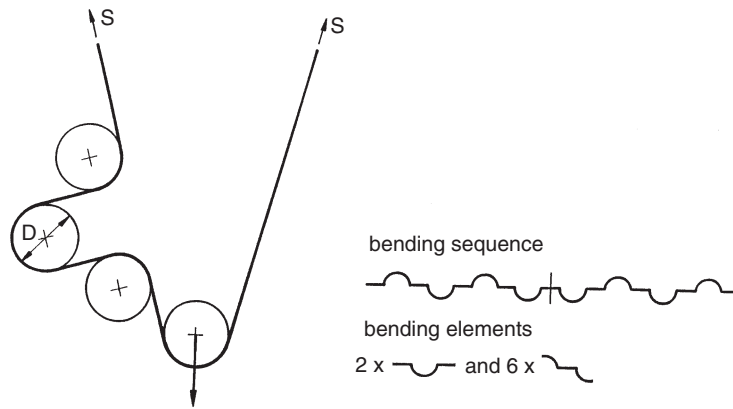


Fig. 3.29. Sheave arrangement to test reverse bending

cycles Z with the help of the Palmgren–Miner–Rule. For the sheave arrangement in Fig. 3.29 the number of reverse bending cycles is

$$N_{\text{rev}} = \frac{6 \cdot Z}{1 - \frac{2 \cdot Z}{N_{\text{sim}}}} \quad (3.40)$$

To evaluate the number of reverse bending cycles N_{rev} (up to rope breakage or discarding) bending tests have to be carried out with the standard test sheave arrangement in Fig. 3.26 to find N_{sim} and with a test sheave arrangement as shown in Fig. 3.29 to find Z and then with (3.40) the number of reverse bendings N_{rev} .

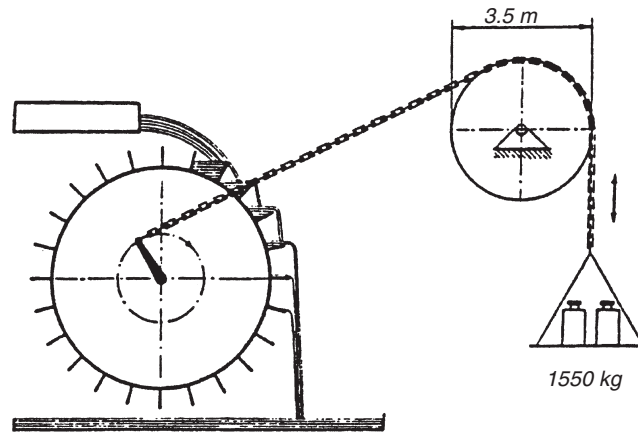


Fig. 3.30. Rope-bending fatigue machine made by Albert in the year 1828, Benoit (1935)

Rope Bending Fatigue Machines

The inventor – or at least the first manufacturer and first user – of wire ropes was Albert and he also constructed the first rope-bending fatigue machine. That also was the first known arrangement for fatigue testing of material on the whole. Figure 3.30 shows his bending machine from 1828 which he originally used to test the endurance of fibre ropes and chains. Albert carried out these tests in order to find out which types had better endurance for use in hoisting apparatus for mines. Later he carried out such bending tests with a certain degree of success on his self-produced ropes made of steel wires. In 1834 he installed the first real stranded wire rope for the hoisting installation in the mining shaft Caroline with a depth of 480 m, Bahke (1984).

The real research work using rope-bending machines began much later in the early 20th century. The researchers construct their own rope-bending machines, usually on the principle shown in Fig. 3.26. Rope-bending-machines have been devised, for example, by Benoit (1915), Scoble (1920), Woernle (1929), Shitkow and Pospechow (1957), Wiek (1976), Jehmlich (1985), Waters and Ulrich (1990) and the latest from Vogel and Nikic (2004).

In most of the rope-bending machines, the rope is guided very simply with a large traction sheave which remains unaltered during all the tests and a rope-testing sheave with a smaller diameter. These test sheaves can normally be replaced easily for testing different ropes and sheave diameters. In the newer machines, the traction sheave and the test sheave are in overhung position. This construction makes it possible just to lay the rope samples on to the sheave and to exchange the test sheave quickly. Even more important,

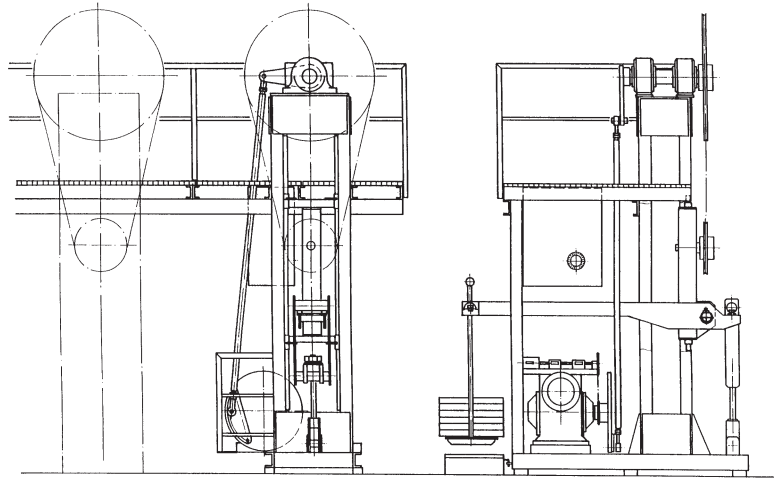


Fig. 3.31. Rope-bending fatigue machine, Feyrer and Hemminger (1983)

however, is the easy accessibility of the rope bending zones, so that the wire breaks can be counted without difficulty during the course of the test.

A rope-bending machine with overhung sheaves is shown in Fig. 3.31. The tensile force per rope is achieved through leverage with weights over a lever arm with knife edge suspension. In rope-bending machines for very large rope tensile forces and in rope-bending machines for testing bending combined with fluctuating tensile forces, the rope forces are induced using hydraulic cylinders.

In all the newer rope-bending machines, the bending frequency and the stroke can be varied. The machines are fitted with two of the usual devices for counting the bending fatigue cycles accurately and a device for measuring the crank revolutions per minute. Additionally, they have a counting device which switches the machine off when the desired number of bending cycles has been reached so that it is possible to estimate the state of the rope, for instance, to count the wire breaks. A built-on lubrication pump enables the ropes to be lubricated under quantity control during the bending test.

In any case, it should be ensured that the rope tensile force together with the force amplitudes vary less than 1% from the required tensile force. To reduce the unintended fluctuating forces, the sheaves are very precisely mounted in bearings in a central position and the test sheaves are very light so that the accelerating forces remain small. With their joint rope bending machine REFMA, built in Turin and Delft, Ciuffi (1976), Meeuse (1976) and Wiek (1976) tried to create a very precise bending machine. With the help of springs, they synchronised the test frequency close to the self-frequency of the rope-sheave-load system so that it was almost possible to avoid a rope force to accelerate or decelerate the test sheave.

Test Conditions

To gain reliable findings from rope bending tests, the test conditions have to be very clearly defined and observed. The OIPEEC Recommendation No. 4 OIPEEC Bulletin 56 (1988) gives these specifications. Table 3.2 lists the minimum specifications defining a wire rope bending fatigue test according to OIPEEC Recommendation No. 4. These specifications are sufficient to ensure that different institutes with different rope-bending machines come to the same result if the specifications are complied with, Feyrer (1990a). Other well-defined bending fatigue tests result in comparable findings, for example from Nabijou and Hobbs (1994) and Costello (1997).

The rope tension is normally given by the rope tensile force S . In the case of tensile rope stress σ_z or the specific tensile force S/d^2 , it must be declared whether they are based on the actual or nominal values of a rope cross-section or rope diameter. For detecting the influence of wire stresses, the real dimensions have to be used. However, for practical endurance tests, test results related to the nominal rope cross-section or the nominal rope diameter are to be preferred because the user normally only knows the nominal values. Moreover, this has the advantage that the deviation of the actual to the nominal rope cross-section and rope diameter is included in the standard deviation of the test results. In the following, the test results are normally related to the specific tensile force S/d^2 with the nominal rope diameter d .

Table 3.2. Minimum specifications for defining a wire rope-bending fatigue test according to the OIPEEC Recommendation No. 4, OIPEEC Bulletin No. 56 (1988)

test specification	kind of bending (simple bending, reverse bending) tensile force S bending length l rope temperature, if more than 50°C deflecting angle α , if less than 30° lateral deflection angle ϑ , if $\vartheta \neq 0$
wire rope specification	wire rope construction and lay direction nominal rope diameter d measured rope diameter d_m nominal wire strength R_0 , rope grade mean measured wire strength R_m minimum breaking force $F_{\min}$ measured breaking force F_m kind of lubrication (lubricant, before and during test)
sheave specification	sheave diameter D (related to the rope axis) or D_0 (groove ground) sheave material and hardness groove form, groove radius r , V -groove angle γ , etc.

The wire rope and its lubrication have to be defined precisely, Table 3.2. In the following bending tests described here, the wire ropes are normally well lubricated. For the tests in the Stuttgart Institute, usually a viscous mineral oil was used as a lubricant, without additives and with a viscosity of $1,370\text{--}1,520\text{ mm}^2/\text{s}$ for 40°C .

The sheave diameter D means the distance from rope centre to rope centre as shown in Fig. 3.32. The diameter ratio D/d is normally related to the nominal rope diameter. If not defined otherwise, the sheave is made of steel with a hardened round groove, a groove radius $r = 0.53d$ (nominal rope diameter) and a groove opening angle $\gamma = 60^\circ$.

The bending test is completed when the rope or at least one strand is broken so that the bending test cannot be continued. The number of rope bending cycles achieved is the so-called breaking number of bending cycles N . In most cases, the number of bending cycles is recorded as well when a discarding criterion such as the discarding number of wire breaks B is detected. This number of bending cycles is called the discarding number of bending cycles N_A .

During a wire rope bending test, the rope-bending machine is stopped several times so that the state of the rope bending zones can be inspected. Any change in the rope bending zones is then recorded, in particular the number of wire breaks is counted and the rope diameter measured. The number of bending cycles at which the machine has to be stopped (by the counting device for the machine revolutions) is normally taken from the Renard row R10. The numbers in this row are: 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1,000 multiplied by 10^x , where x is a whole number.

The number of wire breaks B and, in some special cases, the rope diameter d as a function of the number of bending cycles found during the bending tests is used twice. First of all, for all tests with ropes of the same construction, the number of wire breaks B_A is evaluated to find out if the wire rope should be discarded. The basis for this evaluation is the number of wire breaks B found

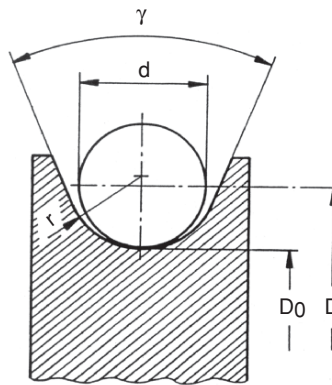


Fig. 3.32. Sheave with round groove

at 80% of the breaking number of bending cycles N . Secondly, the discarding number of bending cycles N_A can be recorded based on the number of wire breaks B_A obtained during the rope-bending test.

3.2.2 Number of Bending Cycles

Tensile Force and Diameter Ratio

The most important influences on the number of bending cycles are the rope tensile force S and the diameter ratio D/d (sheave to rope diameter). In Fig. 3.33, these influences on the breaking number of bending cycles (simple bending cycles) are shown for a Filler rope $8 \times (19 + 6F) - \text{NFC} - \text{sZ}$. The breaking number of bending cycles and the specific rope tensile force are drawn in logarithm scale. In this diagram form the numbers of bending cycles from the tests form very straight lines for constant diameter ratios D/d .

At a certain high tensile force, the number of bending cycles drops abruptly. The limit of the tensile force where the number of bending cycles begins to drop is called the Donandt force. This force, which is the absolute limit of the usable tensile force, will be discussed later on. First of all, a look will be taken at the influence of the tensile force and the diameter ratio in the usable range.

A great number of researchers have tested the influence of the tensile force and the diameter ratio D/d on the number of rope bending cycles. Some

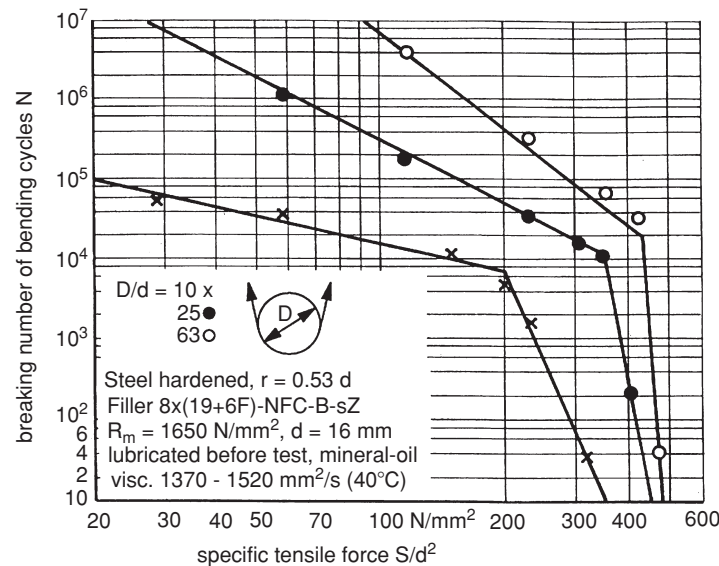


Fig. 3.33. Breaking numbers of bending cycles for one Filler rope

researchers such as Klein (1937), Niemann (1946) and Shitkow and Pospechow (1957) have created early endurance equations to describe their test results.

The different equations for wire rope endurance that can be used for regression calculation belong mainly to the two groups listed in Tables 3.3 and 3.4. The variables in both of these tables have been written in a uniform way so that they can be compared more easily, but the equations are not expressed in the identical way used by the authors.

In the equations listed in Table 3.3, the number of bending cycles is given as a function of a combined constant stress (from the constant tensile force) and fluctuating stress (mostly from the bending and from the pressure). The number of bending cycles is a function of these added stresses. These equations with the added stresses (constant + fluctuating) as only one variable make it

Table 3.3. Number of bending cycles as function of the sum of fluctuating bending and constant stresses

Rossetti (1975)	$\lg N = a_0 + a_1 \cdot \lg \frac{\sigma_z + \sigma_b}{\sigma_0}$	(3.42)
Meeuse (in Tonghini (1980))	$\lg N = a_0 + a_1 \cdot \lg \frac{\sigma_z + \sigma_b/2}{\sigma_0}$	(3.43)
Pantucek (1977) (in Bahke (1984))	$\lg N = a_0 + a_1 \cdot \lg \frac{\sigma_z + \sigma_b + \sigma_p}{\sigma_0}$	(3.44a)
Jehmlich and Steinbach (1980)	$\lg N = a_0 + a_1 \cdot \lg \frac{\sigma_z + \sigma_b + \sigma_p}{\sigma_0}$	(3.44b)
Rope tensile stress σ_z , wire bending stress σ_b , pressure σ_p , unit stress σ_0 in N/mm ²		

Table 3.4. Number of bending cycles as a function of specific tensile force S/d^2 and of diameter ratio D/d

Woernle (1934)	$\lg N = a_0 + a_1 \cdot \lg \frac{D}{d} \text{ for } S/d^2 = \text{const.}$	(3.45)
Drucker and Tachau (1944)	$\lg N = a_0 + a_1 \cdot \left(\lg \frac{S \cdot d_0^2}{S_0 \cdot d^2} + \lg \frac{d}{D} \right)$	(3.46)
Mebold (1961)	$\lg N = a_0 + 1.8 \cdot \left(\lg \frac{S \cdot d_0^2}{S_0 \cdot d^2} + \lg \frac{d}{D} \right)$	(3.47)
Calderale (1960) Giovannozzi (1967)	$\lg N = a_0 + a_1 \cdot \lg \frac{S \cdot d_0^2}{S_0 \cdot d^2} + a_2 \cdot \lg \frac{d}{D}$	(3.48)
Feyrer (1981a,b)	$\lg N = a_0 + a_1 \cdot \lg \frac{S \cdot d_0^2}{S_0 \cdot d^2} + a_2 \cdot \lg \frac{D}{d}$ $+ a_4 \cdot \lg \frac{S \cdot d_0^2}{S_0 \cdot d^2} \cdot \lg \frac{D}{d}$	(3.49)

Rope tensile force S , nominal rope diameter d , sheave diameter D , unit tensile force $S_0 = 1$ N, unit diameter $d_0 = 1$ mm

possible to achieve a result with only a few tests. However, this method offends against the fundamental rules of fatigue strength which means that the results gained by these equations can only be valid for relatively small test ranges of tensile stresses and diameter ratios D/d .

In the equations listed in Table 3.4, the number of bending cycles is given as a function of the constant tensile force (specific tensile force S/d^2) and separate from the fluctuating stresses caused by the diameter ratio D/d . The degree to which the different equations correspond can be proved by the coefficient of determination. The regression calculation for a great number of bending tests results with a great range of tensile stress and of diameter ratio D/d show the highest coefficient of determination for (3.49) with three independent variables, Feyrer (1981a). Just how well the points from the test results correspond to the straight lines from (3.49) is demonstrated in Fig. 3.33.

A good predecessor was created by Calderale (1960) and Giovannozzi (1967) and it results in a smaller coefficient of determination. However, this equation does not consider the different gradients of the lines needed for the diameter ratios.

Only the independent equation from Clement (1980) with its three independent variables – as in (3.49) –

$$\ln N = a_0 + a_1 \cdot \frac{d}{D} + a_2 \cdot \sqrt{\frac{S}{d^2} \cdot \frac{d}{D}} + a_3 \cdot \frac{d}{D} \cdot \sqrt{\frac{S}{d^2} \cdot \frac{d}{D}} \quad (3.50)$$

shows a coefficient of determination close to that of (3.49), Feyrer (1981a).

In the following, only (3.49) will be used. This equation is valid both for the breaking and the discarding number of bending cycles. In (3.49), the unit force $S_0 = 1$ N and the unit diameter $d_0 = 1$ mm have been left out to simplify the overview (knowing that the units N and mm always have to be used). Then (3.49) is

$$\lg N = a_0 + a_1 \cdot \lg \frac{S}{d^2} + a_2 \cdot \lg \frac{D}{d} + a_4 \cdot \lg \frac{S}{d^2} \cdot \lg \frac{D}{d}. \quad (3.49)$$

There have been a great many bending fatigue tests done with most common types wire rope, Feyrer (1981a, 1985a,b, 1988, 1997). From the results of these tests, the constants a_i of (3.49) are evaluated by regression calculation for the breaking number N and for the discarding number N_A of bending cycles.

These constants a_i are related to the

Mean nominal strength $R_0 = 1,770$ N/mm²

Nominal rope diameter $d = 16$ mm

Rope bending length $l = 60$ d.

Lubricated before the test, not relubricated during test.

These constants (changed from a_i – b_i) are listed in Table 3.14a, Sect. 3.4.3, for the extended rope endurance equation which also includes some additional influences.

As (3.49) shows, of the number of bending cycles has normal logarithm distribution. The standard deviation is therefore to be described as $\lg s$. The mean standard deviation for the breaking number of bending cycles of one rope is $\lg s = 0.05$. For several ropes with the same construction, the standard deviation for the breaking number is $\lg s = 0.19\text{--}0.28$ and for the discarding number $\lg s = 0.22\text{--}0.30$. All these standard deviations refer to a bending length $l = 60\text{ d}$.

The number of bending cycles N_{10} at which – with a certainty of 95% – not more than 10% of such wire ropes are broken or have to be discarded, can be calculated by

$$\lg N_{10} = \lg \bar{N} - k_T \cdot \lg s. \quad (3.49a)$$

The constant k_T stands for a mean ratio between $\bar{N}$ and N_{10} that is in reality smaller in the middle of the region being considered and greater at the edges, Stange (1971).

From bending tests with Filler ropes $8 \times (19+6F)\text{-FC-sZ}$, the breaking numbers of bending cycles are drawn in Fig. 3.34 and the discarding number of bending cycles in Fig. 3.35. In addition to the test results (as points), these Figs. 3.34 and 3.35 also include the lines for the calculated number of bending cycles from (3.49) and (3.49a). The thicker lines show the mean number and

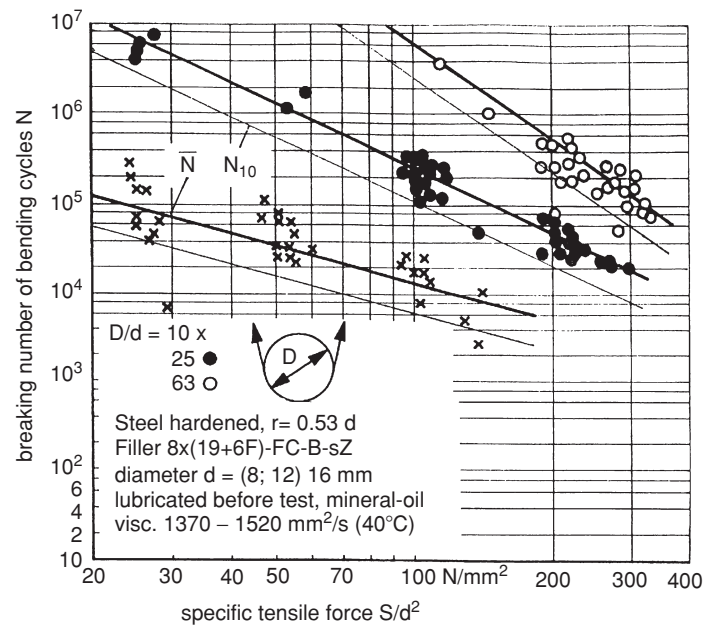


Fig. 3.34. Breaking number of bending cycles for Filler ropes $8 \times (19+6F)\text{-FC-sZ}$ in simple bending, Feyrer (1985a)

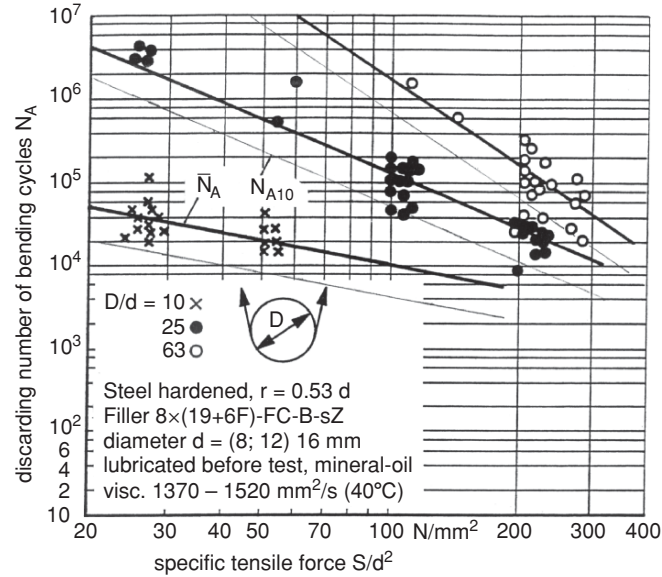


Fig. 3.35. Discarding number of bending cycles for Filler ropes $8 \times (19+6F)\text{--FC--sZ}$ in simple bending, Feyrer (1985b)

the thinner lines the numbers of the 10% limit. As Figs. 3.34 and 3.35 show, the 10% lines provide safe limits for the test results.

Strength

Woernle (1929) and Müller (1966) carried out some bending tests with wire ropes of different strengths. They both found that the rope endurance only increases a little with increased rope strength. For a nominal rope strength $R_0 = 1,370 \text{ N/mm}^2$ up to $R_0 = 1,770 \text{ N/mm}^2$, Shitkow and Pospechow (1957) observed an increase in the numbers of bending cycles that does not continue into the next higher rope strength $R_0 = 1,960 \text{ N/mm}^2$.

Wolf (1987) evaluated a large number of bending tests with wire ropes of different strengths. He found that the endurance increases slightly with the strength. A regression calculation has been derived from a new evaluation of these and other results, Feyrer (1992). For this regression calculation, the tensile force S for ropes with the nominal strength R_0 has been defined in relation to those with the mean nominal strength $1,770 \text{ N/mm}^2$.

$$S = S_{1770} \cdot \left(\frac{R_0}{1,770} \right)^c \quad (3.51)$$

or

$$\lg S = \lg S_{1770} + c \cdot \lg \frac{R_0}{1,770}.$$

S_{1770} is the tensile force in (3.49) and their constants a_i for the mean nominal tensile strength 1,770. Using (3.49) the ratio of the bending cycles N/N_{1770} is given by

$$\lg \frac{N}{N_{1770}} = \left(a_1 + a_4 \cdot \lg \frac{D}{d} \right) \cdot \lg \frac{S}{S_{1770}}. \quad (3.51a)$$

With (3.51) in (3.51a) the ratio of the numbers of bending cycles as a function of the rope strengths is

$$\lg \frac{N}{N_{1770}} = c \cdot \left(a_1 + a_4 \cdot \lg \frac{D}{d} \right) \cdot \lg \frac{R_0}{1,770}. \quad (3.51b)$$

The regression calculation has been based on (3.51b). The mean exponent c found for all bending tests is $c = -0.408$. The part-results for the diameter ratios $D/d = 10$; 25 and 63 for the breaking number of bending cycle is $c = -0.370$; -0.348 and -0.408 and for the discarding number of bending cycles $c = -0.561$; -0.390 and -0.355 . The influence of the constant -0.561 is very high and relatively unsafe because of the small gradient for small diameter ratios D/d . Therefore, the constant has been finally set at $c = -0.4$.

An example is given in Fig. 3.36 showing the ratio of the number of bending cycles for different nominal tensile strengths. A line has been drawn for

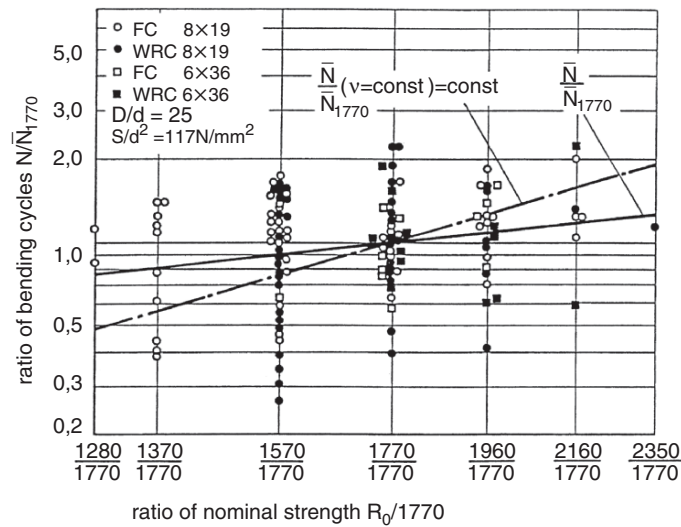


Fig. 3.36. Influence of the nominal tensile strength on the breaking number of bending cycles N for $D/d = 25$ and $S/d^2 = 117 \text{ N/mm}^2$

$c = -0.4$ and for $c = -1$. The line for $c = -1$ means that the ratio of the bending cycle would increase proportionally with the nominal tensile strength, which is apparently not the case.

Rope Diameter, Size Effect

Müller (1966) carried out a lot of bending fatigue tests on cross lay ropes $6 \times 19 - \text{FNC-sZ}$ and zZ with rope diameters between $d = 2.5$ and 24 mm. He found that the endurance of a rope decreases with an increasing rope diameter. As an example Fig. 3.37 shows his results for ordinary ropes bent over a sheave with the diameter ratio $D/d = 25$. By regression calculation with these and other results from Müller (below the yielding strength, Donandt force) the ratio of the numbers of bending cycles is

$$\frac{N}{N_0} = \left(\frac{d}{d_0} \right)^{-0.32}. \quad (3.52)$$

Shitkow and Pospechow (1957) presented a diagram with practically the same result. More recently, Ciuffi and Roccati (1995) and Virsik (1995) confirmed (3.51). A larger exponent can be derived from a small number of bending tests carried out by Costello (1997), but his results are also a good match for the cloud of points collected by Virsik from a very large number of bending fatigue tests. From these, Virsik found again the exponent -0.32 .

For wire ropes with the nominal rope diameter $d = 16$ mm, the number of bending cycles N_{16} was evaluated with (3.49) and the constants a_i . Using (3.52), the number of bending cycles for any nominal rope diameter is

$$\lg N = \lg N_{16} - 0.32 \cdot \lg d + 0.32 \cdot \lg 16. \quad (3.52a)$$

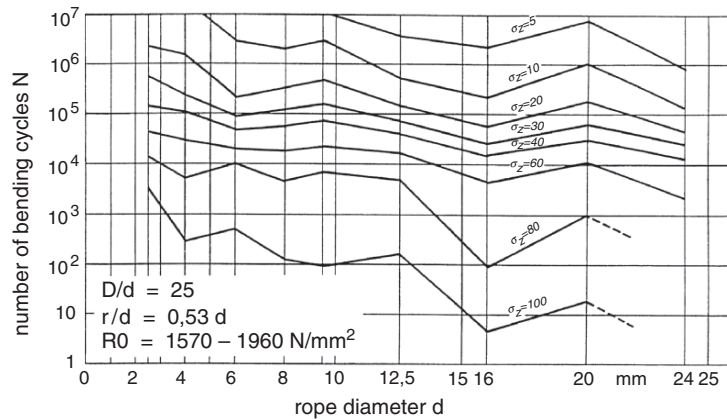


Fig. 3.37. Breaking numbers of bending cycles of cross lay ropes $6 \times 19 - \text{FNC} - \text{sZ}$ of different nominal rope diameters, Müller (1966)

Bending Length, Size Effect

Müller (1961) was the first to carry out a number of bending fatigue tests on a wire rope with different bending lengths. He found that the number of bending cycles N decreases up to the bending length $l = 8d$ and remains more or less constant for greater bending lengths. However, as can be seen in Fig. 3.38, numerous bending tests up to a bending length $l = 1,000d$ show that the number of bending cycles decreases permanently with the bending length. In Fig. 3.38, curves are drawn for the mean number of bending cycles $\bar{N}$ and for the numbers of bending cycles N_{10} and N_{90} as the limit for 10% of the numbers of bending cycles being smaller respectively greater. The curves are calculated from the 13 numbers of bending cycles for the bending length $l = 45d$ using the method described in the following section.

The longer the bending length of the wire rope under consideration is, the higher the probability that it has a length element with low endurance. The survival probability of the rope with the bending length l is

$$R(l) = R_0^{(l-\Delta l)/(l_0-\Delta l)}. \quad (3.53)$$

R_0 is the survival probability of the wire rope with the bending length l_0 . The bending lengths l and l_0 are those defined for bending a limp yarn. On the rope bending length near the sheave, the fluctuating stresses resulting from rope bending stiffness are smaller. To take this into consideration, the small bending length Δl has been introduced. Δl is the small part of the bending length where the radius difference of the rope curvature is smaller than 90% of the total radius difference. This bending length is normally $\Delta l < 2.5d$.

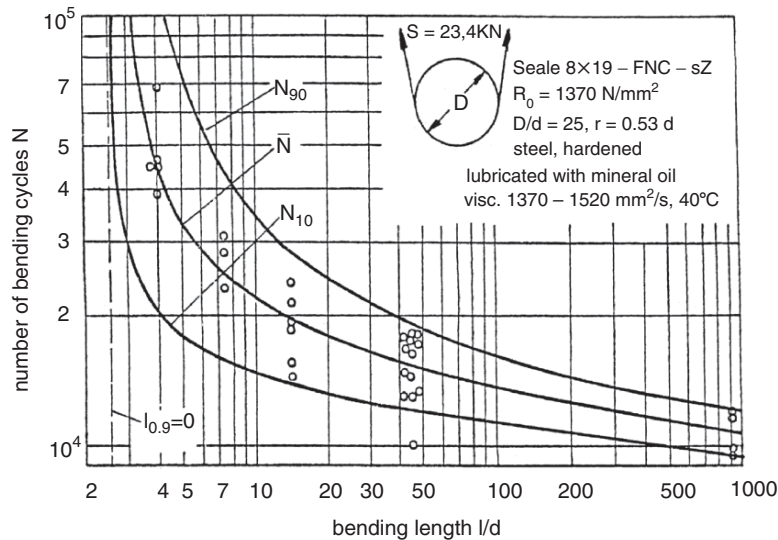


Fig. 3.38. Number of bending cycles for different rope bending lengths, Feyrer (1981)

The survival probability of a wire rope is well represented by a logarithm normal distribution of the numbers of bending cycles. Beside the mean number of bending cycles, the determining value is the standard deviation of the number of bending cycles for a single rope (and not the common standard deviation for all the ropes of a particular rope construction). From 73 bending fatigue tests done in various test series, the mean standard deviation was $\lg s = 0.0518$ for a bending length $l_0 = 60d$. Furthermore the number of bending cycles was evaluated in 26 double tests with 22 different wire ropes (Filler, Seale, Warrington and Warrington-Seale with fibre or steel core) with different tensile stresses and diameter ratios, mostly $\sigma_z = 300 \text{ N/mm}^2$ and $D/d = 25$. The bending length for all these tests was again $l_0 = 60d$. The standard deviation of the ratio of the two numbers of bending cycles from the same rope is $\lg s_V = 0.0669$. For the distribution of single ropes, the standard deviation is $\lg s = \lg s_V / \sqrt{2} = 0.0473$. From the two series of tests, it was decided to use $\lg s = 0.05$ as the standard deviation for the calculation of the number of bending cycles of different bending lengths.

Equation (3.53) is not suitable for practical use. Therefore, on the basis of this equation an approximate equation was developed for the number of bending cycles as a function of the bending length. The mean number of bending cycles is

$$\lg \bar{N} = \lg N_{60} + \frac{1}{1.2 + \lg \frac{l}{d}} - \frac{1}{1.2 + \lg 60} \quad (3.54)$$

and the number of bending cycles that are smaller – with a certainty of 95%, at most 10% – is

$$\lg N_{10} = \lg N_{10,60} + \frac{1}{1.9 + \lg \frac{l}{d}} - \frac{1}{1.9 + \lg 60}. \quad (3.54a)$$

As Fig. 3.38 shows, the influence of the bending length on the number of bending cycles is very well represented on the basis of a logarithm normal distribution. The suggestion of Castillo et al. (1990, 1992) to use a Weibull distribution with three parameters normally fails because the small numbers of test results available mean that the three parameters cannot be evaluated safely enough.

Extended Equation for the Number of Bending Cycles

The number of bending cycles (simple bending cycles) is given in (3.49) as a function of the specific tensile force S/d^2 and of the diameter ratio D/d . With (3.50), (3.51a), (3.53) and (3.54), the influence of the nominal strength R_0 , the nominal rope diameter d and the bending length l can be introduced in (3.49). Furthermore, with (3.49a) it is possible to express the number of bending cycles N_{10} in addition to the mean number of bending cycles $\bar{N}$. According to that, the number of bending cycles (breaking or discarding) for any nominal strength R_0 , nominal rope diameter d and bending length l is

$$\lg N = b_0 + \left(b_1 + b_4 \cdot \lg \frac{D}{d} \right) \cdot \left(\lg \frac{S}{d^2} - 0.4 \cdot \lg \frac{R_0}{1,770} \right) + b_2 \cdot \lg \frac{D}{d} + b_3 \cdot \lg d + \frac{1}{b_5 + \lg \frac{l}{d}}. \quad (3.55)$$

where

d is the nominal rope diameter in mm,

D the sheave diameter in mm

S the rope tensile force in N

R_0 the nominal tensile strength in N/mm²

l is the bending length.

and the constants b can be derived from

$$b_0 = a_0 + 0.32 \cdot \lg 16 - \frac{1}{b_5 + \lg 60} \quad \text{for } \bar{N}.$$

$$b_1 = a_1 + 0.32 \cdot \lg 16 - \frac{1}{b_5 + \lg 60} - k_T \cdot \lg s \quad \text{for } N_{10}.$$

$$b_2 = a_2$$

$$b_3 = -0.32$$

$$b_4 = a_4$$

$$b_5 = 1.2 \quad \text{for } \bar{N}$$

$$b_5 = 1.9 \quad \text{for } N_{10}.$$

The constants b_i for wire ropes with the most important rope constructions are listed in Table 3.14a, Sect. 3.4.3.

3.2.3 Further Influences on the Number of Bending Cycles

Rope Core

For wire ropes with fibre cores, Müller (1966) demands a great mass for the core to prevent an arching of the strands so that the endurance attained by the wire rope will be sufficient. See Sect. 1.6.2 for the dimensioning of the core mass.

Wolf (1987) carried out bending fatigue tests to discover how much influence the mass and the material of the fibre core have on the endurance of the wire rope. All the wire ropes he tested were made with identical strands but with different cores. Fig. 3.39 shows the results of these tests. The mass of the core is drawn as a percentage of the core mass as was required by the former German Rule for Rope Ways BOSeil. The breaking number N and the discarding number N_A of bending cycles increase in all cases with the mass of the fibre core. The wire ropes with cores made of natural fibres and polypropylene fibres reach nearly the same endurance. With polyamide fibre cores, the numbers of bending cycles increased remarkably, probably as a result of the higher durability of the polyamide.

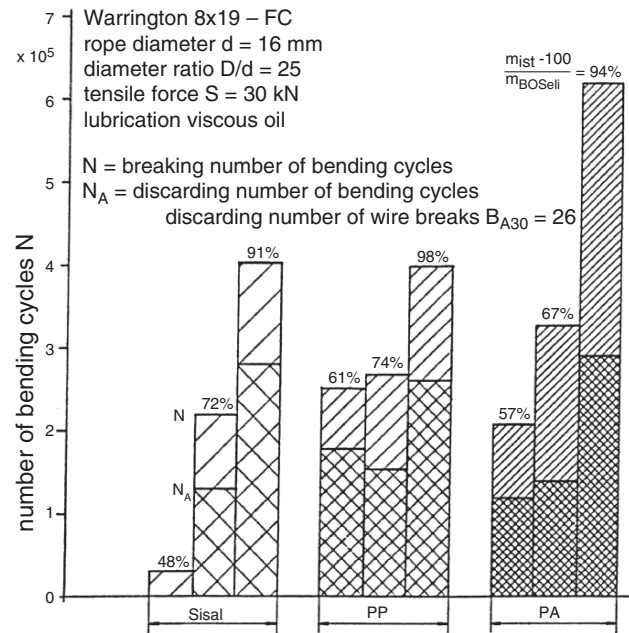


Fig. 3.39. Number of bending cycles N and N_A of a rope with different fibre cores, Wolf (1987)

Unlike the case of ropes with fibre cores, the number of bending cycles decreases with an increasing clearance between the strands for ropes with a steel core IWRC. This was discovered by Müller during a large series of tests, but was not published until after his death by one of his employees, Greis (1979). Wolf (1987) confirmed this result with his own series of bending fatigue tests. The reason for the loss of the number of bending cycles when there is greater strand clearance is that the strands are free to move laterally during bending which induces stresses and wear.

For wire ropes with steel cores enveloped with solid polymer ESWRC or wire ropes with parallel steel core with the outer strands PWRC, the position of the strands is well defined. As for ropes with fibre cores, a lateral movement of the strands is practically impossible. Therefore these special steel cores achieve a greater number of bending cycles. Figure 3.40 shows the breaking number of bending cycles for 6 and 8 strand ropes with different steel cores in relation to those for 8 strand ropes with independent wire rope cores. This relation is also nearly valid for the discarding number of bending cycles.

Zinc Coating

Woernle and Müller have carried out a great number of bending fatigue tests to compare the endurance of wire ropes with bright or zinc coated wires. All

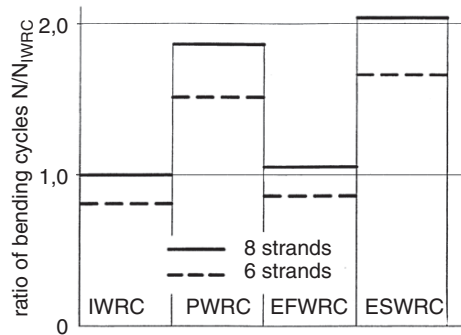


Fig. 3.40. Number of bending cycles N for six and eight strand ropes with different steel cores from Wolf (1987) and others

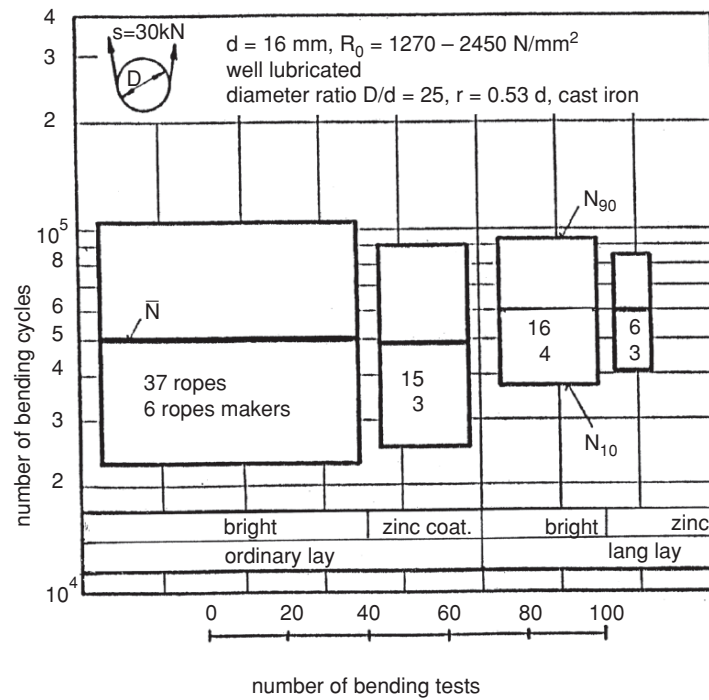


Fig. 3.41. Number of bending cycles $\bar{N}$, N_{10} and N_{90} for wire ropes with bright or zinc coated wires from the results of Woernle and Müller

the wire ropes tested were lubricated before starting the tests. The result of these tests is shown in Fig. 3.41. There was practically no difference recorded between the bright and the zinc coated ropes, irrespective of whether the wire ropes tested had ordinary lay or lang lay. Zimmerman and Reemsnyder (1983) came to the same result.

Lubrication

The lubrication of the wire rope exerts a great influence on the endurance of running ropes. To evaluate this influence, Müller (1966) carried out bending tests with two parallel strand ropes, one with fibre core and one with steel core, to compare the endurance of lubricated and degreased ropes. The results of these tests are shown in Fig. 3.42. In the tests, the endurance of the degreased rope only attained 15–20% of that found for the lubricated rope. For rope lubrication see Sect. 1.4.

Re-lubrication

The lubricant put in the rope during manufacturing is only fully effective for a time or for a certain number of bending cycles. Normally, running wire ropes are only lubricated during manufacturing. However, long life wire ropes need re-lubricating.

Figure 3.43 shows, as an example, the effect of re-lubrication in fatigue bending tests with specimens of one Warrington rope. Six specimens were lubricated before and nine specimens were lubricated before and during the bending fatigue tests. The effect of the re-lubrication was to increase the mean

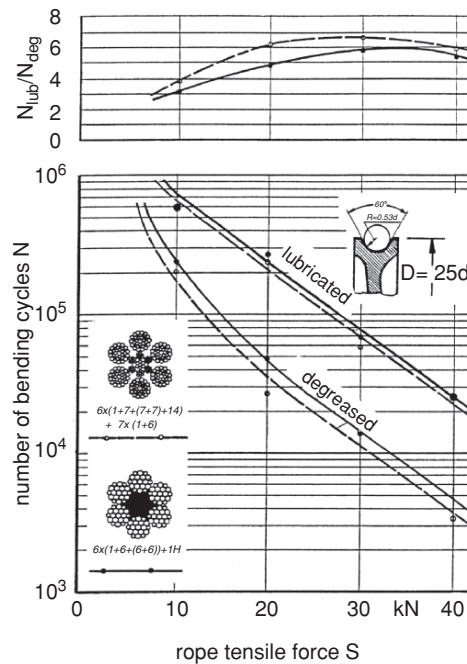


Fig. 3.42. Number of bending cycles of two wire ropes, lubricated and degreased, Müller (1966)

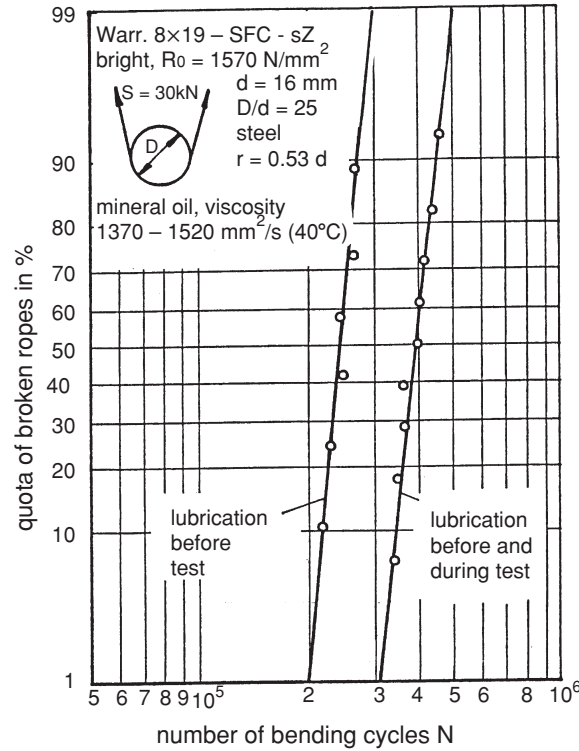


Fig. 3.43. Number of bending cycles of a wire rope with or without re-lubrication

number of bending cycles from $\bar{N} = 246,000$ to $\bar{N} = 392,000$. The standard deviation for logarithm distribution is $\lg s = 0.038$ and $\lg s = 0.047$, based on the bending length $45d$.

A larger number of bending fatigue tests, both with and without re-lubrication, were carried out with seven parallel strand ropes with the rope diameters between 12 and 16 mm, Feyrer (1998). Figure 3.44 shows the ratio of the breaking numbers of bending cycles N_m/N with (index m) and without re-lubrication (no index). By regression calculation, the breaking number of bending cycles with re-lubrication is

$$N_m = 0.0316 \cdot N^{1.307}. \quad (3.56)$$

and the discarding number of bending cycles with re-lubrication is

$$N_{Am} = 0.0682 \cdot N_A^{1.248}. \quad (3.57)$$

Normally, the number of bending cycles will be increased by the re-lubrication. However, if the wire rope without re-lubrication only reaches the numbers of bending cycles $N = 80,000$, respectively, $N_A = 50,000$, there is no increase

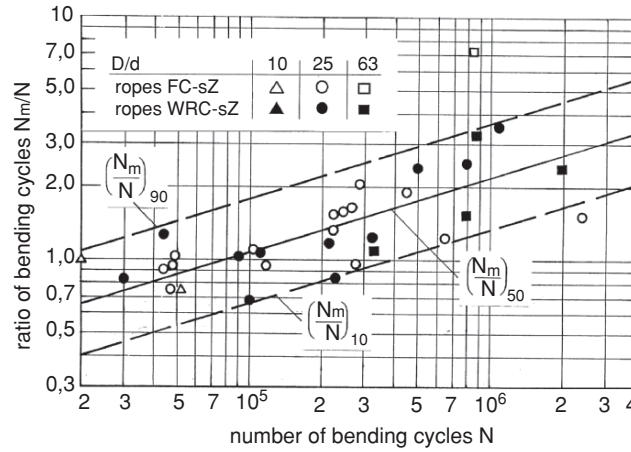


Fig. 3.44. Ratio of the breaking numbers of bending cycles N_m/N with (index m) and without re-lubrication, Feyrer (1998)

in endurance to be expected. Moreover, up to these limits the numbers of bending cycles may be even reduced by re-lubrication, Fig. 3.44.

The reason for this may be that the re-lubrication reduces the friction between the wires and the strands which then results in a reduction of the fluctuating secondary tensile stresses. On the other hand, a broken wire only bears the load again after a certain length which means that the neighbouring wires will be more highly stressed. The result of these findings may show that for smaller stresses the first influence and for higher stresses the second influence is predominant.

In all the bending fatigue tests described, the wire ropes were lubricated before and, in the case of re-lubrication, during the tests with a viscous mineral oil without any additives and with a viscosity 1,370–1,520 cSt, 40°C. The amount of lubricant – transported by a pump and dripped onto the rope – is very small. To find the lowest limit of lubricant needed, the quantity of lubricant was reduced from test to test in a series of bending tests. In the last test where the endurance was not reduced, the amount of lubricant used was only 1.8 g/m for 100,000 bending cycles of the 16 mm wire rope, see Sect. 1.4.

Round Groove

If the radius of the round groove is greater than half the rope diameter, the wire rope will be ovalised and the pressure in the bottom of the groove is high on the wire rope running over the sheave. Therefore the number of bending cycles decreases with the increasing ratio r/d of groove radius r and the rope diameter d .

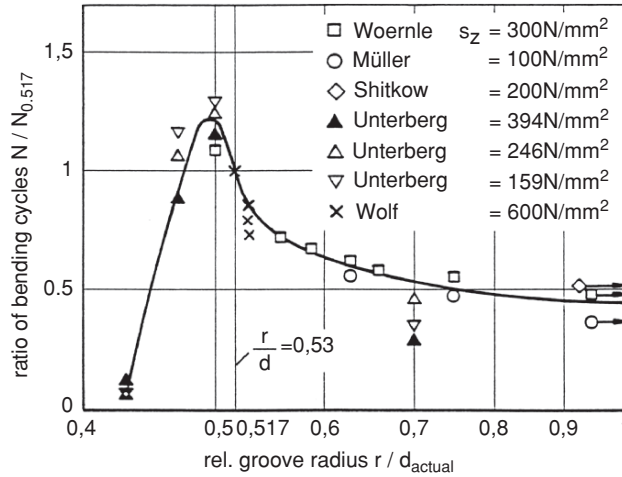


Fig. 3.45. Influence of the groove radius on the breaking number of bending cycles

The ratio of the breaking number of bending cycles as a function of the ratio r/d has been evaluated by Woernle (1929), Müller (1954), Shitkow (1957), Wolf (1987) and Unterberg (1991). Their results are shown in Fig. 3.45. Unterberg's results in particular show that – as expected – the greatest rope endurance has been found for the ratio $r/d_m = 0.5$ of the groove radius and the actual measurement of the rope diameter d_m . Below this ratio, the endurance of the rope is strongly reduced and should be avoided.

Normally, the groove radius is $r = 0.53d$ with d for the nominal rope diameter. As the actual measurement of the rope diameter can be 5% greater than the nominal rope diameter, the wire rope will always be well-bedded and optimal rope endurance can be more or less expected.

Special Form Grooves

Early bending fatigue tests with differently formed grooves have been carried out by Woernle (1934). The results of these tests are shown in his frequently published Fig. 3.46. The cross lay ropes tested are no longer used in elevators and the test conditions are very different from those found in practice with elevators. In Woernle's tests which did not include traction forces, the rope tensile stress σ_z is higher and the diameter ratio D/d is smaller. Furthermore, these tests were carried out until the rope actually broke. However, the rope endurance up to its discarding point is of especial interest for elevators. Figure 3.46 therefore only gives a first impression.

To find out more about the influence of form grooves, Holeschak (1987) investigated rope endurance in existing elevators. His findings for undercut

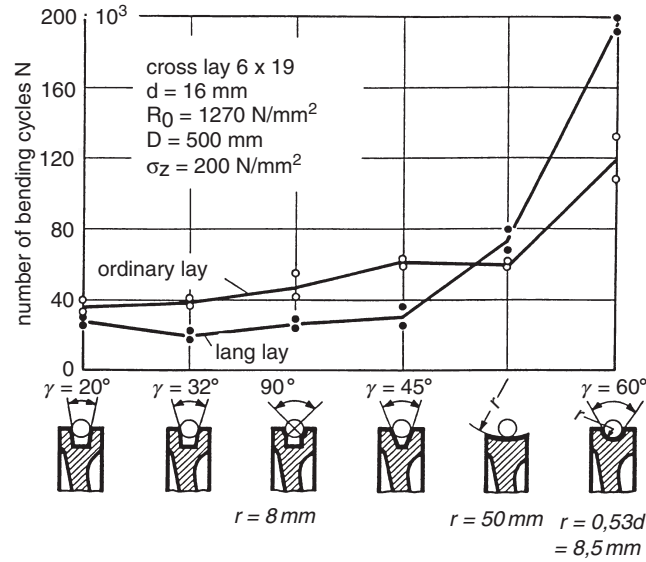


Fig. 3.46. Breaking numbers of bending cycles for a ordinary and a lang lay rope in different form grooves, Woernle (1934)

round grooves and V-grooves are given in Fig. 3.47 as the ratio of the discarding numbers of bending cycles with form grooves and with standard round grooves.

In most of the elevators looked at during Holeschak's research programme, the wire rope runs – in addition to the traction sheave – over one or more deflection sheaves with round grooves. This means that the wire rope will be ovalised in the form groove of the traction sheave and then be ovalised in the opposite direction by running over the deflection sheaves with round grooves. The rope endurance is therefore not only reduced by the additional bending cycles from the deflection sheaves but also by this fluctuating ovalisation. The ratios of the discarding numbers of bending cycles – presented in Fig. 3.47, Holeschak (1987) – are calculated with the Palmgren–Miner-Rule and the known discarding number of bending cycles for round grooves. Therefore, for elevators with deflection sheaves, the number of elevator trips up to the discarding of the rope can be evaluated using Holeschak's results, Fig. 3.47. In the rare cases of elevators where the ropes only run over the traction sheave, the discarding numbers of bending cycles will be somewhat greater than those calculated with the ratios from Fig. 3.47.

Groove Material

Normally sheaves and their grooves are made of steel or cast iron. In some cases the steel grooves are hardened. This hardening does not reduce the

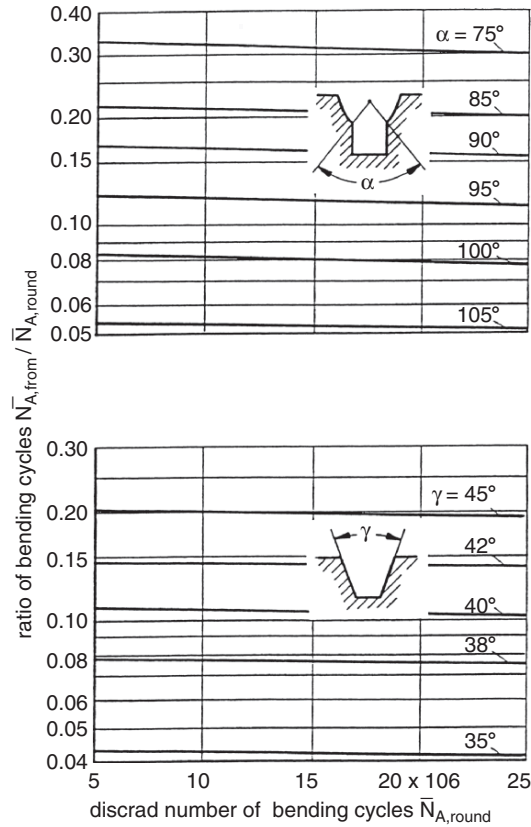


Fig. 3.47. Ratio of the discarding numbers of bending cycles in form and round grooves, Holeschak (1987)

endurance of the wire rope at all, on the contrary it is increased, Bechtloff and Szelagowski (1967) and Eilers and Schwarz (1974). The reason is to be found in the constant form of the groove without change by wear. It is to be recommended that grooves are hardened.

When sheave grooves are made of a soft material (small elasticity module), the wire rope endurance increases. Müller (1961) carried out a series of comparative bending fatigue tests with plastic and cast iron sheaves. The results are shown in Fig. 3.48.

The same tendency is also to be found in the work done by other authors. The ratio of the breaking number of bending cycles with sheave grooves made of plastic material in comparison with those made of steel or cast iron shown in Fig. 3.49 has been taken from all the known series of comparative tests. Only the very large ratios Babel (1980) found in tests with degreased wire

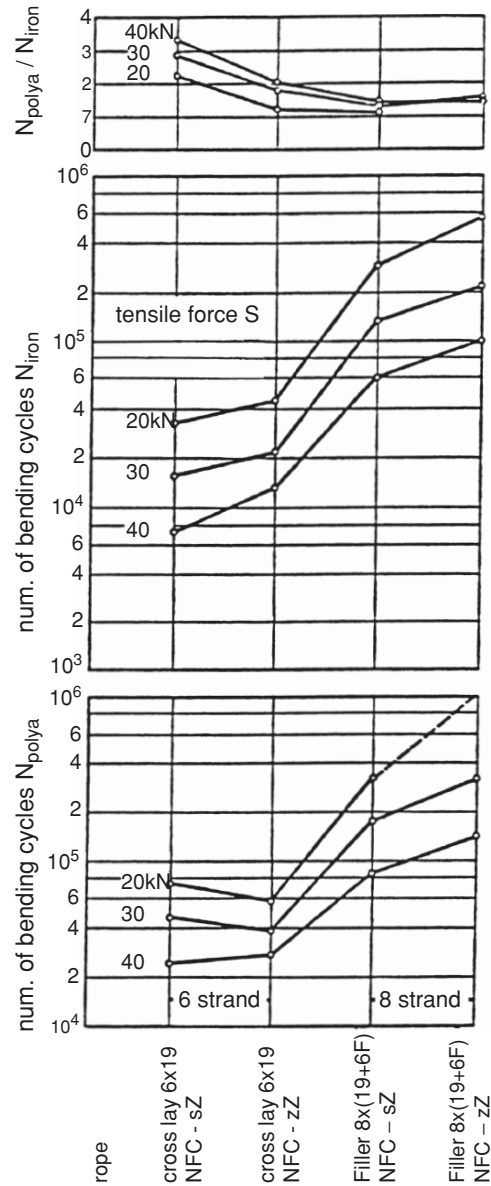


Fig. 3.48. Breaking number of bending cycles for cross lay ropes in sheave grooves out of polyamide and cast iron, Müller (1961). $d = 16 \text{ mm}$; $R_0 = 1,570 \text{ N/mm}^2$; $D = 300 \text{ mm}$; $r = 0.53d$

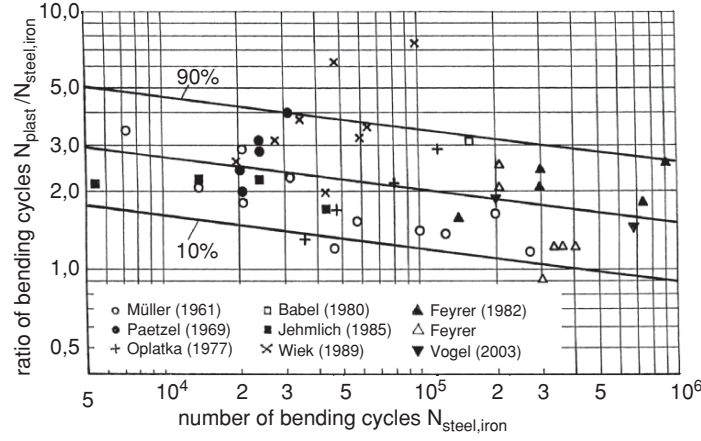


Fig. 3.49. Ratio of breaking numbers with sheave grooves made of plastic and steel or cast iron

ropes have not been included. The mean ratio of the breaking number with plastic grooves N_{pl} and of those with steel or cast iron N_{st} is

$$\left(\frac{N_{pl}}{N_{st}}\right)_{50} = 8.37 \cdot N_{st}^{-0.124} \quad (3.58)$$

or

$$\left(\frac{N_{pl}}{N_{st}}\right)_{50} \approx 0.75 + 0.36 \cdot \frac{S/d^2}{D/d} - 0.023 \cdot \left(\frac{S/d^2}{D/d}\right)^2. \quad (3.58a)$$

The lower the breaking cycles are with steel grooves, the higher the relation of the number of bending cycles with plastic grooves to those with steel grooves is. There is a tendency for rope endurance to be higher with very soft polyurethanes than with the somewhat harder polyamides.

Multi-Layer Rope Spooling

A good overview of the problems occurring during multi-layer rope spooling onto a drum is given by Verreet (2003) and his paper also includes many useful recommendations. Correct spooling can only be achieved with a well designed rope drive with a grooved drum and a suitable wire rope.

Endurance tests with multi-layer rope spooling have been described by Briem (2002) and Weiskopf, Wehling and Vogel (2005). The number of bending cycles for a wire rope under multi-layer spooling N_{spool} is always much smaller than the number of bending cycles N for a wire rope running over a sheave or drum grooves. Assuming the spooling has been done correctly, the ratio of both of these numbers of bending cycles is the multi-spooling factor

$$f_{spool} = \frac{N_{spool}}{N}.$$

In the case of two-layer spooling, for the multi-spooling factor up to the first outer strand breakage Briem (2002) found f_{spool} and also for the rope discard factor $f_{\text{spool,A}}$ (for different ratios of the tensile forces S and the minimum rope breaking force F_{min})

S/F_{min}	0.33	0.14	0.091	0.067
f_{spool}	0.23	0.13	0.10	0.08
$f_{\text{spool,A}}$	0.25	0.12	0.08	0.06

Weiskopf, Wehling and Vogel (2005) found global multi-spooling factors between 0.02 and 0.09; mean 0.027. Their endurance tests were done by alternating the spooling of the wire rope from the second layer to the third layer and back again during each cycle. The results are shown in Fig. 3.50. From Weiskopf, Wehling and Vogel (2005) a regression calculation has been done to evaluate the number of bending cycles on the basis of the equation

$$\lg N_{\text{spool}} = a_0 + a_1 \cdot \lg \frac{S}{d^2} + a_2 \cdot \lg \frac{D}{d} + a_3 \cdot \lg \frac{D}{d} \cdot \lg \frac{S}{d^2} + a_4 \cdot \lg \frac{S_0}{d^2} + a_5 \cdot \lg \frac{z_1 + z_2}{2}.$$

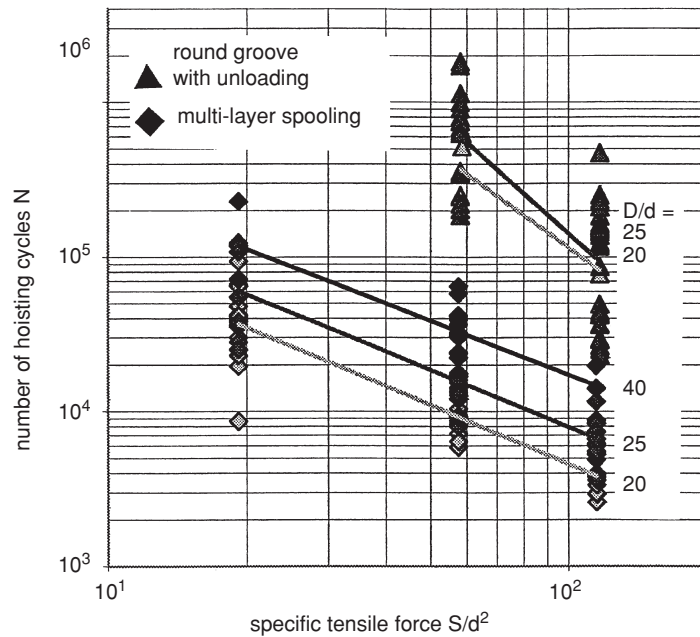


Fig. 3.50. Breaking number of rope bending cycles under multi-layer spooling and in steel sheaves with round grooves $r = 0.53d$, Weiskopf (2005)

In addition to the known symbols, S_0 is the tensile force with which the rope has been wound on the drum before starting the endurance test and z_1 and z_2 are the numbers of the two involved layers. The constants are still unpublished.

Rope Deflection

With Fig. 3.51, Müller (1961, 1966) published a first diagram describing the influence of the deflection angle on the breaking number of bending cycles leading up to the point where the rope breaks. For small deflection angles, the number of bending cycles is very high because in this region the radius of the rope curvature is large and therefore the bending stress is small. A dip in the number of bending cycles exists in this case at a deflection angle $\vartheta_{D,dip} \approx 20^\circ$. As is to be expected, that corresponds to the deflection angle $\Delta\vartheta = 19.7^\circ$ arising out of Häberle's (3.29). The reason for this dip in rope endurance is that the line pressure – which has been more or less concentrated on one point up to this angle – has reached its maximum and the radius of the rope curvature reaches that of half of the sheave diameter producing the full fluctuating bending stresses, see Sect. 3.1.6. The critical deflection angles from Müller's bending fatigue tests and Häberle's pressure measurement tests are nearly the same, but it should be noted here that the ropes used for the two groups of tests did not have the same construction.

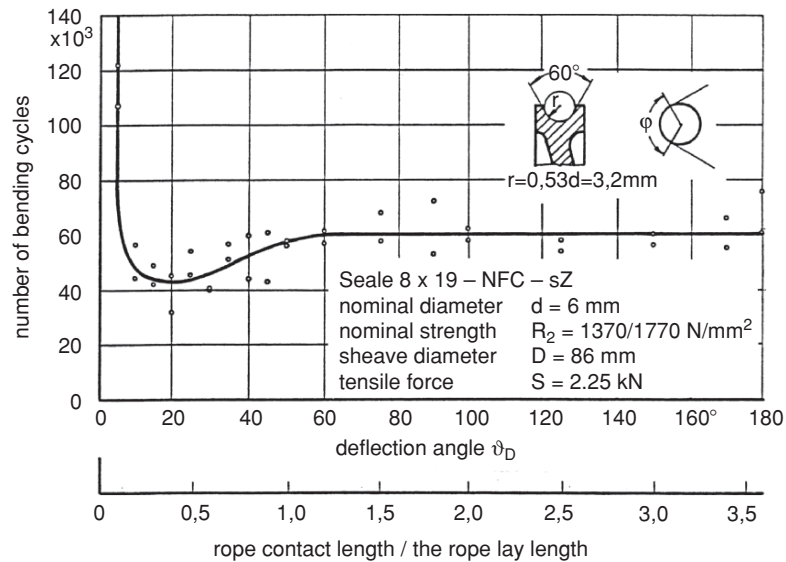


Fig. 3.51. Breaking number of bending cycles of a rope for different deflection angles, Müller (1961)

For bigger deflection angles over about 60° , the number of bending cycles is more or less constant, Fig. 3.51. On looking at the second abscissa in Fig. 3.51, the ratio of the rope contact length and the rope lay length, it is however imaginable that there may be a slight influence on the number of bending cycles. Donandt supposed such an influence to exist as Woernle (1934) reported. Jurk (1973) found such an influence for this ratio which was only slight on the number of bending cycles but great on the number of wire breaks.

Side Deflection of the Rope

It has been well-known for a long time that the side deflection of the rope from the sheave groove reduces the number of rope bending cycles. Therefore the technical rules prescribe the limits of 4° for the angle to be allowed for the stranded ropes and 1.5° for non-rotating and low-rotating ropes.

Matthias (1966, 1970) has described the contact form of a side-deflected wire rope on the groove flank. The first systematic bending fatigue tests with a different side deflection for the rope were carried out by Neumann (1987). Figure 3.52 shows as his results the discard number of bending cycles for side deflection angles between $\vartheta = \psi = 0$ and 4° .

Schönherr (2005) researched the influence of the side deflection on the breaking number of bending cycles between the side deflection angle $\psi = 0^\circ$ and 7° in a great number of bending fatigue tests. The tests were carried

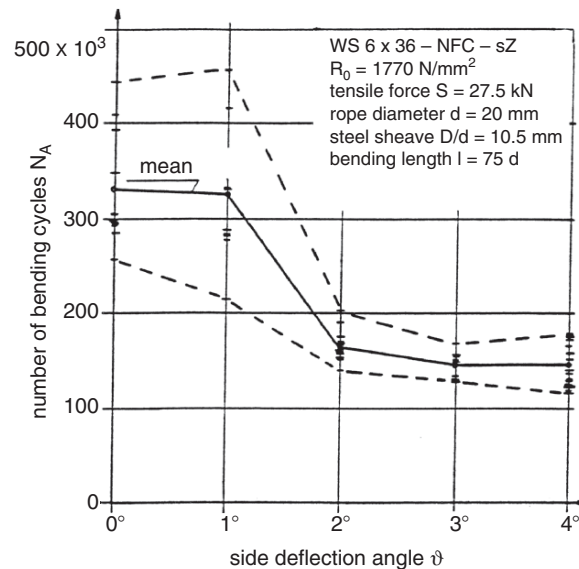


Fig. 3.52. Discarding number of rope bending cycles under a different side deflection of the rope, Neumann (1987)

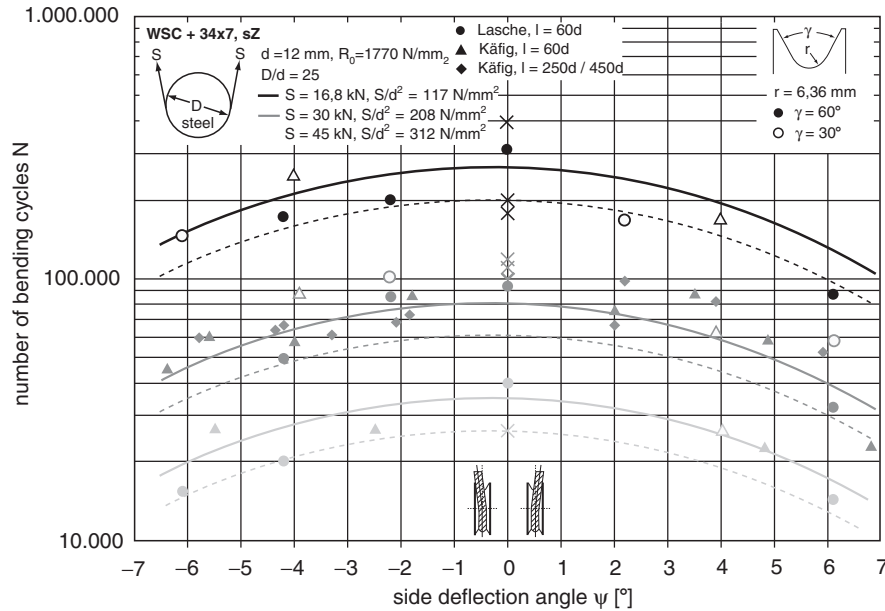


Fig. 3.53. Breaking number of bending cycles for a multi-strand rope, $D/d = 25$, Schönherr (2005)

out on six ordinary lay ropes and multi-strand ropes. The diameter ratio was $D/d = 12.5$ and 25 and the specific tensile force varied between $S/d^2 = 58\text{--}312\text{ N/mm}^2$. As an example, Fig. 3.53 shows the breaking number of bending cycles $\bar{N}$ and N_{10} of a multi-strand rope, with the side deflection in the same (+ sign) or in the opposite (− sign) direction to the lay direction of the outside strands. All the results show that the groove opening angle between $\gamma = 30^\circ$ and 60° has no influence on the number of rope bending cycles.

Schönherr (2005) evaluated her test results by regression calculation. From this, the ratio of the numbers of bending cycles with side deflection to those ropes without side deflection is

$$\frac{N_\psi}{N_0} = 1 - \left(0.00863 + 0.00243 \cdot \frac{D}{d} \right) \cdot \psi - 0.00103 \cdot \psi^2. \quad (3.60)$$

The side deflection angle ψ has to be set in degree, all with a positive sign. That means that the influence of the direction of the side deflection in relation to the lay direction of the outside strands has not been considered in (3.60). As is to be seen in Fig. 3.53, this expected influence is very small and, generally speaking, even undetectable. Neumann's (1987) bending cycles ratios N_ψ/N_0 tend to be smaller than those resulting from Schönherr's equation (3.60), but because of the relatively large standard deviation $s = 0.185$, it lies in the confidence interval.

Equation (3.60) is valid under test conditions. However, there is no great error to be expected if it is used for lang lay ropes, which were not tested, and the discarding numbers of bending cycles.

Because of the friction force acting on the outside, the side deflection leads to a turn of the rope depending on the torsion stiffness and the free length of the rope. Neumann (1987) and Schönherr (2005) observed such rope turns. Oplatka (1990) demonstrated the rope turns in an impressive video.

3.2.4 Reverse Bending

Müller (1961) and Jehmlich (1985) established that about the half the number of bending cycles will be reached for reverse rather than for simple rope bending. Newer tests from Feyrer and Jahne (1991a) show that this result is only valid for a small range of test conditions. These tests were carried out on bending fatigue machines with a test sheave arrangement as shown in Fig. 3.29.

Figure 3.54 shows the ratio of the numbers of reverse and simple bending cycles up to breakage taken from these tests, as well as from tests carried out by Müller (1961) and Jehmlich (1985). However, it has been taken into consideration that Jehmlich's definition of a bending cycle was different from the standard one. From the regression calculation, the number of reverse bending cycles is

$$N_{\text{rev}} = a_0 \cdot N_{\text{sim}}^{a_1} \cdot \left(\frac{D}{d}\right)^{a_2}. \quad (3.61)$$

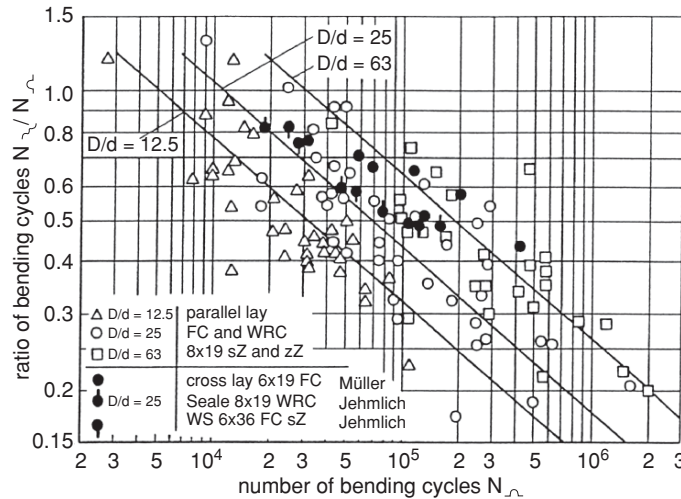


Fig. 3.54. Ratio of breaking number of reverse and simple bending cycles $N_{\text{rev}}/N_{\text{sim}}$, Feyrer and Jahne (1991)

The constants a_i for the numbers of bending cycles up to discard or breakage are listed in Table 3.16 (Sect. 3.4.3). The standard deviation for the ratio of the numbers of bending cycles $N_{\text{rev}}/N_{\text{sim}}$ is $\lg s = 0.132$ and $\lg s_A = 0.084$.

The regression equation with its constants relates to the 12 parallel strand wire ropes 8×19 in ordinary lay and lang lay with fibre cores FC and steel cores WRC used in the tests. The ratio of the number of reverse and simple bending cycles is a little greater for the six strand ropes used in the tests carried out Müller and Jehmlich. The ratio of the numbers of reverse and simple bending cycles $N_{\text{rev}}/N_{\text{sim}}$ increases with the decreasing number of simple bending cycles and with the increasing diameter ratio D/d .

For very small numbers of simple bending cycles, the ratio $N_{\text{rev}}/N_{\text{sim}}$ is partly even greater than 1. That means that the number of reverse bending cycles is greater than those of simple bending cycles. This surprising result is caused mainly by the standard definition of a bending cycle. According to these standard definitions, the reverse bending cycle (bent – straight – reverse bent) is in reality a half-stress cycle. It is only after two reverse bending cycles that the wire rope will have returned to the same condition as at the beginning. In contrast, the simple bending cycle (bent – straight – bent) is a whole stress cycle. Furthermore, the fluctuating pressure works – with the same numbers of cycles in both cases – on one side of the rope in simple bending cycles and half and half on two rope sides in reverse bending cycles.

By definition, the reverse bending cycle means that the axes of the two sheaves involved are parallel. Research into bending fatigue tests with wire ropes running over several sheaves with axes which are not parallel has not been carried out up to now.

3.2.5 Fluctuating Tension and Bending

Two types of fluctuating tension and bending exist: independent and combined. In Fig. 3.55, the course of the longitudinal wire stress with independent

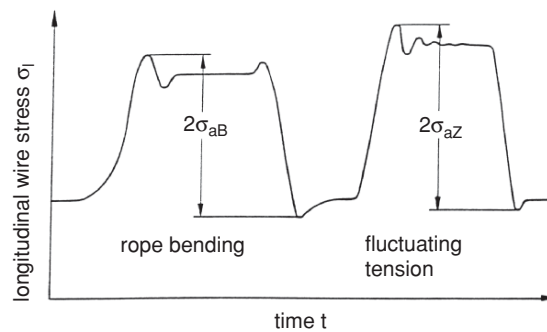


Fig. 3.55. Course of longitudinal stress under independent bending and fluctuating tension

tension and bending is shown. In this type, the stress amplitudes of the two loadings – the fluctuating tension and the bending – do not influence each other. The rope endurance can be calculated for the independent loadings and summarized with the help of the Palmgren–Miner-Rule. This independent tension and bending – which is very rare – will not be considered in the following passage.

Combined fluctuating tension and bending causes increased amplitude in the longitudinal wire stress. The course of the longitudinal wire stress is shown in Fig. 3.56. In the case of combined loadings, the endurance could be evaluated by means of tests. However, because of the numerous possible combinations of tensions and bending cycles, it is necessary to find a method which is a combination of the results of bending tests and a theoretical back-up. From the stresses of the two combined loadings, an equivalent tensile stress σ_{equ} or equivalent tensile force S_{equ}

$$\sigma_{\text{equ}} = f_{S5} \cdot \sigma_z \quad \text{or} \quad S_{\text{equ}} = f_{S5} \cdot S. \quad (3.62)$$

will be derived, with that the number of bending cycles N_{comb} for combined tension and bending can be calculated using (3.55), Feyrer (1993).

As can be seen in Figs. 3.57 and 3.58, the amplitude and the middle of the fluctuating longitudinal stress both increase under combined tension and bending. Based on Sects. 2.1 and 3.1 (and using in particular the work of

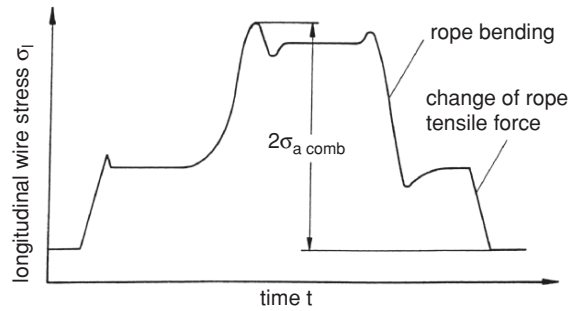


Fig. 3.56. Course of longitudinal stress under combined fluctuating tension and bending

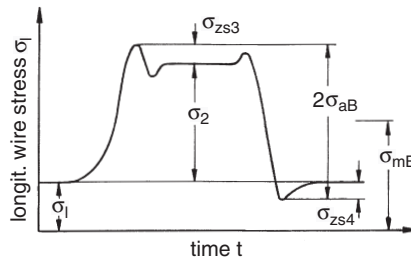


Fig. 3.57. Longitudinal wire stresses as the rope runs over a sheave, Schiffner (1986)

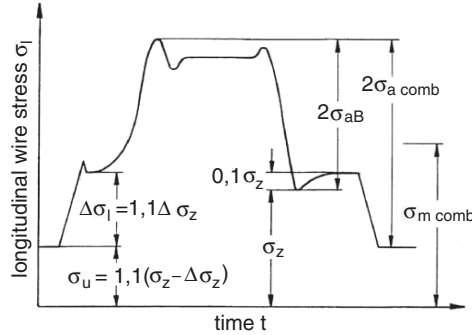


Fig. 3.58. Longitudinal wire stresses under combined fluctuating tension and rope bending

Andorfer (1983), Schiffner (1986) and Wang (1989)), these stresses will be derived with some simplifications for the most interesting wires on the bottom of the sheave. By bending the rope – taking the bending stress, the ovalisation stress and the secondary tensile stress into consideration – the amplitude of the longitudinal stress is

$$\sigma_{aB} = 72,500 \cdot \frac{\delta}{d} \cdot \frac{d}{D} + 300 \cdot \frac{\delta}{d} + 0.1 \cdot \sigma_z \quad (3.63)$$

and the middle stress

$$\sigma_{mB} = \sigma_z + \sigma_{aB}. \quad (3.64)$$

For the combined fluctuating tension and bending, the amplitude of the longitudinal wire stress is

$$\sigma_{a,comb} = \sigma_{aB} + \frac{1.1 \cdot \Delta\sigma_z - \sigma_z}{2} \quad (3.65)$$

and the middle stress is

$$\sigma_{m,comb} = 1.1 \cdot (\sigma_z - \Delta\sigma_z) + \sigma_{a,comb}. \quad (3.66)$$

The middle stresses σ_{mB} and $\sigma_{m,comb}$ are not very different and do not have any great influence on the endurance of the rope. The main influence on rope endurance under combined loading is rather the ratio of the stress amplitudes $\sigma_{a,comb}/\sigma_{aB}$. The factor f_{S5} is therefore based on this ratio supplemented by correcting factor k_S . This correcting factor k_S will be found by comparing bending fatigue tests. With it, the equivalent force factor is

$$f_{S5} = 1 + k_S \cdot \frac{1.1 \cdot \Delta\sigma_z - 0.1 \cdot \sigma_z}{2 \cdot \sigma_{aB}}. \quad (3.67)$$

Bending fatigue tests have been carried out on two wire ropes, Filler 8×19 -SFC-sZ and Warrington 8×19 -IWRC-sZ. The diameter ratios of sheave and rope were $D/d = 12.5, 25$ and 63 . In one type of test, the wire rope was bent, stressed with tensile force S , and reduced by ΔS before and after

bending. In the other type of test constant tensile forces (forces S_{equ}) were used. Correcting factor k_S was found by comparing tensile forces S and ΔS , respectively, S_{equ} and for this the number of bending cycles was the same in the two types of tests. The mean of the correcting factors found – valid for both the breaking and discarding number of bending cycles – has been found by regression calculation to be

$$k_S = 1.31 - 0.0014 \cdot \Delta\sigma_z. \quad (3.68)$$

At $s = 0.19$, the standard deviation for the correcting factor is relatively large. In the regression calculation, some comparative tests carried out by DEMAG and the R: STAHL with Warr.-Seale ropes in their rope hoists have also been included. Dudde (1991) found that the equation can also be used for lang lay ropes for the breaking number of bending cycles.

With (3.63), (3.65), (3.67) and (3.68) and the relation $\sigma_z = a S/d^2$, the equivalent force factor is

$$f_{S5} = 1 + \frac{\left(1.31 - 0.0014 \cdot a \cdot \frac{\Delta S}{d^2}\right) \cdot \left(1.1 \cdot \frac{\Delta S}{d^2} - 0.1 \cdot \frac{S}{d^2}\right) \cdot a}{145,000 \cdot \frac{\delta}{d} \cdot \frac{d}{D} + 600 \frac{\delta}{d} + 0.2 \cdot a \cdot \frac{S}{d^2}}. \quad (3.69)$$

The constants for (3.69) are listed in Table 3.13 (Sect. 3.4.2).

An alternative method of evaluating the number of bending cycles under combined tension and bending based on an equivalent diameter ratio $(D/d)_{\text{equ}}$ instead of an equivalent tensile force is described in Feyrer (1993). However, this method does not provide a better result.

3.2.6 Palmgren–Miner Rule

According to the damage accumulation hypothesis of Palmgren (1924) and Miner (1945), the endurance of specimen ropes under different loads can be calculated using the basic equation

$$\sum \frac{n_i}{N_i} = 1. \quad (3.70)$$

where n_i is the number of cycles under the load i and N_i is the endurance number under the load i .

Dragone (1973) and Rossetti (1975) were the first to carry out bending fatigue tests to check whether the Palmgren–Miner Rule could be used for running ropes. From the results of their bending fatigue tests with different tensile forces, they found that the Palmgren–Miner Rule is fulfilled quite well. Ciuffi (1979) reported about a block load programme that had been done in various institutes. From these bending tests, they found damage sums between 0.8 and 1.2. Wohlrab and Jehmlich (1980) calculated mean damage sums of 0.96 up to discarding and of 0.91 up to breakage of the ropes with only a small standard deviation. All this research shows that the Palmgren–Miner-Rule is valid for running ropes.

3.2.7 Limiting Factors

Donandt-Force

If a certain limiting tensile force is exceeded in a series of wire rope bending fatigue tests, the number of bending cycles drops abruptly. This force, which is the absolute limit of the usable tensile force, is called Donandt force. Schmidt (1965) was the first to do research on this force after taking up an idea coming originally from Donandt. Above the Donandt force, an increasing part of the wires cross-section exceeds the yielding strength which then causes an abrupt breakdown in the number of bending cycles.

In Fig. 3.33, the beginning of the abrupt breakdown of the number of bending cycles (and with that the Donandt force) can clearly be seen. The lines for the usable region of rope endurance have been taken from test results found in the endurance regression calculation. In addition lines have also been drawn for the test results in the yielding region. The intersection of the two lines is known as the Donandt force.

In the case of simple bending, the Donandt forces $S_{D,\text{sim}}$ for ropes of the same construction have been evaluated by regression calculation with the basis equation

$$S_{D,\text{sim}} = q'_0 \cdot F_e + q'_1 \cdot \frac{d}{D} \cdot F_e.$$

To be on the safe side, the evaluated rope breaking force F_e (which is no longer standardised) is evaluated as the sum of the wire breaking forces and can be replaced by the smaller calculated breaking force F_c . Normally, the minimum rope breaking force $F_{\min}$ has to be used today. With the spinning loss factor k , the minimum breaking force is

$$F_{\min} = k \cdot F_c.$$

With that, the constants q related to the minimum breaking force are

$$q = q'/k,$$

which means the Donandt force is then

$$S_{D,\text{sim}} = q_0 \cdot F_{\min} + q_1 \cdot \frac{d}{D} \cdot F_{\min}. \quad (3.71)$$

In the case of reverse bending cycles, the Donandt force is of course smaller than that of the simple bending cycles. The test results have also been evaluated with the basis (3.71). A more or less constant difference has been found by comparing the constants q_i for simple and reverse bending. This means that the Donandt force for reverse bending is, Feyrer and Jahne (1991a)

$$S_{D,\text{rev}} = (q_0 - 0.035) \cdot F_{\min} + (q_1 - 0.25) \cdot \frac{d}{D} \cdot F_{\min}. \quad (3.72)$$

The constants q_i for the different rope constructions are to be found in Table 3.17 (Sect. 3.4.5).

In reality, the transition of the two straight endurance lines marking the Donandt force is rounded. If the number of bending cycles and the tensile force are drawn in a diagram in linear and not in logarithm scale as done by Nabijou and Hobbs (1994) the Donandt force can hardly be detected.

Discard Limiting Force

The discard limiting force S_G is the rope tensile force at which, with sufficient probability, a number of wire breaks of at least $B_{A30} = B_{A30 \min}$ has to be expected. B_{A30} means the discard number of visible wire breaks on a rope reference length of thirty times the rope diameter $L = 30d$. The discard limiting force can be calculated using an equation which is regrouped from (3.83) in Sect. 3.2.9 for a given discarding number of wire breaks. This regrouped (3.94) is presented in Sect. 3.4.5.

Optimal Rope Diameter

For a given tensile force S and a given sheave diameter D , the tensile stress for small rope diameters is high and the bending stress low. In the case of large rope diameters this is vice versa. High tensile stresses as well as high bending stresses reduce the number of bending cycles. Consequently the greatest possible number of bending cycles can be expected when both tensile stress and bending stress are not too high and when they are in a reasonable relation to each other. The rope diameter with which the greatest number of bending cycles can be expected is called the optimal rope diameter d_{opt} .

Müller (1961) had already drawn the attention to the existence of an optimal rope diameter. When preparing the standard DIN 15020, on the basis of his bending tests he proposed using coefficients $c = d/\sqrt{S}$ and diameter ratios D/d for the different groups of rope drives, from which a relatively optimal rope diameter would result. Clement (1981) also derived an optimal rope diameter from his equation (3.50) which was developed to determine the endurance of a rope.

Here the optimal rope diameter is derived on the basis of the rope endurance (3.55). The number of bending cycles N or $\lg N$ reaches a maximum for

$$\frac{\partial \lg N}{\partial \lg d} = 0.$$

The small influence of the bending length can be neglected. Then the optimal rope diameter for simple bending or for combined fluctuating tension and bending (with the actual valid tensile force $S = S_{\text{sim}}$ or $S = S_{\text{equ}}$) is

$$\lg d_{\text{opt}} = \frac{2 \cdot b_1 + b_2 + 0.32}{4 \cdot b_4} - \frac{0.4}{4} \cdot \lg \frac{R_0}{1,770} + \frac{\lg D}{2} + \frac{\lg S}{4}. \quad (3.55a)$$

Deriving the combined (3.55) and (3.61), the optimal rope diameter for reverse bending is

$$\lg d_{\text{opt}} = \frac{2 \cdot a_1 b_1 + a_2}{4 \cdot a_1 \cdot b_4} + \frac{b_2 + 0.32}{4 \cdot b_4} - \frac{0.4}{4} \lg \frac{R_0}{1,770} + \frac{\lg D}{2} + \frac{\lg S}{4}. \quad (3.55b)$$

The constants b_i are listed in Table 3.14a and constants a_i in Table 3.16, Sec. 3.4. The constants c_0 can be added to this for standardised wire ropes and standardised nominal rope strengths. Then the optimal rope diameter is

$$\lg d_{\text{opt}} = \lg c_0 + \frac{\lg D}{2} + \frac{\lg S}{4}.$$

or

$$d_{\text{opt}} = c_0 \cdot \sqrt{D \cdot \sqrt{S}}. \quad (3.73)$$

The constant c_0 is listed in Table 3.19 (Sect. 3.4.5).

In Fig. 3.59, the number of bending cycles of a rope for the tensile force $S = 10 \text{ kN}$ and different diameters D are drawn over the rope diameter d . The optimal rope diameter is shown as a broken line. In reality, the optimal diameter is an economic limit. If a rope diameter larger than the optimal one is used, the disadvantages are that the rope endurance will be lower and the costs higher. The maximum of the number of bending cycles is rather flat which means that there will only be a small change in the number of bending cycles when a minor deviation from the optimal rope diameter occurs. Therefore, the rope diameter can be smaller than the optimal rope diameter with a reasonable percentage without too much of the rope endurance being lost.

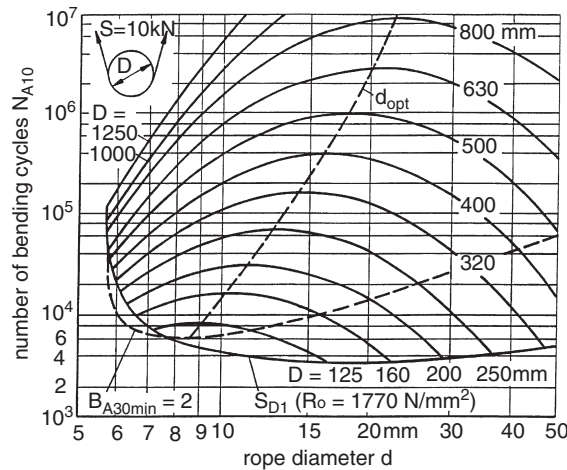


Fig. 3.59. Number of bending cycles N_{A10} of Warrington or Filler rope 8×19 -FC-sZ as function of the rope diameter

3.2.8 Ropes during Bendings

Residual Breaking Force

In the course of time, wire ropes running over sheaves will suffer from increasing damage. Wire breaks and wear which occur reduce the residual breaking force of the wire rope. Woernle (1929) systematically measured the breaking force of wire ropes after different numbers of cycles in bending fatigue tests. His results are drawn in Fig. 3.60 for a cross-lay rope in a series of test with three tensile forces. It is to be seen that the reduction in the wire rope breaking force is weak at first and then gets stronger towards the end of the rope's life. In many cases, the breaking force even increases a little during the first third of the rope's life. As shown in Fig. 3.60, comparable results have been found by Davidson (1955), Arnold and Hackenberg (1971) and Rossetti (1989).

Very high costs are involved in carrying out such research on residual breaking forces as presented here. Many of the bending fatigue tests to find the course of the residual breaking force have to be done for only one rope under only one loading. The bending fatigue tests where the wire rope runs over several sheaves have the advantage that the wire rope being tested has zones with different numbers of bending cycles. Jahne (1992) has carried out such bending fatigue tests with parallel-lay ropes using the test sheave arrangement shown in Fig. 3.29. From these tests, the wire rope zones – loaded by different numbers of bending cycles – has been used to find out the residual breaking force, Feyrer and Gu (1990b). The results which were evaluated by a first regression calculation are shown in Fig. 3.61.

From a newer regression calculation the relative number of bending cycles for different limits γ is

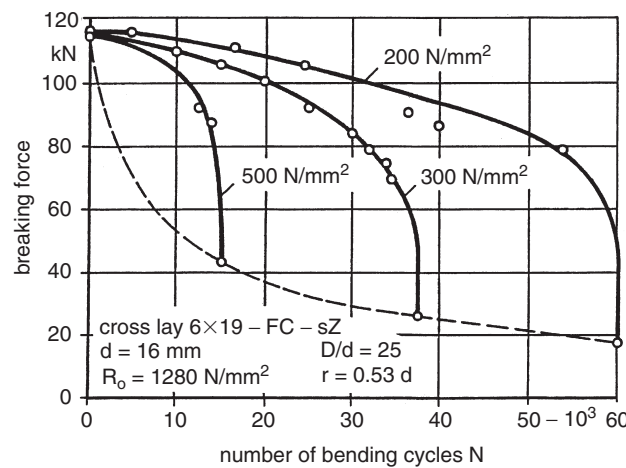


Fig. 3.60. Residual rope breaking force, Woernle (1929)

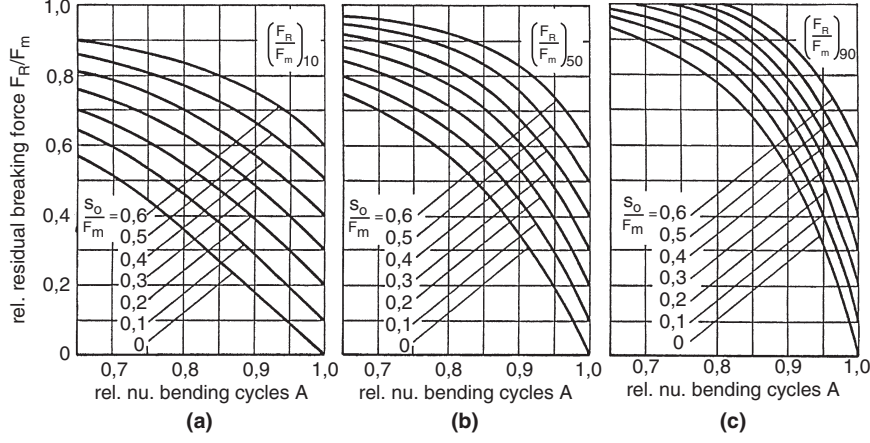


Fig. 3.61. Relative residual rope breaking force, Feyrer and Gu (1990b)

$$A_\gamma = \left(\frac{1 - F_R/F_m}{1 - S_0/F_m} \right)^{a_\gamma}. \quad (3.74)$$

The wire rope force S_0 is the tensile force that is used during the bending fatigue test. In practice the measured rope breaking force F_m , which is unknown, can be replaced on the safe side by the calculated breaking force $F_m \approx F_c = F_{\min}/k$ (minimum breaking force/spinning loss factor). The exponent is $a_\gamma = 0.203 \cdot 1.664^{a_\gamma}$. For the relative number of bending cycles with the limits $\gamma = 10, 50$ and 90% , the exponents are $a_{10} = 0.106$; $a_{50} = 0.203$ and $a_{90} = 0.39$.

Parallel bearing wire ropes (redundant bearing ropes) have the advantage that if one of these ropes breaks, the other wire ropes can survive. The probability that the other rope or the other ropes do not break depends mainly on the relative number of bending cycles A_γ at which the residual breaking force F_R is still the same as the impact force S_S . This impact force occurs in the survival ropes when one of the ropes breaks, Feyrer (1990c).

With equation (3.74) and equations for the impact force and the rope endurance the survive respectively failure probability can be calculated, see Fig. 3.75, Sect. 3.3.

Rope Diameter Reduction

During the life time of a wire rope running over sheaves, the rope diameter will be continually reduced. The relative rope diameters measured in the course of bending fatigue tests with different tensile forces are shown in Fig. 3.62 for a rope with fibre core and in Fig. 3.63 for a rope with steel core. The wire rope diameter in these figures is the rope diameter measured under the tensile force used for the bending fatigue tests.

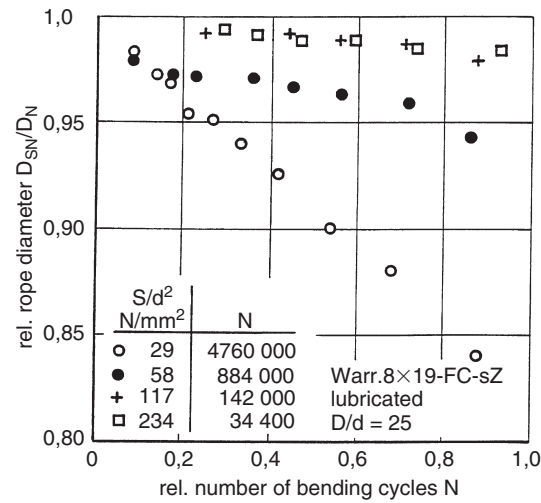


Fig. 3.62. Diameter reduction of a wire rope with fibre core during bending fatigue tests

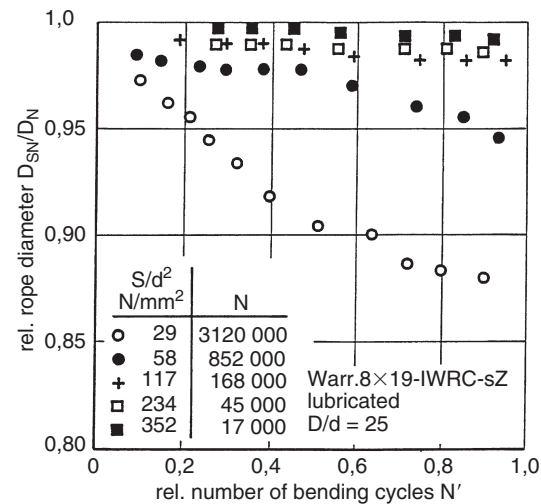


Fig. 3.63. Diameter reduction of a wire rope with steel core during bending fatigue tests

Both of these figures show that a large diameter reduction of about 10% only occurs for the very small specific tensile force $S/d^2 = 29 N/mm^2$ with a very high rope endurance. In practical applications, diameter reduction can be much greater with outside wear. Such a rope diameter loss can cause a severe reduction of the rope breaking force. Therefore, in rope inspections the diameter of the wire rope has to be looked at as a possible discard criterion.

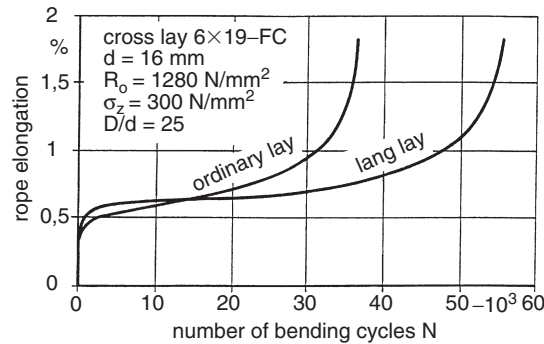


Fig. 3.64. Rope elongation during bending fatigue tests, Woernle (1929)

Wire Rope Elongation

In the course of bending fatigue tests, an elongation of the rope occurs. This elongation has been measured by Woernle (1929), Hankus (1985), Winkler (1988) and others. The historical Fig. 3.64 shows the typical course of rope elongation. During the first bending cycles, the wire rope will be strongly elongated, then over a longer period the rope elongation is small and only close to the end of its working life will the rope be progressively elongated again. In principle, this progressive increase in the elongation shows the imminent breakage of the rope. However, for a variety of reasons, the progressive elongation of the rope often cannot be detected in practice.

3.2.9 Number of Wire Breaks

The endurance of wire ropes running over sheaves is always limited. Fluctuating stresses and increasing wear lead to an increasing number of wire breaks. The number of wire breaks referring to a pre-defined length, the rope reference length is the most important discard criteria for wire ropes.

Growth of the Number of Wire Breaks

The first wire break occurs on a wire rope running over sheaves after several bending cycles. After the first break, the higher the tensile stress σ_z and the smaller the diameter ratio D/d is, the faster further wire breaks occur. An early diagram, Fig. 3.65, shows the numbers of wire breaks observed in bending fatigue tests and their mean curves.

In single bending cycle fatigue tests with relatively small bending lengths, the increase in the number of wire breaks is not uniform. The observed number of outside wire breaks of a Filler rope is shown in Fig. 3.66. Other examples of the increase in outer wire breaks can be found for different tensile stresses and diameter ratios D/d , in Woernle (1929, 1931), Rossetti and Thaon (1977),

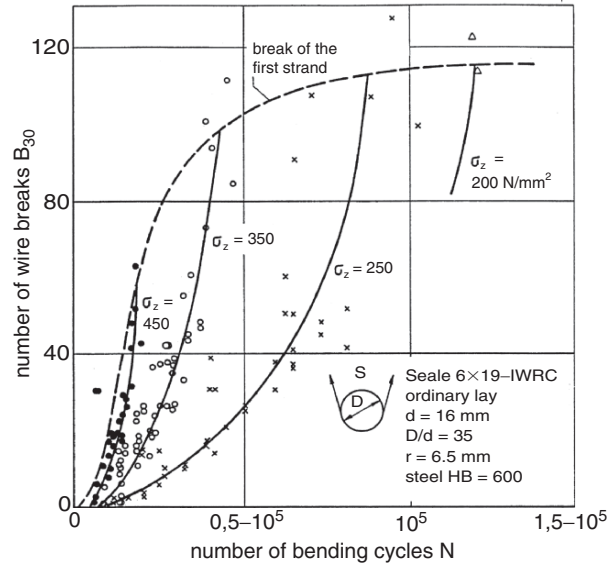


Fig. 3.65. Growth of the number of wire breaks of a Seale rope, Müller (1966)

Gräbner and Schmidt (1979), and Feyrer (1983c) and for outer and inner wire breaks in Oplatka (1969), Babel (1980) and Jahne (1992).

In wire ropes with longer bending zones, the weakening of the rope breaking force is related to the accumulation of wire breaks in critical rope lengths. The critical rope length depends on the rope length where a broken wire once more takes the load more or less completely. Different critical rope lengths occur depending on the kind of load and the construction of the wire rope, Woernle (1929), Herbst (1934), Costello (1997) and Raoof (1992). Based on these findings, the reference rope lengths where the number of wire breaks is important have been standardised. For stranded ropes, the reference lengths are normally $L = 30d$ and $6d$ and for spiral ropes $L = 500d$, $200d$ and $6d$.

Figure 3.66 shows the numbers of wire breaks $B_{6\max}$ and B_{60} in relation to the rope reference length $L = 6d$ and the bending length $l = 2 \times 30d$. The numbers of wire breaks $B_{6\max}$ are the maximum number of wire breaks to be found on the rope bending length $l = 2 \times 30d$ or $l = 60d$. For a big diameter ratio $D/d = 63$, the wire breaks occur at very early stage, i.e. 10 or 20% of the rope endurance (relative number of bending cycles $N' = 10$ or 20%). In contrast, for a small diameter ratio such as $D/d = 10$, the first wire breaks only occur just before the end of the rope life.

For greater bending lengths l , the increase in the number of wire breaks B produces a smoother curve. Such a curve is given in Fig. 3.67 for the growth of the total number of wire breaks on a bending length $l = 360d$. Up to about

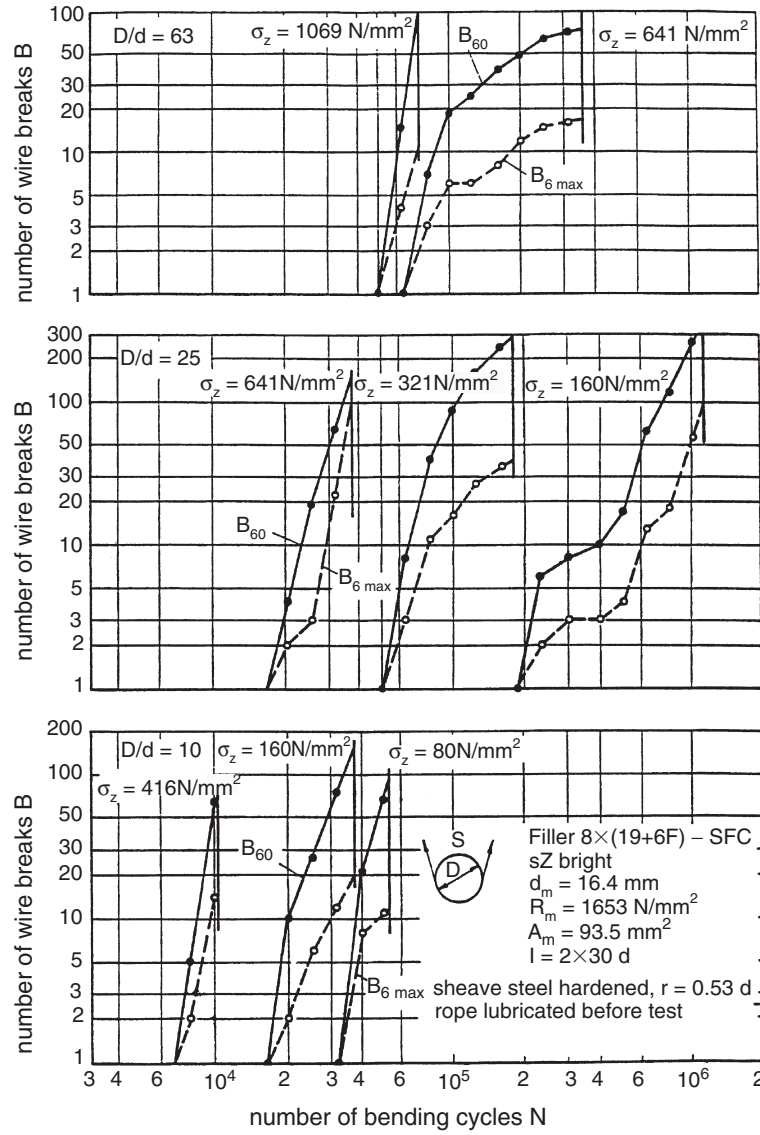


Fig. 3.66. Growth of the number of wire breaks of a Filler rope under different loads, Feyrer (1983c)

the total number of wire breaks $B_{360} = 15$ (whereby, on the reference length $L = 30d$, the mean number of wire breaks is $\bar{B}_{30} = 1.25$) the number of wire breaks increase exponentially with the number of bending cycles

$$\bar{B} = a_0 \cdot e^{a_1 \cdot N}. \quad (3.75)$$

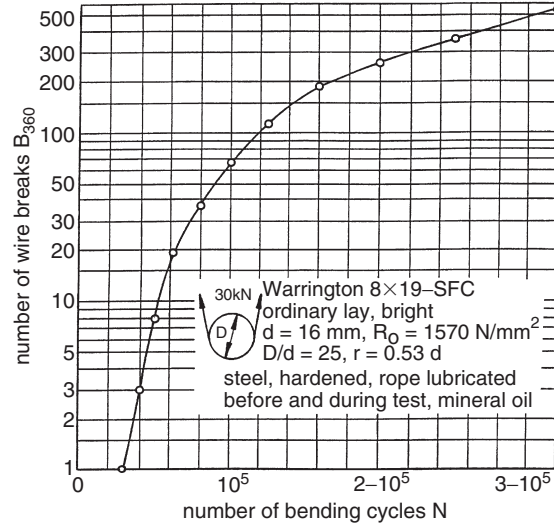


Fig. 3.67. Number of wire breaks B_{360} on a bending length $l = 8 \times 2 \times 22.5 d = 360d$, Feyrer (1983b)

This function of the increase in the number of wire breaks has also been found during the inspection of mining shaft hoists by Daeves and Linz (1941) and Ulrich (1980) and in ropeways by Beck (1992). The fact that this corresponds to (3.75) is possible as for safety reasons the wire ropes in these installations have to be replaced at an early stage with a mean number of wire breaks of about $\bar{N}_{30} = 1$ to 2.

Gräbner (1968) found that the increase in the number of wire breaks could be divided into three phases. In phase I, the total number or, out of this, a mean number of wire breaks on a reference length L as a section of the rope bending length l increases exponentially. In phase II, the increase in the number of wire breaks is reduced. Then in phase III, the number of wire breaks increase again, but progressively. The findings of bending fatigue tests with relatively large rope bending lengths, Ren (1996, 1998) have frequently confirmed this three-phase model.

The same total number of wire breaks B_{360} as in Fig. 3.67 for the bending length $l = 16 \times 22.5 d = 360d$ are now shown in Fig. 3.68 in a double logarithm scaled diagram. Supplementary to this, the maximum numbers of wire breaks counted $B_{22.5, \max}$ on a bending length $L = 22.5d$ have been introduced in Fig. 3.68. The increase in the maximum number of wire breaks is described very well by a straight line in the diagram 3.68, that is, by the equation

$$B_{L, \max} = c_0 \cdot N^{c_1}. \quad (3.76)$$

In her bending fatigue tests with large bending lengths, Jahne (1992) confirmed that the increase in the mean number of wire breaks in the first phase

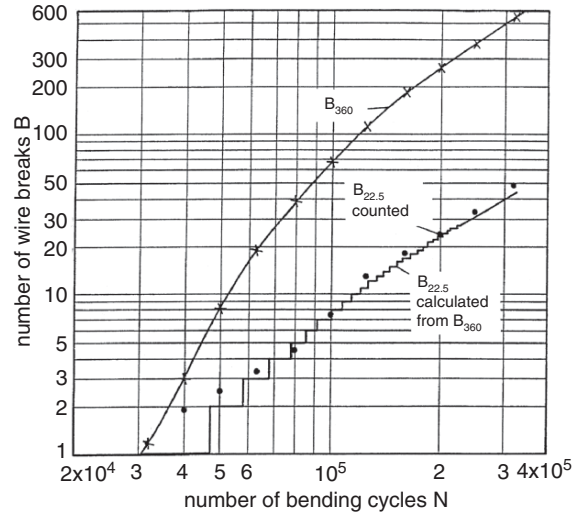


Fig. 3.68. Number of wire breaks B_{360} and the maximum number of wire breaks $B_{22.5, \max}$, Feyrer (1983c)

can normally be described by (3.75) and the maximum number of wire breaks by (3.76).

In Fig. 3.68, there is also a line going up in steps. This stepped line has been calculated with (3.77) from the total number of wire breaks, respectively, the mean number of wire breaks on a reference length. Equation (3.77) requires a Poisson distribution for the wire breaks. Figure 3.68 shows that the maximum number of wire breaks calculated in this manner comes very close to the findings of the bending fatigue tests.

Distribution of Wire Breaks on a Rope

If it is equally probable that all sections of a rope bending length may get the next wire break, so the theoretical distribution of the number of wire breaks in the sections of the rope is the Poisson distribution. This is true if the wire breaks occur accidentally in the sections as the numbers do when throwing dice. The following conditions have to be fulfilled for the validity of the Poisson distribution, Feyrer (1983b):

- The wire breaks have to occur independently of each other
- The whole bending length should be divided into a lot of sections, $l/L \geq 10$
- The probability that a wire break is found in a unit length should be low, which means that the wire break rate $\lambda = B_l/l$ should be low.
- The reference length L should be greater than the length unit of about $\Delta l = d$ or $d/2$ and the mean number of wire breaks on the sections $\bar{B}_L = \lambda \cdot L$ is finite. Of course, this latter condition is always fulfilled.

Here

B_L is the number of wire breaks on the reference length L

B_l , the number of wire breaks on the fatigue stressed rope length l (bending length l)

$\bar{B}_L = B_l \cdot L/l$, the mean number of wire breaks on the reference length L

$B_{L,\max}$, the maximum number of wire breaks on a reference length L

B_{AL} , the discarding number of wire breaks on a reference length L

l , the whole bending length (fatigue stressed length)

L , the reference length

Δl , the step length and

z is the number of steps.

The probability w of the Poisson distribution that the number of wire breaks $B_L = 0, 1, 2, 3$, etc. exists on the reference lengths L is

$$w = \frac{\bar{B}_L^{B_L}}{B_L!} \cdot e^{-\bar{B}_L}. \quad (3.77)$$

The variance is

$$V = \sigma^2 = \bar{B}_L. \quad (3.78)$$

The probability that the number of wire breaks is smaller or equal to B_L is

$$p(B_L) = \sum_0^{B_L} \frac{\bar{B}_L^{B_L}}{B_L!} \cdot e^{-\bar{B}_L}. \quad (3.79)$$

The whole bending length l of the rope is divided into sections with the reference length L as shown in Fig. 3.69. If, for example, a wire rope was stressed by bending cycles in the same way over its complete length, this does not mean that course there would be the same number of wire breaks in every section of the rope. This is to be seen in Fig. 3.70 where the observed numbers of wire breaks are shown in 2×5 sections. The whole bending length is $l = 60d$ and the section length is $L = 6d$. The first line shows the number of wire breaks B_L after the number of bending cycles $N = 32,000$, the second after 40,000 and the third after 50,000. Thus the relative life of the rope is $N' = 58.3\%$ and so on.

The maximum number of wire breaks $B_{L,\max}$ normally does not occur in one of the ten sections. The real maximum number of wire breaks is rather to be found in a length L in between. This can be ascertained by moving a window with an opening of the length L in steps Δl over the whole bending length and by counting the wire breaks in the window at every step (window

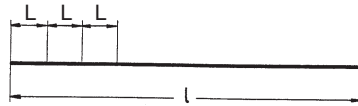


Fig. 3.69. Reference lengths L on the bending length l

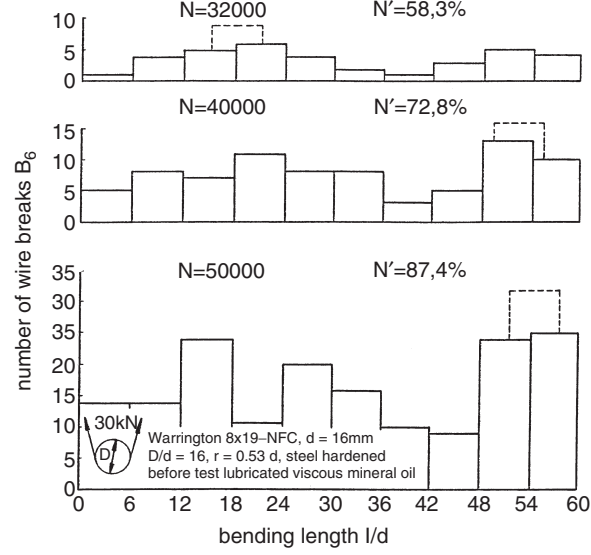


Fig. 3.70. Number of wire breaks in the reference lengths L

method). The result of this counting is shown in Fig. 3.70 by broken lines, one for the left five sections and one for the right.

The number of steps is

$$z = \frac{l - L}{\Delta l} + 1. \quad (3.80)$$

or – for wire ropes which are not too short – it can be simplified by considering the rope length as a ring and then the number of steps is

$$z = \frac{l}{\Delta l}. \quad (3.81)$$

As wire breaks are relatively rare, too small a length should not be chosen for the step Δl . A realistic length is $\Delta l = d$ for visible outside wire breaks and $\Delta l = 6d$ for magnetic inspection.

The probable maximum number of wire breaks $B_{L,\max}$ on the reference length L is given by

$$z \cdot (1 - p(B_{L,\max} - 1)) \geq 1 > z \cdot (1 - p(B_{L,\max})). \quad (3.82)$$

For the Poisson distribution is

$$z \cdot \left(1 - \sum_{B_L=0}^{B_{L,\max}-1} \frac{\bar{B}_L^{B_L}}{B_L!} \cdot e^{-\bar{B}_L}\right) \geq 1 > z \cdot \left(1 - \sum_{B_L=0}^{B_{L,\max}} \frac{\bar{B}_L^{B_L}}{B_L!} \cdot e^{-\bar{B}_L}\right). \quad (3.82a)$$

The probable maximum number of wire breaks $B_{L,\max}$ depends only on the mean number of wire breaks $\bar{B}_L$ and the number of steps z .

Example 3.6: Distribution of the number of wire breaks**Data:**

Rope diameter $d = 24$ mm
 Bending length $l = 30$ m
 Total number of wire breaks $B_l = 150$
 Step length $\Delta l = 1d$
 Reference length $L = 30d$

Results:

Mean number of wire breaks $\bar{B}_L = \frac{B_l \cdot L}{l} = 150 \cdot 30 \cdot 24/30,000 = 3.6$

Number of steps $z = l/\Delta l = 30,000/(1 \times 24) = 1,250$

$e^{-\bar{B}_L} = e^{-3.6} = 0.02732$.

The probability w that B_L occurs and the probability p that B_L or smaller occurs

B_L	$w =$	$w =$	$p =$
0:	$3.6^0/0! \cdot 0.02732 = 1/1 \cdot 0.02732$	$= 0.02732;$	0.02732
1:	$3.6^1/1! \cdot 0.02732 = 3.6/1 \cdot 0.02732$	$= 0.09837;$	0.1257
2:	$3.6^2/2! \cdot 0.02732 = 12.96/2 \cdot 0.02732$	$= 0.1771;$	0.3027
3:	$3.6^3/3! \cdot 0.02732 = 46.66/6 \cdot 0.02732$	$= 0.2125;$	0.5152
and so on.			

According to (3.82a), the probable maximum number of wire breaks is $B_{L,\max} = 11$.

Figure 3.71 shows the distribution of the wire breaks observed in a bending fatigue test and the calculated Poisson distribution with (3.79) drawn as smooth curve. This figure shows just how closely the observed number of wire breaks corresponds to the Poisson distribution close to the end of the rope life.

Very often the Poisson distribution is valid only for the smaller relative number of bending cycles. An example for this is to be seen in Fig. 3.72 where the observed numbers of wire breaks already deviate from the Poisson distribution for the relative number of bending cycles $N' = 55\%$. However these observed numbers of wire breaks can be explained as a birth-distribution introduced by Ren (1996).

For the birth-distribution, the wire breaks continue to occur by chance but prefer those sections already weakened by wire breaks. For the birth-distribution, Ren (1996) defined the variance factor $\nu = V/\bar{B}_L$ where the variance is greater than the mean number of wire breaks $V > \bar{B}_L$. Remember, $\nu = 1$ for the Poisson-distribution.

The probability $w(B_L)$ of the birth-distribution that the number of wire breaks $B_L = 0, 1, 2, 3$, etc. exists on the reference length L is according to Ren (1996)

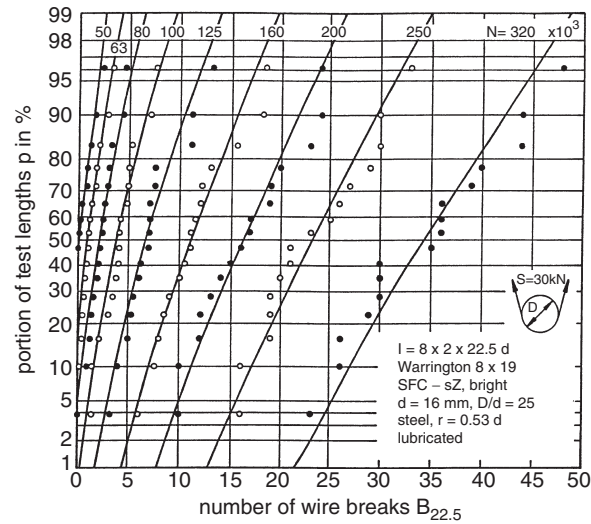


Fig. 3.71. Number of wire breaks distribution and Poisson distribution

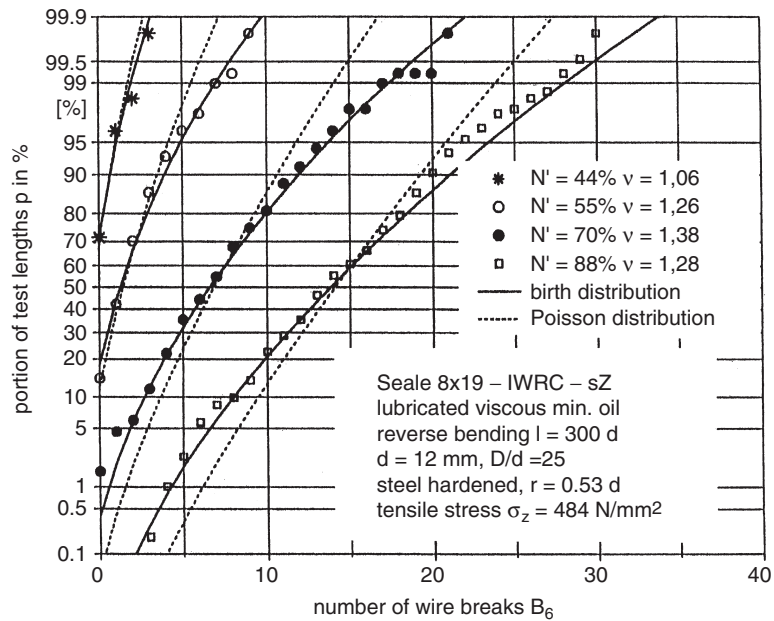


Fig. 3.72. Comparison of wire breaks distribution with Poisson and birth distribution, Ren (1996)

$$w(B_L) = \frac{1}{\beta(B_L, \frac{\bar{B}_L}{\nu-1}) \cdot B_L} \cdot \left(\frac{\nu-1}{\nu} \right)^{B_L} \cdot \frac{-\bar{B}_L}{\nu^{\nu-1}}. \quad (3.77a)$$

with the Beta-function expressed by the better known Gamma-function Γ

$$\beta\left(B_L, \frac{\bar{B}_L}{\nu-1}\right) = \Gamma(B_L) \cdot \Gamma\left(\frac{\bar{B}_L}{\nu-1}\right) / \Gamma\left(B_L + \frac{\bar{B}_L}{\nu-1}\right).$$

The probability that the number of wire breaks is smaller or equal to B_L is

$$p(B_L) = \sum_0^{B_L} w(B_L), \quad (3.79a)$$

and the probable number of wire breaks $B_{L,\max}$ on the reference length L is given again by (3.82), now with $p(B_L)$ from (3.79a).

The observed numbers of wire breaks, which can be explained through the Poisson-distribution or the birth-distribution, are regarded as normal since they are the result of the natural process of the occurrence of wire breaks, Ren (1996). Those breaks which are not able to be explained, even by the birth-distribution, are described with the term “dangerous break concentrations”. These weak points can be discovered at an early stage. Figure 3.73 shows the percentage of the wire break distributions in a series of bending fatigue tests that Ren (1996) was able to explain by either the Poisson or, at least, by the birth-distribution.

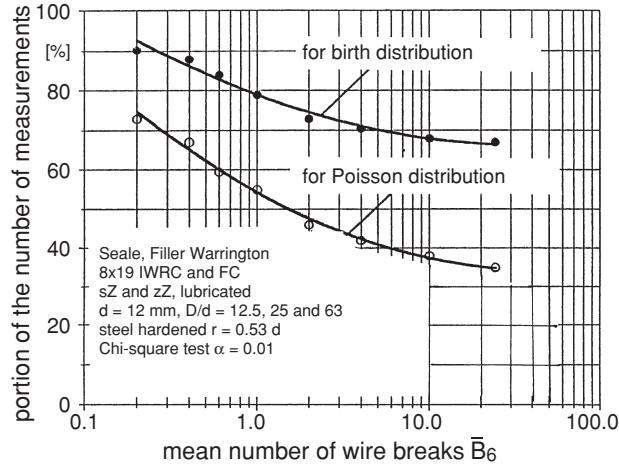


Fig. 3.73. Percentage of the wire break distributions explained by Poisson- or by birth-distribution, Ren (1996)

Discarding Number of Wire Breaks

As shown in Fig. 3.66, the number of wire breaks on a reference bending length $L = 30d$ were recorded during a number of series of bending fatigue tests. The results of these counts show that the number of wire breaks B_{30} found by interpolation was reached at 80% of the rope endurance. From these numbers of wire breaks found for different groups of wire ropes of the same construction, the discarding number of wire breaks has been evaluated by regression calculation and reasonable limits to be

$$B_{A30} = g_0 - g_1 \cdot \left(\frac{S}{d^2}\right)^2 - g_2 \cdot \left(\frac{d}{D}\right)^2 - g_3 \cdot \left(\frac{S}{d^2}\right)^2 \cdot \left(\frac{d}{D}\right)^2. \quad (3.83)$$

where S is the rope tensile force in N, d is the rope diameter in mm, and D is the sheave diameter in mm.

The discarding number of wire breaks on the small reference length $L = 6d$, chosen in order to detect concentrations of breaks, is given by definition as

$$B_{A6} = 0.5 \cdot B_{A30}. \quad (3.84)$$

The constants g_i for the ropes with different constructions listed in Table 3.18 (Sect. 3.4.5) are based on more safe limits than for the first form, Feyrer (1984). Furthermore, in Jahne's test results (1992) there is no longer any trace of the small difference between Seale ropes and Warrington and Filler ropes.

The discarding numbers of wire breaks from (3.83) and (3.84) are valid for outside visible wire breaks on wire ropes running in simple bending over sheaves made of steel or cast iron. For wire ropes with reverse bending cycles, Jahne (1992) recommended calculating the discard number of wire breaks also with (3.83) but with a $\Delta S/d^2 = 50 \text{ N/mm}^2$ higher specific tensile force than really works.

Figure 3.74 shows the discard number of wire breaks B_{A30} for parallel lay ropes (Seale, Warrington and Filler) $8 \times 19\text{-FC-sZ}$ from (3.83). For wire ropes with steel cores the discard number of wire breaks is always greater than for wire ropes with fibre cores.

The lower the discard number of wire breaks from (3.83) and the shorter the rope bending length, the less chance there is of detecting a dangerous rope situation by counting the wire breaks on the rope. Rope drives with a low discard number of wire breaks should not be used in hoisting applications with safety requirements.

In the case of lang lay ropes, multi-strand ropes and all kinds of ropes running over sheaves out of soft material (small elasticity module), it is not possible to detect the moment safely when a wire rope should be discarded by counting the number of outside visible wire breaks, because wire breaks of those ropes more often happen inside the rope. However lang lay ropes and multi-strand ropes show outside damage as discard criteria if the ropes run over sheaves with undercut grooves or V-grooves or if the ropes are wound in multi-layer spooling on drums. Table 3.5 summarizes possible methods of rope

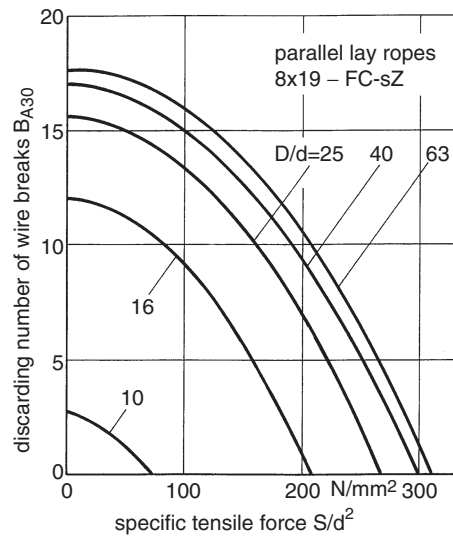


Fig. 3.74. Discarding number of wire breaks B_{A30} for parallel lay ropes $8 \times 19 - FC - sZ$

Table 3.5. Inspection methods to detect wire breaks in running ropes

	sheave groove out of steel or cast iron	sheave groove out of plastic material
ordinary lay rope	visual and tactual	magnetic methods
lang lay rope	magnetic methods (exception: visual and tactual for form grooves* and multi-layer spooling on drums)	
multi-strand rope		

*Form grooves = V-groove and undercut round grooves, undercut angle $\alpha \geq 90^\circ$

inspection. If the numbers of wire breaks are detected by magnetic devices, the discard number of wire breaks from (3.83) can be used for all types of wire rope and all kinds of sheaves.

It may be remarked here briefly that, together with the magnetic detection of wire breaks in special cases,

Metallic cross-section loss, Rieger (1983)

Diameter loss, Fuchs (1989)

Change of lay length, Briem (1996).

can be measured when the rope is running through the testing devices. By combining the results of these measurements it is possible to improve the diagnosis of the safety condition of the rope, Briem (1996).

3.3 Rope Drive Requirements

3.3.1 General Requirements

The dimensions of wire ropes in rope drives have to be calculated so that the ropes have sufficient endurance, definite discard criteria, fluctuating wire stresses not exceeding the yielding strength as well as the ability to resist extreme forces which may rarely arise. In addition, for economic reasons, the rope diameter should be smaller than the so-called optimal rope diameter. In short, the limits are:

- Rope endurance
- Discard criteria
- Donandt force
- Extreme forces
- Optimal rope diameter.

Each wire rope has to meet all these requirements independently from each other.

Rope Endurance

In most cases, the rope drive will be defined by the rope endurance required, mainly as a combination of safety factor ν and diameter ratio D/d . Rope endurance can be calculated, see Sect. 3.4. In order not to disappoint expectations, rope endurance should be regarded as the number of bending cycles that at most 10% of the ropes do not survive. For rope drives where ropes can be used up to the breaking point of the rope (sun blind, sliding door, etc.), the breaking number of bending cycles is N_{10} and for rope drives with safety requirements (crane, elevator, etc.), the discarding number of bending cycles is N_{A10} .

For safety reasons it is also important the rope has sufficient endurance. The rope endurance should last long enough for the rope to be inspected several times before the rope life comes to an end so that any dangerous rope condition can be detected in time.

Discard Criteria

The second most important aspect is to have a rope with safe discard criteria before a dangerous situation arises. Technical regulations list the discard criteria required. The most important discard criterion is the number of wire breaks on reference rope lengths. Discard numbers of wire breaks based on test results can be calculated, see Sect. 3.4. These numbers are more detailed and generally stricter than those found in technical regulations.

Furthermore, the calculated discard number of wire breaks shows the degree of safety at the time the rope is due to be discarded. If the calculated

discard number of wire breaks is smaller than $B_{30} < 2$, the rope drive should not be used for lifting appliances at all and if $B_{30} < 10$, the rope drive should not be used in cases where a load cannot be prevented from moving over people safely (Passenger lifting installations, see Sect. 3.3.2).

It is not easy to detect discard criteria in small bending lengths. Therefore, small zones of the rope which are bent frequently should be avoided.

It is only if there is no other subsequent severe damage to be expected when the rope breaks (for example in a window awning or a sliding door) that the second limiting factor “discard criteria” may be disregarded. In such cases, the broken wire rope simply has to be replaced without any other consequences.

Donandt Force

For small diameter ratios D/d , the usable tensile force can be limited by the yielding of the rope wires. This limit is given by the Donandt force, first investigated by Schmidt (1965). To be on the safe side, the rope tensile force should be

$$S < S_{D1},$$

where S_{D1} is the Donandt force that at most in one percent of the cases does not exceed the yield force.

The Donandt force S_{D1} can be calculated, see Sect. 3.4.

Extreme Force

To prevent wire rope breakage by extreme forces which occur only rarely, technical regulations normally require the minimum breaking force $F_{\min}$ to be several times stronger than the nominal rope tensile force S

$$F_{\min} \geq \nu \cdot S.$$

The so-called rope safety factor ν takes the increase of the tensile force by possible impact forces, acceleration forces and overloading into consideration (by means of impact factor f_{imp}) as well as the weakening of the wire rope breaking force over time due to fatigue, wear or corrosion (by means of residual breaking factor f_R). The rope safety factor ν is a combination of the impact factor f_{imp} and the residual breaking factor f_R

$$\nu = \frac{f_{\text{imp}}}{f_R}.$$

The decreasing rope breaking force over time is characterised by the residual breaking factor $f_R = F_R/F_{\min}$, whereby F_R is the residual breaking force of the rope. The weakening of the wire rope breaking force under the effect of rope drives, first investigated by Woernle (1929), will be rated as sufficient with the residual rope breaking force $F_R = 2/3 F_{\min}$ or with the residual breaking factor $f_R = 2/3$.

The impact factor f_{imp} varies widely. A minimum impact factor can be set at $f_{\text{imp}} = 1.67$. Then the minimum rope safety factor will be

$$v_{\text{min}} = 2.5.$$

Due to the impact force, however, in some cases the safety factor has to be much greater. For example, in a high speed elevator operating with safety gear or a buffer, the impact force can be more even than five times the normal rope tensile force; that means the impact factor is $f_{\text{imp}} > 5$, Feyrer (1977), Vogel (1996).

As far as extreme forces are concerned, the rope terminations only have to be considered.

Optimal Rope Diameter

The optimal diameter for a wire rope is the diameter which enables the wire rope to reach an optimum for the number of bending cycles. If the wire rope diameter is smaller or larger, the endurance of the wire rope is reduced. For economic reasons, the diameter of the wire rope should not be larger than the optimal rope diameter. The disadvantage of using a rope diameter larger than the optimal one is that the rope endurance is reduced and costs increased.

3.3.2 Lifting Installations for Passengers

There are technical regulations governing the requirements for important passenger installations using rope drives and these include not exceeding a given tensile force and not falling short of a given diameter ratio D/d of sheave and rope diameters. These technical regulations are also used for installations needing comparable safety requirements. The general design requirements for lifting installations for passengers listed in Table 3.6 also represent to a great extent the various current technical regulations.

The so-called rope safety factor in Table 3.6 varies considerably depending on the installation being considered although, of course, in all cases the passengers have to be provided with the same degree of safety. The required rope safety factors guarantee that extreme rope forces – which occur only rarely – in any case will be surely surpassed by the rope breaking force. However, the predominant criterion that defines the required rope safety factors is a

Table 3.6. General requirements for the rope drives in passenger lifting installations

installation	min. safety factor ν	min. diameter ratio D/d	max. pressure
elevator	12	40	9 N/mm ² ^{a)}
mine hoisting	8	80	2 N/mm ² ^{b)}
ropeway	3.8–4.5	80	–

^aSpecific rope pressure, k

^bGlobal rope pressure, p

reasonable rope endurance allowing sufficient time for distinct indications of an approaching rope discard to be produced and detected safely. Therefore, in combination with the required diameter ratio of sheave and rope and the different kinds of sheave grooves, an elevator with the rope safety factor $\nu = 12$ is not safer than a ropeway with the smaller factor $\nu = 4$ and the higher diameter ratio D/d .

The different requirements for these two parameters: rope safety factor ν and diameter ratio D/d for elevator, mine hoisting and rope way have practical reasons. In elevators, the diameter of sheaves and traction sheaves need to be small because of the space available and also to keep the price for the gear down. On the other hand, the rope safety factor for ropeways has to be small so that the suspension bow can be kept low. In between, the wire ropes for mine hoistings which are worked intensively should not be too heavy.

The technical regulations for lifting installations for passenger transportation always require that a combination of several methods be used to prevent a car from plunging downwards. The required methods are shown in Table 3.7. The most important method of protection involves the detection – both visually and magnetically; Table 3.5 – of wire breaks, of cross-section loss as well as other failures so that the wire rope can be replaced before a dangerous situation occurs. Installations should be designed to facilitate the inspection of the wire ropes.

Ropeways and mine hoistings must be permanently supervised by a responsible manager so that any change in the state of the installation will be noticed in time.

It is advisable to have redundant bearing ropes installed as they make it much easier to inspect the state of the wire rope. However, the main function of these redundant ropes – if this is at all possible for the given safety factor – is that in case of rope breakage there is another surviving rope or ropes. The probability of one of these other ropes breaking depends on the arrangement of the ropes. The effect on the failure probability of different arrangements of two ropes is shown for example in Fig. 3.75.

Table 3.7. Safety methods preventing car plunge in passenger lifting installations

safety method	elevator	hoisting mining shaft	rope way
visual and tactual inspection	yes	yes	yes
magnetic inspection	partly	yes	yes
permanent supervision		yes	yes
redundant bearing ropes	yes	partly	
safety gear	yes		partly

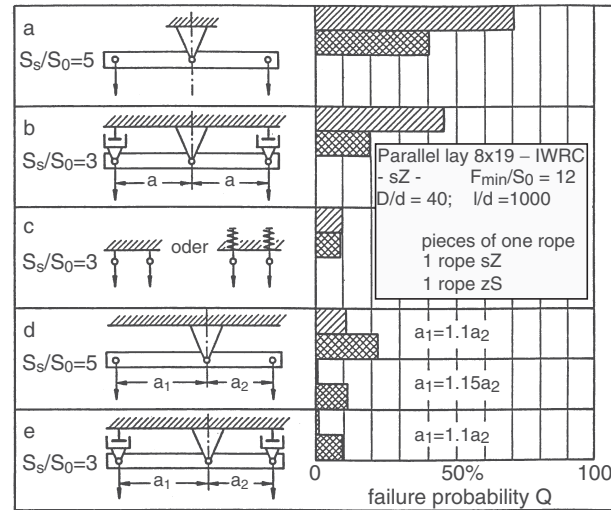


Fig. 3.75. Failure probability Q and relative impact force S_s/S_0 of two parallel bearing wire ropes (example)

The beneficial effect of having redundant ropes can be demonstrated with elevator ropes. Firstly, it is much easier to inspect the ropes with the result that only one of the redundant bearing wire ropes breaks in about one million elevator years. The second effect of the redundant rope is that even if one of these very rare rope breakages occurs, as far I know there has never been another rope breakage found. The failure probability of having a further rope breakage for multi-bearing wire ropes in elevators therefore lies theoretically between about 3 and 5%, Feyrer (1991b).

For installations where the cars have to overcome a very great difference in height, special rotation-resistant ropes should be used. If the difference in height is not too great, for many years now wire ropes with fibre cores have been regarded as acceptable. Wire ropes with steel cores in normal construction with relatively large torque should only be used for relatively low heights because of the great fluctuating stresses found especially in the steel core and the possible total loosening of the strands on the upper rope end. However wire ropes with special steel core can be used for larger differences of height for that they are qualified by good experience or by calculated relative small stresses from rope twist.

3.3.3 Cranes and Lifting Appliances

Because of the wide range of use for cranes and lifting appliances, there are very different requirements concerning rope endurance. Cranes and lifting appliances have therefore been classified according to the conditions of usage as laid down in ISO 4301/1. Table 3.8 gives the values for c (for the rather

Table 3.8. Cranes and lifting appliances, ISO 4308/1 (1986)

class of mechanism	safety factor	c -value for $R_0 = 1,570$ N/mm^2	Sheave D/d	Drum D/d	specific force S/d^2	number of hoisting cycles Z_{A10}
	ν					
M1	3.15	0.085	12.5	11.2	138	790
M2	3.35	0.087	14	12.5	132	1,100
M3	3.55	0.090	16	14	123	1,700
M4	4	0.095	18	16	111	2,500
M5	4.5	0.100	20	18	100	4,400
M6	5.6	0.112	22.4	20	80	9,100
M7	7.1	0.125	25	22.4	64	19,300
M8	9	0.140	28	25	51	43,200

unusual nominal rope strength $R_0 = 1,570 \text{ N/mm}^2$) and the diameter ratio D/d which is used for the eight classes of mechanism in order to achieve the minimum requirements of ISO 4308/1 (1986). Parameter c corresponds to the most frequently used specific tensile force in the following

$$S/d^2 = 1/c^2.$$

In the last column of Table 3.8, the rope endurance is listed as the number of hoisting cycles Z_{A10} . This number of hoisting cycles Z_{A10} , at which with a certainty of 95% at most 10% of the ropes have to be discarded, has been calculated for

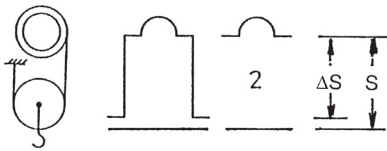
- A full load in every hoisting cycle
- A Warrington-Seale rope $6 \times 36 - \text{IWRC} - \text{sZ}$, $d = 16 \text{ mm}$, $R_0 = 1,570 \text{ N/mm}^2$
- A drum + 1 sheave; that means 1 combined fluctuating tension and bending on the drum + 2 simple bendings on the sheave
- Bending length $l = 500d$.

Wire ropes with fibre core cannot be used for either of the lowest classes of mechanism M1 and M2 as the Donandt force is too small. For the same reason, wire ropes even with a steel core cannot be used for class M1 if the ropes have a reverse bending.

The same number of hoisting cycles can be obtained with very different specific tensile forces S/d^2 and diameter ratios D/d . As an example, Table 3.9 shows these parameters for a row of numbers of hoisting cycles Z_{A10} (under full load) that with a certainty of 95% at most 10% of the ropes have to be discarded. The data are the same as in Table 3.8, however with the more usual nominal rope strength $R_0 = 1,770 \text{ N/mm}^2$ and the drum and the sheave have the same diameter ratio.

In any case, non-rotating wire ropes have to be fastened freely rotating over a swivel and other wire ropes have to be fastened to prevent any rotation.

Table 3.9. Hoisting cycles Z_{A10} for different rope tensile forces S/d^2 and diameter ratios D/d (under full load)



Lifting appliance with 1 drum + 1 sheave
 $\Delta S = 0.8S$; Warr.-Seale rope $6 \times 36 - \text{IWRC}$
 – Sz; nominal strength $R_0 = 1,770 \text{ N/mm}^2$;
 min. breaking force $F_{\min}/d^2 = 630 \text{ N/mm}^2$;
 rope diameter $d = 16 \text{ mm}$; bending length $l = 500d$;
 Z_{A10} = number of hoisting cycles at which with a certainty of 95%; at most 10% of the ropes have to be discarded

		specific rope tensile force S/d^2										
Hoisting cycles	$D/d =$	12.5	14	16	18	20	22.4	25	28	31.5	35.5	40
Z_{A10}												
800	153	175	205	230								
1,600	100	118	140	160	176	195	211	230				
3,200	65 ^a	79	96	112	125	140	154	170	184	199	213	
6,300	42 ^a	53 ^a	66	79	90	102	115	127	140	152	163	
12,500		36 ^a	46 ^a	57	65	75	85	96	106	117	127	
25,000			31 ^a	38 ^a	46	54	63	72	81	90	99	
50,000				27 ^a	33 ^a	39	46	54	61	69	77	

^aUneconomic because of $d > d_{\text{opt}}$

3.4 Calculation of Rope Drives

Rope drives should be dimensioned so that they comply with safety and economic requirements. For economic reasons, the endurance of the component parts, especially of the wire ropes, should be adjusted according to their practical use.

The basic information for dimensioning the rope drive can be evaluated with the calculation method described:

- The number of working cycles (trips) of the rope drive up to the point the rope has to be discarded or breaks and
- The limits of the tensile force with respect to wire yielding, the discard number of wire breaks and the optimal rope diameter.

This method of calculation is mainly based on the results of the fatigue bending tests presented in Sect. 3.2. Extreme tensile forces should be considered separately.

The number of working cycles obtained by calculation agrees to a great extent with the numbers reached by wire ropes in practical usage. The results of comparisons made between calculated and real wire rope endurance for a variety of cranes, elevators, ropeways, mine hoistings, etc. correspond to a great extent, Feyrer (1988), Beck and Briem (1993, 1995), Verschoof (1993), Briem and Jochem (1998) and Briem (2001). This method of calculation has been also used in very special cases, such as computerised maintenance management

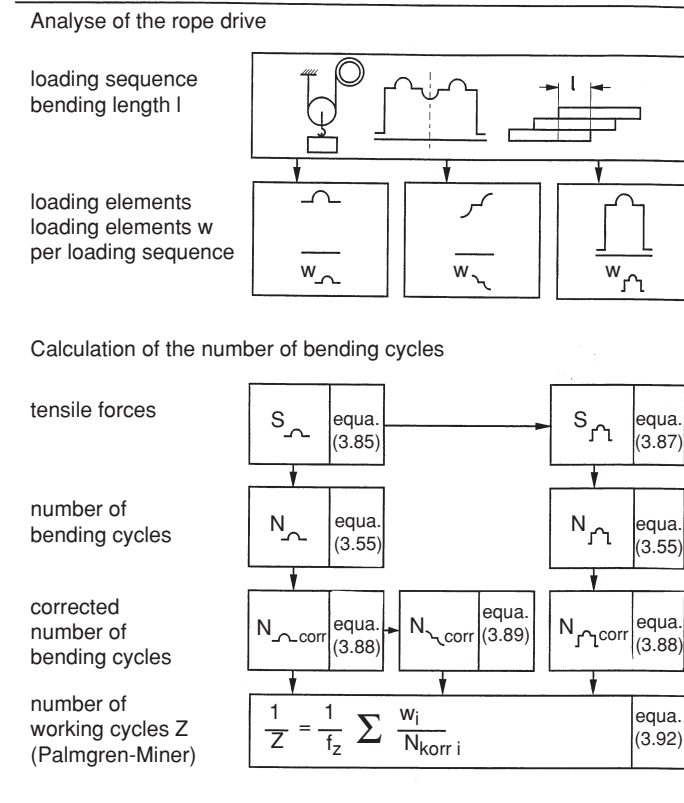


Fig. 3.76. Course of calculations for the number of working cycles

by Wiek (1997) or for rope tension equalizer for floating drilling rigs by Bradon and Chaplin (1997).

The number of working cycles up to the rope being discarded or breaking is calculated in stages. First the loading sequence and the bending length are derived by analysing the rope drive. Then the rope tensile force and the course of the force as well as the load collective are evaluated. The numbers of bending cycles can be calculated from this and combined with the Palmgren-Miner rule to find out the number of working cycles. The sequence of calculations for the number of working cycles is shown in Fig. 3.76.

The calculation for the wire rope drives is completed by calculating the limiting forces needed for safe and cost-effective operation.

3.4.1 Analysis of Rope Drives

Loading Sequence and Loading Elements

In the most rope drives, the wire rope is stressed by a number of bendings (e.g. when running over several rope sheaves) and by changes of the tensile

force as in a loading sequence during a hoisting cycle, a working cycle or a trip. A hoisting cycle is defined by an up and down movement. A working cycle is defined by a rope movement forwards and backwards so that in the end the original condition at the start of the working cycle is once again reached. A working cycle can result from one hoisting cycle or, in the case of cranes, out of two hoisting cycles – one with a load and one with no load going back to the starting position. A trip is defined by a rope movement in one direction. Working cycles or hoisting cycles are used especially for cranes and comparable lifting appliances. The rope endurance in elevators, ropeways and mine hoistings is normally counted in trips.

The endurance calculation starts by establishing the loading sequence for the wire rope zone (bending length) with the greatest stresses and separating this loading sequence into loading elements. The symbols for the loading elements are shown in Table 3.10. The loading elements A, B, C and D are the standard loading elements.

For wire ropes stressed by one of these standard elements, the rope endurance can be calculated directly. The calculation method for elements A, B and D is described in Sect. 3.4. Element C which only has fluctuating tensile force is very rarely found in rope drives and the rope endurance for this can be calculated according to Sect. 2.6. The three last loading elements in Table 3.10, of which F and G are very rare, can be converted into standard elements A, B, C and D. Loading element E always occurs in rope drives with traction sheaves or in those with drums if the load is changed while the rope is being wound onto the traction sheave or the drum. In both cases, it is to be supposed that the whole bending take place under the higher tensile force.

Normally the loading sequence during one working cycle (respectively during one trip) is composed of the numbers of standard loading elements (bendings) per loading sequence (working cycle or trip):

Simple bendings per working cycle $w_{\curvearrowright} = w_{\text{sim}}$

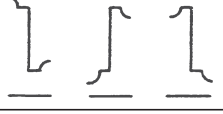
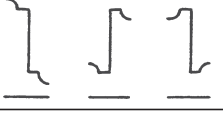
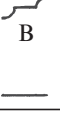
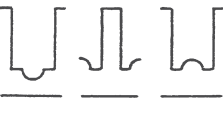
Reverse bendings per working cycle $w_{\curvearrowleft} = w_{\text{rev}}$ and

Combined fluctuating tensions and bendings per working cycle $w_{\curvearrowright\curvearrowleft} = w_{\text{com}}$.

The symbols from Table 3.10 are used as an index to characterize the standard loading element. The indices sim, rev and com are suggested for the case that the characterising symbols are not available for printing.

The bending length l is evaluated together with the numbers of the different loading elements during one working cycle (trip). This wire rope bending length represents the length of the wire rope zone which is stressed by the highest number of loading elements (bendings). The bending length can be evaluated most simply by marking off the distances a_i from the end of the rope to the sheaves in both end positions, which can be easily obtained from CAD. As an example, the bending length and the numbers of the loading elements (bendings) for a wire rope drive with three sheaves is shown in Fig. 3.77. For every sheave, the stroke is drawn as a band beginning from both ends of the

Table 3.10. Symbols for loading elements

	name	symbol	other symbols for the same loading	substitute loading
A	simple bending			
B	reverse bending			
C	fluctuating tension			
D	fluctuating tension and bending			
E	simple bending, while changing tension			
F	reverse bending, while changing tension			
G	bending and increased tension			change of tension for the rope zone on: a) free length  +  b) sheave 

sheave contact bow. The reverse bendings are marked by a thick line between the bands of both sheaves involved.

The rope length with the highest loading is the bending length l . The highest loading represents the number of bendings with respect to the reverse bendings. In the example Fig. 3.77 the small bending length is

$$l = h - (u_1 + a_2 + u_2).$$

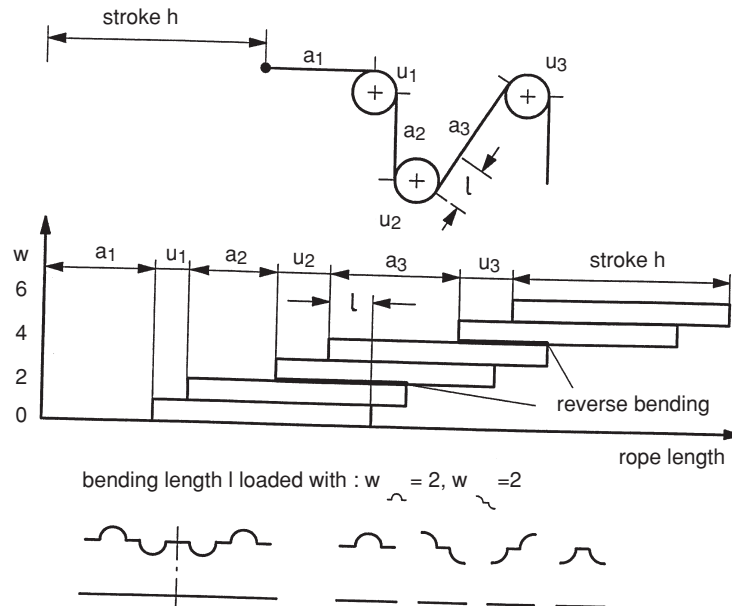


Fig. 3.77. Loading sequence and bending length

The loading sequence and the loading elements separated for this bending length l in a cycle forwards and backwards are also drawn in Fig. 3.77.

For the hoisting cycle of a crane, the loading sequence and the loading elements of the wire rope can be seen in Fig. 3.78. In this case, the stroke is so great that a rope piece moves over both sheaves and the drum. At the start of the loading sequence – lifting the load – the rope is exposed to increasing tensile force which is again reduced at the end of the sequence – when the load is set down. To understand the loading sequence, it is useful to start thinking from point P . The rope moves with the double load–stroke $2h$ over the stationary sheave and the drum.

Simplified Analysis

Because of the stochastically distributed working distances of most cranes, elevators and other applications, the number of loading elements (bendings) w in a working cycle and the bending length l can only be estimated.

An estimate of the effective numbers of bendings w and the relative bending length l/h for cranes based on the available load stroke h are listed in Table 3.11. From the theoretically possible number of bendings w_{th} , the simple bendings first have to be subtracted to determine the effective number of bendings. Measurements have been made of the movements in a special crane, Grolik and Hartung (1990) and Wünsch (1991).

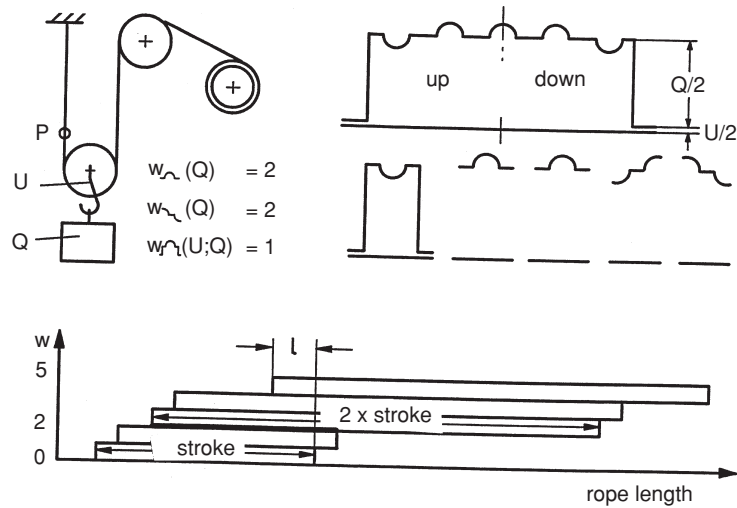


Fig. 3.78. Loading sequence and bending length of a crane rope

Table 3.11. Estimation of the effective number of loading elements w and the relative bending length l/h in cranes

theoretical number of bendings w_{th}	effective number of bendings w	relative bending length l/h
1	1	1.0
2	2	0.92
3	3	0.86
4	4	0.79
5	4	0.72
6	5	0.66
7	5	0.60
8	6	0.55
9	6	0.50
10	7	0.45
11	7	0.41
12	8	0.37

For elevators in multi-storied buildings, most car trips start or end on the ground floor or another main floor. Therefore the most stressed rope piece (bending length) is that running over the traction sheave and the deflection sheaves when the car moves between the ground floor and one or two floors above. The number of trips up to the rope being discarded has to be calculated for this rope piece. The ratio between the number of trips Z_G that start or

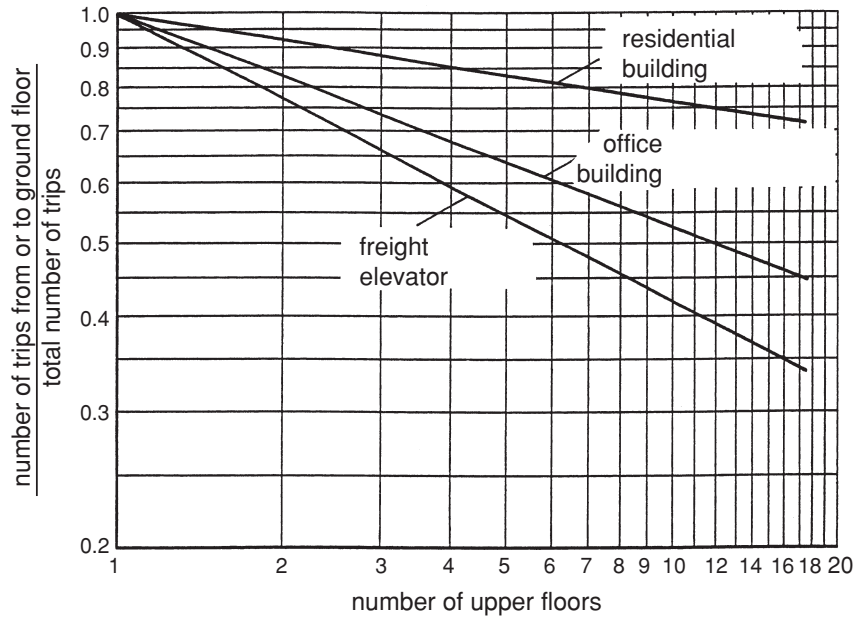


Fig. 3.79. Ratio of the number of trips Z_G/Z in elevators, Holeschak (1987)

end on the ground floor and the total number of trips Z , Holeschak (1987), has been evaluated by regression calculation from his records. He found the ratio Z_G/Z for different types of elevators and this is shown in Fig. 3.79.

3.4.2 Tensile Rope Force

Simple Bending and Reverse Bending

When calculating the number of bending cycles, it is necessary to know the effective rope tensile force as precisely as possible. If no more precise information is available, the effective rope tensile force S for lifting appliance can be evaluated from the load Q , the number of bearing wire ropes n_T , the acceleration g due to gravity and the global rope force factors f_S . For calculating the number of simple bending cycles as well as the number of reverse bending cycles, the tensile force is

$$S = \frac{Q \cdot g}{n_T} \cdot f_{S1} \cdot f_{S2} \cdot f_{S3} \cdot f_{S4}. \quad (3.85)$$

Table 3.12 lists the force factors $f_{S1} - f_{S4}$ which increase the wire rope force. The load guidance factor f_{S1} , the rope efficiency factor f_{S2} and the factor for parallel arrangements of the wire ropes f_{S3} can be applied very simply and do not need any further explanation. For two parallel ropes, the force factor f_{S3} has been estimated. However, for several parallel ropes the factor f_{S3} comes

Table 3.12. Force factors

loading	force factor
friction from the load guidance	
roller guidance	$f_{S1} = 1.05$
sliding guidance (greater for excentrical suspension)	$f_{S1} = 1.10$
rope efficiency η calculation of η in Sect. 3.5	$f_{S2} = \frac{1}{2} \cdot \left(1 + \frac{1}{\eta}\right)$
parallel bearing ropes	
separate sheaves, with whip or compensating sheave	$f_{S3} = 1.0$
separate sheaves, without whip or compensating sheave	$f_{S3} = 1.10$
common sheave, two ropes	$f_{S3} = 1.15$
common sheave, more than two ropes	$f_{S3} = 1.25^a$
acceleration, deceleration	
load speed $\nu \leq 0.3$	$f_{S4} = 1.05$
$0.3 < \nu \leq 0.8$	$f_{S4} = 1.10$
$0.8 < \nu \leq 1.6$	$f_{S4} = 1.12$
$\nu > 1.6$ m/s	$f_{S4} = 1.15$

^aJanovsky (1985), Holeschak (1987) and Aberkrom (1989)

from the measurements in elevators of Janovsky (1985), Holeschak (1987) and Aberkrom (1989).

With force factor f_{S4} , the increase in the wire rope force due to the acceleration of the load has been taken in account. Only in cases involving very high speeds, as for example in special elevators, in ropeways and mine hoistings, a rope piece is under acceleration or deceleration while passing more than one sheave.

Normally the bending length under acceleration or deceleration is very small. To avoid too many positions in the Palmgren-Miner equation for all numbers of bendings w (for any case if $v \leq 1.0$ m/s), the diminished force factor f_{S4w} can be used to replace the force factor f_{S4} . This diminished factor is

$$f_{S4w} = 1 + \frac{w_g \cdot (f_{S4} - 1)}{w}. \quad (3.86)$$

In the approximating (3.86) w is the number of bendings and w_g is the number of bendings considered under acceleration. For hoisting cycles $w_g = 2$ (with $w \geq 2$) and for trips $w_g = 1$.

Combined Fluctuating Tension and Bending

For combined fluctuating tension and bending, the equivalent tensile force S_{equ} is

$$S = S_{\text{fl}} \cdot f_{S5}. \quad (3.87)$$

Table 3.13. Constants for calculating the force factor f_{S5} , (3.69)

rope		d/δ	$a = 4/(\pi \cdot f)$				
			FC	IWRC	PWRC	EFWRC	ESWRC
six strand	Filler	16	2.55	2.20	1.97	2.38	2.13
	Seale	12.5	2.60	2.24	2.02	2.42	2.17
	Warr.	14	2.60	2.24	2.02	2.42	2.17
	W:-Seale	18	2.55	2.20	1.97	2.38	2.13
eight strand	Filler	20	2.86	2.17	1.95	2.52	2.10
	Seale	15	2.93	2.22	2.00	2.57	2.15
	Warr.	17	2.93	2.22	2.00	2.57	2.15
	W.-Seale	22	2.86	2.17	1.95	2.52	2.10
spiral round	18×7	15			2.31		
strand rope	34×7	21			2.33		

FC: fibre core; IWRC: independent wire rope core; PWRC: parallel-closed rope (parallel steel core with strands); EFWRC: wire rope core enveloped with fibres; ESWRC: wire rope core enveloped with solid polymer

For the loading element combined fluctuating tension and bending, the equivalent force factor f_{S5} is according to Sect. 3.2.5

$$f_{S5} = 1 + \frac{\left(1.31 - 0.0014 \cdot a \cdot \frac{\Delta S}{d^2}\right) \cdot \left(1.1 \cdot \frac{\Delta S}{d^2} - 0.1 \cdot \frac{S}{d^2}\right) a}{145,000 \cdot \frac{\delta}{d} \cdot \frac{d}{D} + 600 \frac{\delta}{d} + 0.2 \cdot a \cdot \frac{S}{d^2}}. \quad (3.69)$$

In (3.69)

S is the higher tensile force in N

ΔS the tensile force difference in N and

d is the nominal rope diameter in mm.

The constants a and δ/d are listed in Table 3.13.

If $1.1\Delta S \leq 0.1S$, the factor is $f_{S5} = 1.0$.

If a practically unloaded wire rope has to accelerate a load abruptly to reach a speed v_A determined by the rope speed, then, in order to consider the resulting impact force, an additional $0.5v_A$ can be supplemented to factor f_{S5} in (3.69). This addition $0.5v_A$ (in m/s) is derived from the results of Heptner (1971), Roos (1975) and Franke (1991).

3.4.3 Number of Bending Cycles

Simple Bending and Combined Fluctuating Tension and Bending

With the constant tensile forces S and $S_{\text{equ}} = S_{\wedge}$ the number of simple bending cycles $N_{\wedge}$ and the number of combined fluctuating tension and simple bending cycles $N_{\wedge}$ can be calculated with the equation (3.55) in Sect. 3.2

$$\lg N = b_0 + \left(b_1 + b_4 \cdot \lg \frac{D}{d} \right) \cdot \left(\lg \frac{S}{d^2} - 0.4 \cdot \lg \frac{R_0}{1770} \right) + b_2 \cdot \lg \frac{D}{d} + b_3 \cdot \lg d + \frac{1}{b_5 + \lg \frac{l}{d}}. \quad (3.55)$$

With d is the nominal rope diameter in mm, D , the sheave diameter in mm, S , the rope tensile force in N, R_0 , the nominal tensile strength in N/mm² and l is the bending length for $l > 15d$.

The constants b_i are listed in Table 3.14a and 3.14b. The numbers of bending cycles calculated with this are valid for up to a few million bending cycles under the following conditions:

- The wire rope is well-lubricated with viscous oil or Vaseline
- The sheaves have steel grooves, $r = 0.53d$

Table 3.14a. Constants for calculating the number of bending cycles, (3.55)

wire rope	b_0 for $\overline{N}$		b_0 for N_{10}		b_1	b_2	b_4	b_5	
	sZ	zZ	sZ	zZ				$\overline{N}$	N_{10}
cross lay 6 × 19 FC	−0.760	−	−1.225	−	0.875	6.480	−1.850	1.2	1.9
	−	−0.609	−	−1.019	0.562	6.430	−1.628		
Seale 8 × 19	−1.900	−1.677	−2.166	−1.943	1.280	8.562	−2.625		
Filler 8 × (19+8F) FC	−1.679	−1.456	−1.945	−1.722	1.280	8.562	−2.625		
Warr. 8 × 19	−1.679	−1.456	−1.945	−1.722	1.280	8.562	−2.625		
Warr.Seale 8 × 36	−0.858	0.966	0.592	0.700	0.096	7.078	−1.920		
Seale 8 × 19	−1.723	−1.663	−2.018	−1.958	1.290	8.149	−2.440		
Filler 8 × (19+6F) IWRC	−1.635	−1.575	−1.930	−1.870	1.290	8.149	−2.440		
Warr. 8 × 19	−1.635	−1.575	−1.930	−1.870	1.290	8.149	−2.440		
Warr.Seale 8 × 36	−1.327	1.381	1.032	1.086	0.029	6.241	−1.613		
spiral round- strand rope	18 × 7	−2.492	−2.724		1.566	9.084	−2.811		
	34 × 7	−1.014	−1.461		1.351	7.652	−2.485		

Table 3.14b. Discarding number of bending cycles N_A

wire rope	b_0 for $\overline{N}_A$		b_0 for N_{A10}		b_1	b_2	b_4	b_5	
	sZ	Zz	sZ	zZ				$\overline{N}_A$	N_{A10}
Seale 8 × 19	−2.611	−2.388	−2.927	−2.704	1.887	8.567	−2.894	1.2	1.9
Filler 8 × (19 + 6) FC	−2.476	−2.253	−2.792	−2.569	1.887	8.567	−2.894		
Warr. 8 × 19	−2.476	−2.253	−2.792	−2.569	1.887	8.567	−2.894		
Warr.Seale 8 × 36	−1.302	−1.194	−1.618	−1.510	1.322	8.070	−2.649		
Seale 8 × 19	−2.148	−2.088	−2.534	−2.474	1.588	8.056	−2.577		
Filler 8 × (19 + 6) IWRC	−2.015	−1.955	−2.401	−2.341	1.588	8.056	−2.577		
Warr. 8 × 19	−2.015	−1.955	−2.401	−2.341	1.588	8.056	−2.577		
Warr.Seale 8 × 36	0.633	0.687	0.247	0.301	0.377	6.232	−1.750		
spiral round	18 × 7	−2.772	−3.102		1.834	8.991	−2.948		
strand rope	34 × 7	−1.383	−1.679		1.619	7.559	−2.622		

The discarding numbers of bending cycles for lang lay ropes and for spiral round strand ropes are valid

- if the ropes will be inspected by magnetic methods
- or if the for the considered rope have been established by tests that outside wire breaks occur indicating the discard.

- There is no side deflection
- It is in a dry environment.

The following can be calculated with (3.55) and using the constants listed in Table 3.14a and 3.14b:

- The mean breaking number of bending cycles $\bar{N}$
- The breaking number of bending cycles N_{10} at which with 95% certainty, not more than 10% of the wire ropes break.
- The mean discarding number of bending cycles $\bar{N}_A$
- The discarding number of bending cycles N_{A10} at which with 95% certainty not more than 10% of the wire ropes have to be discarded.

These numbers of bending cycles are:

- Simple bendings $N_{\curvearrowright}$, calculated with the tensile force S , (3.85) and
- Combined fluctuating tension and bendings $N_{\curvearrowright\cup}$, calculated with the equivalent tensile force S_{equ} (3.87).

The breaking number of bending cycles N is reached if the rope or at least one strand has broken. The discarding number of bending cycles is reached if the number of wire breaks B_{30} or B_6 – not defined by Technical Rules but according to (3.83) and (3.84) – have been found on a reference length of the rope. For lang lay ropes and spiral round strand ropes, there is no reliable appearance of outside wire breaks so that the discard numbers of wire breaks generally have to be detected with magnetic devices. For cross lay ropes, which are rarely used today, no discarding numbers of bending cycles have been evaluated.

Correction of the Number of Bending Cycles

The number of bending cycles calculated with (3.55) can be corrected (adjusted) for other conditions with the help of endurance factors f_{Ni} . With these endurance factors listed in Table 3.15 the adjusted number of bending cycles is

$$N_{\text{cor}} = N \cdot f_{N1} \cdot f_{N2} \cdot f_{N3} \cdot f_{N4}. \quad (3.88)$$

f_{N1} is the endurance factor for the influence exerted by the rope lubrication. The endurance of wire ropes which are not lubricated is greatly reduced. According to Müller (1966) bending tests for ropes without lubrication, the endurance factor is about $f_{N1} = 0.2$.

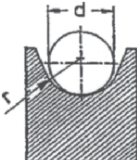
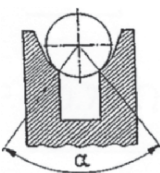
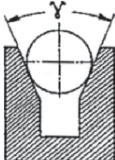
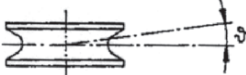
If the wire ropes will be re-lubricated during the rope life, the endurance factor is according (3.56) and (3.57) for the breaking number of bending cycles

$$f_{N1} = 0.0316 \cdot N^{0.307}. \quad (3.56a)$$

and for the discarding number of bending cycles

$$f_{N1} = 0.0682 \cdot N^{0.248}. \quad (3.57a)$$

Table 3.15. Endurance factors f_N

Rope lubrication			
rope well lubricated			$f_{N1} = 1.0$
rope without lubrication, Müller (1966)			$f_{N1} = 0.2$
Rope construction		core	8 strands 6 strands
– Fibre core		FC	$f_{N2} = 1.0$ $f_{N2} = 0.94$
– Steel core		IWRC	$f_{N2} = 1.0$ $f_{N2} = 0.81$
		PWRC	$f_{N2} = 1.86$ $f_{N2} = 1.51$
		ESWRC	$f_{N2} = 2.05$ $f_{N2} = 1.66$
		EFWRC	$f_{N2} = 1.06$ $f_{N2} = 0.86$
Sheave groove			
– Steel round groove	groove radius (Chapter 3.2.2)	$r/d = 0.53$ $r/d = 0.55$ $r/d = 0.60$ $r/d = 0.70$ $r/d = 0.80$ $r/d = 1.00$	$f_{N3} = 1.00$ $f_{N3} = 0.79$ $f_{N3} = 0.66$ $f_{N3} = 0.54$ $f_{N3} = 0.51$ $f_{N3} = 0.48$
			
– Undercut grooves	undercut angle Holeschak (1987)	$\alpha = 75^\circ$ $\alpha = 80^\circ$ $\alpha = 85^\circ$ $\alpha = 90^\circ$ $\alpha = 95^\circ$ $\alpha = 100^\circ$ $\alpha = 105^\circ$	$f_{N3} = 0.40$ $f_{N3} = 0.33$ $f_{N3} = 0.26$ $f_{N3} = 0.20$ $f_{N3} = 0.15$ $f_{N3} = 0.10$ $f_{N3} = 0.066$
			
– V grooves	angle Holeschak (1987)	$\gamma = 35^\circ$ $\gamma = 36^\circ$ $\gamma = 38^\circ$ $\gamma = 40^\circ$ $\gamma = 42^\circ$ $\gamma = 45^\circ$	$f_{N3} = 0.054$ $f_{N3} = 0.066$ $f_{N3} = 0.095$ $f_{N3} = 0.14$ $f_{N3} = 0.18$ $f_{N3} = 0.25$
			
– Plastic round groove (chapter 3.2.2)			$f_{N3} = 8.37 \cdot N_{st}^{-0.124}$
or	$f_{N3} \approx 0.75 + 0.36 \cdot \frac{S/d^2}{D/d} - 0.023 \cdot \left(\frac{S/d^2}{D/d} \right)$		
Side deflection. Schönherr (2005), Neumann (1987)			
	$f_{N4} = 1 - \left(0.00863 + 0.00243 \cdot \frac{D}{d} \right) \cdot \vartheta - 0.00103 \cdot \vartheta^2$		
angle of side deflection ϑ in $^\circ$			

In (3.56a) and (3.57a), the number of bending cycles N and N_A should be used after corrected by other involved endurance factors.

f_{N2} is the endurance factor for the influence exerted by different rope constructions. With the factor f_{N2} , the numbers of bending cycles of parallel lay ropes can be adjusted for ropes with special cores and for six strand ropes.

f_{N3} is the endurance factor for the influence exerted by the sheave groove. Based on round steel grooves with the radius $r = 0.53d$, the number of bending cycles can be adjusted with endurance factor f_{N3} for other groove radii, for grooves made of synthetic and for shaped grooves. The factors for the formed grooves (undercut and V grooves) come from investigations carried out by Holeschak (1987) on existing elevators. For evaluating his factors, he supposed that the car load was 75% of the nominal load. No great error will arise if this factor is used for a standard loading of 50% of the nominal car load.

f_{N4} is the endurance factor for the influence exerted by side deflection. The equation in Table 3.15 for side deflection has been derived by Schönherr (2005) from his large-scale research into the breaking number of bending cycles. Schönherr's equation also represents the few first endurance factors from Neumann (1987) up to ropes being discarded and can therefore also be used for adjusting the discard numbers of bending cycles.

The endurance factors f_N have been evaluated in simple bending fatigue tests. Therefore they are only really valid for calculating the number of simple bending cycles. These factors are only conditionally qualified for reverse bending cycles and fluctuating tension and bending cycles. For calculating the number of reverse bending cycles $N_{\sim}$, it would be useful first of all to adjust the number of simple bending cycles $N_{\wedge}$ with endurance factor f_N to $N_{\wedge, \text{cor}}$ and then calculate the number of reverse bending cycles according to (3.89). For fluctuating tension and bending cycles, the number of combined cycles N_{Ω} can also be calculated using endurance factors f_N . An exception is the factor f_{N2} for sheaves with grooves made of synthetic materials and here the endurance factor should be set $f_{N2} = 1.0$ for this case of fluctuating tension and bending.

Reverse Bending

Following (3.61) from Sect. 3.2.4, the number of reverse bending cycles can be calculated from the number of simple bending cycles with the equation

$$N_{\sim, \text{cor}} = N_{\text{rev}, \text{cor}} = a_0 \cdot N_{\text{sim}, \text{cor}}^{a_1} \cdot \left(\frac{D}{d} \right)^{a_2}. \quad (3.89)$$

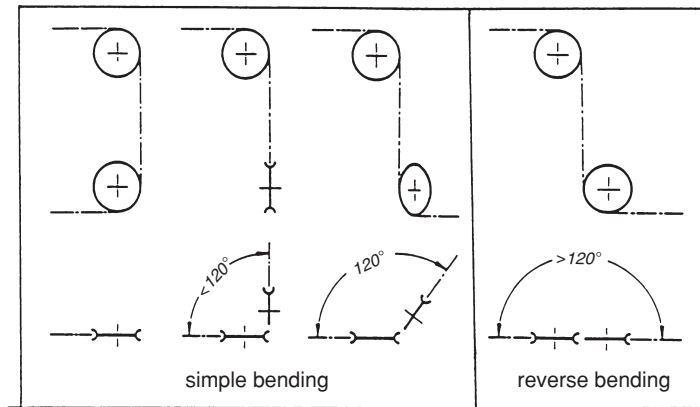
The constants a_i are listed in Table 3.16. If the two related sheave diameters are not the same, a substitute diameter D_m can be used

$$D_m = \frac{2 \cdot D_1 \cdot D_2}{D_1 + D_2}. \quad (3.90)$$

and for different grooves the substitute endurance factor is

Table 3.16. Constants for calculating the number of reverse bending, (3.89)

factor	breaking number		discarding number	
	$\bar{N}$	N_{10}	$\bar{N}_A$	N_{A10}
a_0	9.026	6.680	3.635	2.670
a_1	0.618	0.618	0.671	0.671
a_2	0.424	0.424	0.499	0.499

**Fig. 3.80.** Definition of simple and reverse bending for angles between the sheave axis, DIN15020

$$f_{N3m} = \frac{2 \cdot f_{N3,1} \cdot f_{N3,2}}{f_{N3,1} + f_{N3,2}}. \quad (3.91)$$

By definition, reverse bending means that the axes of the two sheaves involved are parallel. Up to now, there are no known fatigue tests for wire ropes running over sheaves where the axes are not parallel. For cases where the sheave axes are not parallel, the definition from DIN 15020 can be used, Fig. 3.80.

If there are greater distances between the sheaves, the wire rope can be turned – especially in the case of side deflection – so that reverse bending can probably be avoided. Beck and Briem (1993) found a rope turn of 50° for Warrington ropes with fibre cores in an elevator with a sheave distance of $165d$. In current rope endurance calculations it is not possible to assume such a rope turn.

3.4.4 Palmgren–Miner Rule

With the help of the cumulative damage hypothesis of Palmgren (1924) and Miner (1945), it is possible to evaluate the number of working cycles (number of loading sequences) Z . The basic equation (Sect. 3.2.6) is

$$\sum \frac{n_i}{N_i} = 1. \quad (3.70)$$

In this, n_i is the number of cycles under the load i , the wire rope is stressed and N_i is the endurance number under the load i .

In one working cycle or trip the wire rope is often stressed by numbers of the three standard loading elements w_i ($w_{\text{sim}} = w_{\text{sim}}$, $w_{\text{rev}} = w_{\text{rev}}$ and $w_{\text{com}} = w_{\text{com}}$). For the tensile force S_j , the number of working cycles Z_j can then be calculated using the following equation

$$\frac{1}{Z_j} = \sum_{i=1}^3 \frac{w_i}{N_i} = \frac{w_{\text{sim}}}{N_{\text{sim}}} + \frac{w_{\text{rev}}}{N_{\text{rev}}} + \frac{w_{\text{com}}}{N_{\text{com}}}. \quad (3.92)$$

In very rare cases where a running wire rope is also loaded by fluctuating tension cycles, the number of tension cycles can be calculated using the equations in Sect. 2.6 (up to now, only for ropes Warr.-Seale 6×36 -IWRC). With these, the further quotient $w_{\text{tens}}/N_{\text{tens}}$ can be included in (3.92), though at present there are no test results known which make allowances for this addition of damage ratios.

If the wire ropes are loaded by a collective of forces S_j with the proportion of the numbers of working cycles $a_j = Z_j/Z$, then the number of working cycles is

$$Z = \frac{1}{\sum_{j=1}^k \frac{a_j}{Z_j}} = \frac{1}{\sum_{j=1}^k a_j \cdot \sum_{i=1}^3 \frac{w_i}{N_{ij}}}. \quad (3.93)$$

3.4.5 Limits

The following limits have to be respected for rope drives:

- The rope tensile force must be smaller than the Donandt force $S < S_D$.
- In case safety requirements have to be met, the expected number of wire breaks should be greater than the given limiting number of wire breaks or alternatively the rope tensile force should be smaller than the discard limiting force $S < S_G$.
- For economic reasons, the rope diameter should be smaller than the optimal rope diameter $d < d_{\text{opt}}$.
- Extreme forces (only for special applications)

Donandt-Force

The method of endurance calculation presented here is limited by the Donandt force at which point the yielding stress of the wires is reached. For simple bending, the Donandt force is according to Sect. 3.2.7

$$S_{D,\text{sim}} = q_0 \cdot F_{\text{min}} + q_1 \cdot \frac{d}{D} \cdot F_{\text{min}} \quad (3.71)$$

Table 3.17. Constants for calculating the Donandt force, (3.71) and (3.72)

Rope		q_0 mean Donandt force		Donandt force that with a certainty of 95% for at most 1% of the ropes will be smaller		q_1
		$\bar{S}_D$		S_{D1}		
		sZ	zZ	sZ	zZ	
FC	6×19	0.787	0.824	0.619	0.656	−4.10
	8×19	0.796	0.826	0.624	0.654	−4.20
	6×36	0.781	0.798	0.608	0.625	−4.20
	8×36	0.782	0.782	0.605	0.605	−4.30
WRC	6×19	0.809	0.849	0.653	0.693	−3.77
	8×19	0.852	0.886	0.686	0.719	−4.02
	6×36	0.802	0.821	0.642	0.661	−3.86
	8×36	0.835	0.835	0.664	0.664	−4.12
WSC	18×7	0.693		0.492		−3.02
	34×7	0.715		0.537		−3.34

and the Donandt force for reverse bending is

$$S_{D,rev} = (q_0 - 0.035) \cdot F_{\min} + (q_1 - 0.25) \cdot \frac{d}{D} \cdot F_{\min}. \quad (3.72)$$

The constants q_i for the different rope constructions are listed in Table 3.17 for the mean Donandt force $\bar{S}_D$ and for the Donandt force S_{D1} where with a certainty of 95% it is smaller for at most 1% of the ropes. As for the simple and the reverse bendings, for the combined fluctuating tension and bendings, the tensile force S – and not the equivalent tensile force S_{equ} – should be smaller than the Donandt force S_{D1} . The Donandt force S_{D1} has to be used as the wire rope tensile force limit, but there is no danger if in a few cases the tensile force reaches the mean Donandt force.

Discarding Limit

(a) Discarding Number of Wire Breaks

The evaluation of the discarding numbers of wire breaks has been presented in Sect. 3.2.8, as an example refer to Fig. 3.74. This discarding number of wire breaks on a reference rope length $L = 30d$ is for simple bendings

$$B_{A30} = g_0 - g_1 \cdot \left(\frac{S}{d^2}\right)^2 - g_2 \cdot \left(\frac{d}{D}\right)^2 - g_3 \cdot \left(\frac{S}{d^2}\right)^2 \cdot \left(\frac{d}{D}\right)^2. \quad (3.83)$$

For reverse bendings the number of wire breaks has to be calculate with a $\Delta S/d^2 = 50 \text{ N/mm}^2$ higher specific tensile force, Jahne (1992).

The discarding number of wire breaks on the small reference length $L = 6d$ – which has been chosen to detect break concentrations – is given by definition as

$$B_{A6} = 0.5 \cdot B_{A30}. \quad (3.84)$$

It should be noticed that these discarding numbers of wire breaks generally differ from those in existing technical regulations.

(b) Discard Limiting Force

As an alternative design limit to the discarding number of wire breaks, the wire rope tensile force S must be smaller than the discard limiting force S_G . This discard limiting force S_G can be calculated by an equation regrouped from (3.83) for a required discarding number of wire breaks $B_{A30,\min}$. The discard limiting force is then

$$S_G = d^2 \cdot \sqrt{\frac{-B_{A30,\min} + g_0 - g_2 \cdot \left(\frac{d}{D}\right)^2}{g_1 + g_3 \cdot \left(\frac{d}{D}\right)^2}}. \quad (3.94)$$

In (3.83) and (3.94)

S is the rope tensile force in N

d , the rope diameter in mm, and

D is the sheave diameter.

The constants g_i for ropes running in simple bendings over sheaves with grooves made of steel or cast iron are listed in Table 3.18. For reverse bendings the specific discard limiting force S_G/d^2 has to be reduced with $\Delta S/d^2 = 50 \text{ N/mm}^2$.

Wire ropes running over grooves made of synthetic material have to be inspected with magnetic methods, Table 3.5. There is one exception for ropes in elevators running over traction sheaves with V grooves or undercut round grooves with an undercut angle $\alpha, \geq 90^\circ$ the discard number of wire breaks $B_{A30} = 26$ for sZ and $B_{A30} = 13$ for zZ ropes can be used.

The rope drives should be designed in such a way that an unsafe condition can be reliably detected by a high discarding number of wire breaks. If not laid down by the standards given, the minimum discarding number of wire breaks should be:

- For pull drives $B_{A30} \geq 2$
- For lifting appliances $B_{A30} \geq 8$
- For lifting appliances with loads probably moving over persons $B_{A30} \geq 15$.

For lifting appliances containing dangerous loads or even persons, additional safety methods and installations are required, see Table 3.7.

Table 3.18. Constants for calculating the discarding number of wire breaks or the discard limiting force, (3.83) or (3.94)

rope		g_0	g_1	g_2	g_3
Filler Warr. Seale	FC-8 × 19-sZ ^{a,b} zZ ^{b,c}	18	0.000174	1,550	0.026
	WRC-8 × 19-sZ ^{a,b} zZ ^{b,c}	33.3	0.000184	1,830	0.0447
Warr.- Seale	FC-8 × 38-sZ ^{a,b} zZ ^{b,c}	29	0.000271	2,400	0.0403
	WRC-8 × 36-sZ ^{a,b} zZ ^{b,c}	44.5	0.000222	2,200	0.0536
spiral round strand rope	18 × 7 ^{b,c}	14	0.000160	−350	0.0350
	34 × 7 ^{b,c}	20	0.000230	−500	0.0500

For reverse bendings the number of wire breaks has to be calculated with a specific tensile force, increased by $\Delta S/d^2 = 50 \text{ N/mm}^2$. For six strand ropes the number of wire breaks 0.75 of that for the eight strand ropes.

^aOutside visible wire breaks

^bMagnetic detected wire breaks

^cHalf of the calculated number of wire breaks as visible wire breaks if confirmed by bending fatigue tests for the considered wire rope

Optimal Rope Diameter

The optimal wire rope diameter is the diameter at which the wire rope reaches an optimum for the number of bending cycles with a given tensile force S and a given sheave diameter D . If the wire rope diameter is smaller or bigger, the wire rope endurance is reduced.

According to Sect. 3.2.7, the optimal rope diameter is

For simple bendings

$$d_{\text{opt,sim}} = c_{0,\text{sim}} \cdot \sqrt{D \cdot \sqrt{S}} \quad (3.73a)$$

For combined fluctuating tension and bendings

$$d_{\text{opt,com}} = c_{0,\text{sim}} \cdot \sqrt{D \cdot \sqrt{S_{\text{equ}}}} \quad (3.73b)$$

and for reverse bendings

$$d_{\text{opt,rev}} = c_{0,\text{rev}} \cdot \sqrt{D \cdot \sqrt{S}}. \quad (3.73c)$$

In (3.73), S is set in N and d and D in mm. The constants c_0 – listed in Table 3.19 – are calculated according to Sect. 3.2.7 for the discarding number

Table 3.19. Constants for calculating the optimal rope diameter, (3.73)

loading	wire rope		constant c_0 for nominal strength R_0 (N/mm ²)			
			1570	1770	1960	2160
Simple bending or Fluctuating tension a. bending	F,W u S	FC + 8 × 19	0.0816	0.0806	0.0798	0.0790
	WS	FC + 8 × 36	0.0920	0.0909	0.0900	0.0891
	F,W u S	WRC + 8 × 19	0.0767	0.0757	0.0750	0.0742
	WS	WRC + 8 × 36	0.0915	0.0904	0.0895	0.0886
	spiral round-strand rope	18 × 7 34 × 7	0.0803 0.0882	0.0793 0.0871	0.0785 0.0862	0.0777 0.0854
reverse bending	F,W u S	FC + 8 × 19	0.0703	0.0695	0.0688	0.0681
	WS	FC + 8 × 36	0.0783	0.0774	0.0766	0.0758
	F,W u S	WRC + 8 × 19	0.0649	0.0641	0.0635	0.0629
	WS	WRC + 8 × 36	0.0717	0.0708	0.0701	0.0694
	spiral round-strand rope	18 × 7 34 × 7	0.0694 0.0749	0.0686 0.0740	0.0679 0.0732	0.0672 0.0725

The constants for eight strand ropes are also valid for six strand ropes

of bending cycles. For the breaking number of bending cycles, the constants are a little greater.

For rope drives with different numbers of loading elements w_i , during one working cycle a common optimal rope diameter can be calculated

$$d_{\text{opt}} = \frac{w_{\text{sim}} \cdot d_{\text{opt,sim}} + w_{\text{com}} \cdot d_{\text{opt,com}} + w_{\text{rev}} \cdot d_{\text{opt,rev}}}{\sum w}. \quad (3.95)$$

To be cost-effective, the wire rope diameter should not be bigger than the optimal rope diameter. Using a rope diameter bigger than the optimal one has the disadvantage of getting lower rope endurance for higher costs. The maximum of the number of bending cycles is rather flat which means that with only a minor deviation from the optimal rope diameter the number of bending cycles does not change very much. Therefore, the rope diameter can be smaller than the optimal rope diameter by a reasonable percentage without suffering too great a loss of endurance.

3.4.6 Rope Drive Calculations, Examples

Example 3.7: Numbers of bending cycles

Data:

Filler-rope 6 × (19 + 6F) – ESWRC – sZ, well lubricated

Nominal rope diameter $d = 16$ mm

Nominal strength $R_0 = 1,960$ N/mm²

Sheave diameter $D = 400$ mm

Steel groove radius $r = 0.55d$

Tensile force $S = 30$ kN
 Rel. force difference $\Delta S/S = 0.8$
 Bending length $l = 2.4$ m.

Number of simple bending cycles:

According to (3.55) the discard number of bending cycles is

$$\lg N_{A10} = 4.5890 \longrightarrow N_{A10} = 38,800.$$

The endurance factor is $f_N = f_{N2} \cdot f_{N3} = 1.66 \cdot 0.79 = 1.31$ according to Table 3.15. Then the adjusted number of simple bending cycles at which with 95% certainty not more than 10% of the ropes have to be discarded is

$$N_{A10 \cap \text{cor}} = N_{A10 \text{sim,cor}} = N_{A10} \cdot f_N = 38,800 \cdot 1.31 = 50,900.$$

This and the other discarding and breaking numbers of bending cycles are

$$N_{A10 \cap \text{cor}} = 50,900 \quad \bar{N}_{A \cap \text{cor}} = 139,000.$$

$$N_{10 \cap \text{cor}} = 123,000 \quad \bar{N}_{10 \cap \text{cor}} = 272,000.$$

Number of combined fluctuating tension and bending cycles.

The force factor is $f_{S5} = 1.445$ according to (3.69).

Then the equivalent tensile force for the fluctuating tension and bending is

$$S_{\text{equ}} = f_{S5} \cdot S = 1.445 \cdot 30 = 43.35 \text{ kN}.$$

with this equivalent force the discard number of bending cycles is

$$\lg N_{A10 \text{com}} = 4.2887 \quad \text{and} \quad N_{A10 \text{com}} = 18,500.$$

Then the adjusted (with the endurance factor $f_N = 1.31$) discard number of combined fluctuating and bending cycles is

$$N_{A10 \cap \text{cor}} = N_{A10 \text{com,cor}} = N_{A10 \text{com}} \cdot f_N = 18,500 \cdot 1.31 = 24,200.$$

This and the other discarding and breaking numbers of combined bending cycles are

$$N_{A10 \cap \text{cor}} = 24,200; \quad \bar{N}_{A \cap \text{cor}} = 66,200$$

$$N_{10 \cap \text{cor}} = 56,200; \quad \bar{N}_{10 \cap \text{cor}} = 125,000.$$

Number of reverse bending cycles:

According to (3.89) and Table 3.16 the adjusted discard number, at which with 95% certainty not more than 10% of the ropes have to be discarded, is

$$N_{A10 \sim \text{cor}} = N_{A10 \text{rev,cor}} = a_0 \cdot N_{A10 \text{sim,cor}}^{a_1} \cdot \left(\frac{D}{d} \right)^{a_2}.$$

$$N_{A10 \sim \text{cor}} = N_{A10 \text{rev,cor}} = 2.670 \cdot 50,900^{0.671} \cdot 25^{0.499} = 19,200.$$

This and the other discarding and breaking numbers of reverse bending cycles are

$$\begin{aligned} N_{A10 \sim, \text{cor}} &= 19,200; & \bar{N}_{A \sim, \text{cor}} &= 51,200. \\ N_{10 \sim, \text{cor}} &= 36,500; & \bar{N}_{\sim, \text{cor}} &= 80,700. \end{aligned}$$

Example 3.8: Limits

Data:

from Example 3.7

Donandt force S_{D1} :

According to Table 1.10, the minimum breaking force is $F_{\min} = 698d^2 = 179 \text{ kN}$.

With (3.71) and Table 3.17 for simple bending the Donandt force is

$$\begin{aligned} S_{D \sim} &= S_{D, \text{sim}} = q_0 \cdot F_{\min} + q_1 \cdot \frac{d}{D} \cdot F_{\min}. \\ S_{D \sim} &= S_{D, \text{sim}} = 0.653 \cdot 179 - \frac{3.77}{25} \cdot 179 = 89.9 \text{ kN}. \end{aligned}$$

and according to (3.72) for reverse bending the Donandt force is

$$\begin{aligned} S_{D \sim} &= S_{D, \text{rev}} = (q_0 - 0.035) \cdot F_{\min} + (q_1 - 0.25) \cdot \frac{d}{D} \cdot F_{\min}. \\ S_{D \sim} &= S_{D, \text{rev}} = (0.653 - 0.035)179 + (-3.77 - 0.25) \cdot \frac{1}{25} \cdot 179 = 81.8 \text{ kN}. \end{aligned}$$

Both of the Donandt forces are – as they should be – greater than the rope tensile force $S = 30 \text{ kN}$.

Discard limit:

a) Discarding number of wire breaks

According to (3.83) and using the constants listed in Table 3.18 and the factor 6/8 for the six strand rope, the discard number of wire breaks for simple bendings is

$$\begin{aligned} B_{A30} &= \frac{6}{8} \cdot \left[g_0 - g_1 \cdot \left(\frac{S}{d^2} \right)^2 - g_2 \cdot \left(\frac{d}{D} \right)^2 - g_3 \cdot \left(\frac{S}{d^2} \right)^2 \cdot \left(\frac{d}{D} \right)^2 \right]. \\ B_{A30} &= \frac{6}{8} \cdot \left[33.3 - 0.000184 \cdot (117.2)^2 - 1,830 \cdot \left(\frac{1}{25} \right)^2 - 0.0447 \cdot (117.2)^2 \cdot \left(\frac{1}{25} \right)^2 \right]. \\ B_{A30} &= 20. \end{aligned}$$

For reverse bending, the specific rope tensile force has to be set $S_{\text{rev}}/d^2 = S/d^2 + 50 \text{ N/mm}^2$. Then the discard number of wire breaks for reverse bendings is

$$B_{A30, \text{rev}} = 17 \text{ and } B_{A30, \text{com}} = 17.$$

According to (3.84), the discard number of wire breaks on the small reference length $L = 6d$ is

$$B_{A6} = 0.5 \cdot B_{A30} = 0.5 \cdot 20 = 10 \quad \text{and} \quad B_{A6, \text{rev}} = 9.$$

b) Discard limiting force (alternatively)

According to (3.94) the discard limiting force is

$$S_G = d^2 \cdot \sqrt{\frac{-B_{A30} + g_0 - g_2 \cdot \left(\frac{d}{D}\right)^2}{g_1 + g_3 \cdot \left(\frac{d}{D}\right)^2}}.$$

With the constants listed in Table 3.18, the factor $6/8$ for the six strand rope and a required number of wire breaks $B_{A30} = 15$, the discard limiting force for simple bendings is

$$S_G = 16^2 \cdot \sqrt{\frac{-\frac{8}{6} \cdot 15 + 33.3 - 1,830 \cdot \left(\frac{1}{25}\right)^2}{0.000184 + 0.0447 \left(\frac{1}{25}\right)^2}} = 51,600 \text{ N} = 51.6 \text{ kN}.$$

For reverse bending, the discard limiting force $S_{G, \text{rev}}$ is $\Delta S = 50d^2$ smaller $S_{G \sim} = S_{G, \text{rev}} = S_G - 50 \cdot d^2 = 51,600 - 50 \cdot 16^2 = 38,800 \text{ N} = 38.8 \text{ kN}$.

Both of the discard limiting forces are – as they should be – greater than the rope tensile force $S = 30 \text{ kN}$.

Optimal rope diameter:

According to the equations (3.73) and the constants c_0 listed in Table 3.19, the optimal rope diameter is

$$d_{\text{opt}} = c_0 \cdot \sqrt{D \cdot \sqrt{S}}.$$

simple bending

$$d_{\text{opt}, \sim} = d_{\text{opt}, \text{sim}} = 0.075 \cdot \sqrt{400 \cdot \sqrt{30,000}} = 19.7 \text{ mm}.$$

combined fluctuating tension and bending

$$d_{\text{opt}, \sim} = d_{\text{opt}, \text{com}} = 0.075 \cdot \sqrt{400 \cdot \sqrt{43,350}} = 21.6 \text{ mm}.$$

reverse bending

$$d_{\text{opt}, \sim} = d_{\text{opt}, \text{rev}} = 0.0635 \cdot \sqrt{400 \cdot \sqrt{30,000}} = 16.7 \text{ mm}.$$

A common optimal rope diameter can be calculated using (3.95) for the different numbers of loading elements w .

Example 3.9: Crane according to Fig. 3.78**Data:**

max. load $Q = 5,000 \text{ kg}$
 max. load stroke $h = 10 \text{ m}$
 nominal strength $R_0 = 1,960 \text{ N/mm}^2$
 drum diameter $D = 20d$
 hook block $G = 300 \text{ kg}$
 Warr.-Seale-6 $\times$ 36-IWRC-sZ
 nom. rope diameter $d = 16 \text{ mm}$
 sheave diameter $D = 22.4d$

Analysis:

In this example, the working cycles are calculated for the maximal load and the maximal load stroke.

The loading elements w (bending cycles) per one loading sequence (hoisting cycle) are drawn in Fig. 3.78. They are

$$w = 5 \rightarrow w_{\nearrow\searrow}(20d) = 1; \quad w_{\nearrow\searrow}(22.4d) = 2; \quad w_{\searrow\searrow}(22.4) = 2;$$

rope bending length l : $l = 0.15 h = 1.5 \text{ m}$

Force factors:

no guidance $f_{S1} = 1.0$
 rope efficiency from Table 3.21: $\eta = 0.99$ and $\eta_L = 0.995$,
 $f_{S2} = 0.5 \left(1 + \frac{1}{0.99 \cdot 0.995} \right) = 1.0075$.
 no parallel bearing ropes $f_{S3} = 1.0$.
 acceleration for the speed $v = 0.3 \text{ m/s}$ according to Table 3.12: $f_{S4} = 1.05$,
 $f_{S4w} = 1 + \frac{w_g \cdot (f_{S4} - 1)}{w} = 1 + \frac{2 \cdot (1.05 - 1)}{5} = 1.02$.

Rope tensile forces:

tensile force, (3.85)

$$S = \frac{(Q + G) \cdot g}{2} \cdot f_{S2} \cdot f_{S4w} = \frac{(5,000 + 300) \cdot 9.81}{2} \cdot 1.0075 \cdot 1.02 = 26,720 \text{ N.}$$

and with no load

$$S_0 = \frac{G \cdot g}{2} \cdot f_{S2} \cdot f_{S4w} = \frac{300 \cdot 9.81}{2} \cdot 1.0075 \cdot 1.02 = 1,510 \text{ N.}$$

difference of tensile forces $\Delta S = S - S_0 = 25,210 \text{ N}$, $\Delta S/S = 0.9435$.

force factor f_{S5} according to (3.69) $f_{S5} = 1.584$.

equivalent tensile force $S_{\nearrow\searrow} = f_{S5} \cdot S = 39,920 \text{ N}$.

Number of bending cycles:

According to (3.55) and (3.89) and (with $f_{N2} = 0.81$ for the six strand rope) the numbers of bending cycles are

full load	no load (but with the hook block)
$N_{A10\downarrow\downarrow}(22.4) = 0.81 * 36,420 = 29,500$	$N_{A10\downarrow\downarrow}(22.4) = 0.81 * 10,950,000 = 8,866,000$
$N_{A10\sim\downarrow}(22.4) = 0.81 * 14,500 = 12,600$	$N_{A10\sim\downarrow}(22.4) = 0.81 * 714,000 = 578,400$
$N_{A10\uparrow\downarrow}(20) = 0.81 * 11,800 = 10,700$	$N_{A10\uparrow\downarrow}(20) = 0.81 * 6,272,000 = 5,080,000$

Hoisting cycles, Working cycles:

According to the Palmgren (3.92) the numbers of hoisting cycles – at which with a 95% certainty not more than 10% of the ropes have to be discarded – are:

$$Z_{A10full} = \frac{1}{\left(\frac{2}{29500} + \frac{2}{12600} + \frac{1}{10700}\right)}; Z_{A10no} = \frac{1}{\left(\frac{2}{8,866,000} + \frac{2}{578,400} + \frac{1}{5,080,000}\right)}$$

$$Z_{A10full} = 3,125 \quad Z_{A10no} = 334,000$$

Then the number of working cycles – at which with a 95% certainty not more than 10% of the ropes have to be discarded – (one hoisting cycle with full and one with no load):

$$W_{A10} = \frac{1}{\left(\frac{1}{Z_{A10full}} + \frac{1}{Z_{A10no}}\right)} = \frac{1}{\left(\frac{1}{3,125} + \frac{1}{334,000}\right)} = 3,100.$$

The mean number of working cycles is $\bar{W}_A = 8,400$.

As to be expected the influence of the small hook block on the number of working cycles can normally be neglected so that the number of hoisting cycles can be taken as the number of working cycles.

Limits:

Donandt force, reverse bending, (3.72)	$S_{D1} = 76.4 \text{ kN} > S = 26.85 \text{ kN}$
optimal rope diameter, (3.95)	$d_{0pt} = 20.0 \text{ mm} > d = 16 \text{ mm}$
discard number of wire breaks (reverse bendings), (3.83)	$B_{A30} = 23 \text{ from } S_{equ}$

With these, all the limits have been taken into consideration.

Example 3.10: Crane with load collective**Data:**

The data in Example 3.9 is used here; however the crane is now loaded with the following load collective

quota $a_1 = 10\%$	load $Q_1 = 5,000 \text{ kg}$
$a_2 = 40\%$	$Q_2 = 3,000 \text{ kg}$
$a_3 = 50\%$	$Q_3 = 1,000 \text{ kg}$

and with the hook block as permanent load $G = 300 \text{ kg}$.

Rope tensile forces:

For the given loads, the tensile forces are –calculated as in Example 3.9 –

$$\begin{array}{lll} S_1 = 26,720 \text{ N} & S_0 = 1,520 \text{ N} & \Delta S/S_1 = (26,720 - 1,520)/26,720 = 0.9435 \\ S_2 = 16,630 \text{ N} & & \Delta S/S_2 = (166,30 - 1,520)/16,630 = 0.9086 \\ S_3 = 6,550 \text{ N} & & \Delta S/S_3 = (6,550 - 1,520)/6,550 = 0.7680. \end{array}$$

Number of hoisting and working cycles:

The numbers of hoisting cycles Z and working cycles W for the different tensile forces are

$$\begin{array}{ll} Z_{A10,1} = 3,125 & W_{A10,1} = 3,100 \\ Z_{A10,2} = 7,010 & W_{A10,2} = 6,870 \\ Z_{A10,3} = 30,700 & W_{A10,3} = 28,100 \end{array}$$

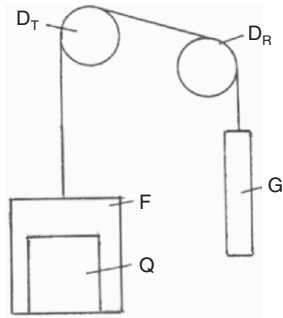
According to (3.93) the number of working cycles – at which with 95% certainty not more than 10% of the ropes have to be discarded – for the collective load is

$$W_{A10} = \frac{1}{\sum_{j=1}^k \frac{a_j}{W_{A10,j}}} = \frac{1}{\frac{0.1}{3,100} + \frac{0.4}{6,870} + \frac{0.5}{28,100}}.$$

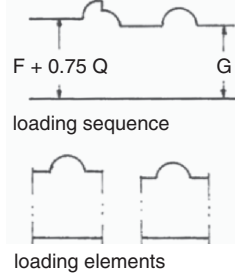
$$W_{A10} = 9,240.$$

and the mean discard number of working cycles is

$$\bar{W}_A = 24,900.$$

Example 3.11: Elevator for residential building**Data:**

Nominal load $Q = 800 \text{ kg}$
 Car mass $F = 1,000 \text{ kg}$
 Wire rope Warrington $6 \times 19 - \text{IWRC} - \text{sZ}$
 Nominal strength $R_0 = 1,570 \text{ N/mm}^2$
 Rope diameter $d = 10 \text{ mm}$
 Number of bearing ropes $n = 6$
 Rope bending length $l = 6 \text{ m}$
 Diameter of traction sh. $D_T = 400 \text{ mm}$
 Diameter of deflection sh. $D_R = 450 \text{ mm}$
 Speed $v = 1 \text{ m/s}$.

**Analysis:**

Most trips made by the car come or go to the ground floor. Therefore the sections of rope running over the sheaves determine rope endurance in elevators.

For one trip from or to the ground floor the loading element (bending cycle) for the traction sheave is

$$w_{\curvearrowright T} = 1.$$

and for the deflection sheave

$$w_{\curvearrowright R} = 1.$$

Rope tensile forces:

For traction sheaves with form grooves Holeschak (1987) evaluated the endurance factor f_{N3} (Table 3.15) under the supposition of a cabin permanently loaded (force factor $f_{S5} = 1$) with 75% of the nominal load. Under the same condition the rope tensile force for the traction sheave is

$$S_T = \frac{(F + 0.75 \cdot Q) \cdot g}{n} \cdot f_{S1} \cdot f_{S2} \cdot f_{S3} \cdot f_{S4w}$$

and for the deflection sheave with the counterweight $G = F + 0.5Q$

$$S_S = \frac{(F + 0.5 \cdot Q) \cdot g}{n} \cdot f_{S1} \cdot f_{S2} \cdot f_{S3} \cdot f_{S4w}$$

Force factors from Table 3.12

For sliding guidance

$$f_{S1} = 1.1$$

For rope efficiency

$$f_{S2} \approx 1$$

For unequal forces in the parallel bearing ropes

$$f_{S3} = 1.25$$

For acceleration or deceleration

$$f_{S4} = 1.12$$

$$f_{S4w} = 1 + 1 \cdot (1.12 - 1) / 2 = 1.06.$$

With these, the rope tensile forces are

$$S_T = \frac{1,600 \cdot 9.81}{6} \cdot 1.1 \cdot 1 \cdot 1.25 \cdot 1.06 = 3,810 \text{ N.}$$

$$S_S = \frac{1,400 \cdot 9.81}{6} \cdot 1.1 \cdot 1 \cdot 1.25 \cdot 1.06 = 3,340 \text{ N.}$$

Number of bending cycles:

With these forces for the traction sheave and with the endurance factor $f_{N3} = 0.1$ for the undercut groove $\alpha = 100^\circ$, the discard numbers of bending cycles are

$$N_{A10,T} = 0.1 \cdot 1,730,000 = 173,000$$

and for the deflection sheave with a normal steel groove $r = 0.53d$ and practically no side deflection

$$N_{A10,S} = 3,910,000.$$

Number of trips:

Then, according to (3.92), the number of trips from or to the ground floor is

$$Z_{A10} = \frac{1}{\frac{1}{173,000} + \frac{1}{3,910,000}} = 166,000.$$

According to Fig. 3.79 for residential buildings with seven floors, the number of trips from or to the ground floor is expected to be 80% of the total number of trips. Then the total number of trips, at which not more than 10% of the wire ropes have to be discarded, is

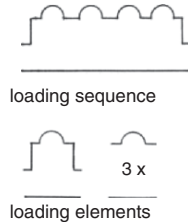
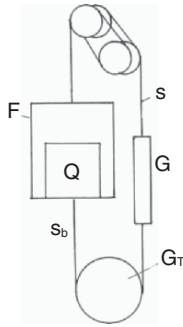
$$Z_{A10,tot} = \frac{Z_{A10}}{0.8} = 207,000$$

and calculated in the same way the mean total number of trips is $\bar{Z}_A = 549,000$.

It is not necessary to consider the force limits for elevators.

Example 3.12: Elevator for high office building**Data:**

Nominal load	$Q = 1,600 \text{ kg}$
Car mass	$F = 2,500 \text{ kg}$
Counterweight	$G = 3,300 \text{ kg}$
Tension sheave mass	$G_T = 400 \text{ kg}$
Hoisting distance	$H = 150 \text{ m}$
Warrington 8×19 - FC - sZ	
Nominal strength	$R_0 = 1,570 \text{ N/mm}^2$
Nominal rope diameter	$d = 16 \text{ mm}$
Number of ropes	$n = 5$
Rope mass	$s = 653 \text{ kg}$
Balance rope mass	$s_b = 653 \text{ kg}$
Speed	$v = 5 \text{ m/s}$
Traction sheave diameter	$D = 40d$
Deflection sheave diam.	$D = 40d$
Steel, groove radius	$r = 0.53d$

**Analysis:**

The load sequence is shown for a rope piece running over the sheaves when the car drives from or to the ground floor. Instead of the unknown rope force changing from both of the traction sheaves at the beginning and the end of the trip, the smallest occurring tensile force will be set in.

As shown in the figure on the left, the loading elements (bendings per trip) is

$$w_{\text{F}\cap\text{L}} = 3 \quad \text{and} \quad w_{\text{F}\cap\text{L}} = 1.$$

The rope bending length is

$$l = 6.4 \text{ m}.$$

Rope tensile forces:

Force factors from Table 3.12

For roller guidance $f_{S1} = 1.05$

For rope efficiency $\eta = 0.995^4 = 0.98$ (Table 3.21 $f_{S2} = (1 + 1/\eta)/2 = 1.01$

For unequal forces in the parallel bearing ropes $f_{S3} = 1.25$

For acceleration or deceleration (on the whole bending length) $f_{S4} = 1.15$.

When the car starts from the ground floor, the rope tensile force (for a mean loading of half of the nominal load) in the rope piece running over the sheaves is

$$S = \frac{(F + 0.5 \cdot Q + s) \cdot g}{n} \cdot f_1 \cdot f_2 \cdot f_3 \cdot f_4$$

$$+ \frac{G_T \cdot g}{2 \cdot n} = \frac{3953 \cdot 9.81}{5} \cdot 1.524 + \frac{400 \cdot 9.81}{2 \cdot 5}. \quad (3.96)$$

$$S = 12,220 \text{ N.}$$

When the counterweight is standing at its lowest position later on, the smallest tensile force in the same rope piece is

$$S_0 = \frac{G \cdot g}{n} + \frac{G_T \cdot g}{2 \cdot n} = \frac{3,300 \cdot 9.81}{5} + \frac{400 \cdot 9.81}{2 \cdot 5}$$

$$S_0 = 6,870 \text{ N.}$$

The difference in tensile force is $\Delta S = S - S_0 = 5,350 \text{ N.}$

$$\Delta S/S = 0.43782.$$

According to (3.69), the force factor is $f_{S5} = 1.2362$

and the equivalent tensile force is $S_{f_{\wedge}} = f_{S.5} \cdot S = 15,100 \text{ N}$

Number of bending cycles and car trips:

The discarding number of bending cycles in simple bending with the endurance factor $f_{N4} = 0.93$ for the side deflection of 0.65° is

$$N_{A_{10\wedge}} = 0.93 \cdot 1,251,000 = 1,163,000.$$

$$N_{A_{10\wedge}} = 0.93 \cdot 698,000 = 649,000.$$

Then the number of trips from or to the ground floor is

$$Z_{A10} = \frac{1}{\frac{3}{1,163,000} + \frac{1}{649,000}} = 243,000.$$

According to Fig. 3.79 for office buildings with 20 possible stops, the number of trips from or to the ground floor is expected to be 43% of the total number of trips. Then the total number of trips at which not more than 10% of the wire ropes has to be discarded is

$$Z_{A10,\text{tot}} = \frac{Z_{A10}}{0.43} = 565,000$$

and the mean total number of trips

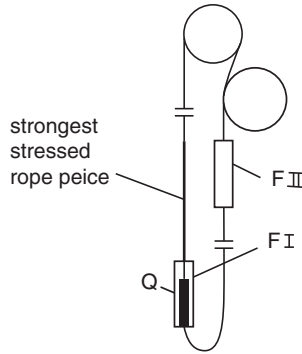
$$\bar{Z}_{\text{Atot}} = 1,284,000.$$

The loss of endurance by the rope twisting is still unknown. For the present twisting distance a relative small endurance loss has to be expected.

For elevators it is not necessary to consider the force limits.

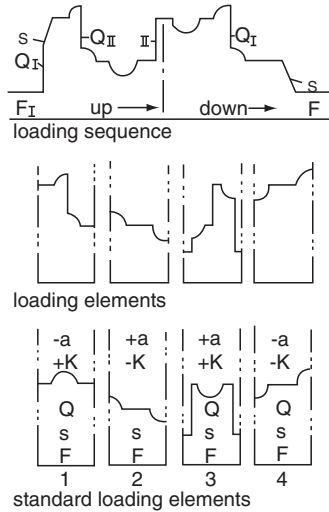
Example 3.13: Mining installation with traction sheave

Data:



Mass of pay load $Q = 10,000 \text{ kg}$
 Mass of cage of attachment $F = 7,000 \text{ kg}$
 Force of moving resistance $K = 8 \text{ kN}$
 Hoisting distance $L = 1,000 \text{ m}$
 Warr.-Seale $6 \times 36 - \text{FC} - \text{sZ}$
 Rope diameter $d = 60 \text{ mm}$
 Number of ropes $n = 1$
 Mass of the rope $s = 13,200 \text{ kg}$
 Mass of the balance rope $s_b = 13,200 \text{ kg}$
 Nominal strength $R_0 = 1,770 \text{ N/mm}^2$
 Speed $v = 15 \text{ m/s}$
 Acceleration $a_0 = 1.5 \text{ m/s}^2$
 Diameter of traction sheave $D_T = 6,000 \text{ mm}$
 Diameter of deflection sheave $D_R = 6,000 \text{ mm}$
 Bending length $l = 40,000 \text{ mm}$.

Analysis:



The rope pieces above the cage I and II are those stressed most strongly. Under the action of acceleration or deceleration, these pieces may have a bending length $l = 40 \text{ m}$. The loading sequence shown in the figure on the left is taken from the rope piece above cage I. The rope pieces not running over a sheave and the attachment are counted as part of the cage mass F .

The loading sequence shown in the figure on the left is taken from the piece above cage I. The loading sequence has been subdivided into loading elements in a most unfavourable way. They are finally transformed into the standard loading elements and marked with symbols for the acting forces (from the load mass Q ; the cage mass F , the mass of rope s or the balance rope s_b ; the moving resistance force K and the acceleration or deceleration a_0).

Tensile forces:

The tensile forces of the rope during bending are

$$S_1 = (F + Q + s) \cdot (g - a_0) + K = 266\,000 \text{ N.}$$

$$S_2 = (F + s) \cdot (g + a_0) - K = 230\,000 \text{ N.}$$

$$S_3 = (F + Q + s) \cdot (g + a_0) + K = 359\,100 \text{ N} \quad S_0 = F \cdot g = 68\,700 \text{ N.}$$

$$(S_3 - S_0)/S_3 = 0.8087 \quad f_{S5} = 2.246 \quad S_{3\text{equ}} = S_3 \cdot f_{S5} = 806\,400 \text{ N.}$$

$$S_4 = (F + Q + s) \cdot (g - a_0) - K = 250\,000 \text{ N}$$

Number of bending cycles:

According to (3.55) for the mean discarding numbers of simple bending cycles and together with (3.89) for reverse bending cycles, the bending cycles are

$$\begin{array}{ll} \bar{N}_{A \curvearrowright 1} = & \bar{N}_{A \curvearrowright 1} = 11\,520\,000 \\ \bar{N}_{A \curvearrowright 2} = 20\,530\,000 & \bar{N}_{A \curvearrowright 2} = 2\,918\,000 \\ \bar{N}_{A \curvearrowright 3} = & \bar{N}_{A \curvearrowright 3} = 149\,000 \\ \bar{N}_{A \curvearrowright 4} = 14\,740\,000 & \bar{N}_{A \curvearrowright 4} = 2\,337\,000 \end{array}$$

(Because of the large numbers of simple bending cycles, the synthetic material of the groove of the traction sheave does not exert any influence on the endurance of the rope.).

Number of working cycles:

According to the Palmgren-Miner (3.92), the 10%-limit and the mean number of working cycles is

$$Z_{A10} = 55\,000 \quad \bar{Z}_A = 125\,000$$

and the load trips for cages I and II up to the ropes being discarded is

$$T_{A10} = 2 \cdot Z_{A10} = 110\,000 \quad \bar{T}_A = 2 \cdot \bar{Z}_A = 250\,000$$

The rope mass investment for one load trip is $s/\bar{T}_A = 0.0528 \text{ kg/load trip}$. (The theoretical minimum rope mass investment for one trip $s/\bar{T}_A = 0.0458 \text{ kg/trip}$ can be found for the rope diameter $d = 72 \text{ mm}$.)

The acceleration has a remarkable influence on rope endurance. For a slightly reduced acceleration to $a_0 = 1 \text{ m/s}^2$, the mean number of trips is $\bar{T}_A = 306\,000$.

The loss of endurance by the rope twisting is still unknown. For the present hoisting distance a relative small endurance loss has to be expected.

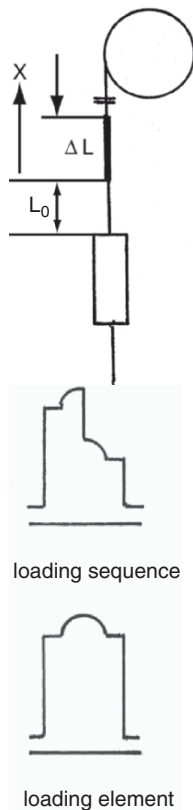
Alternative construction:

In the present example, a traction sheave installation with one bearing rope has been chosen for direct comparison with the drum installation in the following Example 3.14. It is preferable to use parallel bearing ropes for new traction sheave installations.

An installation with four parallel bearing ropes with the rope diameter $d = 30$ mm and the sheave diameters $D = 3,000$ mm produces the same stresses as the present one rope installation. Because of the size effect, the rope endurance of this four rope installation would have slightly better endurance than of a one rope installation, if – in the not very realistic case – the four ropes had the same tensile force. If the rope with the greatest stresses only has an 8% higher tensile force than the mean of the four ropes, then the endurance of this rope will be about the same as the rope in the one rope installation.

Example 3.14: Mining installation with drum

The wire rope is wound in only one layer on the drum in a fitted groove. The drum winder is moved in the direction of the drum axis, strictly geared in relation to the drum rotation in such a way that the wire rope always runs in the same position in relation to the shaft on or off the drum. Therefore virtually no side deflection of the rope exists, Feyrer (2002).



Data:

The data are the same as in Example 3.13.
 The bending length is again $l = 40$ m.
 The rope balance mass ratio is $c = s_b/s$.
 Acceleration zone $\Delta L = v_0^2/2a_0$
 Rope length mass factor $W = 0.367/100$
 The rope piece L_0 not running onto the drum and the attachment is counted as part of the cage mass.

Analysis:

As can be seen in the figure on the left, for one working cycle the wire rope is only stressed by one loading element. This loading element (with changing the tensile force in between the bending while the rope lying on the drum) is transformed in the standard combined fluctuating tension and bending

$$w_{\sigma\epsilon} = 1.$$

Tensile forces:

The wire rope is subject to different tensile forces over the entire length when running onto the drum. When running the rope onto the drum, the tensile force in the rope element at the distance x to the cage is

$$S = (F + Q + W \cdot d^2 \cdot x + W \cdot d^2 (L - x) \cdot c) \cdot (g + a) + K.$$

In this equation, a is to be put in for ΔL (the acceleration and braking distance) and for L (the hoisting distance).

$$a = a_0 \quad \text{for} \quad L - \Delta L < x \leq L$$

$$a = 0 \quad \text{for} \quad \Delta L < x \leq L - \Delta L$$

$$a = -a_0 \quad \text{for} \quad 0 < x \leq \Delta L.$$

The minimum tensile force in a rope element at the distance x to the cage exists when the cage is standing in the lower station

$$S_0 = (F + W \cdot d^2 \cdot x) \cdot g.$$

While accelerating the cage from the lower station, the rope piece $x = L - \Delta L$ to L winding onto the drum is stressed by the highest tensile force. However the tensile force difference $\Delta S = S - S_0$ for this rope piece is relatively small. Therefore, this piece of the rope does not have the lowest endurance if the rope mass ratio here is $c \geq 0.865$.

The wire rope piece most stressed while bent is situated above the cage and above the decelerating rope zone $x > \Delta L$. The distance between the cage and the zone of rope with the highest stresses is $x = \Delta L + l/2$ and $a = 0$.

$$S = 304,400 \text{ N}$$

and the lower tensile force

$$S_0 = 81,000 \text{ N}.$$

The difference between the tensile forces is

$$\Delta S = S - S_0 = 223,400 \text{ N and } \Delta S/S = 0.734.$$

The force factor according to (3.69) is $f_{S5} = 2.057$ and the equivalent force is $S_{\text{equ}} = f_{S5} \cdot S = 626,000 \text{ N}$

Load trips:

The 10%-limit and the mean number of load trips (working cycles) up to the rope discard is (for drum installations the number of load trips $T = Z$).

$$T_{A10} = Z_{A10} = N_{A10\%} = 170,000; \quad \bar{T}_A = \bar{Z}_A = \bar{N}_{A\%} = 383,000.$$

The rope mass investment for one trip is $s/\bar{T}_A = 0.0349 \text{ kg/trip}$.

Alternative:

For $c = 1.0$; $d = 78$ mm:

The rope mass investment is $s/\bar{T}_A = 22,330/857,000 = 0.0261$ kg/trip

For $d = 60$ mm; $c = 0.9$:

The rope mass investment is $s/\bar{T}_A = 13,200/480,000 = 0.0275$ kg/trip.

Further information about ropes in mining installations can be found in Briem (1998, 2001), Damien and Terriez (1995), Fuchs (1988), Fuchs and Spas (1993, 1995), Rebel (2001) and Verreet (2001).

3.5 Rope Efficiency

3.5.1 Single Sheave

When a rope moves over a sheave, a loss of energy occurs due to the friction in the rope itself, the deformation in the contact zone of rope and sheave, and the sheave bearing. The tensile force S on the pulling side is higher than on the other side where the tensile force is $S - \Delta S$. As can be seen schematically in Fig. 3.81, the lever arms for these acting forces correspond in such a way that the momentum is the same on both sides.

The rope efficiency is

$$\eta = \frac{S - \Delta S}{S}. \quad (3.96)$$

The force loss is decisive for rope efficiency. Using their test results, Rubin (1920) and others were among the first to evaluate equations for calculating the loss of force. Their equations have the disadvantage that the specific force loss depends on the diameter of the rope in contradiction to the similarity rule.

Schraft (1997) measured the force loss for wire ropes with different constructions moving over sheaves made of steel. He used tensile forces in a wide

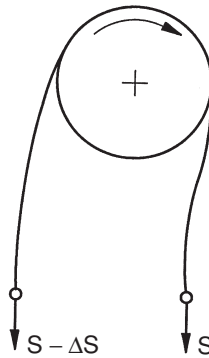


Fig. 3.81. Wire rope moving over a sheave

range and sheave diameters $D = 10d - 100d$ (definition of D according to Fig. 3.32). From his test results and some results from Rubin (1920) and Hecker (1933), he derived the equation for the loss of the specific rope force

$$\frac{\Delta S}{d^2} = \left(\frac{D}{d}\right)^{-1.33} \cdot \left(c_0 + c_1 \cdot \frac{S}{d^2}\right). \quad (3.97)$$

The constants c_0 (in N/mm^2) and c_1 (dimensionless) are evaluated from Schraft and are listed in Table 3.20 for standardised round strand ropes with fibre or steel core and bright or zinc coated. From different types of the two spiral round strand rope constructions 18×7 and 34×7 mean constants have been calculated and included in Table 3.20. The constants relate to well-lubricated ropes. For degreased ropes, the constants c_0 have to be enlarged with 1.56 N/mm^2 and the constants c_1 with 0.084.

The constant c_0 , representing the force loss of wire ropes for the tensile force $S = 0$, has a big standard deviation $s_0 = 1.73 \text{ N/mm}^2$. This big standard deviation depends to a great extent on manufacturing conditions especially the pre-forming of the strands. The constant c_1 , which depends mainly on dimension deviations, has the relatively small standard deviation $s_1 = 0.031$.

The constants in Table 3.20 are only valid for a loss of force due to rope friction. The entire force loss from rope friction, bearing friction and seal friction is

$$\frac{\Delta S_{\text{tot}}}{d^2} = \frac{\Delta S}{d^2} + \frac{\Delta S_{\text{bear}}}{d^2} + \frac{\Delta S_{\text{seal}}}{d^2}. \quad (3.98)$$

The force loss from the bearing is

$$\frac{\Delta S_{\text{bear}}}{d^2} = \frac{\mu_{\text{bear}} \cdot d_{\text{axle}}}{D} \cdot \frac{S}{d^2} \cdot 2 \cdot \sin \frac{\vartheta_D}{2}. \quad (3.99)$$

Table 3.20. Constants c_0 in N/mm^2 and c_1 (dimensionless) for calculating the loss of the specific rope force, (3.97), steel sheave groove, Schraft (1997)

wire rope	bright				zinc coated			
	FC		WRC		FC		WRC	
	c_0	c_1	c_0	c_1	c_0	c_1	c_0	c_1
6×7	0.81	0.189	0.00	0.154	2.59	0.213	1.27	0.178
8×7	1.11	0.198	0.00	0.163	2.89	0.223	1.57	0.188
6×19	1.27	0.211	0.00	0.176	3.05	0.235	1.73	0.200
8×19	1.73	0.228	0.41	0.193	3.51	0.252	2.19	0.217
6×35	1.88	0.240	0.56	0.205	3.66	0.265	2.34	0.230
8×36	2.60	0.270	1.28	0.235	4.38	0.294	3.06	0.259
18×7	2.80	0.251	1.48	0.217	4.58	0.276	3.26	0.241
34×7	5.28	0.329	4.14	0.294	7.06	0.353	5.74	0.318

Standard deviation for c_0 is $s_0 = 1.73$. Standard deviation for c_1 is $s_1 = 0.031$

In that μ_{bear} is the friction coefficient, d_{axle} the axle diameter and ϑ_D the deflection angle of the rope. According to the SKF catalogue for standard roller bearings, the friction coefficient is $\mu_{\text{bear}} \leq 0.0024$. With the general assumption that the axle diameter is five times the rope diameter and that the deflection angle is 180° , the rope force loss from one or two roller bearings is

$$\frac{\Delta S_{\text{bear}}}{d^2} = 0.024 \cdot \frac{d}{D} \cdot \frac{S}{d^2}. \quad (3.100)$$

For rarely used sheaves with slide bearings, the friction coefficient is much greater. The sliding bearings in (3.99) can be set to $\mu_{\text{bear}} = 0.1$ with an additional charge of 50% for tearing loose.

The rope force loss for two seals of roller bearings is

$$\frac{\Delta S_{\text{seal}}}{d^2} = \frac{2 \cdot \pi \cdot q \cdot d_{\text{axle}}^2}{D \cdot d^2}. \quad (3.101)$$

For the seals of roller bearings, the friction resistance is $q = 0.01 \text{ N/mm}$ according to the SKF catalogue. Then again with $d_{\text{axle}} = 5d$, the specific rope force loss for two seals of roller bearings is

$$\frac{\Delta S_{\text{seal}}}{d^2} = \frac{1.6}{D} \text{ in N/mm}^2. \quad (3.102)$$

Müller (1990) found a specific friction force between $q = 0.07$ and 0.20 N/mm for standard sliding lip seals with spring pressing.

Simplified Calculation

A simplified calculation of the rope force loss is possible with (3.97) if the constants c_0 and c_1 have been supplemented to take into account the resistance caused by the roller bearings and the sliding seals of roller bearings.

To take the roller bearing resistance into account, the rope force loss is set the same as in (3.97) with $c_0 = 0$ and (3.100)

$$c_{1\text{bear}} \cdot \left(\frac{D}{d}\right)^{-1.33} \cdot \frac{S}{d^2} = 0.024 \cdot \frac{d}{D} \cdot \frac{S}{d^2}.$$

Then the constant for the influence of the sheave bearing is

$$c_{1\text{bear}} = 0.024 \cdot \left(\frac{D}{d}\right)^{0.33} \approx 0.075. \quad (3.103)$$

To take the seals of roller bearings into account, the rope force loss is set the same as in (3.97) with $c_1 = 0$ and (3.102)

$$c_{0\text{seal}} \cdot \left(\frac{D}{d}\right)^{-1.33} = \frac{1.6}{d} \cdot \frac{d}{D}.$$

Then the constant is

$$c_{0\text{seal}} = \left(\frac{D}{d}\right)^{0.33} \cdot \frac{1.6}{d} \approx \frac{5}{d} \quad (3.104)$$

and to be on the safe side for $d > 5$ mm, the constant for the rope force loss in (3.97) is on the safe side $c_{0\text{seal}} = 1.0$.

Then the constants for the wire rope force loss for a steel sheave with one or two roller bearings and two roller bearing seals with c_0 and c_1 from Table 3.20 and $c_{0\text{seal}}$ and $c_{1\text{bear}}$ from (3.103) and (3.104) are

$$c_{0\text{tot}} = c_0 + c_{0\text{seal}} \text{ and}$$

$$c_{1\text{tot}} = c_1 + c_{1\text{bear}}.$$

With a certainty of 95%, in not more than 10% of the cases the rope force loss is greater than that calculated with the constants

$$c_{0\text{tot},10} = c_0 + 1.5 \cdot s_0 + c_{0\text{seal}} \text{ and}$$

$$c_{1\text{tot},10} = c_1 + 1.5 \cdot s_1 + c_{1\text{bear}}.$$

According to (3.97), with these constants the rope force loss is

$$\frac{\Delta S_{\text{tot}}}{d^2} = \left(\frac{D}{d}\right)^{-1.33} \cdot \left(c_{0\text{tot}} + c_{1\text{tot}} \cdot \frac{S}{d^2}\right). \quad (3.105)$$

and according to (3.96), the rope efficiency is

$$\eta = 1 - \left(\frac{c_{0\text{tot}}}{S/d^2} + c_{1\text{tot}}\right) \cdot \left(\frac{D}{d}\right)^{-1.33}. \quad (3.106)$$

In Table 3.21 the rope efficiency (that with a certainty of 95% is higher in not more than 10% of the cases) is listed for the stiff wire rope 34×7 - FC

Table 3.21. Wire rope efficiency in %

D/d	$S/d^2 = \text{in N/mm}^2$					
	10	50	100	150	200	250
10	92.8	96.8	97.3	97.4	97.5	97.6
12.5	94.6	97.6	98.0	98.1	98.2	98.2
16	96.1	98.3	98.5	98.6	98.7	98.7
20	97.1	98.7	98.9	99.0	99.0	99.0
25	97.9	99.0	99.2	99.2	99.3	99.3
32	98.5	99.3	99.4	99.5	99.5	99.5
40	98.9	99.5	99.6	99.6	99.6	99.6
50	99.2	99.6	99.7	99.7	99.7	99.7
63	99.4	99.7	99.8	99.8	99.8	99.8

Rope efficiency is in 90% of the cases even for the unfavourablest rope construction higher than that of the table for ropes lubricated with diameter $d \geq 5$ mm; for not too deep temperature; for metallic sheaves with roller bearings and sliding seals.

lubricated, in room temperature, zinc coated, rope diameter $d \geq 5$ mm for different specific tensile forces and diameter ratio of metallic sheaves. The sheave has roller bearings with sliding seals. Under these conditions, the constants c using the rope efficiency in Table 3.21 has been calculated with (3.106) using the constants

$$c_{0\text{tot}} = 7.06 + 1.5 \cdot 1.73 + 1 = 10.66 \text{ and}$$

$$c_{1\text{tot}} = 0.353 + 1.5 \cdot 0.031 + 0.075 = 0.475.$$

The rope efficiency is in reality mostly higher than that taken from Table 3.21. It can therefore be generally used as it is on the safe side.

However, in very low temperatures, wire ropes are much less efficient than those listed in Table 3.21. Bendix and Sommerfeld (1979) took measurements at temperatures between 20 and -50° C with various lubricants. They found that appropriate lubricants had less than a double rope force loss in contrast to room temperature. However, when inappropriate lubricants were used, a multiple rope force loss had to be expected. No research has yet been done on the rope force loss for sheaves with grooves made of synthetic material and for rope side deflection.

3.5.2 Rope Drive

Stationary Sheaves

For a rope running over n stationary sheaves (that means with the same peripheral speed) the efficiency is

$$\eta_{\text{stat}} = \eta^n. \quad (3.107)$$

Hanging Sheave

The efficiency for a rope running over a hanging sheave – as seen in Fig. 3.82 – is by definition the ratio of the half force Q and the rope force S

$$\eta_{\text{h}} = \frac{Q/2}{S_1} = \frac{S_1 + S_2}{2 \cdot S_1}.$$

For $S = S_1$ and $S_2 = S \cdot \eta$ the efficiency for the hanging sheave is

$$\eta_{\text{h}} = \frac{1 + \eta}{2}. \quad (3.108)$$

Tackle Block

The forces in a tackle block – as seen in Fig. 3.83 – are

$$S_1 = S$$

$$S_2 = \eta \cdot S$$

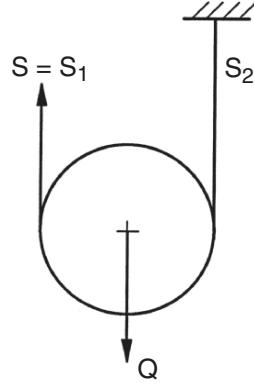


Fig. 3.82. Hanging sheave

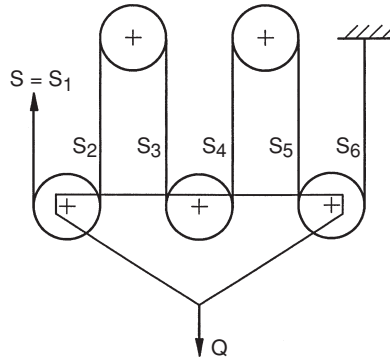


Fig. 3.83. Tackle block

$$S_3 = \eta^2 \cdot S$$

$$S_z = \eta^{z-1} \cdot S.$$

The sum of all rope forces S_i is equal to the force from the hanging load

$$Q = \sum_{i=1}^z S_i$$

or

$$Q = S \cdot (1 + \eta + \eta^2 + \dots + \eta^{z-1}).$$

The sum for this geometrical row is

$$Q = S \cdot \frac{1 - \eta^z}{1 - \eta}. \quad (3.109)$$

Now the efficiency of the tackle block is by definition

$$\eta_{\text{tac}} = \frac{Q}{z \cdot S}.$$

Then with (3.109) the efficiency of a tackle block with z bearing ropes is

$$\eta_{\text{tac}} = \frac{1}{z} \cdot \frac{1 - \eta^z}{1 - \eta}. \quad (3.110)$$

Rope Drive Efficiency

For a rope drive with n stationary sheaves and in addition a tackle block with z bearing wire rope falls, the total rope drive efficiency is

$$\eta_{\text{tot}} = \eta_{\text{stat}} \cdot \eta_{\text{tac}}$$

or

$$\eta_{\text{tot}} = \eta^n \cdot \frac{1}{z} \cdot \frac{1 - \eta^z}{1 - \eta}. \quad (3.111)$$

Example 3.15: Efficiency of a rope drive

Data:

Crane with pay load 30 t
 Force from load $Q = 30,000 \times 9.81 = 294,000 \text{ N}$
 Number of rope falls $z = 8$
 Number of stationary sheaves $n = 2$
 Spiral round strand rope 34×7+WSC, bright, lubricated, $W = 0.904 \text{ kg/m}$
 Nominal rope diameter $d = 15 \text{ mm}$
 Sheave diameter $D = 20 \text{ d}$, cast iron
 Roller bearings with sliding seals

Constants:

For the rope force loss that is greater in not more than 10% of the cases, using the data from Table 3.20, the constants c_{tot} are

$$c_{0\text{tot}} = 04.14 + 1.5 \cdot 1.73 + 5/15 = 7.07 \text{ and}$$

$$c_{1\text{tot}} = 0.294 + 1.5 \cdot 0.031 + 0.075 = 0.416$$

Rope efficiency for one sheave:

With these constants, according to (3.106) the wire rope efficiency is lower in not more than 10% of the cases for the specific tensile force $S/d^2 = Q/(z \cdot d^2) = 164 \text{ N mm}^{-2}$ for the rope running over one sheave

$$\eta_{90} = 1 - \left(\frac{7.07}{164} + 0.416 \right) \cdot 20^{-1.33} = 0.9914.$$

Efficiency of the rope drive:

According to (3.111), the efficiency for the rope drive – once again it is lower in not more than 10% of the cases – is

$$\eta_{\text{tot}90} = 0.9914^2 \cdot \frac{1}{8} \cdot \frac{1 - 0.9914^8}{1 - 0.9914} = 0.9538.$$

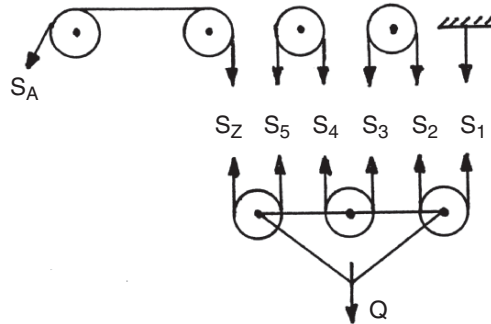


Fig. 3.84. Rope drive of a crane, $Z = 6$ and $n = 2$

3.5.3 Lowering an Empty Hook Block

For lowering an empty hook block, the weight force of the empty hook block should be greater than the weight force of the hanging wire rope (between the drum and the first sheave) and the resistance forces from the wire rope efficiency. The weight force of the hook block must be so great that it moves down if the wire rope is loosened by the drum. The necessary minimum weight force of the hook block will be derived for the rope drive according to Fig. 3.84. It will be presupposed that the stationary sheaves are positioned at about the same height and that the hanging sheaves of the hook block in the highest position will not be any great distance from the stationary sheaves. That means that the weight force of the wire rope pieces in between is small and can be neglected. It will be further presupposed that all the sheaves have the same diameter.

When the hook block is lowered, the greatest rope tensile force S_1 is on the fixed point of the rope. When running over a sheave, the rope tensile force will be reduced by a small force difference ΔS_i . The rope tensile forces in the tackle block are then

$$S_1$$

$$S_2 = S_1 - \Delta S_1$$

$$S_3 = S_1 - \Delta S_1 - \Delta S_2.$$

$$S_z = S_1 - \sum_{i=1}^z \Delta S_i. \quad (3.112)$$

where

z is the number of bearing falls of wire ropes, and

ΔS_i is the loss of rope tensile force between the fall of wire rope i and $i+1$.

The loss of tensile force ΔS depends partly on the rope tensile force and partly not. The rope tensile forces are very low when the empty hook block is lowered. Therefore, the loss of tensile force which is dependent on the rope

tensile force is very small and the loss of rope tensile force can be used as constant

$$\Delta S_i = \Delta S.$$

With this, (3.112) will be simplified to

$$S_z = S_1 - (z - 1) \cdot \Delta S. \quad (3.113)$$

The weight force $Q_H = m_H \cdot g$ of the hook block will be borne by the rope forces

$$Q_H = S_1 + S_2 + S_3 + \cdots + S_z = \sum_{i=1}^z S_i. \quad (3.114)$$

or

$$Q_H = S_1 + S_1 - \Delta S + S_1 - 2\Delta S + S_1 - 3\Delta S + \cdots + S_1 - (z - 1)\Delta S.$$

and summarized

$$Q_H = z \cdot S_1 - \frac{z-1}{2} \cdot z \cdot \Delta S. \quad (3.115)$$

When the rope leaves the tackle block, using this and (3.113), the rope tensile force is

$$S_z = \frac{Q_H}{z} - \frac{z-1}{2} \cdot \Delta S. \quad (3.116)$$

This equation was also developed by Matthias (1972).

The rope tensile force will be reduced further when the wire rope runs over n stationary sheaves. The rope tensile force is then

$$S_A = S_z - n \cdot \Delta S. \quad (3.117)$$

and using (3.116)

$$S_A = \frac{Q_H}{z} - \frac{z-1}{2} \cdot \Delta S - n \cdot \Delta S. \quad (3.118)$$

From this equation, the minimum weight force of the hook block is

$$Q_H = z \cdot \left(S_A + \frac{z-1}{2} \cdot \Delta S + n \cdot \Delta S \right). \quad (3.119)$$

The mean loss of the rope tensile force is calculated for the mean rope force S (as the average of the smallest $S_A + \Delta S$ and the biggest S_1). The mean rope force is

$$S = \frac{S_A + \Delta S + S_1}{2}. \quad (3.120)$$

or with (3.113) and (3.117)

$$S = S_A + \frac{z-1+n+1}{2} \cdot \Delta S = S_A + \frac{z+n}{2} \cdot \Delta S. \quad (3.121)$$

With (3.105), (3.119) and (3.121) and by eliminating S and ΔS , the minimum mass force of the empty hook block is

$$Q_H = z \cdot S_A + z \cdot \left(\frac{z-1}{2} + n \right) \cdot \frac{c_{0\text{tot}} \cdot d^2 + c_{1\text{tot}} \cdot S_A}{\left(\frac{D}{d} \right)^{+1.33} - c_{1\text{tot}} \cdot \frac{z+n}{2}}. \quad (3.122)$$

In this equation, d is once again the nominal rope diameter, D the sheave diameter both in mm, n the number of sheaves outside the tackle block and z the number of bearing rope falls.

The rope tensile force S_A on the first sheave has an important influence on the minimum weight force of the hook block. This rope tensile force depends on the arrangement of the drum and the sheaves. In the simplest case – the drum is situated vertically below the first sheave – the rope tensile force S_A is then equal to the weight force of the rope piece between the drum and the first sheave. Thereby, as mentioned before, it is presupposed that all the stationary sheaves are located at the same height.

Example 3.16: Lowering an empty hook block

The data of the crane are the same as for example 3.15 with, in addition, at the first stationary sheave for the weight force of the vertical rope piece with a length $h = 20$ m above the drum, the rope tensile force is $S_A = W g h = 177$ N.

According to (3.122), the minimum weight force of the empty hook block is

$$\begin{aligned} Q_{H90} &= 8 \cdot 177 + 8 \left(\frac{8-1}{2} + 2 \right) \frac{7.07 \cdot 15^2 + 0.416 \cdot 177}{20^{+1.33} - 0.416 \cdot \frac{8+2}{2}} \\ &= 1,416 + 1,417 = 2,833 \text{ N.} \end{aligned}$$

The minimum weight of the hook block is then 289 kg.

For very low temperatures with an appropriate lubricant, the force loss by friction may be doubled. Then the minimum mass force is

$$Q_{H90\text{cold}} = 1,416 + 2,834 = 4,250 \text{ N.}$$

and the minimum weight of the hook block is 433 kg.

References

- Aberkrom, P.: Seilzugkräfte in Treibscheibenaufzügen. Lift-Report 15 (1989) 2, S15–20
- Andorfer, K.: Die Zugkraftverteilung in schwingend beanspruchten geraden Drahtseilen. Diss Techn. Universität Graz, 1983
- Apel, G.: Beitrag zur Optimierung sicherheitsrelevanter Eigenschaften korrosionsgeschützer, hochfester Stahldrahtseil. Diss. Universität, Wuppertal, 1981

- Arnold, H. and Hackenberg, W.: Seilatlas für Zugseile und Streckenförderanlagen. Essen: Verlag Glückauf GmbH, 1971
- Babel, H.: Metallische und nichtmetallische Futterwerkstoffe für Aufzugtreibscheiben. Diss. TH Karlsruhe, 1980
- Bach, von K.: Die Maschinenelemente. Stuttgart: Bergsträsser Verlag, 1881
- Bahke, E.: 150 Jahre Lebensdaueruntersuchung von Seildraht und Drahtseil. Int: Kolloquium 150 Jahre Drahtseil. Techn. Akademie Esslingen, 13–14th September 1984
- Bechtloff, G. and Szelagowski, P.: Höhere Lebensdauer von Drahtseilen durch verbesserte Seilrollen. Fördern und Heben 17 (1967) 6, S337–342
- Bechtloff, G.: Die Beanspruchung des Drahtseilquerschnittes unter ebener Querpressung. Drahtwelt 55 (1969) 2, S80–87; 3, S147–158 and 4, S196–203
- Beck, W.: The increase of wire breaks in haul ropes of aerial tramways and lifts. 5. OITAF-NACS Symposium in Steamboat Springs, Colorado, USA, 1992
- Beck, W. and Briem, U.: Correlation between endurance prediction and service life of running ropes. OIPEEC Round Table Delft, Sept. 1993, ISBN 90-370-0091-6
- Beck, W. and Briem, U.: Correlation between the estimated and actual (service life of running ropes. WIRE 45 (1995) 6, pp. 333–335
- Bendix, H. and Sommerfeld, H.-U.: Einfluß tiefer Temperatur auf die. Berechnung und konstruktive Ausführung von Seilflaschungen., Hebezeuge und Fördermittel 19 (1979) 6, S169–173
- Benoit, G.: Die Drahtseile. Karlsruhe, Verlag F. Gutsch, 1915
- Benoit, G.: Zum Gedächtnis an W. A. J. Albert and die Erfindung seines Drahtseiles. Berlin: VDI-Verlag GmbH, 1935
- Bock, E.: Die Bruchgefahr der Drahtseile. Diss. TH Hannover, 1909
- Bradon J. E. and Chaplin, C. R.: The prediction of fatigue endurance of riser tensioner ropes for floating drilling rigs as a function of location. OIPEEC Round Table Reading 1997, pp. 237–256
- Briem, U.: Verbesserung der Ablegereifeerkennung laufender Drahtseile durch Kombination von Ablegekriterien. Universität Stuttgart Diss., 1996
- Briem, U. and Jochem, M.: Berechnung der Lebensdauer von Seilen in Schachtförderanlagen. Glückauf-Forschungshefte 56 (1998) 3, S87–92
- Briem, U.: Correlation between predicted and actual service life of ropes in mining installations. OIPEEC Bulletin 82 (2001), pp. 13–25
- Briem, U.: Seillebensdauer bei Mehrlagenspulung. Hebezeuge und Fördermittel 42 (2002) 11, S566–567
- Calderale, P. M.: Festigkeitsversuche mit Drahtseilen und analytische Auswertung. DRAHT 11 (1960) 8, S407–415
- Casey, N. F.: The fatigue endurance of wire ropes for mooring offshore structures. OIPEEC Round Table, Delft, Sept. 1993, pp. I.21–I.49
- Castillo, E., Fernandez-Canteli, A., Ruiz-Tolosa and Sarabia, J. M.: (1990) Statistical models for analysis of fatigue life of long elements. Transactions

- of the ASCE Journal of Engineering Mechanics 116 (1990) 5, pp. 1036–1049, Paper 24618
- Castillo, E. and Fernandez-Canteli, A.: Statistical models for fatigue analysis of long elements. Introductory Lectures of the IABSE- Workshop, “Length Effect on the Fatigue of Wires and Strands”. Madrid, Sept. 1992
- Chaplin, C. R.: Bending-Tension Fatigue in Offshore Applications. OIPEEC Bulletin 52, Torino, Dec. 1986, pp. 9–22
- Chaplin, C. R.: Prediction of Offshore Mooring Ropes. OIPEEC Round Table, Delft, Sept. 1993, pp. I.50–I.75
- Ciuffi, R. and Wiek, L.: Report of the working party for the reference machine. OIPEEC Bulletin 28, Torino, March 1976, pp. 26–27
- Ciuffi, R.: Report on B. L. P. (block load program). OIPEEC Bulletin 36 (1979), pp. 38–52
- Ciuffi, R. and Roccati, G.: Wire rope and size effect. OIPEEC Technical Meeting Stuttgart 1995, Proceedings, pp. 10–1 to 10–10
- Clement, P.: Fatigue d’un cable passant sur une poulie; interpretation des resultats d’essais. OIPEEC Bulletin 37, Torino Mai 1980, pp. 8–29
- Clement, P.: Fatigue d’un cable en flexion sur unepoulie. OIPEEC (1981) Bulletin 40, Torino, Sept. 1981, pp. 26–56
- Clement, P.: Etude de l’endommagement subi par un cable en flexion alternee sur poulies. OIPEEC Bulletin No. 45, Torino, July 1983, pp. 67–90
- Costello, G. A.: Large deflection of helical spring due to bending. Journal of the Engineerings Mechanical Division (ASCE) 103 (1977) 3, pp. 479–487
- Costello, G. A. and Miller, R. E.: Lay effect of wire rope. Journal of the Engineering Mechanics Division, ASCE, 105 (1979), No. EM5, pp. 597–608
- Costello, G. A. and Butson, G. J.: Simplified bending theory for the wire rope. Journal of the Engineerings Mechanical Division (ASCE) 108(1982) 8, pp. 219–227
- Costello, G. A.: Theory of wire rope. Second Edition. New York Springer, 1997, ISBN 0–387–98202–7
- Czitary, E.: Seilschwebbahnen. Wien: Springer, 1951
- DIN 15020: Grundsätze für Seiltriebe, Feb. 1974
- Daeves, K. and Linz, P.: Die Beanspruchung und Entwicklung von Förderseilen für hohr Förderdichten. Glückauf 77 (1941) 43, S601–606
- Damien, J.-P. and Terriez, J.-M.: Endurance analysis of wire ropes as used in deep extraction installations. OIPEEC Technical Meeting, Stuttgart, Sept 1995 and OIPEEC Bulletin 70, Dec. 1995
- Davidson: Investigation and calculation of the remaining tensile strength in wire ropes with broken wires. Ingeniörs-Ventenskaps Akademiens Handlingar No. 214, Stockholm, 1955
- Dietz, P.: Ein Verfahren zur Berechnung ein- und mehrlagig bewickelter (1971), Seiltrommeln. Diss. TH Darmstadt, 1971
- Donandt, H.: Über die Berechnung von Treibscheiben im Aufzugbau. Diss. TH Karlsruhe, 1927

- Donandt, H.: Zur Beanspruchung der Drahtseile. Z. VDI 78 (1934) 52, S1497–1498
- Donandt, H.: Zur Dauerfestigkeit von Seildraht und Drahtseil. Archiv für das Eisenhüttenwesen 21 (1950) 9/10, S284–292
- Dragone, G.: Spectres de charges et l'étude de l'endurance des cables par la method charge variable. OIPEEC Table Ronde, Milano, 1973
- Drucker, D. C. A. and Tachau, H.: A new design criterion for the wire rope. Annual meeting of the American Society of Mechanical Engineers, New York, 27 Nov.–1. Dec. 1944
- Dudde, F.: Gleichschlagseile unter schwellender Zug- und Biegebeanspruchung. Diplomarbeit Inst. für Fördertechnik, Universität Stuttgart, 1991
- Eilers, R. and Schwarz, W.: Dauerversuche an Seilrollen und Seilen beweisen die Vorteile gehärteter Rillen. Maschinenmarkt 80 (1974) 70, S1359–1361
- Ernst, H.: Beitrag zur Beurteilung der behördlichen Vorschriften für die Seile von Personenschwebbahnen. Diss. TH Danzig, 1933
- Eshkenazy, R., Weiss M. P. and Elata, D.: Torsion and bending stresses in wires of non-rotating tower crane ropes. OIPEEC Technical, Meeting Lenzburg, Sept. 2003, pp. 77–99
- Feyrer, K.: Der Fangvorgang bei Treibscheibenaufzügen. Fördern und Heben 27 (1977) 2, S129–135 and 3, S216–220
- Feyrer, K.: Die Biegewechselzahl von Standardseilen beim Einfachbiegeversuch. DRAHT 32 (1981a) 10, pp. 556–558 and 11, pp. 603–606
- Feyrer, K.: Effect of bending length on endurance of wire ropes. Wire World 23 (1981b), July/August, pp. 115–119
- Feyrer, K.: The number of visible wire breaks as discarding criterion of wire ropes. OIPEEC Bulletin 40, Torino, Sept. 1981c, pp. 12–25
- Feyrer, K.: The polygonal deflection of a wire rope. WIRE 32 (1982), 1, pp. 23–29, DRAHT, Le Trefile, Alambre
- Feyrer, K. and Hemminger, R.: New-rope-bending-fatigue-machine constructed in the traditional way. OIPEEC Bulletin 45 (1983a), Torino, pp. 59–66
- Feyrer, K.: The distribution of wire breaks on a wire rope. OIPEEC Bulletin 46 (1983b), pp. 48–52, Torino 1983
- Feyrer, K.: Die Drahtbruchentwicklung von laufenden Seilen. DRAHT 34 (1983c) 5, S245–249
- Feyrer, K.: Ablegedrahtbruchzahl von Parallelschlagseilen. DRAHT 35 (1984) 12, S611–615
- Feyrer, K.: The number of bending cycles to breakage of parallel lay wire ropes. (WIRE 35 (1985a) 5, pp. 198–202 (also in German, French, Spain, Italien)
- Feyrer, K.: Ablegiebiegewechselzahl von Parallelschlagseilen. DRAHT 36 (1985b) 3, S127–131
- Feyrer, K.: Pressure between tape and pulley. OIPEEC Bulletin 52 (1986a), pp. 23–31

- Feyrer, K.: The optimal rope diameter. WIRE 36 (1986b) 2, pp. 66–68; DRAHT 36 (1985) 6, Le Trefile 36 (1986) 2, Alambre 35 (1985) 6, Il Filo Metallico 33 (1986) 2
- Feyrer, K.: The application of a new method of wire rope calculation WIRE 38 (1988) 1, pp. 44–47 and OIPEEC-Bulletin 55, (1988), pp. 35–51
- Feyrer, K.: Discard Number of Wire Breaks for Stranded Ropes. (OIPEEC – Round – Table 1989 in Zürich, pp. 6.1–6.11; from DRAHT 35 (1984) 12, S611–615
- Feyrer, K.: Number of bending cycles of wire ropes in different bending machines. OIPEEC Bulletin 59, Torino, (1990a), pp. 5–12
- Feyrer, K. and Gu, B.: Die Restbruchkraft von Drahtseilen nach wiederholtem Lauf über Seilrollen. Draht 41 (1990b) 10, S953–957
- Feyrer, K.: Increased safety with two parallel ropes. Lift-Report 16 (1990c) 5, pp. 21–30 and Wire Industry 59 (1992), pp. 631–637
- Feyrer, K. and Jahne, K.: Seillebensdauer bei Gegenbiegung. DRAHT 42 (1991a) 6, S433–438. LE TREFILE 42 (1992) 1, S68–73. ALAMBRE 42 (1992) 1, S64–70. IL FILO METALLICO 38 (1991) 4, S274–279
- Feyrer, K.: Sicherheitszuwachs durch mehrere parallele Seile. Fördern und Heben 41 (1991b) 12, S1036–1040
- Feyrer, K.: Einfluß der Drahtfestigkeit auf die Biegewechseizahl von Drahtseilen. DRAHT 43 (1992) 7/8, S663–666
- Feyrer, K. and Vogel, W.: High-strength fibre rope running over rope pulleys. WIRE 42 (1992) 5, S455–458 and DRAHT 42 (1991) 11
- Feyrer, K.: Wire ropes under fluctuating tension and bending. WIRE 43 (1993) 1, pp. 48–52. Little shortened in OIPEEC – Bulletin 63, Reading May 1992, pp. 27–40
- Feyrer, K.: Einfache Berechnung der Seiltriebe von Hebezeugen. Hebezeuge and Fördermittel 36 (1996) 4, S186–188 and 5, S242–243
- Feyrer, K.: Biegewechselzahl und Ablegereife von Spiral-Rundlitzenseilen. Fördern und Heben 47 (1997) 5, S356–360
- Feyrer, K.: Nachschmierung von laufenden Drahtseilen. DRAHT 49 (1998) 1, S40–46
- Feyrer, K.: Absenken der leeren Hakenflasche. Fördern und Heben 50 (2000) 3, S216–218
- Feyrer, K.: Wire rope endurance in mining installations for very deep shafts. OIPEEC Bulletin No. 84. Reading, Dec 2003, pp. 11–19. Seillebensdauer in Schachtförderanlagen für sehr große, Teufen. Glückauf-Forschungshefte 63 (2002) 1, S16–19
- Franke, K.P.: Feder-Dämpferkoppelemente, Dr.-Ing. Diss. Universität der Bundeswehr Hamburg (1991)
- Fuchs, D.: Veränderte Beanspruchung von Förderseilen mitzunehmender Teufe. Glückauf 124 (1988) 23/24, S1262–1268
- Fuchs, D.: Andere zerstörungsfreie Seilprüfungen. In: Laufende Drahtseile, Expert Verlag 7044 Ehningen (1989)

- Fuchs, D. and Spas, W.: A method of calculating the hoisting cycles of a rope as a function of stress to the point of discarding. OIPEEC Round Table, Delft, Sept. 1993, 1-91 to 1-102. TU Delft, ISBN 90-370-0091-6
- Fuchs, D. and Spas, W.: The influence of fluctuating tensile stress on the endurance of hoisting ropes on Koepe haulage systems. OIPEEC Technical Meeting, Stuttgart. Sept. 1995 and OIPEEC Bulletin 70, Dec. 1995
- Gareis, C.: Aus der Arbeit des Deutschen Aufzugausschusses (DAA1992). Lift-Report 18 (1992) 1, S36-37
- Giovannozzi, R.: Report on the research of the fatigue of wire ropes. OIPEEC Bulletin 8, Torino (1967), pp. 9-10
- Giovannozzi, R.: Report on REFMA. OIPEEC Bulletin 36, Torino Nov. 1979, pp. 53-60
- Gräbner, P.: Dauererprobung von Stahldrahtseilen. Hebezeuge und Fördermittel 8 (1968) 11, S339-341
- Gräbner, P. and Schmidt, U. Beitrag zur Ermittlung der Ablegereife Wiss. Zeitschrift der Hochschule für Verkehrswesen Dresden 26 (1979) 5, S883-890
- Greis, P.: Untersuchung der Lebensdauer von Drähten und Seilen für Krananlagen. Stahl und Eisen 99 (1979) 10, S518-527
- Grolik, H. and Hartung, R.: Zur Ermittlung von Biegewechselzahlen an Seiltrieben von Fördermaschinen. Hebezeuge und Fördermittel 30 (1990) 9, S260-263
- Häberle, B.: Pressung zwischen Seil und Seilrille. Diss. Universität Stuttgart, 1995
- Haibach, F. and Fuchs, D.: Zur Betriebsfestigkeit von Förderseilen in Koepe-Anlagen. DRAHT 36 (1985) 5, S232-236
- Hankus, J.: Fatigue tests on hoisting wire ropes of diameters above 50 mm. OIPEEC Round Table, East Kilbride Glasgow June 1985, pp. 4.2.1-4.2.26
- Hecker, H. W.: Über den Biegewiderstand von Drahtseilen. Diss. TH, Hannover 1933
- Heptner, K.: Dynamische Seilkräfte bei Elektro-Hebezeugen. Fördern und Heben 21 (1971) 11, S691-694
- Herbst, H.: Zur Bewertung von Drahtbrüchen für die Sicherheit von Förderseilen. Der Bergbau 47 (1934) 15, S215-220
- Hobbs, R. E. and Nabijou, S.: Changes in wire curvature as a wire rope (1995) is bent over a sheave. Journal of strain analysis 30 (1995) 10, S271-281
- Holeschak, W.: Die Lebensdauer von Aufzugseilen und Treibscheiben im praktischen Betrieb. Diss. Universität Stuttgart, 1987. Kurzf.: Lift-Report 14 (1988) 1, S6-9
- Hymans F. and Hellborn, A. V.: Der neuteitliche Aufzug mit Treibscheibenantrieb. Berlin Heidelberg New York: Springer, 1927
- Isaachsen, J.: Die Beanspruchung von Drahtseilen. Z. VDI 51 (1907) 17, S652-657
- Jahne, K.: Zuverlässigkeit des Ablegekriteriums Drahtbruchzahl beilaufenden Seilen. Diss. Universität Stuttgart 1992, Kurzf.: DRAHT 44 (1993a) 7/8, S427-434

- Jahne, K.: Reliability of the number of wire breaks as discarding (1993b) criterion for steel wire ropes running over sheaves. OIPEEC Bulletin 66, Nov. 1993
- Janovsky, L.: Verteilung der Zugkräfte in Aufzugseilen. Lift-Report 11 (1985) 5/6, S35–39
- Jehmlich, G. and Steinbach, G.: Einsatz und Überwachung von Drahtseilen, Dimensionierung von Seiltrieben. Hebezeuge und Fördermittel 20 (1980) 12, S360–362
- Jehmlich, G.: Anwendung und Überwachung von Drahtseilen. Berlin, VEB Verlag Technik, 1985
- Jurk, G.: Einfluß des Umschlingungsbogens auf die Lebensdauer von Drahtseilen. Wiss. Zeitschrift der TH Karl-Marx-Stadt XV (1973) 4, S727–735
- Klein, E. H. L.: Die Berechnung der Drahtseile. Fördertechnik XXX (1937) 16, pp. 321–328
- Leider, M. G.: Untersuchungen über die Zusatzspannungen von Drahtseilen. DRAHT 24 (1973) 5, S247–255
- Leider, M. G.: Die Änderung der Zugkraft in den einzelnen Drähten bei der Biegung von mehrlagigen Litzen. DRAHT 25 (1974) 9, S497–501
- Leider, M. G.: Die Bestimmung der Zusatzspannungen bei der Biegung von Drahtseilen und ihr Einfluss auf die Seillebensdauer. Diss. Universität Karlsruhe, 1975
- Leider, M. G.: Krümmung und Biegespannungen von Drähten in gebogenen Drahtseilen. DRAHT 28 (1977) 1, S1–8
- Malinovsky, V.: Bending stiffness of steel wire ropes. OIPEEC Round Table Delft 1993. Part II, pp. 11–35–11–49
- Malinovsky, V. A.: Additional forces arising in steel rope elements at (1999) bending. OIPEEC Bulletin 78, Reading, Dec, 1999, pp. 45–65
- Mancini, G. and Rossetti, U.: Sur l'analyse des contraintes et des deformations des cables flechis. OIPEEC Table Ronde, Milano publisher OIPEEC Torino, 1973
- Matthias, K.: Zur Seilablenkung an Seiltrommeln. Hebezeuge und Fördermittel 6 (1966) 2, S46–49
- Matthias, K.: Geometrische Betrachtungen zur Berührung von Rotationskörpern. Hebezeuge und Fördermittel 10 (1970) 6, S173–179
- Matthias, K.: Eine einfache Flaschenzug-Formel. Hebezeuge und Fördermittel 12 (1972) 10, S305
- Mebold, R.: Drahtseilforschung. Drahtwelt 47 (1961) 3, pp. 202–205
- Meeuse, G. C.: The reference machine REFMA. OIPEEC Bulletin 29 (1976) Torino, June 1976, pp. 12–16 Miner, M. A.: Cumulative damage in fatigue. J. Appl.-Mech. (1945) Trans. ASNE 67 (1945), pp. 159–164
- Molkow, M.: Die Treibfähigkeit von gehärteten Treibscheiben mit Keilrillen. Diss. (1982), Universität Stuttgart 1982. Kurzfassung dhf 29 (1983) 7/8, S209–217
- Müller, H.: Versuche an Drahtseilen und ihre Ergebnisse für den Betrieb. Fördern und Heben 2 (1952) 9, S229–234

- Müller, H.: The properties of wire rope under alternating stresses. *Wire World* 3 (1961) 5, pp. 249–258
- Müller, H.: Beziehungen zwischen Seilbeanspruchung und Seil-konstruktion. Vortrag Drahtseilvereinigung 23, Nov. 1962
- Müller, H.: Fragen der Seilauswahl und der Seilbemessung anTurmdrehkranen. *Technische Überwachung* 4 (1963) 2, S62–66
- Müller, H.: Drahtseile im Kranbau. VDI-Bericht Nr. 98und dhf 12 (1966) 11, S714–716 and 12, S766–773
- Müller, H. K.: Abdichtungen bewegter Maschinenteile. Waiblingen: Medien-verlag u. Müller 1990, ISBN 3-920484-00-2
- Nabijou, S. and Hobbs, R. E.: Fatigue of wire ropes bent over small, sheaves. *Fatigue* 16 (1994), S453–460
- Neumann, P.: Untersuchungen zum Einfluß tribologischer (1987) Beanspruchung auf die Seilschädigung. Diss. TH, Aachen 1987
- Niemann, G.: Umdruck Nr 23, Inst. Maschinenelemente T; H. Braunschweig 1946 and Hütte IIA, 28. Aufl. Berlin: Verlag W. Ernst & Sohn, 1954
- Nussbaum, J.-M.: Zur Erkennbarkeit von Drahtbrüchen in (1999) Drahtseilen durch Analyse des magnetischen Störstellenfeldes, Universität Stuttgart, Diss. 1999
- OIPEEC Recommendations, OIPEEC-Bulletin 56. Torino, Nov. (1988), pp. 25–30
- Oplatka, G.: Die zeitliche Folge von Drahtbrüchen in auf Wechsel-biegung beanspruchten Drahtseilen. 3. Int. Seilbahnkongress. Luzern, 1969
- Oplatka, G. and Roth, M.: Repeated bending tests with Seale ropes. OIPEEC Round Table, Luxemburg 4.–5th October 1977, pp. 1–6
- Oplatka, G.: Seileinlauf in Seilscheibe. Video, Institut für Bau und Transport-maschinen ETH Zürich, 1990
- Oplatka, G. and Vaclavik, P.: Nachschmierung von laufenden Drahtseilen. *DRAHT* 46 (1995) 11, S570–572
- Paetzel, D.: Seilbeanspruchung und Verschleiss bei plastgefütterten Rollen. *Hebezeuge und Fördermittel* 9 (1969) 6, S. 161–169
- Palmgren, A.: Die Lebensdauer von Kugellagern. *Z. VDI* 68 (1924), S339–341
- Pantucek, P.: Pressung von Seildraht unter statischer und dynamischer Beanspruchung. Diss. Universität Karlsruhe, 1977
- Raoof, M.: Wire recovery length in steel cables. Proc. 11th Int. Conf: on Offshore Mecanics and Arctic Engineering. ASM, Calgary, Canada, Vol. III, part B, June 1992, pp. 449–458
- Rebel, G., Chaplin, C.R and Borello, M.: Depth limitation in the use of trian-gular strand ropes for mine hoisting. OIPEEC Bulletin 81, 42–62, University of Reading, June 2001, ISSN 1018–8819
- Recknagel, G.: Untersuchungen an Aufzugtreibscheiben mit Sitzrillen unter Verwendung von Drahtseilen verschiedener Litzenzahl. Diss. Universität Karlsruhe, 1972

- Ren, Guojun: Drahtbruchprozeß bei laufenden Drahtseilen. Diss. Universität Stuttgart, 1996. Kurzfassung: Techn. Überwachung TÜ 39 3, S37–40. Engl.: Lift-Report 24 (1998) 4, pp. 10–14
- Reuleaux, F.: Der Konstrukteur. 1. Aufl., Braunschweig: Vieweg, 1861
- Rieger, W.: Ein Beitrag zur magnetinduktiven Querschnitts-messung von Drahtseilen. Universität Stuttgart, Diss. 1983
- Ridge, I.M.L., Zheng, J. and Chaplin, C.R.: Measurement of cyclic bending strains in steel wire rope. *J. Strain Anal.* 35 (2000) 6, pp. 545–558
- Roos, H. J.: Ein Beitrag zur Formalisierung der inneren dynamischen Vorgänge in Kransystemen während des Hubspiels. Diss. TH, Darmstadt 1975
- Rossetti, U.: Nouvelle methode d'interpretation des resultats des essais de fatigue sur cables. *OIPEEC Bulletin* 26, Torino, June 1975
- Rossetti, U. and Thaon di Revel, M.: On the discarding criterion based on the external wire breakages. *OIPEEC Round Table Luxembourg*, Oct. 1977
- Rossetti, U.: Fatigue tests for the evaluation of the residual resistance of ropes. *OIPEEC Bulletin* 58, Torino, Nov. 1989
- Rubin, A.: Untersuchung über die Steifigkeit von Kran- und Aufzug-seilen. Diss. TH Karlsruhe, 1920
- Schiffner, G.: Spannungen in laufenden Drahtseilen. Diss. Universität, Stuttgart 1986
- Schiffner, G.: EN 81–1 Annex N: Evaluating the safety factor for suspension ropes. *Lift-Report* 26 (2000) 2, pp. 56–58
- Schmidt, K.: Die sekundäre Zugbeanspruchung der Drahtseile aus der Biegung. Diss. TH Karlsruhe, 1965
- Schönherr, S.: Einfluss der seitlichen Seilablenkung auf die Lebensdauer von Drahtseilen beim Lauf über Seilscheiben. Diss. Universität Stuttgart, 2005
- Schraft, O.: Die Biegesteifigkeit und der Wirkungsgrad laufender Rundlitzen-seile. Diss. Universität Stuttgart, 1997. Kurzfassung: *Fördern und Heben* 48 (1998) 7, S527–531
- Scoble, W. A.: First report of wire rope research committee. *Proceedings of the Institution of Mechanical Engineers* 115 (1920), pp. 835–868
- Shitkow, D. G. and Pospechow, I. T.: Drahtseile. VEB Verlag Technik (1957), Berlin 1957 (German translation from Russian)
- Stange, K.: *Angewandte Statistik*. 2. Teil, Mehrdimensionale Probleme. Springer-Verlag Berlin, New York (1971)
- Tonghini, M.: On some formulae of the study of the endurance of wire ropes. *OIPEEC Bulletin* 37 Torino, 1980
- Ulrich, E.: Schädigung durch den Betrieb bei Förderseilen großer Durchmesser in Treibscheibenanlagen. *DRAHT* 31 (1980) 1, S3–6
- Unterberg, H.-W.: Die Lebensdauer und die Auftriegszeit von 8litzigen Drahtseilen. *DRAHT* 34 (1983) 3, S95–97
- Unterberg, H.-W.: Der Einfluß der Rillenform auf die Lebensdauer von laufenden Drahtseilen. *DRAHT* 42 (1991) 4, S233–234

- Verreet, R.: The influence of wire rope fatigue research on crane standards and crane performance. OIPEEC Round Table Reading 1997, pp. 59–65. ISBN 0 7049 1181 7
- Verreet, R.: Steel wire ropes with variable lay length for mining application. OIPEEC Bulletin 81, 53–70. University of Reading, June 2001, ISSN 1018–8819
- Verreet, R.: Wire rope damage due to bending fatigue and drum crushing. OIPEEC Bulletin 85, 27–46. University of Reading, June 2003, ISSN 1018–8819
- Verschoof, J.: Cranes, rope reeving systems: Lifetime, wear and tear. OIPEEC Round Table Delft 1993. Part 1, Paper 7, pp. 1–103–1–120, ISBN 90 370 0091 6
- Virsik, K.: Einfluß des Seildurchmessers auf die Seillebensdauer. Diplomarbeit, Institut für Fördertechnik, Universität Stuttgart, 1995
- Vogel, W.: Zur Dimensionierung von hydraulischen Puffern für (1996) Treibscheibenaufzüge. Diss. Universität Stuttgart, 1996
- Vogel, W.: Ropes for lock gates: Components in steel construction for hydraulic engineering. OIPEEC Bulletin 86, Dec. 2003, pp. 57–66
- Vogel, W. and Nikic, I.: Neue Seildauerbiegemaschine des IFT für kleine Seildurchmesser. Euroseil 123 (2004) 2, SES29–30
- Wang, N.: Spannungen in einem geraden Rundlitzenseil. Studienarbeit. Inst. Fördertechnik, Universität Stuttgart, 1989
- Waters, D. and Ulrich E.: Three machines for the bending-tension fatigue of ropes. OIPEEC Bulletin 59, Torino, April 1990, pp. 30–42
- Weiskopf, U., Wehling, K.-H. and Vogel, W.: Kranseile in der Mehrlagenwicklung. Fördern und Heben (2005) 4, S184–187
- Wehking, K. H.: Endurance of high-strength fibre ropes running over pulleys. OIPEEC Round Table Reading 1997, pp. 207–213, ISBN 0 7049 1181 7
- Wehking, K.-H., Vogel, W. and Schulz, R.: Dämpfungsverhalten von Drahtseilen. Fördern und Heben 49 (1999) 1/2, S60–61
- Wehking, K.-H.: Zukunftsausrichtung des IFT im Bereich der Seiltechnik. 1. Internationaler Stuttgarter Seiltag, 2002, pp. 1–14
- Wehking, K.-H.: Life time and discard for multilayer spooling in cranes. OIPEEC Technical Meeting, Lenzburg, Switzerland, Sept. 2003
- Wiek, L.: Facts and figures of stresses in ropes. OIPEEC Round Table Milano Sept. 1973, S94–111 and Wiek, L. DRAHT 26 (1975) 6, S283; 8, S387 and 10, S484
- Wiek, L.: The reference machine, the characteristics. OIPEEC Bulletin 29, Torino, June 1976
- Wiek, L.: Contact pressure and steel wire endurance. OIPEEC Bulletin 43, Torino July 1982a, pp. 18–24
- Wiek, L.: The distribution of the contact fores on steel wire ropes. OIPEEC 44, Torino Nov. 1982b, pp. 10–25
- Wiek, L.: Nylon sheaves and rope discarding. OIPEEC Round Table Conference, Zürich, 1989

- Wiek, L.: Computerised maintenance management and endurance prediction of steel wire ropes on cranes. OIPEEC Round Table Reading 1997, pp. 229–236. ISBN 0 7049 1181 7
- Woernle, R.: Zur Beurteilung der Seilschwebbahnen zur Personenbeförderung. Habilitationsschrift TH Karlsruhe, 1913
- Woernle, R.: Ein Beitrag zur Klärung der Drahtseilfrage. Z. VDI 72 (1929) 13, pp. 417–426
- Woernle, R.: Drahtseilforschung. Z. VDI 75 (1931) 49, S1485–1489
- Woernle, R.: Drahtseilforschung. Z. VDI 78 (1934) 52, pp. 1492–1498
- Wohlrab M. and Jehmlich, G.: Ermüdungsuntersuchungen im Mehrstufenversuch zur Prüfung der Anwendbarkeit von Schädigungstheorien. Hebezeuge und Fördermittel 20 (1980) 11, S326–330
- Wolf, E.: Seilbedingte Einflüsse auf die Lebensdauer laufender (1987) Drahtseile. Diss. Universität Stuttgart 1987. Shortened: DRAHT 39 (1988) 11, S1088–1093
- Wünsch, D., Liesenfeld, G. and Schlecht, B.: Seiltriebe praxismäßig überwachen. Fördern und Heben 41 (1991) 4, S301–398
- Wyss, Th.: Stahldrahtseile der Transport- und Förderanlagen. Zürich; Schweizer Druck- und Verlagshaus AG, 1956
- Zemmrigh, G.: Dynamische Beanspruchung von Brückenkränen beim Heben von Lasten. Diss. TU Braunschweig, 1968
- Zimmerman, Z. and Reemsnyder, H. S.: Bend-over-sheaves fatigue testing of 2 inch diameter marine ropes. OIPEEC Bulletin 45, Torino, July 1983, pp. 121–151
- Zweifel, O.: Biegebeanspruchung und Pressung von Drahtseilen bei gefütterten Lauf- und Tradrollen. Schweizer Bauzeitung 87 (1969) 36, S665–671

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